



NATIONAL GUIDELINES FOR COMPREHENSIVE HIV PREVENTION, CARE AND TREATMENT

February 2022

Table of Contents

Forward	i
Acknowledgement	ii
Acronyms and Abbreviations	iii
Summary of the changes made in these guidelines	v
CHAPTER ONE - INTRODUCTION	1
1.1. Background and Context	1
1.2. Rationale	2
1.3. Objectives of the Guidelines	2
1.4. Target Audience	3
1.5. Guiding Principles	3
CHAPTER TWO: HIV PREVENTION	4
2.1 HIV Combination Prevention	4
2.2. Condom programing for KPP and general populations	8
2.3. Pre Exposure Prophylaxis	8
2.4. Prevention and management of Gender Based Violence (GBV)	11
2.5. Post Exposure Management including Prophylaxis	12
2.5.1. Management of Occupational Exposure to HIV	12
2.5.2. Prevention of HIV Transmission after Sexual Assault	16
2.6. Undetectable=Un-transmittable (U=U) (P=P)	17
2.7. Screening and treatment of Sexually Transmitted Infections	18
2.8. Voluntary Medical Male Circumcision (VMMC)	18
2.9. Care of HIV Exposed Infants (HEI)	18
CHAPTER THREE: HIV CASE FINDING	23
3.1. HIV Testing Services	23
3.1.1. Core Principles	23
3.1.2. Guiding Principles for HIV counseling and testing	24
3.2. Demand creation for HTS	25
3.3. HIV Testing Service Provision Settings	26
3.3.1. Facility Setting	26
3.3.2. Community Setting	26

3.4. Approaches of HTS	27
3.4.1. Voluntary Counseling and Testing (VCT)	27
3.4.2. Provider Initiated Testing and Counseling (PITC)	27
3.4.2.1. Use of HIV Risk Screening Tool (HRST) to Enhance PITC	29
3.4.2.2. Eligible clients for HIV Testing and counseling while providing PITC includes:	29
3.4.3. Index Case Testing	29
3.4.4. Social Network Strategy (SNS)	33
3.4.5. HIV Self Testing	34
3.4.6. Recency Testing	36
3.5. Procedures of HTS delivery	37
3.6. HIV Testing in infants and children	43
3.7. Retesting	44
3.8. Quality assurance and Quality Improvement for HTS	46
3.9. Linking People Diagnosed with HIV Infection	48
3.10. Ethical and Legal Recommendations for HIV Testing Service	50
CHAPTER FOUR: CARE AND TREATMENT OF PEOPLE LIVING WITH HIV INFECTION	52
4.1. General Care Packages for PLHIV	52
4.2. Advanced HIV Disease	53
4.3. Preparing People Living with HIV for ART	55
4.4. When to Start ART	56
4.4.1. When to Start ART in Adults and Adolescents	56
4.4.2. When to Start ART in Pregnant and Breast-feeding Women	56
4.4.3. When to Start ART in Children	57
4.4.4. When to start ART in adults, adolescents, and children with TB	57
4.4.5. When to start ART in adults, adolescents, and children with drug resistant TB	57
4.4.6. When to start ART in HIV/HBV co-infected patients	57
4.4.7. When to start ART in HIV/Cryptococcal meningitis co-infected patients	57

4.5. What ART regimen to start with (first-line ART)	58
4.5.1. First-line regimen for adults and adolescents	59
4.5.2. First line ART for children	59
4.5.3. Consideration for first line ART regimen for PLHIV on TB treatment	59
4.5.4. Consideration for alternative and special circumstance first line ART regimens	59
4.6. Monitoring response to ART	59
4.6.1. What to expect in the first months of ART and how to manage them	59
4.6.2. Clinical and laboratory monitoring	61
4.6.3. Monitoring drug toxicities and substitution of ARV	67
4.6.4. Drug interactions	71
4.7. Re-engaging with care after ART interruption	75
4.8. Diagnosis and management of antiretroviral treatment failure	75
4.9. Management of treatment failure	80
4.9.1. Management of first-line treatment failure (switching to second-line ART)	80
4.9.2. Management of second-line treatment failure (switching to third line ART)	81
4.9.3. Second line ART service in health centers	84
CHAPTER FIVE: PREVENTION, SCREENING, AND MANAGEMENT OF COMMON CO-INFECTIONS AND COMORBIDITIES	85
5.1. Co-trimoxazole preventive therapy (CPT)	86
5.2. Tuberculosis	88
5.3. Pneumonia	108
5.3.1. Bacterial Pneumonia	108
5.3.2. Pneumocystis pneumonia	110
5.3.3. Lymphoid Interstitial Pneumonitis	111
5.4. Hepatitis B and C	111
5.4.1. HBV/HIV co-infection management	112
5.4.2. HCV HIV co-infection management	113
5.5. Sexual Transmitted Infection and Cervical Cancer	114
5.6. Non-Communicable Disease (NCD)	117
5.7. HIV and Mental Health Disorders	117

5.8.HIV related skin and oral conditions	121
5.8.1. Etiological classification of skin disorders in HIV disease	121
5.8.2. Pruritic Papular Eruption (PPE)	124
5.8.3. Dysphagia and odynophagia	124
5.9.HIV related Neurological conditions	125
5.9.1.Toxoplasma gondii encephalitis	126
5.9.2.Cryptococcal infection	127
5.9.3. Peripheral neuropathies	131
5.10. Visceral leishmaniasis	131
CHAPTER SIX: SERVICE DELIVERY	133
6.1. Differentiated HIV service delivery	134
6.1.1 Less intensive DSD models	137
6.1.1.1 Appointment Spacing Model (ASM/6MMD)	137
6.1.1.2 Three Months ARV Dispensing (3MMD)	137
6.1.1.3. Fast Track ARV Drugs Refill Model	137
6.1.1.4. Health Extension Professional Managed Community ART refill group (HEP_CAG)	138
6.1.1.5. Peer lead community based ART distribution/Group (PCAD/G)	138
6.1.2 More intensive DSD models	138
6.1.2.1. Health care worker managed DSD Model for adolescent living with HIV (DSD for ALHIV)	138
6.1.2.2 DSD for key population (for FSWs)	139
6.1.2.3 DSD for Advanced HIV Disease and PLHIV at high risk Disease Progression	139
6.1.2.4 MCH _DSD	140
6.2. Recommendations to strengthen the continuum of treatment and care	140
6.2.1 Disclosure	140
6.2.1.1 Disclosure in Children	141
6.2.1.1 Planning for disclosure	143
6.2.1.2 Stages of disclosure	143
6.2.1.3 Post –disclosure assessments and follow-up	144
6.2.1.3 Disclosure in adults	144

6.2.2. Retention	146
6.2.2.1 Good practices to optimize retention	146
6.2.2.2 Specific population considerations	147
6.2.3. Adherence to ART	150
6.2.3.1. Barriers to adherence	150
6.2.3.2. Supportive interventions	151
6.2.3.3 Monitoring adherence to ART in routine program and care settings	152
6.2.4. Task shifting for HIV treatment and care	153
6.2.5 Decentralizing HIV treatment and care	154
6.2.6 Integrating and linking services	154
6.2.7. Adolescent-friendly health services	155
6.2.8. Improving the quality of HIV care services	156
6.3. Pharmaceuticals Supply Management System	157
6.3.1. Supply Chain Management	157
6.3.2. ART Pharmacy Services	159
6.4. Laboratory and diagnostic services	162
CHAPTER SEVEN - GUIDANCE FOR PROGRAM MANAGERS AND LEADERS	167
7.1 Guiding principles	167
7.2. Overview of the HIV/AIDS National Strategic Plan 2021-2025	168
7.4. HIV response in the context of COVID 19	170
7.5. Key parameters for decision-making	171
7.6. Roles and responsibility	172
7.7. Coordination mechanisms	174
CHAPTER EIGHT - MONITORING AND EVALUATION	175
8.1. Introduction to HIV/AIDS Program Monitoring & Evaluation (M&E)	175
8.2. Key Indicators	176
8.3. Data reporting, data flow and quality assurance	197
8.3.1. Data quality assurance	197
8.4. HIV/AIDS Program Monitoring	198
8.4.1. Supportive supervision	
8.4.2. program Performance Review	198

8.5. Other Monitoring Considerations	198
8.6. Recording and reporting Tools in HIV/ AIDS Program	199
CHAPTER NINE - ANNEXES	200
Annex 1: KP Minimum Service Package	200
Annex 2: Immunization schedule for HIV exposed infants	204
Annex 3: Growth curves	205
Annex 4. Instruction for MUAC measurement	211
Annex 5a. HIV Risk assessment tool for children <15 years of age	212
Annex 5b. HIV Risk assessment tool for adults and adolescents > 15 years of age	214
Annex 6a. Clinical Conditions that make eligible for HIV Testing Service in adults and adolescents	215
Annex 6b. Clinical Conditions that make eligible for HIV Testing Service in children	216
Annex 6b. Clinical Conditions that make eligible for HIV Testing Service in children	217
Annex 7: Dosage of antiretroviral drugs for adults and adolescents	218
Annex 8: Dosage of antiretroviral drugs in children	219
Annex 9: Pediatric ARV drug formulations, side effects and special considerations in children	221
Annex 10: Grading of toxicity in adults and adolescents	226
Annex 11: Grading of adverse events in children	227
Annex 12. Laboratory grading of adverse events in adults and adolescents (ACTG)	229
Annex 13: Grading toxicities in children by selected laboratory findings	230
Annex 14: Pediatric TB screening tool	231
Annex 15: Adult TB screening tool	233
Annex 16: Follow-up form for Clients Whose Viral Load Result >50 copies/ml	235
Annex 17: Cohort register for clients with unsuppressed viral load(>50copies/ml)	238
Annex 18: Screening with VIA testing using see and treat approach	239
Annex 19: Screening with HPV testing followed by VIA triage algorithm	240
Annex 20: Brief mental health disorders symptom screening tool and referral tool for PLHIV	241
Annex 21: DSD model of HIV care client classification tool/Assessment form	243

Forward

Antiretroviral treatment (ART) began in 2003 and free ART was launched in Ethiopia in 2005. According to the 2020 spectrum estimate, 622,326 Ethiopians are living with human immunodeficiency virus (HIV) and all of them require ART. However, only 80.5 % of adults and 40% of children and adolescents are currently taking antiretroviral (ARV) drugs.

Recognizing the need for antiretroviral treatment, the Government of Ethiopia (GOE) issued the first ARV guidelines in 2003, which were revised in 2005, 2008, 2014 and 2018 for provision of quality prevention, care and treatment services at all levels. With continued evidence based updates, the Ministry of Health has revised the national guideline to scale up and improve the quality of service at all levels.

Expansion and strengthening HIV prevention, care and treatment activities at regional, zonal, woreda and community levels through targeted social mobilization emphasizing Key and Priority Populations and active community participation are expected to enhance the HIV epidemic control.

These consolidated guidelines on preventing and treating HIV infection bring together a series of recommendations to promote the highest quality, person-centered delivery of care for people living with and affected by HIV. Amid COVID 19 pandemic, program adaptations are required to ensure continuity of the services including service delivery approaches put forward with the aim to promote self-management.

The Ministry of Health believes that these guidelines, along with other national guidelines and training manuals, provide the much needed framework and impetus to move towards universal access for HIV services and the agenda for ending AIDS by 2030 as a key national health strategic objective.

Dereje Duguma, MD, MIH
State Minister of Health,
Federal Democratic Republic of Ethiopia

Acknowledgement

The Ministry of Health expresses its appreciation for the institutions participated in the revision of these consolidated HIV prevention care and treatment guidelines and partners, which supported the revision workshops expense. Special thanks goes to UNICEF for supporting the design and printing of the document. The ministry also recognizes the following experts for their contribution in the revision of the guidelines.

Name	Organization
Mrs. Mirtie Getachew	MOH
Sr Seble Mamo	MOH
Mrs. Alemtsehay Abebe	MOH
Mr.Dejene Mulatu	MOH
Dr. Zerihun Hika	MOH/ICAP
Mrs.Genet Getachew	MOH
Dr. Getnet Bantayehu	MOH/ICAP
Mr. Mulu Legesse	MOH/GHSC-PSM
Mr.Teklu Lemessa	MOH
Mr. Mesfin Bekele	MOH
Mrs. Aregash Mola	MOH
Dr. Zebyideru Zewdie	MOH/ICAP
Mr. Kifle Mitiku	FHAPCO
Mr. Netsanet Haniko	FHAPCO
Mr Getnet Hailu	EPHI
Mrs. Kidist Zealias	EPHI
Mrs. Tigist Kassahun	EPSA
Dr. Mesfin Shimelis	CDC
Dr. Ghion Tirsite	WHO

Name	Organization
Dr. Ambachew Tefera	Project Hope
Dr. Miftah Kemal	ICAP
Dr. Mesfin Girma	ICAP
Dr. Seblewongel Abate	WHO
Dr. Yayeh Negash	UNICEF
Dr. Fana Tefera	CDC
Dr. Solomon Ahmed	CDC
Mr Asmamaw Silesh	CHAI
Dr. Minesh Shah	CDC
Mr Yalemsew Derib	PSM
Dr. Fahmi Ahmed	WHO
Dr. Chanie Temesgen	CDC
Mr. Tamene Tadesse	Project Hope

Acronyms and Abbreviations

3TC	Lamivudine
ABC	Abacavir
AFB	Acid fast bacilli
AIDS	Acquired Immune Deficiency Syndrome
ANC	Antenatal Care
ARV	Antiretroviral
ART	Antiretroviral Therapy
AZT/ZDV	Zidovudine
CBC	Complete Blood Count
CD4 cells	Type of T-lymphocyte, white blood cells
CMV	Cytomegalovirus
CPT	Cotrimoxazole Preventive Therapy
CrAg	Cryptococcal antigen
DBS	Dried Blood Spot
DTG	Dolutegravir
DRV	Darunavir
DHS	Demographic and Health Survey
DNA	Deoxyribonucleic acid
DPV-VR	Dapivirine vaginal ring
DOTS	Directly Observed Therapy Short Course
EFV	Efavirenz, also abbreviated as EFZ
FBO	Faith-based organization
FDC	Fixed dose combination
FHAPCO	Federal HIV/AIDS Prevention and Control Office
MOH	Ministry of Health
HAART	Highly active antiretroviral therapy
HBV	Hepatitis B Virus
HCV	Hepatitis C Virus
HIV	Human Immunodeficiency Virus
HTS	HIV Testing Services
IP	Infection Prevention
IPT	INH Preventive Therapy
IPLS	Integrated Pharmaceuticals Logistics System
IRIS or IRS	Immune Reconstitution Inflammatory Syndrome also called Immune Reconstitution Syndrome (IRS)

INSTI	Integrase strand transfer inhibitor (also known as integrase inhibitor)
LFT	Liver Function Test
LPV	Lopinavir
MTCT	Mother-To-Child Transmission (of HIV)
MD	Medical Doctor
NFV	Nelfinavir
NGO	Non-governmental Organization
NNRTI	Non-nucleoside reverse transcriptase inhibitor
NRTI	Nucleoside Analogue Reverse Transcriptase Inhibitor
NVP	Nevirapine
OIs	Opportunistic Infections
PCR	Polymerase chain reaction
PEP	Post-exposure prophylaxis
PI	Protease Inhibitor
PITC	Provider Initiative Testing and Counselling
PLHIV	People living with HIV
PMTCT	Prevention of mother-to-child transmission (of HIV)
PrEP	Pre-exposure prophylaxis using ARVs before HIV exposure
RMU	Rational medicine use
RNA	Ribonucleic acid
RTV, r	Ritonavir
PI/r	Ritonavir boosted Protease Inhibitor
RAL	Raltegravir
RFT	Renal function test
RT	Reverse transcriptase
STI	Sexually Transmitted Illnesses
TB	Tuberculosis
TDF	Tenofovir
U/A	Urine analysis
UNAIDS	The Joint United Nations Program on HIV/AIDS
UP	Universal Precautions
WHO	World Health Organization
ZDV	Zidovudine (also abbreviated as AZT)

Summary of the changes made in these guidelines

The following list of recommendations consists of either introducing new initiatives or/and modifications of previous recommendations, as adopted from the updated WHO 2021 guidelines, and other relevant working documents.

HIV Prevention

- For adults and adolescents TDF + 3TC are recommended as the preferred backbone regimen for HIV PEP and Dolutegravir (DTG) is recommended as the preferred third drug for HIV PEP and when available, ATV/r, DRV/r and LPV/r may be considered as an alternative third drug options for PEP.
- In ten years and younger, AZT + 3TC is recommended as the preferred backbone regimen for HIV PEP and ABC + 3TC or TDF + 3TC can be considered as alternative regimens and DTG is recommended as the preferred third drug for HIV PEP with approved DTG dosing and when available.
- Undetectable=Untransmittable (U=U) is contextualized in Ethiopia setting by naming “የማይታይ መጠን = የተገታ መተላለፍ (የ=የ)” (“Yemaytay Meten = Yetegeta Metelalef”) focusing on demand creation for viral load testing, improving adherence to ART that result in durable undetectable viral load resulted preventing sexual transmission of HIV.

- All HIV Exposed Infants will undergo through DNA-PCR antigen test at 4-6 weeks and repeat DNA-PCR test at 9 months for those who tested negative.

HIV Case Finding

- HIV Risk Screening Tool (HRST) should be implemented for all individuals (children, adolescents and adults) at service delivery units to improve HIV case detection and yield of provider initiated testing.
- Universal HIV testing will be implemented for all pregnant women at the first ANC visit, however retesting will be done based on their risk.
- Retesting for unknown or HIV-negative post-partum women, can be considered for key populations or for those who have virally unsuppressed HIV positive partners.
- The age limit of biological children of PLHIV for ICT service increased from 15 to 19.
- The window period has been changed from 12 weeks to 6 weeks since third generation kit can detect infection of 4 to 6 weeks.
- Recency testing is implemented to detect recent HIV infection less than one year.

- Provider Initiated Testing and Counseling workflow has been modified to follow those who decline testing
- The HIV testing algorithm is revised considering the current implementation challenges and the available recommendations.

HIV care and treatment

- ART should be initiated for all individuals (children, adolescents, and adults) living with HIV rapidly, preferably same day, (within an hour for laboring mother) after confirming HIV diagnosis, regardless of WHO clinical stage and CD4 cell count except for TB and cryptococcal meningitis.
- Start ART in all TB patients living with HIV as soon as possible within 2 weeks following initiation of anti-TB treatment regardless of their CD4 count except when there is TB meningitis. If a patient has TB meningitis, delay ART for at least 4 weeks and initiate within 8 weeks after treatment of TB meningitis is initiated.
- ART should be delayed by 4-6 weeks of ART following initiation of treatment for cryptococcal meningitis. Earlier ART is associated with more severe adverse event and increased mortality with cryptococcal meningitis

- DTG containing regimen is the preferred first line regimens for all children age >4 weeks and/or weight ≥ 3 kg and the preferred first-line regimen for all adults and adolescents and children >30 kg is TDF+3TC+DTG as a once-daily dose. Accordingly the optimized treatment regimen sequencing is modified.
- Diagnosis of advanced HIV disease is done through CD4 testing of clients at baseline (re-engaging with care after a period of interruption for >28 days) and targeting those who have interrupted ART treatment and with persistently high Viral load (>1000 copies per ml).

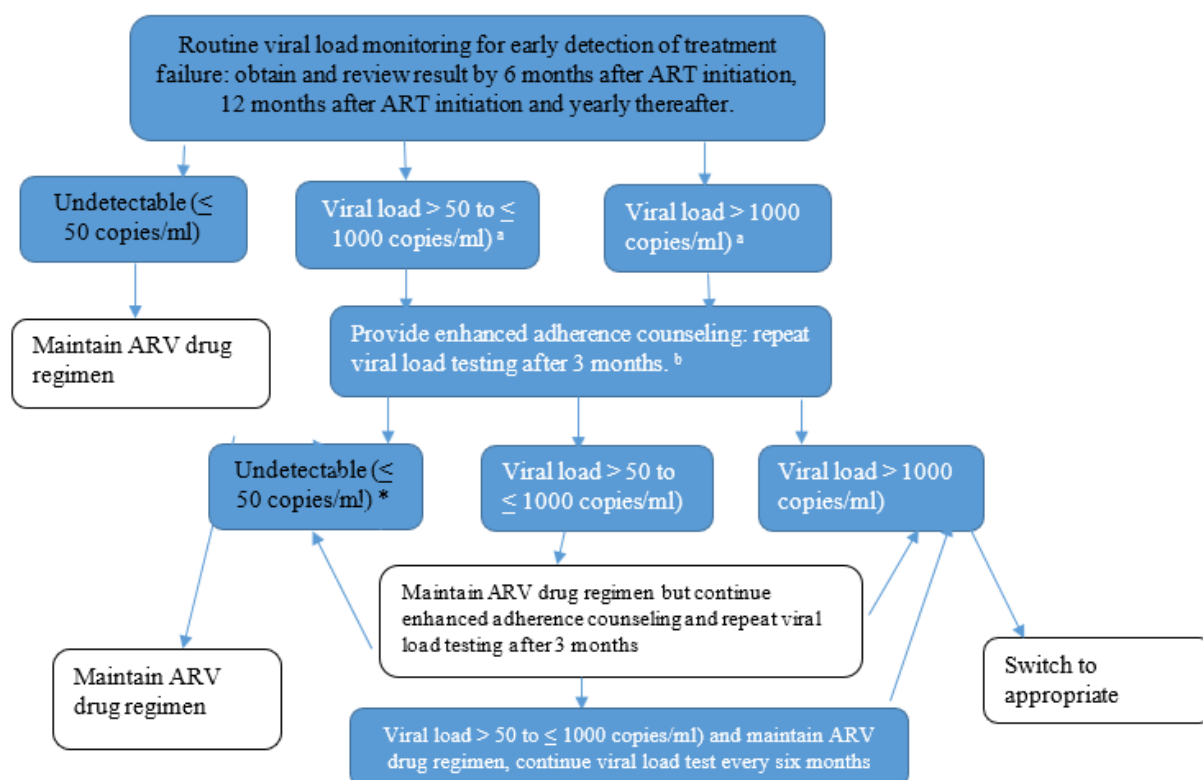
Revised treatment monitoring with viral load testing

- 1. Viral suppression:** is a viral load that is undetectable, equal to or less than 50 copies/ml.
- 2. Low-level viraemia:** is one or more viral load results that are detectable (more than 50 copies/ml) but equal to or less than 1000 copies/ml.
- 3. Virological failure** Viral load above 1000 copies/ml based on two consecutive viral load measurements in 3 months, apart with enhanced adherence support following the first viral load test.

- Treatment failure threshold should remain at 1000 copies/ml. The treatment monitoring algorithm is revised based on these new recommendations.
- Whenever possible, use same-day point-of-care testing (POC) for viral load testing of pregnant and breastfeeding women to expedite the return of results and clinical decision-making. If this is not available, viral load specimens and results for pregnant and breastfeeding women should be given priority across the laboratory referral process.
- The viral load testing schedule is revised for pregnant and breast feeding women.

a **Unsuppressed viral load results should be immediately communicated**

b Conduct same-day testing using point-of-care viral load testing for a repeat viral load test, where available, to expedite the return of results. If not available, viral load specimens and results for a repeat viral load should be given priority across the laboratory referral process (including specimen collection, testing and return of results).



TB/HIV

- LF-LAM is included for active TB diagnosis in PLHIV with TB symptoms, serious illness and/or Advanced HIV Disease
- TB Preventive Therapy (TPT) is the use of Isoniazid, rifapentine or other medications to sterilize latent TB infection. The preferred regimen for adult PLHIV is a combination of INH and rifapentine to be taken once per week for three months (3HP).

Cervical cancer Screening

- Cervical cancer screening is an important test in women living with HIV to prevent significant morbidity and mortality associated with HPV. Cervical cancer is a preventable disease and is curable if diagnosed and treated early. Women living with HIV have a higher risk of pre-cancer and invasive cervical cancer.

Service Delivery

- Differentiated service delivery for HIV treatment is based on four building blocks. In any given differentiated service delivery model for HIV treatment, the building blocks need to be defined separately for clinical consultations, ART refills and psychosocial support.
- The diversity of models and eligibility criteria are modified based on the new recommendations. Eligibility criteria to be considered as established on ART (Stable) are those clients which are above 5 years old of age, on ART for >6 month, evidence of treatment of success, adherence of the client should be >95% and no current illness.

CHAPTER ONE: INTRODUCTION

1.1. Background and Context

HIV infection is one of the global public health issues. In 2020, more than 37.7 million [30.2 million–45.1 million] people were living with HIV, and 1.5 million [1.0 million–2.0 million] people acquired HIV. Nearly 45% of the people newly infected with HIV live in sub-Saharan Africa (UNAIDS; 2021).

The first evidence of HIV epidemic in Ethiopia was detected in 1984. Since then, HIV/AIDS has claimed the lives of millions and has left behind hundreds of thousands of orphans. The government of Ethiopia took several steps in preventing further disease spread, and in increasing accessibility to HIV care, treatment and support for persons living with HIV.

The HIV epidemic in Ethiopia is characterized as mixed, with wide regional variations and concentrations in urban areas, including some distinct hotspot areas driven by key and priority populations. According to the EDHS done in 2016, the national adult (15-49) HIV prevalence is 0.96 %; the urban prevalence was 2.9%, which is seven times higher than that of the rural (0.4%). National HIV Related Estimates and Projections (2020), also shows that the HIV prevalence varies from region to region ranging from less than 0.15% in Ethiopia Somali to 4.13% in Gambella.

The progress towards achieving the first 95 target has been far behind the track; only 79% of PLHIV know their HIV status. To accelerate

the performance of the HIV case identification in Ethiopia for closing the gaps to treatment and achieve epidemic control, the remaining 21 % of PLHIV need to be reached. (EDHS 2016)

Free ART service was launched in January 2005 and public hospitals start providing free ART in March 2005. Currently ART service is being available in more than 1,500 health facilities. According to the program data in 2020, around 465,457 adults and 17,670 children under the age of 15 are taking ARV, with ART need of 578,188 for adults and 42,971 for children under 15 years of age. Based on the new spectrum estimate for 2020, ART coverage for adults (age >15) has reached 80.5 % but the coverage remains low (40.03%) for children (age <15) living with HIV. According to EPHI 2020/21 report, the national viral load testing coverage was 74.5%, with a suppression rate of 95%.

The National Guidelines for Comprehensive HIV Prevention, Care and Treatment was last revised in 2018. Since then, new information as well as evidence-based best practices has become available to make HIV prevention, testing and treatment more effective and accessible, ensuring the continuity of HIV prevention, treatment and care specially focusing on KPPs amid COVID-19 pandemic which created the need to revise the existing guidelines. Hence, this guideline is revised taking into consideration of the current recommendations released by WHO in 2021.

1.2. Rationale

- Guidance on the implementation of national epidemic control roadmap through geographic and population prioritization, targeted interventions focusing on cost effective and high impact prevention, care and treatment services.
- Guidance on using combination HIV prevention services like targeted SBCC and demand creation focusing on KPPs, condom promotion and distribution, PEP, PrEP, STI, VMMC and prevention and management of GBV.
- Guidance on using innovative HIV testing approaches like targeted HIV testing by HIV risk screening tools (HRST) for all age groups, Index Case Testing (ICT), social network strategy (SNS) among FSWs and their sexual network members, HIV self-testing and recency testing.
- Guidance on using optimized ARV drugs presented within the context of the continuum of HIV-related prevention, treatment, and care. These guidelines address updates on the newly recommended ARV regimen in all population groups since the introduction of DTG.
- Guidance on the consistency of approaches and linkage across the various settings in which HIV Prevention, Case identification, Care and Treatment related services provided.

- These guidelines enable key clinical, operational, and programmatic implications of new science and emerging practice in HIV programs to be comprehensively reviewed periodically across populations, age groups and settings.

1.3. Objectives of the Guidelines

- To provide evidence based update on combination prevention interventions targeting KPP and general populations.
- To provide update on new initiatives of HIV case identification and Care and treatment implementation strategies and standardize service delivery models in the context of the continuum of HIV care in the comprehensive HIV/AIDS service delivery setting.
- To provide guidance on key operational and service delivery issues that need to be addressed to increase access to HIV services, to improve quality, strengthen the continuum of HIV care and further integrate the provision of HIV prevention, care and treatment into health systems; and
- To serve as a reference material for health service providers and program managers.

1.4.Target Audience

The guideline is intended to be used by:

- Health care providers at health facilities and communities such as medical and public health specialists, physicians, health officers, nurses, pharmacy personnel, laboratory technologist and case managers providing care to people infected and affected with HIV and others.
- HIV/AIDS program managers, health experts, health care program planners and researchers
- Organizations involved in HIV related commodities procurement, supply management, and service delivery.
- Public and private health sector, Civic Society Organizations including PLHIV associations, community-based and faith-based organizations, universities, health science colleges and other key stakeholders working on HIV/AIDS programs.
- **Public Health Approach:** the guidelines are based on a public health approach to scale up the use of HIV Testing, and ARV drugs along the continuum of HIV prevention, treatment, and care.
- **Participatory:** Adapting and implementing the guidelines should realize the rights and responsibilities, Greater and meaningful involvement of people living with HIV principles.
- **Accessibility:** The recommendations in the guideline should be implemented at all levels to strengthening broader health systems for ensuring the provision of universal health care.
- **Human Right:** Implementation of the guidelines needs to be accompanied by efforts to promote and protect the human rights of people who need HIV services, including ensuring informed consent, preventing stigma and discrimination in the provision of services, and promoting gender equity.

1.5.Guiding Principles

- **Effectiveness:** the guidelines should facilitate effectiveness of the national program by contributing and expediting the achievement of key global targets and the goals for the National HIV/AIDS Strategic Plan for 2021-2025 and Health Sector Transformation Plan 2020/21- 2024/25(HSTP II) to realize the Sustainable Development Goals.
- **Evidence-Based:** Implementation of the recommendations in these guidelines should be informed by local context, including HIV epidemiology, availability of resources and comorbidities, the organization and capacity of the health system and anticipated cost-effectiveness.

CHAPTER TWO:

HIV PREVENTION

2.1 HIV Combination Prevention

HIV prevention approach based solely on one element does not work and can hinder the HIV response. There is no single magic bullet for HIV prevention. Hence, use of a growing number of combination interventions have shown promise in effectively protecting against HIV transmission and acquisition. These include behavioral risk reduction, knowledge of Sero-status, Pre-Exposure Prophylaxis (PrEP), condoms, medical male circumcision, screening, and management of sexually transmitted infections, and use of antiretroviral medications. Therefore, it is recommended to use a mix of behavioral, biomedical, and structural HIV prevention actions and tactics which suit with the country's epidemic.

HIV Combination prevention is likely to be most effective when implemented at different points in the "transmission cycle," when the transmission rate is impeded, and structural barriers are addressed.

Core programmatic components

HIV Combination prevention approach includes three types of mutually reinforcing interventions:

1. Behavioral interventions include a range of social behavior change communication and demand creation programs that use various communication channels and platforms (e.g.

mass media, community level, small group level and inter personal) to disseminate behavioral messages designed to encourage people to reduce behaviors that increase risk of transmission. Condom promotion is an essential component for the behavioral intervention.

2. Biomedical interventions are those that directly influence the biological system through which the virus infects a new host, such as condom use, PrEP, PEP, voluntary medical male circumcision, Screening and management of STIs, HTS and ART. Condoms effectively reduce sexual transmission of HIV, if used consistently and correctly. Voluntary medical male circumcision reduces acquisition of infection for men by up to 60% as compared to uncircumcised men.

3. Structural interventions address the critical social, legal, political, and environmental enablers that contribute to the national HIV response and measures to reduce stigma and discrimination, the promotion of gender equality and prevention of gender-based violence, economic empowerment, and access to education.

A package of combination HIV prevention intervention is recommended for KP to standardize the services (Refer Annex 1)

Recommended HIV Combination Prevention Interventions/Activities

Behavioral interventions/Activities

- SBCC intervention: Peer education sessions (small group and one to one), risk reduction counseling and preparation and Distribution of SBCC materials
- Scale-up comprehensive prevention interventions addressing key and priority populations.
 - Targeted outreach activities (like targeted mobilization and demand creation)
- Condom promotion: demand creation and skill building among KPPs and general populations.
- Harm reduction for people who inject drugs (PWID): needle and syringe programs through social marketing and CSO channels and opioid substitution therapy.
- Promotion of health care seeking behaviors through existing services like health education.
- Strengthen community-based HIV prevention interventions to address the general population through: -
 - Enhancement of adaptive community conversation (CC) and dialogue and integrate with existing community structures.
 - Develop and disseminate HIV prevention messages using print, social and electronic media.
- Strengthen workplace HIV prevention interventions through mainstreaming.
- Strengthen intra- and extracurricular school HIV education programs.
 - Conduct peer education programs in schools, higher education institutes and Technical, Vocational Education and Training (TVET).
 - Conduct life-skill education in schools, higher education institutes and TVET.
 - Integrate HIV/AIDS into school curriculum.
 - Train teachers on management of school HIV/AIDS programs.
 - Develop and disseminate targeted SBCC message in schools, higher education institutes and TVET.
 - Strengthen youth leadership development programs.
 - Develop an HIV intervention strategy for school and higher education.
 - Strengthen anti-AIDS clubs in schools, higher education institutes and TVETs.
- Strengthen out-of-school youth HIV prevention programs.
 - Life skill education, Peer Education, and youth dialogue to address children, adolescent, and youth
 - Strengthen Youth Center to provide comprehensive HIV SRH services

- Integration of HIV Prevention with Voluntarism
- Targeted outreach mobilization for prevention, testing, and linkage services
- Intensify HIV prevention in development schemes including new development corridors.
 - Map development corridors, mega project areas like industrial park, construction sites, military, refugees campus and private development schemes.
 - Integrate HIV prevention in the project proposals of development schemes.
 - Develop and disseminate targeted HIV/AIDS messages.
 - Conduct peer education, life skill education and canteen-based education targeting workers (permanent and seasonal).
 - Provide VCT, STI and ART/PMTCT services at worksite clinics or through outreach or referral and linkages with nearby health facilities.
 - Ensure the provision of HIV prevention services to the surrounding communities of development corridors.
- Scale-up HIV prevention among population groups with special needs.
 - Integrate SBCC interventions for people with disability in their service provision centers.

- Develop and disseminate SBCC materials for people with disability.

Biomedical interventions/Activities

- Ensure availability, acceptability and affordability of quality condoms using the total marketing approach (free condoms, social marketing, and private sector)
 - Identify universal need of condom and forecasting condom quantity.
 - Ensure quality condom procurement, storage, and distribution.
 - Monitor appropriate storage, availability and accessibility of condoms.
 - Promote use of condom and build skill on correct and consistent use of condom.
 - Conduct targeted condom distribution, particularly for key and priority populations.
- Ensure access and enhance uptake of targeted HIV testing and counseling services to eligible clients and address the special need of people with disability.
- Strengthen Pre Exposure prophylaxis (PrEP) service for eligible population groups
- Ensure the provision of PEP service for the eligible population groups.
- Increase availability and utilization of STI prevention, screening and treatment services

- Create strong leadership for STI programs.
- Intensify health education to improve treatment seeking behavior and utilization of STI services.
- Promote and implement active STI partner notification system to identify and link those in the sexual network to STI management and HTS.
- Ensure availability of STIs kits/drugs in all health facilities.
- Train health care workers on syndromic approaches STI case diagnosis and management with consideration of people with disability.
- Accelerate voluntary medical male circumcision in areas where needed.
- Ensure access and enhance uptake of PMTCT services.
- Treatment with ART and having undetectable viral load.
- Strengthen community mobilization platforms including health extension program, family health team (FHT), CSOs, CBOs and FBOs.
- Improve access to health services through.
 - Decentralization of service provision
 - Provision of differentiated service delivery
 - Ensure affordability and acceptability of health care services
 - Ensure provision of user friendly services
- Address socio-cultural factors like address harmful traditional practices that fuel HIV/AIDS
- Address stigma and discrimination through:
 - Education, advocacy, and communication of the public through religious and community leaders.
 - Empowerment and engagement of PLHIV and their associations in advocacy and communication.

Structural interventions/Activities

- Strengthen Legal and policy environment (legal support system).
 - Revise the national HIV policy
 - Develop capacity for legal support system at different levels
- Strengthen partnership with concerned sectors and organization
 - Empowerment of HIV Positive adolescents and youths using adolescent psychosocial support programs.
 - Engagement of media in communication and advocacy.
 - Monitoring and enforcement of anti-discriminatory laws and regulations.

- Reduce economic vulnerability:
 - Strengthen the provision of income generating activity (IGA) support to vulnerable women and high risk adolescent girls and young women.
 - Care and support program for OVCs and PLHIVs (for those who are less empowered in economy, psychosocial, education etc.)
- Promote gender equality and prevention and management of gender-based violence.
- Supportive interventions designed to enhance referrals, adherence, retention, and community mobilization.
- Mainstream HIV/AIDS prevention activities into sectors and workplaces.

2.2. Condom programing for KPP and general populations

For the achievement of HIV prevention and family planning services, condom programming has been playing a pivotal role. To address gaps in areas of program coordination, supply chain management and access to quality condoms, four major strategic objectives have been identified and should be implemented:

- Enhance supportive environment and leadership for the implementation of coordinated and sustainable condom programming.
- Ensure the availability and accessibility of quality condoms in sustainable manner.

- Increase demand and enhance correct and consistent use of condoms for different segments of population; and
- Generate strategic information and ensure an integrated, effective monitoring and evaluation system for condom programming.

2.3. Pre Exposure Prophylaxis

Pre-exposure prophylaxis (PrEP) of HIV is the use of ARV drugs by individuals who are not infected with HIV but at a substantial risk to block the acquisition of HIV. Substantial risk of HIV infection is provisionally defined as an incidence of HIV higher than 3 per 100 person-years in the absence of pre-exposure prophylaxis (PrEP) and risk behavior like inconsistence condom use. PrEP should be offered as an additional prevention choice for people at substantial risk of HIV infection.

The target beneficiaries for PrEP service in Ethiopia are HIV Negative FSWs and HIV negative partners of Sero- discordant couples. The nationally recommended PrEP drug is a fixed dose combination that contains Tenofovir 300mg and Lamivudine 300mg once daily for the identified target groups with substantial risk for HIV infection. In the context of COVID 19 pandemic, a three Multi Month Dispensing (3MMD) PrEP drugs for all clients is recommended. While taking PrEP, clients should have follow up facility visit for prescription refills, counseling on risk reduction, counseling on correct and consistent use of condoms; routine screening of STIs, family planning service HIV testing, and assessments of adherence and retention as part of combination HIV prevention package.

Nationally developed oral PrEP demand creation materials using flip chart, poster and brochures should be utilized both at facility and community levels through active engagements of PrEP service providers and peer service providers for improving PrEP service uptake and PrEP-retention.

WHO also recommended that the Dapivirine vaginal ring (DPV-VR) for women every 28 days and injectable PrEP every two months for all eligible clients at substantial risk of HIV infection, as part of combination prevention approaches.

The following are the criteria used to identify participants eligible for PrEP service among target populations

Table 2.1: Eligibility Criteria for PrEP

HIV Negative FSWs	HIV Negative Partners in Sero-Discordant Couples
■ HIV negative using a rapid antibody test as per the National HIV testing algorithm on the day of PrEP initiation. ■ No clinical suspicion of acute HIV infection. ■ Self- identifying FSWs. ■ No contraindications to PrEP medicines (TDF/3TC)	■ HIV negative using a rapid antibody test as per the National HIV testing algorithm on the day of PrEP initiation. ■ No clinical suspicion of acute HIV infection. ■ Substantial risk of HIV infection (any ONE of the following in the past six months): ■ Has a known HIV positive sexual partner(s) who is not on ART or ■ On ART less than six months, or not yet achieved undetectable viral load < 50 ml/copies or ■ No contraindications to PrEP medicines (TDF/3TC)

Exclusion criteria and contraindications for PrEP

- Finding of HIV infection (existing HIV infection should be ruled out by testing using the national algorithm on the same day of PrEP initiation).
- Finding of Signs/symptoms of acute HIV infection, with probable recent exposure to HIV.
- Estimated creatinine clearance of less than 60 ml/min (if known).
- Client reported allergy or there is contraindication to any medicine in the PrEP regimen.
- **Known Hepatitis B infection.**
- Unwillingness or not being ready to use PrEP as prescribed and/or give detail information required for monitoring.

When do we stop PrEP?

Starting PrEP does not mean staying on PrEP for life. At every follow up visit, review patient reported risk behavior and evaluate the need to continue PrEP as a component of HIV prevention. The following criteria are reasons to stop PrEP:

- HIV Sero-conversion while on PrEP
- Sustained elevation of creatinine clearance (eCRCL) < 60ml/min.
- Clients who have side effects from the medicine that interfere with quality of life.
- Finding of Hepatitis B infection (clients with HBsAg Positive test result)

- Clients with poor adherence for two consecutive months (Missing five and above pills per month) to the prescribed dosing regimen despite efforts to improve daily adherence.
- Clients on PrEP whose HIV positive partner achieved undetectable viral load while on ART.
- No longer at substantial risk (e.g., no longer engaged in sex work)

How to safely discontinue PrEP

Clients should be informed how to safely discontinue and restart PrEP. This includes the need to contact health care provider and to continue PrEP for 28 days post-high risk

behavior. Before discontinuing PrEP, the health care providers should conduct the following activities.

- Perform HIV test to confirm whether HIV infection has occurred.
- If HIV positive, establish linkage to HIV care & treatment.
- If negative, appoint the client after 12 weeks for HIV testing and ensure that the client continues to take PrEP for 28 days after the high-risk event.
- In addition, discuss with the clients on alternative methods to reduce the risk of acquiring HIV such as correct and consistent use of condoms, desired behavioral change to avoid risk.

How to safely restart PrEP

Any person who wishes to resume taking PrEP medications after having stopped should undergo all the same pre-prescription evaluation as a person being newly prescribed PrEP, including an HIV test to establish that they are still without HIV infection. In addition, an open discussion with clients is important to clarify the changed circumstances since discontinuing medication that indicate the need to resume medication, and the commitment to take it.

2.4. Prevention and management of Gender Based Violence (GBV)

Gender inequalities and gender based violence place girls and women particularly at increased risk of HIV infection. Young women with disabilities face even higher risks. The span and scope of addressing gender inequalities and gender-based violence is broader than just within the health sector. It requires multi-sectoral responses and investments and should include gender responsive programming and budgeting in the HIV response.

The following interventions will be addressed to prevent and manage GBV:

- Build capacity of health facilities to provide comprehensive GBV services including training of health workers, availing provider tools (Job aids, SOPs, etc.) and reporting tools at site level
- Provide comprehensive services in health facilities for survivors of GBV that includes screening and management of intimate partner violence, medico legal examination, HIV testing, STIs screening and management and pregnancy testing, PEP, emergency contraception, counseling, referral for social and legal services.
- Strengthen schoolgirls clubs, make youth centers and health clinics gender sensitive and girls' friendly. Provide integrated services including psychosocial support, HIV, SRH and GBV related services.

- Undertake community dialogue on promotion of gender equality and prevention of GBV by integrating into the health extension program.
- Ensure referral system is in place to medico legal services for survivors
- Strengthen youth friendly clinics at facility levels to provide comprehensive GBV services.

2.5. Post Exposure Management including Prophylaxis

2.5.1. Management of Occupational Exposure to HIV

- Health care workers and supporting staff have a low but measurable risk of HIV infection after accidental exposure to infected blood or body fluid.
- Compliance with infection prevention recommendations is the backbone in prevention of occupational HIV infection. The priorities therefore must be to train health personnel in infection prevention and provide them with necessary materials and protective equipment.
- Risk of HIV infection after a needle stick or cut exposure to HIV-infected blood is estimated to be 0.3% (3 in 1000). The risk of HIV infection after exposure of mucous membranes to HIV-infected blood is estimated to be 0.1% (1 in 1000). However, risk could vary depending on severity of injury and viral load in the source patient.

- Antiretroviral treatment immediately after exposure to HIV can reduce risk of infection by about 80%.

Support for post exposure management in health facilities

- Regular prevention education for employees (health workers, cleaners, and other staff) involved in institutional care for PLHIV.
- Ensure availability of control mechanisms for effective observation of standard precaution.
- Establish system for post exposure management to ensure urgent attention for victims who have sustained accidental blood exposure.

Minimum package for PEP sites/facilities

1. Assign one trained physician / Health Officer / nurse as PEP focal person for the facility.
2. The contact address of the facility PEP focal person and the facility ART nurse or any other second person assigned to coordinate PEP activities in the facility should be posted in all outpatient and inpatient departments within the health facility.
3. PEP ARV drugs should be made available in designated sites inside the health facility which may be accessible to all staff, 24 hours, and 7 days a week.
4. Provider support tools like algorithm for determination of the severity of exposure (Exposure Code) and PEP register should be available in the facility.

Steps to manage potential HIV exposed person

1. Treat the exposure site /immediate measures.

- Percutaneous injury or injury to non-intact skin:
 - Wash the exposed site with soap and water as soon as possible, without scrubbing.
 - Avoid using antiseptics.
 - Allow free bleeding but do not squeeze the wound.
- Exposed mucous membranes:
 - Irrigate copiously with clean water or saline.

2. Report the exposure:

- To the PEP focal person or the ART physician or nurse in the facility immediately.

3. The PEP focal person who needs to do:

- a) Clinical evaluation, counseling and testing of the exposed person and complete the exposure reporting form.
- b) Do risk assessment and determine the exposure code (EC) and source HIV status code (HIV SC) using the PEP algorithm.
- c) Using the EC and HIVSC determine whether PEP is warranted for the exposed HCW.

d) If the HCW is warranted to take PEP: Choose appropriate PEP regimen, counsel about the ARVs and prescribe according to PEP algorithm.

e) Document properly on the PEP follow-up register.

f) Appoint the exposed person and follow.

g) Conduct follow up HIV testing at 6 weeks.

Assessment of exposure risk:

Low-risk exposure:

- Exposure to small volume of blood or blood contaminated fluids
- Following injury with a solid needle
- Asymptomatic source patient

High-risk exposure:

- Exposure to a large volume of blood or potentially infectious fluids.
- Exposure to blood or potentially infectious fluids from a patient with clinical AIDS or acute HIV infection or known positive with high viral load.
- Injury with a hollow needle.
- Needle used in source patient's artery or vein.
- Visible blood on device.
- Deep and extensive injury.

Table 2.2: Interpretation of exposure code (severity of exposure).

	Exposure Code	Type of exposure
1	EC 1	Is a minor muco-cutaneous exposure to small volume of blood for short period (few seconds to minutes)
2	EC 2	Is a major muco-cutaneous exposure to large volume of blood for longer duration (several minutes), or mild percutaneous exposure (with solid needle or superficial scratch or injury).
3	EC 3	Severe percutaneous exposure (large bore hollow needle, deep puncture, visible blood on device, needle used in patient artery/vein).

Table 2.3: Interpretation of the HIV status of the source patient.

	HIV Source Code (SC)	The HIV status and severity of the illness in the source patients
1	HIV SC 1	The source patient is HIV positive but is asymptomatic and has reasonably good immune status.
2	HIV SC 2	The source patient is HIV positive and is symptomatic, may have AIDS or has other evidence of advanced illness (low CD4 or high viral load).
3	HIV SC unknown	The HIV status of the source patients is unknown (either the patient has refused HIV testing or died or discharged before HIV testing) or the source patient is unknown (e.g., unlabeled blood sample in a laboratory).

Table 2.4: Recommended PEP based on risk assessment.

Status code	Exposure code		
	EC 1	EC 2	EC 3
SC 1	Basic 2 drug PEP	Basic 2 drug PEP	Expanded 3 drug PEP
SC 2	Basic 2 drug PEP	Expanded 3 drug PEP	Expanded 3 drug PEP
SC unknown	Consider basic 2-drugs PEP.		
HIV negative	No PEP warranted	No PEP warranted	No PEP warranted

Table 2.5: Recommended ARVs for PEP and administration guide

Category	Regimen	ARV drug regimen	Dose	Frequency	Duration
Adults and adolescents	Two drug regimen	2-Drug Regimen: Tenofovir (TDF) + Lamivudine(3TC)	TDF 300mg 3TC 300mg	Once daily	28 days
	Three drug regimen	3-Drug Regimen: Triple FDC Tenofovir (TDF) + Lamivudine (3TC) + Dolutegravir (DTG) Or Tenofovir(TDF) + Lamivudine (3TC) + Efavirenz (EFV) OR Lopinavir/ritonavir (LPV/r) Or Atazanavir/ritonavir (ATV/r)	Triple FDC (TDF 300mg,3TC 300mg,DTG 50mg) (TDF300mg, 3TC300mg, EFV 600mg)	Once daily	
			LPV/r400mg /100mg	Twice daily	
			ATV/r300mg/100mg	Once daily	
children	Two drugs	Zidovudine (AZT) +3TC	Based on weight (refer dosing chart)	Twice daily	
	Three drugs	AZT +3TC +DTG or EFV or LPV/r Or ATV/r	Based on weight (refer dosing chart)	AZT+ 3TC and LPV/r Twice daily and For DTG /EFV/ ATV/r Once daily	28 days

Timing of initiation of prophylaxis:

To be effective, PEP should be initiated as soon as possible (within 1-2 hours). The maximum delay for initiation of treatment which would prevent infection is not known in humans. Do not consider PEP beyond 72 hours post exposure. Prophylaxis is to be given for 28 days.

Testing and monitoring after occupational exposure:

- Testing source: rapid HIV test is done after counseling and consent has been secured. If the source patient is negative, there is no need of further assessment of the exposed health care worker. If the result is positive the health care worker needs to be tested.

- Testing of health care worker: HIV testing should be performed immediately after exposure. If result is positive there is no need for PEP, but if negative you should administer PEP as soon as possible as outlined above and then repeat HIV testing at 6 weeks.
- Remember to initiate PEP immediately after exposure until test result confirms the HIV status of the victim. Stop PEP if the health worker is positive for HIV antibodies.
- Following HIV exposure there is a need for psychosocial support.
- Any health care workers presenting to a health facility with potential exposure to HIV should be strictly counseled and examined by trained health care worker about the potential risk of HIV infection and ensure there is no misuse of PEP.

2.5.2. Prevention of HIV Transmission after Sexual Assault

1. Any person presenting to a health facility after potential exposure to HIV during sexual assault should be strictly counseled and examined by trained health care worker about the potential risk of HIV infection and ensure there is no misuse of PEP.
2. Parents/guardian of traumatized children should be counseled and informed on the risk of HIV infection after sexual assault.
3. The following points should be covered in the counselling:
 - a) The exact risk of transmission is not known, but it exists.
 - b) It is important to know the victim's HIV status prior to any antiretroviral treatment.

Recommended regimen for PEP

Dolutegravir (DTG) 50 mg daily in combination with tenofovir disoproxil Lamivudine (3TC) 300 mg daily as the preferred regimen in healthy adults and adolescents for 28 days

Alternatively, AZT or TDF+3TC+EFV for 28 days or

Boosted Lopinavir OR boosted Atazanavir can substitute EFV.

DTG is approved for all children older than 4 weeks weighing more than 3kg and available with dispersible tablets that can be easily administered for all children weighing less than 20kg. For children weighing more than 20 kg, 50 mg adult film-coated tablets can be use.

PEP is not recommended.

- a) If victim presents more than 72 hours after exposure.
- b) Following condom leak or tear.

c) It is the patient's choice to have immediate HIV testing or, if s/he prefers, this can be delayed until 72 hours post examination visit.

d) PEP is not recommended after 72 hours following sexual assault. Patients should be counseled about risk of infection and the possibility of transmitting infection during sero-conversion. They should be instructed to return at 6 weeks post sexual assault for voluntary counseling and HIV testing.

4. It is strongly recommended that the implementation of post-rape prophylaxis should be carefully monitored and evaluated for:

- Pregnancy test
- Emergency contraceptives
- Psychosocial Support
- Legal support
- Screening for conventional STIs and follow-up management
- Drug side effects
- Sero-conversion

2.6. Undetectable=Un-transmittable (U=U) (P=P)

When HIV positive person taking ART achieves undetectable viral load levels (<50 copies/ml) and maintained at least for 6 months, the risk of transmitting HIV through sex is significantly minimized. Globally, this concept is called Undetectable = Untransmittable (U=U) and Ethiopia has contextualized its naming in to “የማይታይ መጠን = የተገታ መተላለፍ (P=P).

Creating dialogue for U=U/“P=P”/ in clinical settings is vital because clients perceive their healthcare providers as a trusted source, and providers can improve client's understanding of this information.

Promoting U=U in clinical settings also gives healthcare providers an opportunity to reinforce and support adherence to antiretroviral medicines, ensure continuity of treatment and regular clinical monitoring, encourage viral load testing, and provide complimentary services such as sexual and reproductive health.

At a population level, U=U messages have the power to reduce HIV stigma and discrimination by sharing the robust scientific evidence that PLHIV with durably undetectable viral load prevents sexual transmission HIV. Therefore, appropriate U=U/“P=P” messages should be campaigned and communicated to the communities using various media outlets and community platforms.

2.7. Screening and treatment of Sexually Transmitted Infections

Active screening and treatment of STIs using syndrome approach will be provided to KPPs and their partners integrated through community and health facility level service delivery outlets. Currently, a syndromic approach is being used to screen and treat STIs although consideration for rapid and laboratory STI testing and same day/early treatment for individuals presenting with STIs may be considered in the future. There is a need to build the capacity of HCPs on syndromic management. The STIs program ownership, management and monitoring and evaluation need to be strengthened.

2.8. Voluntary Medical Male Circumcision (VMMC)

Voluntary Medical Male Circumcision (VMMC) services should be offered as part of a combination HIV prevention effort to reduce the incidence of HIV in high HIV and low Male Circumcision (MC) prevalence settings. MOH in collaboration with partners started implementing the VMMC service since 2009 in the Gambella host population, with the objective of reaching 90% among uncircumcised males aged between 10-49 years with a special focus on 15-29 years by 2022.

A minimum package of services, including targeted information and education on safer sex, condom promotion and distribution, offering HIV testing service and management of sexually transmitted infections, must be delivered along with the male circumcision procedure. However, the current national strategic plan includes some woredas of SNNP region at target areas for VMMC.

The national VMMC strategic document 2020-2022 has been developed with the aim of attaining this goal; MOH has identified the following four major strategic objectives:

- Improve communities' understanding on the benefits and risks of MC and build positive attitudes towards utilizing VMMC services.
- Expand the service delivery at all healthcare facilities to keep pace with the created demand of VMMC in specific population groups.
- Promote Early Infant Male Circumcision into health service delivery system.
- Strengthen program sustainability and ownership (develop sustainability strategy).

Moreover, roles and responsibilities have been assigned for key stakeholders and monitoring and evaluation system for VMMC has been incorporated.

2.9. Care of HIV Exposed Infants (HEI)

Infants born to HIV positive pregnant women by definition are HIV exposed and these infants can be infected with HIV during pregnancy, labor or after birth through breast feeding. All HEI will undergo through DNA-PCR antigen test at six weeks and repeat DNA-PCR test at 9 months for those who tested negative. While the child with HIV infection can often be identified during the two months of life, HIV infection often cannot be excluded until after 18 months of age particularly in breast feeding babies.

Pediatric HIV disease can progress very rapidly and may require treatment before a positive diagnosis can be confirmed. HIV infected infants are susceptible to many opportunistic infections including pneumocystis pneumonia (PCP), TB and other bacterial infections that are associated with high rates of mortality.

Components of clinical care for the HEI

1. History

2. Physical examination

3. Growth assessment

- Growth is the most sensitive clinical indicator of HIV infection in infants and young children.
- Children with HIV infection are at high risk for poor growth.
- Growth should be monitored closely for all HIV exposed and infected infants.

4. Developmental assessment: Use developmental check list to assess growth and development.

5. Infant feeding: Nutrition and feeding history should be assessed regularly. Ongoing counseling on infant feeding practice should be done, according to the national recommendation.

- Safer infant feeding practices, either exclusive breastfeeding for the first 6 months or exclusive replacement feeding for the first six months. Avoid mixed feeding.

- Introduce appropriate complementary food after six months and continue breastfeeding as the general population.

6. Immunization: All HEI should be immunized according to expanded program on immunization (EPI) recommendations. (Refer annex 2)

7. ARV prophylaxis for HIV exposed infants

Considering all HIV Exposed Infants (HEIs) as high-risk infants, enhanced postnatal prophylaxis (ePNP) is recommended for all of them. ePNP is providing NVP and AZT prophylaxis for the first 6 weeks and continuing only NVP prophylaxis for an additional 6 weeks. Infant prophylaxis should begin within one hour at birth or as soon as HIV exposure is recognized postpartum.

- NVP and AZT Prophylaxis syrups shall be started within 1 hour of birth to the HEI up to 72hrs.
- For HIV exposed infants identified after birth (through infant or maternal HIV antibody testing)
 - Infants on breastfeeding: - Initiate ART for the mother - Provide NVP and AZT for 6 weeks and continue only NVP for additional 6 weeks. - Collect specimen for DNA PCR testing at 6 weeks of age and repeat DNA PCR at 9 months of age for those who tested negative at 6 weeks of age.

- If the infant is brought within 72 hours of birth provide AZT and NVP prophylaxis for 6 weeks.
- Infant identified as HIV exposed after 72 hours after birth (through infant or maternal HIV antibody testing) and is NOT Breastfeeding ,
 - Initiate maternal ART
 - No prophylaxis needed
 - Do DNA PCR testing accordance with national recommendations on early infant diagnosis
 - Initiate treatment if the infant is infected

For babies delivered at home:

- If a mother is known to be HIV-positive, ARV prophylaxis should be administered to the newborn even if the mother did not receive ARVs during pregnancy or labor.
- If the mother's HIV status is unknown, offer HIV testing and counseling and, if the mother tests positive, give the baby NVP once daily and AZT twice daily; assess the mother, initiate ART and provide appropriate care.

Table 2.6: Dosage of AZT and NVP syrup for infant prophylaxis for different age groups

Infant age	NVP daily dosing (10mg/ml)		AZT daily dose (10mg/ml)	
	Dose in mg	Dose in ml	Dose in mg	Dose in ml
Birth to 6 weeks:				
Birth weight <2000g	2mg/kg, once daily	0.2ml/kg, once daily	4mg/kg per dose, twice daily	0.4 ml/kg per does, twice daily
Birth weight 2000-2499 g	10mg, once daily	1 ml, once daily	10mg, twice daily	1ml, twice daily
Birth weight >2500 g	15mg, once daily	1.5 ml, once daily	15 mg twice daily	1.5ml twice daily
> 6 weeks to 12 weeks				
	20mg, once daily	2 ml, once daily or half a 50 mg tablet, once daily	No dose established for prophylaxis for this age group	

Note:

- AZT and NVP concentration are 50mg/5ml.
- AZT and NVP syrups should be started at birth and provide for all HEI
- AZT and NVP syrups should be started at birth and provided for six weeks
- NVP alone should be continued for additional 6 weeks
- Follow the manufacturer's instruction for the duration of use following opening. The bottle should be labeled with the date on which it was 1st opened.

- NVP and AZT Infant doses: The oral syringe should not be placed directly into the bottle. Infant dose should be measured by pouring a small amount of syrup into a cup, and then draw the actual dose with oral syringe. Discard the leftover suspension in the cup.

8. Co-trimoxazole preventive therapy (CPT) Using pediatric co-trimoxazole in all HIV exposed infants significantly reduces the rate of PCP and other bacterial infections and in turn reduces infant morbidity and mortality rates. Start co-trimoxazole to all HEI from 6 weeks of age and continue until the child is confirmed not to have HIV infection using antibody test after 18 months of age.

Table 2.7: Dose of CTX for prevention of PCP in infants and children

Age	Suspension Per 5ml (200/40mg)	Pediatric tablet (100/20mg)	Single Strength adult tablet (400/80mg)	Double Strength adult tablet (800/160mg)
< 6mo	2.5ml	1 tablet	1/4 tablet	--
6 mo- 5yrs	5 ml	2 tablets	1/2 tablet	--
6-14 yrs	10 ml	4 tablets	1 tablet	1/2 tablet
>14 yrs	--	--	2 tablets	1 tablet

9. TB risk assessment

At each visit the infant should be evaluated for Tuberculosis. We need to ask for household exposure with an adult who has tuberculosis and symptoms suggestive of the disease and chest radiography if clinically indicated.

10. Determination and evaluation of infection status

One of the goals of follow-up of HEI is to identify and treat the HIV infected ones early. All HEI should have DNA-PCR testing at 6 weeks of age and repeat DNA-PCR test at 9 months.

11. Current assessment and plan

At each visit based on the findings on history, physical examination (that includes growth and development assessment) and/or laboratory investigations, we need to have the assessment of the infant and we should plan our next steps in their management and follow-up.

Follow-up visits and schedule

Follow-up schedule: At birth, 6 weeks, then monthly for 6 months, then every 3 months until HIV infection is excluded. But also advise the mother to bring the infant at any time, if there are new signs and symptoms. HIV-exposed infants with health problems may need more frequent follow-up visits than other infants.

Table 2.9.3. Follow-up visit schedule for HIV exposed infants.

Age in weeks/months	at birth	6 wk	10 wk	14 wk	5 m	6 m	9 m	12 m	15 m	18 m
History	x	X	x	x	X	x	X	x	X	x
Physical exam	x	X	x	x	X	x	X	x	X	x
Growth Assessment	x	X	x	x	X	x	X	x	X	x
Developmental assessment	x	X	x	x	X	x	X	x	X	x
Infant feeding Counseling	x	X	x	x	X	x	X	x	X	x
Determination of HIV status		*DNA PCR	Do DNA PCR if the test is not done at 6 weeks** *Repeat DNA PCR at 9 months for all HEI DNA PCR negative at 6 weeks of age and Do DNA PCR test at any time if the child is sick (sign and symptoms suggestive of HIV infection) with negative test result at 6 weeks					Perform rapid antibody test at least 6 weeks after cessation of breastfeeding.		
AZT + NVP for the first 6 weeks	X									
NVP alone for additional 6 weeks		X	X							
CPT		X	Continue until HIV is excluded, and infant is no longer at risk from breastfeeding.							
TB Risk Assessment	At each visit									
Immunizations	x	x	x	x			x		x	x
Adherence counseling	x	x	x	x	x	x	x	x	x	x
Vitamin A						x		x		x
Note: This is the minimum; children should be seen more frequently if clinically indicated.										

CHAPTER THREE:

HIV CASE FINDING

3.1. HIV Testing Services

HIV testing services refer to the full range of services that should be provided with HIV testing, including counselling (pre-test information and post-test counselling); linkage to appropriate HIV prevention, treatment, care and other clinical services and the delivery of accurate results.

HIV testing services (HTS) should be provided to eligible clients who are at high risk of HIV infection. HTS need to focus on high-risk individuals who remain undiagnosed need to be tested and linking them to treatment and care services as early as possible. People who are HIV-negative but with an ongoing risk also need to be re-tested and provided appropriate prevention package of services.

As we move closer to epidemic control, case finding will become more and more difficult hence HIV testing services should utilize HIV risk screening tools for both adults and children, enhanced index case testing response for recent infections, etc.

Goals of HIV testing services

- Identify people living with HIV by providing high-quality testing services for individuals, couples and families,
- Effectively link individuals and their families to HIV treatment, care and support and to HIV prevention services, based on their status,

- Support the scale-up of high-impact interventions to reduce HIV transmission and HIV related morbidity and mortality.

HTS service quality should not be compromised, hence standard operating procedures (SoPs), protocols, and other necessary job aides must be followed and regularly monitored.

Targeted HIV Testing Service

Targeted HTS is a process whereby individuals who are at risk of acquiring of HIV infection are tested for HIV if found eligible based on HIV risk screening tool. The focus of targeted testing is towards identifying of new HIV positive cases and proper utilization of HRST is important to implement targeted testing and achieve the first 95 of the UNAIDS goals.

3.1.1. Core Principles

Effective and efficient HTS service delivery models and approaches should focus on:

1. Reaching the largest number of individuals with HIV who remain undiagnosed with higher HIV risk
2. Increasing acceptability, equity and demand to reach those left behind, including key populations.
3. Prioritizing approaches that are most cost effective and efficient

4. Achieving national program targets (95-95-95)

5. Ensure linkage to treatment for individuals who are diagnosed HIV positive and providing appropriately tailored prevention for those who test HIV negative

3.1.2. Guiding Principles for HIV counseling and testing

HTS should always be voluntary. Protecting and maintaining client confidentiality is important, especially when offering ICT. An enabling environment that removes barriers such as stigma, discrimination, IPV is important for increasing access to and uptake of HTS. Therefore, the WHO guiding principles (also called 5Cs) should be applied in all HTS sites.

Consent: People receiving HIV counseling and testing must give informed verbal consent to be tested and counseled. Written consent is not required. They should be informed of the process for HIV counseling and testing and their right to decline testing. In Ethiopia, for pediatric age group (less than 15 years of age), the parents or guardian need to consent verbally. Mature minors (13-15 years old who are married, pregnant, commercial sex workers, street children, heads of families, or who are sexually active) can give verbal consent by themselves.

Unconscious or patient who is not in status of providing self-consent, should not be tested for HIV unless the clinician determines it is necessary to establish diagnosis and make treatment decisions. Consent of kin should be obtained during counseling.

- **Confidentiality:** HTS are confidential, meaning that what the HIV counseling and testing provider and the person discuss will not be disclosed to anyone else without the expressed consent of the person being tested. Counselors should raise, among other issues, whom else the person may wish to inform and how they would like this to be done. Shared confidentiality with partner or family members and trusted others and with health care providers is often highly beneficial.
- **Counseling:** HIV counseling and testing services must be accompanied by appropriate and standardized pre-test information and post-test counseling.
- **Correct:** HIV counseling and testing providers should strive to provide standardized testing services to reach to correct diagnosis.
- **Connection:** Connections to prevention, care and treatment services should include the provision of effective referral to appropriate follow-up services as indicated, including long-term prevention care and treatment services.

Client's Rights for HTS

Clients' rights during testing and counseling. Clients have the right to:

- Confidentiality
- Privacy
- Refuse testing
- Be treated with respect and
- Asking information or question.

3.2. Demand creation for HTS

Demand creation and mobilization strategies include activities intended to directly improve an individual's knowledge, attitudes, motivations, and intentions to test and to inform the decision to obtain HIV testing services.

Good practice approaches for demand creation

Demand creation to increase HIV testing service uptake and engage those in greatest need of services is a valuable tool for mitigating stigma, discrimination and IPV. Priorities may need to be set among demand creation approaches, depending on the setting, target population, available resources, and geographic area with high burden of HIV as part of a strategy to reach PLHIV who do not know their status and have high risk to HIV infection.

Many demand creation strategies have been assessed for how they affect HIV testing uptake and the proportion of people living with HIV diagnosed, including the following:

1. Evidence-informed platforms for delivering demand creation include:

- Peer-led demand creation interventions for KPPs
- Digital platforms, such as short-prerecorded videos encouraging testing at health facility waiting areas

2. Approaches that have showed evidence of increasing demand for HTS include:

- Advertising specific attributes of HIV testing services: on Radio, TV, social media, print and electronic media.
- Brief key messages and counselling by providers

- Messages during couples counselling that encourage testing,
- Messages related to risk reduction, to encourage testing among high-risk individuals,
- Motivational messages: through PLHIVs, religious leaders, (it could be speech at gatherings, written, picture or graphic message),

3.3. HIV Testing Service Provision Settings

HIV testing and counseling services can be provided in two major settings:

3.3.1. Facility Setting

Currently, the HIV testing and counseling services in health facilities are:

1. Client initiated HIV counseling and testing (VCT)
2. Provider initiated HIV testing and counseling (PITC), which is provided by opt-out approach using HIV risk screening tool at all clinical service delivery points except during first ANC testing.
3. Mandatory HIV testing

Mandatory HIV testing can only be performed for specific reasons with individuals or groups when requested by the court. HIV is a blood-borne pathogen readily spread by blood transfusion or tissue/organ transplantation: therefore, it is mandatory to test blood or tissue for HIV before transfusion, transplantation or grafting. Mandatory screening of donated blood/

organ/tissue is required prior to all procedures involving transfer of body fluids or body parts, such as artificial insemination, corneal grafts, and organ transplant. Donors should be specifically informed about HIV testing of donated blood, organ or tissue and link those with HIV positive test results to posttest counselling, care, and treatment services.

Mandatory HIV testing is required:

- HIV testing that will be done by the court order
- Testing of blood after donation, and
- Testing of organs before organ transplant

3.3.2. Community Setting

Refers to HTS offered in the community, outside of a health facility. It can be delivered in many ways and in different settings and venues. These include HTS at fixed locations in the community, mobile outreach in hotspots and community sites such as bars, youth centers, workplaces, and home based. It can also be delivered either alone or in combination with testing and screening for other infectious diseases such as TB, hepatitis and STIs.

As for any HTS, linkage to appropriate services after community-based testing is critical. While providing HTS services in the community, providers should use nationally approved protocols as appropriate.

Workplace HTS:

HTS in the workplace is an effective strategy for reaching high risk individuals such as mining, the transport and logistics sector, mega projects, large farming areas, the military and other uniformed services.

Targeted mobile outreach

Targeted mobile outreach focuses on high-risk populations to avail HTS as mobile outreach in areas of low coverage and poor accessibility. HTS services can also be provided to high-risk populations such as widows, divorced and homebased female sex workers.

Home-based HTS:

HTS using ICT approach can be offered in the home has the potential to reach undiagnosed partners of index cases and eligible biological children. It can effectively reach undiagnosed individuals if offered at timings outside of work hours.

3.4. Approaches of HTS

A strategic mix of different HIV testing approaches are needed for an effective and efficient national HTS program depending on the epidemiology and resources available. Differentiated HIV testing service delivery approaches are recommended to address the needs of a variety of population groups, contexts, and epidemic settings.

The following HTS approaches are recommended to reach the 2030 global target to end HIV/AIDS epidemic control:

- Client Initiated: Voluntary Counseling and Testing (VCT)
- Provider Initiated Testing and Counseling (PITC)
- Index Case Testing
- Social Network based HIV testing (SNS)
- HIV self-testing (HIVST)

The implementation of the VCT, PITC approaches will use the respective protocols. However, if couples would like to get HTS together, couple counseling and testing (CHCT) protocol will be used. Each protocol has different components having tasks and scripts.

3.4.1. Voluntary Counseling and Testing (VCT)

VCT is initiated by clients seeking to know their HIV status. VCT is the oldest approach of testing that gives client an opportunity to confidentially explore his/her HIV risks to learn his/her HIV test result. The VCT intervention is “client-focused” to the extent that you focus on the client’s unique issues and circumstances related to HIV risk. The intervention is based on a risk reduction model designed to reduce, not necessarily eliminate risk.

3.4.2. Provider Initiated Testing and Counseling (PITC)

This refers to HIV testing, and counselling recommended by a provider during a visit to a health care facility. The health care provider should use HIV risk screening tool to determine whether the client is eligible for the HIV testing

or not. A brief counselling or pretest information should always accompany testing even for diagnostic purposes and patients should never be forced to undergo testing against their will. Clients presenting with symptoms or signs of illness possibly attributable to HIV; it is a basic responsibility of health care providers to recommend HIV testing and counselling as part of routine clinical management.

PITC is an “Opt-out” approach. With the “opt-out” approach, individuals may specifically decline the HIV test having received pre-test information, without affecting their care. If a client declines HIV testing service, a provider should try to understand the reasons for the refusal. If a client is convinced, HTS can be provided. But if not, the client will be advised to reconsider his decision and return to get HTS at another time.

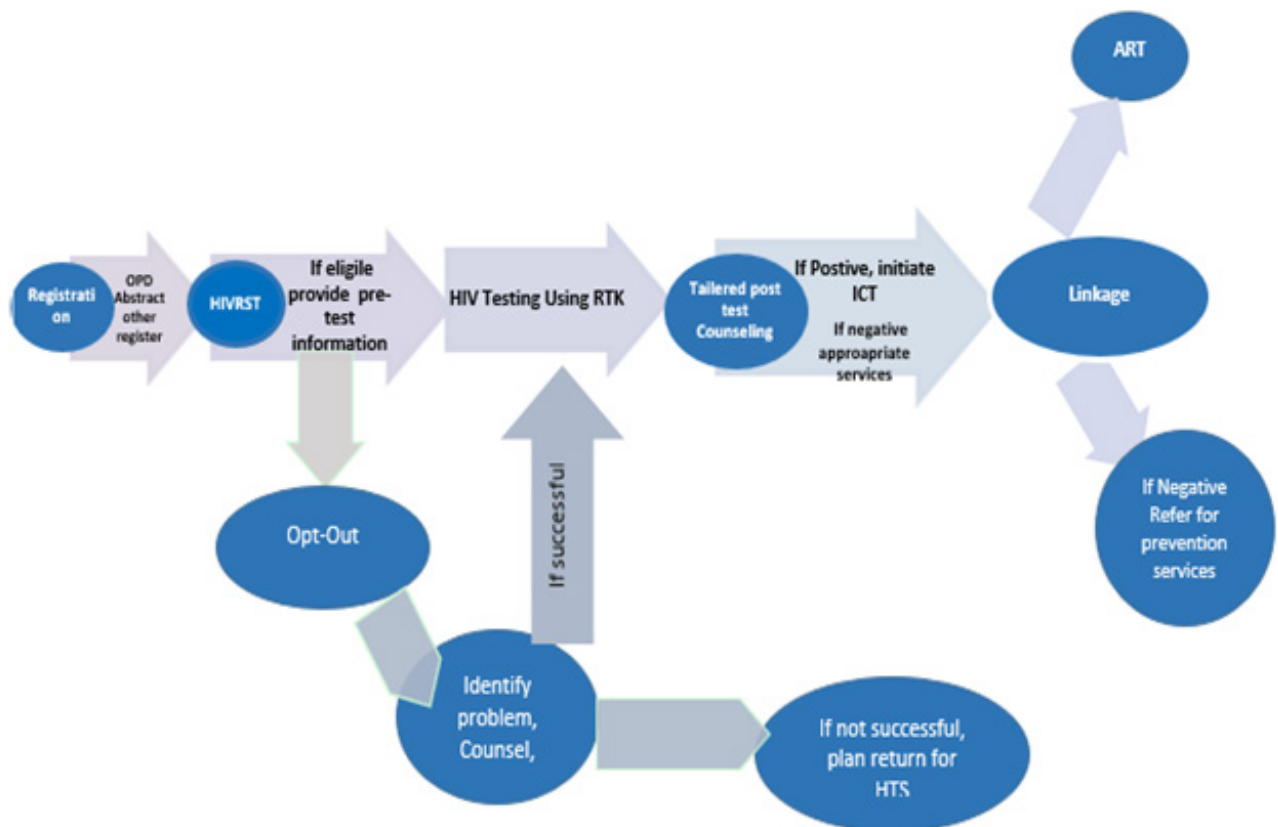


Figure 3.1. PITC Flow Chart

3.4.2.1. Use of HIV Risk Screening Tool (HRST) to Enhance PITC

HIV risk screening tool enables service providers to identify high risk clients. It gives opportunity for targeted HIV testing. HIV risk screening tool is a tool, having a set of questions, used to identify clients with specific risks for HIV. HRST helps to determine a client's HIV risk and whether the client is eligible for HIV testing or not based on the below category:

- Client's HIV status (is the client a known HIV positive case or not?)
- HIV risk behavior (practicing unprotected sex, having concurrent multiple sexual partners, etc.)
- Clinical symptoms or signs of HIV
- Occupational risk

Benefits of Using HIV Risk Screening Tool

HRST reduces over testing, improves case finding and subsequently increase yield. Risk screening tools have been utilized in different countries to identify clients who need to be tested and maximize HIV case detection and increase efficient utilization of limited resources including RTKs. In our setup, risk screening tool can be applied at all service delivery points with the exception for the first ANC/labor/postpartum visit. (Refer annexed 3a and b)

3.4.2.2. Eligible clients for HIV Testing and counseling while providing PITC includes:

- All pregnant at first ANC visit, laboring, and postpartum women with unknown HIV status.
- Partners of HIV positive pregnant/postpartum women
- Sexual partners of index cases and all under 19 children of PLHIV; biological siblings and biological parents.
- Female sex workers with unknown HIV status.
- All TB patients and presumptive TB cases with unknown HIV status
- All patients with Sexually transmitted infections (STIs) and their partners
- Children orphaned by AIDS.
- Patients with clinical signs or symptoms of HIV/AIDS visiting health facilities (Annex 6a, 6b and 6c).

3.4.3. Index Case Testing

An index case is defined as an individual who is HIV positive. When counsellors identify index cases whose sexual partners or eligible biological children are not tested for HIV, they should immediately provide partner services including notification and contact tracing.

To improve case-finding, countries need to use

a mix of innovative testing strategies to identify positive individuals. Index case testing is one of the most important case finding approaches to identify HIV positive individuals.

ICT is a high yield, targeted testing approach for identifying and linking new HIV infected individuals to treatment services. This approach needs to be optimally utilized for case detection and to break HIV transmission cycle.

Some of the PLHIV might have not yet disclosed their HIV status to their partners while others have partners with ongoing risk. Clients' concerns should be addressed to improve disclosure and testing service uptake among index partners and HIV exposed biological children. If there is/are children who are not biological children of the index case but is/are exposed to HIV, HTS will be provided though this approach is not part of ICT.

A critical function of ICT is partner elicitation, a process through which index patients are interviewed to elicit information about their sexual partners, counseled and assisted on disclosure and continuously supported to bring their partner for HIV testing. Partner services are voluntary, based on the decision of the index patient, and are provided confidentially in a patient-centered framework. The role of the service providers will be providing counseling about the risk of HIV transmission and importance of disclosure, assisting on disclosure, educating on the benefits of early treatment and better outcome.

Principles of ICT

- Client centered and focused
- Confidential
- Voluntary and non-coercive
- Non-judgmental
- Culturally, linguistically, appropriate
- Accessible and available to all
- Comprehensive and integrative

Approaches of ICT Services

1. Client Referral - Index clients tell partners about infection and encourage him or her to come to testing center for testing. A trained health care provider encourages the index clients to disclose his /her status to his/her partners. In this approach, the index client takes responsibility for encouraging partner(s) to seek HTS. This is often notified verbally by the index client.

2. Contract referral: The index client enters a "contract" with the health care provider whereby he or she agrees to disclose their HIV status to all partner(s) and or to refer them for HTS within two weeks.

If partner(s) do not access HTS within this period, health care providers contact the partner(s) directly to notify them that they may have been exposed to HIV without any disclosure of the index client after getting permission from the index client. Health care providers offer

voluntary HTS to partner(s) and other family members as appropriate while maintaining the confidentiality of the index client.

3. Dual Referral - Index clients and service provider will notify his/her partner together. The provider offers voluntary HIV testing to the index client and partner(s) for the potential exposure to HIV and encourage them to seek HIV testing together without any disclosure of the index client HIV status.

4. Provider Referral (Optional) - Health care provider will call or send text message to his/her partner of potential exposure of HIV without disclosing the index client name and HIV status to visit HIV testing sites without telling them the client's name (this will be done anonymously). In this approach, the index client has no role and involvement in the process.

Steps of Index Case Testing

Step 1: Introduce the concept of index testing during pretest session or PMTCT/ART visit

Step 2: Offer index testing as a voluntary service to all clients testing HIV-positive

Step 3: If client accepts participation, obtain consent to inquire about their partner(s) and biologic child (ren)

Step 4: Obtain a list of partners in last 12 months and biological children < 19 with unknown HIV status.

Step 5: Conduct an intimate partner violence (IPV) risk assessment for each named partner.

- Has [partner's name] ever hit, kicked, slapped, or otherwise physically hurt you?
- Has [partner's name] ever threatened to hurt you?
- Has [partner's name] ever forced you to do something sexually that made you feel uncomfortable?

Step 6: Determine the preferred method of partner notification or child testing for each named partner/child. Offer HIVST if a client prefers HIVST and provide necessary support including provision of HIVST kit.

Step 7: Contact all named partners in addition biological children less than 19 years old biological children and provide conventional HIV testing at the health facility and community level.

Step 8: Record outcomes of ICT

Step 9: Provide appropriate services for children and partner(s) based on HIV status.

Step 10: Follow-up with client to assess for any adverse events associated with index testing

N.B: IPV screening MUST be administered to all sexual contact of the index client. From the three asked questions, if the answer is "yes" to one of the questions, ICT service should not be initiated to avoid adverse outcomes.

Adverse Event Monitoring and Reporting System

Adverse Event is defined in the context of ICT, as an incident that results in harm to the client as a result of their participation in index testing services.

Site level adverse event monitoring, response and reporting system forms need to be available at the facility or community testing sites for service providers to document and monitor consent, IPV, and frequency of adverse events. Actively monitor reasons for refusing index testing services, prevalence of IPV and other adverse events (e.g., confidentiality breaches, stigmatization, coercive tactics, etc.) for improvement.

Service providers should routinely ask index clients if they experienced any adverse events following participation in index testing services.

NB: Unintended negative outcome as result of HIV status disclosure could still occur in the future and, as such, follow up should occur while all contacts are being traced. All reports of adverse events should be properly documented. They should also be informed of their ability to make a complaint if these rights are violated. This can be done through posters in waiting/examination rooms, patient handouts, and other educational materials. Index clients should be provided multiple pathways for issuing concerns or complaints regarding index testing services. These include suggestion boxes within health facilities and community testing sites.

Phone Counseling for Index Case Testing

Index case testing is not a onetime activity. It requires follow up counseling, relationship

and trust building between the client and the provider. To this effect, ongoing in person counseling has been the main stay used by ICT counselors. However, due to changes in service delivery following the COVID-19, providing in person counseling has been challenged and several clients are being put on multi month dispensing (MMD). Virtual Phone counseling is counseling strategy to use in the context of COVID-19 where clients prefer to stay away from sites. Phone counseling integrated to ICT helps to reach and counsel index cases who are on MMD but have contacts not yet tested/ not elicited, newly diagnosed but have not yet disclosed and required follow up counseling. It is a tool to support sites to provide phone-based counseling to elicit contacts of index clients ensuring communication privacy, support indexes virtually to select contact referral approaches, and provide exposure notification mechanisms for partners with testing options including the HIV self-test.

Index Case Testing (ICT) minimum standards

Index testing should be client-centered and focused on the needs and safety of the index client and his or her partner(s) and children. Further, sites offering index testing services must ensure appropriate systems are in place for testing service providers to identify and respond to clients who disclose their fear or experience with Intimate Partner Violence (IPV) from partner(s).

The minimum standards for a site to provide safe and ethical ICT services are:

- Providers trained on HTS including index testing procedures, IPV screening, adverse event monitoring, 5Cs, and ethics.

- Adherence to 5C's (consent, confidentiality, counseling, correct test results, and connection to prevention/treatment)
- IPV risk assessment and provision of first line response, including safety check and referrals to clinical and non-clinical services (if not provided on site)
- Secure environment to store patient information, and
- A site level adverse event monitoring and reporting system.

3.4.4. Social Network Strategy (SNS)

Social network strategy (SNS) is a case-finding strategy that uses social network connections to locate individuals engaging sexual partner/s and it can also be applied for FSWs, drug injecting partners and social contacts of key population members with HIV and of those who are HIV-negative and at ongoing risk.

SNS has been used to find new HIV cases among key populations who have difficulty in accessing HIV testing services. It has also been used to find previously diagnosed individuals who were out of care, and who may then be linked to ART for the first time or re-engaged in ART services. SNS is a peer-driven HIV testing approach by addressing people's confidentiality concerns and broadening the reach to social contacts, social network-based HIV testing approaches can improve the acceptability of partner services among key populations and so reach more people who may not otherwise test for HIV.

The underlying assumption of SNS is that people in the same social network share similar risk behaviors that predispose them for HIV. This is not limited to current behaviors but includes past behaviors. The overall approach is to enlist HIV-positive and high-risk HIV-negative persons to recruit individuals from their social, sexual, and drug injecting networks for HIV testing services. Once recruited and tested, network members become recruiters and refer members of their networks for testing. This process continues, creating chains of recruitment that can penetrate hidden networks.

This strategy uses inclusion and exclusion criteria to identify the appropriate target/recruitment process and enhances success of the SNS.

Inclusion criteria for recruiters/seeds include:

- HIV positive and/or high-risk negative
- Willing to refer network members to HTS
- Comfortable talking about HIV
- Knowledgeable about HTS and testing locations

Exclusion criteria for recruits include:

- Persons who might pose a risk for violence,
- Persons with significant mental illness who might interfere with appropriate participation,
- Known HIV positive cases,

There are four major phases to a social networks program. These phases are

1. Recruiter Enlistment: In this phase, HIV-positive or HIV-negative high-risk persons who are willing to recruit individuals at risk for HIV infection from their social, sexual, or drug-using networks are enlisted into the program. To identify recruiters, approach HIV positive clients and identify additional people through the existing HTS or ART/PMTCT services.

2. Engagement (Orientation, Interview, and Coaching): After recruiters are enlisted into the program, they are provided with an orientation session that explains the nature of the program and the social network techniques that might be used to approach their associates and discuss HIV testing with them.

3. Recruitment of Network Associates: Next, recruiters will refer individuals for testing who they have identified as being at risk for HIV infection. All individuals should be approached by the recruiter alone, without the provider).

4. HTS: The next phase involves providing HTS to the network associates identified through the social network strategy.

3.4.5. HIV SelfTesting

HIV self-testing (HIVST) refers to a process in which a person collects his or her own specimen (oral fluid or blood) and then performs an HIV test and interprets the result, often in a private setting, either alone or with someone he or she trusts. HIV self-testing should be offered as an additional approach to HIV testing services.

As with all approaches to HIV testing, HIVST should always be voluntary, not coercive, or mandatory.

Delivery of HIV Self Testing

HIVST can be delivered in two ways depending on the level and type of support provided.

Directly assisted HIVST refers to trained/ oriented providers, or peers giving individuals an in-person demonstration before or during HIVST on how to perform the test and interpret the test result.

Unassisted HIVST refers to when individuals self-test for HIV and only use an HIVST kit with manufacturer-provided instructions for use.

HIV Self-Testing, as a key strategy in the context of COVID-19, should be utilized optimally as additional testing strategy to reach clients undiagnosed otherwise and those who could not be reached timely because of MMD and COVID-19 situation.

Both directly assisted and unassisted HIVST may include additional support:

- Manufacturer's instructions and brochures.
- Brief in-person demonstration (one-on one) before testing.
- In-person assistance during procedure.
- Hotline contacts for HIVST information.
- Virtual support via telephone and text to track based on clients preference

Demand creation strategy of HIVST

To create awareness and increase uptake of HIVST, advocacy and communication strategies should aim to emphasize:

- Correct use of the self-test kits, and ensure correct interpretation of results and linkage for confirmatory testing,
- Demand creation should be tailored to high-risk target populations. It will be most successful when developed with communities in local setting.
- Demand creation will be conducted at health facility and community platforms. Clients should get adequate information to increase their knowledge and decision-making ability to test themselves

Methods of demand creation

Clear messages are needed to ensure that users understand what to do after a reactive self-test result, including where to go to access conventional HTS for confirmation of results, treatment, care, and other support. The messages can be delivered through:

- One to one (health care providers, health extension professionals, peer educators, volunteers).
- Print media (banners, poster, fliers)
- Audio visual
- Networks of people living with HIV

- Community-based organizations

NB: Providers and users should be aware that HIVST is not recommended for people with a known HIV status, as this may lead to an incorrect self-test result.

Standards and procedure

Directly Assisted HIVST: HIVST must be conducted using the nationally approved HIV rapid self-test kit(s). The kit(s) will include instructions in English and local language as well as pictorial diagrams to aid ease of use and correct interpretation of results. All service delivery points distributing HIV self-test kits should display instructions on HIVST procedures. In addition, all outlets should show those who will assist clients how to perform the test. The assistant will provide pretest information, demonstration, and interpretation of the result.

If the HIVST result is reactive, the assistant should link the self-tester to conventional HIV testing for confirmation, where the approved national HIV testing algorithm is utilized. And the assistant will also follow whether the confirmation test performed and whether the client is enrolled to ART if test result turns out to be positive.

For individuals with non-reactive self-test results, the assistant should advise the self-tester to retest in 3-6 months if the client has an ongoing risk to HIV infection.

Unassisted HIVST

Refers to when individuals self-test for HIV using a self-test kit using an instruction provided with the kit. Service providers/ trained peers who distribute HIVST kits should demonstrate the instructions on HIVST procedures. Users should be informed about links or contact details to access additional support, such as telephone hotlines or instructional videos.

The kit will include instructions in local languages and pictorial diagrams to aid simplicity of use and correct interpretation of result.

Quality assurance of HIVST

Guidance on ensuring the quality of HIV self-testing test kits and testing processes include:

- Monitor the quality of HIVST through integrating with the existing HTS QA program
- Include HIVST in mentoring and supportive supervision activities
- Ensure provision of HIVST instructions with pictorial illustrations in local language on how to conduct self-testing and obtain the correct results.
- All clients should be informed to follow manufactures instructions in the test kits insert.
- All clients must also be aware of correct practices to minimize biosafety risks the need to confirm any reactive test results as per the national HIV testing algorithm.

- The safety, quality, and performance of HIVST should be further verified upon delivery and before distribution to the clients.
- Adequate orientation and demonstration should be given to the clients to verify the presence of control line irrespective the result.

3.4.6. Recency Testing

Recent HIV infection is referred to as HIV infection acquired in the past 12 months. Recency testing distinguishes between recent (HIV infection acquired in the past 12 months) and long-term HIV infection. Recent HIV infections are presented with the following manifestations as high viral load, immature and weak immune response, continued high risk behavior, High probability of ongoing transmission (40%-60% of transmissions). Recency testing provides opportunity for interruption of transmission by contact tracing and it increase yield of HIV testing.

Probable recent infection case is a confirmed newly diagnosed HIV positive individual who tested positive for recent infection.

Confirmed recent infection is a confirmed newly diagnosed HIV positive individual who tested positive for recent infection and has high viral load.

Purposes of Recency Testing

The identification of newly infected individuals and the presence of recent infection will support the national HIV program to rapidly

respond to sub-populations and sites where high levels of HIV transmissions are detected. Following identification of cases, responding to individuals with probable recent infection or groups of HIV-infected persons with sexual partners and social networks is a critical step toward bringing the nation closer to the goal of no new infections.

Case reporting and HIV recency data should be used to guide an enhanced response at the health facility/site level and at the cluster/above site level. As the country is getting closer to epidemic control, it is recommend conducting recency testing for all newly diagnosed HIV cases.

A. Site Level Response

Healthcare providers are required to document all probable recent infections and risk factor information about the newly identified HIV positive cases to provide enhanced response and timely monitoring.

Site level response includes:

- All newly diagnosed HIV positive individuals should be linked and start ART within the same day.
- ICT services should be provided for sexual partners and eligible biological children

B. Above-Site level response

Above site response is primarily based on the identification and analysis of clusters based on the CBS data. The type and level of public health response needed for each identified cluster

is guided by the magnitude and the spread of transmission. Some clusters may require routine public health actions such as ICT Services and linkage to care and treatment services, while other clusters may require enhanced response activities (e.g. targeted demand creation and testing as well as strengthening partnership of relevant stakeholders).

3.5. Procedures of HTS delivery

During provision of testing service, providers should follow specific procedures as outlined below. While following the procedures, providers need to utilize the national guideline, protocol, cue-card, and job-aids.

a. Client registration

At service delivery points, clients will be registered using code numbers, names, MRN (Medical record number). Conduct HIV risk screening: If eligible based on the HRST, provide pre-test information

b. Pre-test information

Pre-test information should be provided by health care providers or counselors. Couples should be encouraged to receive results together. Pre-test information for PITC can be provided in the form of individual or group information sessions. The relevant information that should be provided includes but not limited to:

- Exposure to HIV infection and implications of undiagnosed HIV infection.
- The clinical and prevention benefits of

HIV testing individuals, sexual partners, and eligible biological children (less than 19 years).

- Benefits of early ART and the fact that people with HIV who achieve and maintain an undetectable viral load cannot transmit HIV sexually to their partners.
- The meaning of an HIV-positive diagnosis and of an HIV-negative diagnosis.
- The importance of disclosing known HIV status to the provider to minimize repeat testing of a known case. There is a possibility for false negative result if a person who is already on ART is tested for HIV using an antibody test.
- Verbal consent should be obtained before conducting HIV testing from:
 - Adult client including mature minors,
 - Guardian or next of kin if a client has an altered consciousness or a child under 15 years of age
- The confidentiality of the test result and any information shared by the client
- Discuss any concern or questions through availing more time for the client
- The client's right to refuse testing and that declining testing will not affect the client's access to HIV-related services or general medical care.

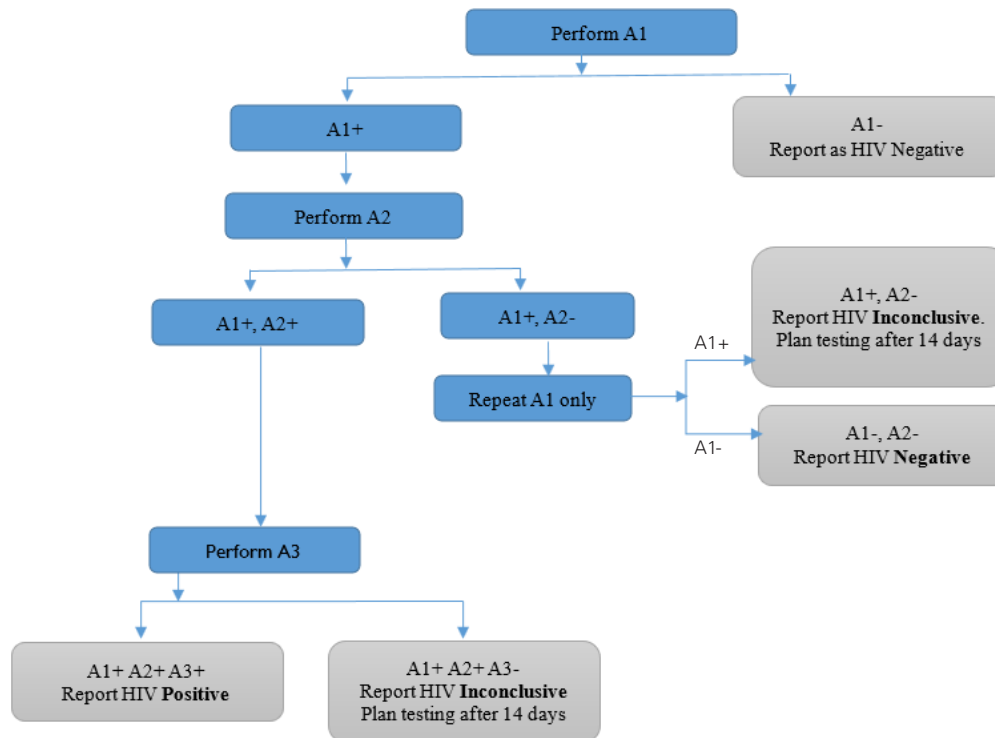
c. Offer HIV Testing:

- If a client accepts HIV testing when offered, proceed to HIV testing.
- If a client decline HIV testing, identify the problem and proceed to HIV testing if successful.
- If a client declines and the counseling is not successful, plan return for testing

d. Conduct HIV Testing

Ethiopia is implementing the low prevalence (<5%) testing strategy. Currently we are using serial algorithm (three test algorithms instead of tiebreaker). In a serial testing algorithm, the samples are tested by a first test. The results of the first test determine whether additional testing is required. To improve the acceptability, uptake and quality of service delivery for HTS in all settings, rapid diagnostic tests (RDTs) should be used. These testing strategies have been developed assuming that all HIV assays used have a sensitivity of at least 99% and a specificity of at least 98%, resulting in an overall positive predictive value of 99%. The HIV testing must be done using nationally accepted RDTs following national HIV testing strategy as shown below. Same day results should always be respected irrespective of the type of delivery approach.

Figure 3.2: Recommended HIV testing strategy for Ethiopia.



- All individuals are tested on Assay 1 (A1). Anyone with a non-reactive test result (A1-) is reported HIV negative.
- Individuals who are reactive on Assay 1 (A1+) should then be tested on a separate and distinct Assay 2 (A2).
- Individuals who are reactive on both Assay 1 and Assay 2 (A1+; A2+) should then be tested on a separate and distinct Assay 3 (A3).
- Report HIV-positive if Assay 3 is reactive (A1+; A2+; A3+)
- Report HIV-inconclusive if Assay 3 is non-reactive (A1+; A2+; A3-). The individual should be asked to return in 14 days for additional testing.
- Individuals who are reactive on Assay 1 but non-reactive on Assay 2 (A1+; A2-) should be repeated on Assay 1
- If repeat Assay 1 is non-reactive (A1+; A2-; repeat A1-), the status should be reported as HIV negative;
- If repeat Assay 1 is reactive (A1+; A2-; repeat A1+), the status should be reported as HIV-inconclusive, and the individual asked to return in 14 days for additional testing.

e. Providing HIV test results

Test results should be declared in person maintaining privacy (not by telephone, email, or letter). Sites should not provide written HIV negative test results to clients to avoid misuse of results. Clients with HIV positive results requesting or requiring referral to other facility should be referred to the appropriate institution with pertinent information.

f. Provide Post-test counseling

All clients undergoing HIV testing should be provided with post-test counseling in person as individual or couple. The messages and information included in post-test counselling and their intensity need to be tailored according to the test result. For those with negative results, this will further depend on whether they face ongoing HIV risk. Messages should always be clear and concise and client-centered, acceptable to the client and delivered in a safe, confidential, and non-judgmental manner. Post-test counseling should be provided by health care providers or counselors.

Post-test counseling service: HIV negative result

Using cue card, provide posttest counseling for HIV negative client as below:

- **Provide HIV test result in person:** Inform the meaning of HIV negative test result. Explain window period in case of recent exposure or ongoing risk and recommend repeat testing based on the client's level of recent exposure and/or ongoing risk.
- **Conduct risk reduction plan:** Clients who do have sex with a person with unknown status can protect themselves by using condoms correctly and consistently every time they have sex.
- **Identify support for risk reduction plan:** Identify a person to whom the client feels comfortable to disclose the risk reduction plan.
- **Provide information on HIV prevention interventions and how to access them:** Condoms, PrEP methods, and harm-reduction services for people who inject drugs
- **Address disclosure and partner referral:** This HIV test result only shows your current HIV status, and it cannot tell us about your sexual partner's HIV status

Posttest counseling services: HIV positive result

Using counseling support tools provide posttest counseling for HIV positive clients as below:

- Provide the HIV test result clearly and simply explain the meaning of test result in person. Focus on the client's level of understanding and acknowledge that the test result may be difficult to hear, however a provider needs to explain to the client and have confidence in client's ability to adjust and cope.
- Discuss immediate concerns and help the client to identify who could provide immediate support (individual or organizations in the community).

- Assess current health conditions and link to services as appropriate, e.g., TB, OI, STI, etc.
- Assess the risk of intimate partner violence (IPV) and discuss possible steps to ensure safety of clients, particularly women, who are diagnosed HIV-positive.
- Assess the risk of suicide, depression, and other mental health issues and discuss positive living.
- Reassure the client about maintaining confidentiality of the test result. Clients might be concerned about others knowing their HIV status, and they will need time to figure out who to disclose to and how to manage their situation. Also explain about the importance of shared confidentiality.
- Advise to use condom consistently and correctly all the time. Assess clients' knowledge on proper use of condoms and conduct condom demonstration. Inform clients about the importance of preventing transmission of HIV or other STI to their partners /or re-infection with different strains of HIV. The client may inform you that his/her partner(s) has already been tested. If the partner tested positive, note that protection from other strains of HIV and STI is important for both individuals. If the partner tested negative, provide counseling to the client to protect the negative partner.
- Provide disclosure support and do partner elicitation. Assisted partner notification and referral services are voluntary at the choice of the client and are provided confidentially in a client-centered framework
- Stress the importance of same day/early ART initiation including prophylaxis for OI if required. If a client is a pregnant woman, advise about the need for PMTCT service.
- Discuss if there are barriers to same day initiation, arrange a follow-up visit for clients who are unable to start ART.
- Retesting is required before ART initiation using the existing testing algorithm by the ART or PMTCT provider.
- Assess the need for any additional health care services including family planning and provide the service or refer.
- Encourage and provide time for the client to ask additional questions.
- Link clients from SDPs to ART clinic through accompanied referral for further counseling, confirmatory testing and subsequent services.

Disclosure Support

All clients, positive or negative, should be empowered to inform their sexual partner/s of their test result. When HIV-positive clients are reluctant or fearful to disclose their results, the counselor should provide additional counseling to help the client to disclose the test result and bring the partner for testing.

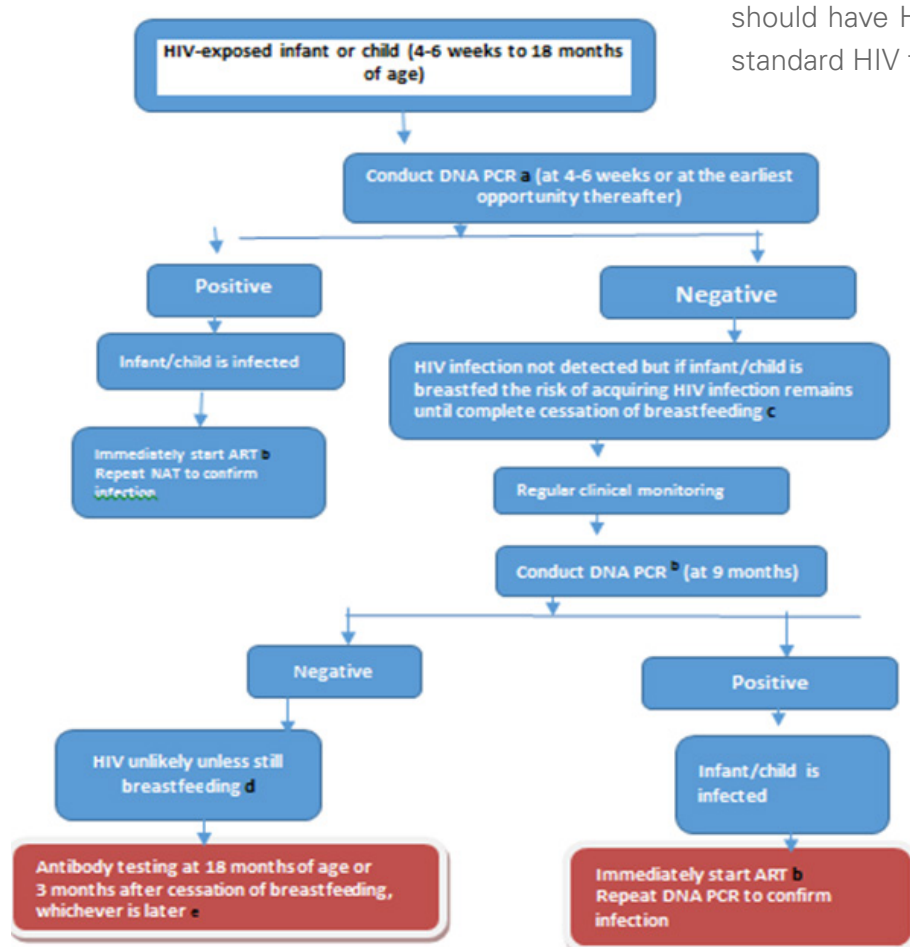
- Counselor should discuss the benefits and risks of disclosing HIV-positive status to partner(s) and support individuals and couples with disclosure.
- For couples, mutual disclosure has many benefits. People with HIV who can share their results with a trusted partner will often find it easier to cope with their diagnosis and to adhere to ART.
- If a client fails to disclose after two documented counseling sessions and the counselor feels that the partner is at risk of infection, he/she should consult the supervisor or health facility head for further action including revealing the result. This will help to share responsibility of handling a difficult case with the health care provider and will protect safety of providers.
- Disclosure and risk of intimate partner violence. Planning for disclosure should include steps to maximize client's safety. Counselling for index client who are considering voluntary or mutual disclosure of HIV-positive status should include discussion about the challenges of their situation, the potential risk of violence and ways to disclose more safely. Options when a client fear of violence or are experiencing violence include mediated, delayed disclosure. If there is risk of severe violence, advice not to disclose.
- Disclosing HIV status to children is a process. Counselors should be encouraged to answer children's questions truthfully from early age. Information should be given in a way a child can understand at a pace she/he can cope with according to their cognitive and emotional maturity. For details refer chapter six

3.6. HIV Testing in infants and children

Mortality is very high among untreated infants infected with HIV in the first year of life, making early HIV testing, prompt return of results and rapid initiation of treatment is essential. Definitive diagnosis at the end of the risk period of mother-to-child transmission (breast feeding period) should be ensured. For children 18 months of age and older (who are not being breastfed or who stopped breastfeeding for at least six weeks), standard HIV serological tests such as rapid diagnostic tests can be used to reliably determine HIV infection status.

Recommendations

- HIV testing and counseling should be offered to under five children visiting health facilities if eligible based on HIV risk screening tool.
- All HIV-exposed infants must have HIV virological testing at six weeks of age or at the earliest opportunity thereafter using conventional PCR testing.
- For infants with an initial positive virological test result, on the same day ART should be started and a second specimen needs to be collected to confirm the initial positive virological test result.
- Children 18 months of age or older with suspected HIV infection or HIV exposure should have HIV serological testing using a standard HIV testing algorithm.



Note:

- ^aPoint-of-care DNA PCR can be used to diagnose HIV infection as well as to confirm positive results.
- ^bStart ART without delay. At the same time, retest to confirm infection. As maternal treatment is scaled up and MTCT transmission rates decrease, false-positive results are expected to increase: retesting after a first positive DNA PCR is hence important to avoid unnecessary treatment, particularly in settings with lower transmission rates. If the second test is negative, a third DNA PCR should be performed before interrupting ART.
- ^cFor children who were never breastfed, additional testing following a negative DNA PCR at 4-6 weeks is included in this algorithm to account for potential false-negative DNA PCR results.
- ^dThe risk of HIV transmission remains as long as breastfeeding continues. If the 9-month test is conducted earlier than 3 months after cessation of breastfeeding, infection acquired in the last days of breastfeeding may be missed. Retesting at 18 months or 3 months after cessation of breastfeeding (whichever is later) should be carried out for final assessment of HIV status.
- ^eIf breastfeeding extends beyond 18 months, the final diagnosis of HIV status can only be assessed at the end of breastfeeding. If breastfeeding ends before 18 months, the final diagnosis of HIV status with antibody testing can only be assessed at 18 months. Antibody testing should be undertaken at least 3 months

after cessation of breastfeeding (to allow for development of HIV antibodies). For infants younger than 18 months of age **DNA PCR** should be performed to confirm infection. If the infant is older than 18 months, negative antibody testing confirms that the infant is uninfected; positive antibody testing confirms infant is infected.

- If a child tested negative for **DNA PCR** becomes symptomatic suggestive of HIV infection, do PCR test.

3.7. Retesting

Retesting refers to a situation where additional testing is performed for an individual after a defined period for explicit reasons. Retesting is always performed on a new specimen and may or may not use the same assays (tests) as the one at the initial test visit. There is a need to reduce unnecessary retesting among persons who have previously been tested and learnt their results. Most people do not require retesting to validate an HIV-negative result. However, it is important to accurately identify persons who do require retesting. Such persons include those whose initial test results were indeterminate, those who tested negative but are at ongoing risk for acquiring HIV (e.g., due to high-risk behaviors) and those who may be in the early stages of infection and have not yet developed a sufficient level of antibodies that can be detected by rapid antibody ('window period').

Retesting will be conducted for:

- A. Individuals with ongoing risk
- B. Before ART initiation

A. Retesting for individuals with ongoing risk

Retesting is recommended for persons who tested negative but have an ongoing risk:

1. Occupational exposure or sexually assaulted client who started post exposure prophylaxis (PEP): retest at 6 weeks, 3 months, and 6 months.

2. A pregnant during the third trimester, laboring, or postpartum period will be eligible for re-testing if:

- She has unknown or HIV-negative status and in sero-discordant relationships,
- Pregnant women whose partner has an updated but unsuppressed viral load result,
- She has ongoing HIV risk in late pregnancy (unprotected sex, STI, etc.).

3. People with STI, viral hepatitis, presumptive TB, clinical manifestations of HIV: retest at 6 weeks.

4. Female sex workers and PWID consider retesting every six months.

5. Have specific incidents of known HIV exposure at 6 weeks.

6. Discordant couple: retest after 6-12 month or even earlier based on risk of infection to a discordant partner such as HVL.

7. Individuals receiving PrEP every quarter

B. Retesting before initiation of ART

It is required that all HIV positive clients linked to care and treatment services need to be retested before treatment is initiated. Retesting aims to rule out possible technical or secretarial errors, including specimen mix-up through mislabeling and transcription errors, as well as random error either due to the provider or the test device.

Retesting a person diagnosed to be HIV positive to verify the diagnosis should include:

- Taking a new specimen for each newly and previously diagnosed individual, preferably conducted by a different provider using the current testing algorithm, prior to initiation of ART.
- Retesting is preferably conducted at a different site/unit, ideally the site where the decision about ART/PMTCT initiation will be made or another unit designated and monitored by the ART/PMTCT clinic.
- If the retesting result is negative (the previous test result is positive and the current test result is negative), different provider should repeat the test either using the same kit or different batch of test kit if available.

Retesting people on ART is not recommended. The effect of ART in suppressing viral replication may extend to suppression of the immune response and therefore reduction of antibody production. Once a person is started on ART, low antibody titers make it challenging to discern whether an individual is indeed HIV positive, and this will lead to potential risks of incorrect diagnosis (false negative result).

C. Retesting 14 days later after an HIV-inconclusive status: Perform retest using the same assay and different/new sample

Repeat testing

When the same specimen is tested again on the same assay when the initial result is reactive, or test results are discordant. The assay is repeated to rule out biological false reactivity. For assays that utilize capillary whole blood, another prick may be needed to collect adequate specimen volume, but it must be in the same testing event.

3.8. Quality assurance and Quality Improvement for HTS

Quality can be defined as accessible services that meet the need of clients and providers, in an equitable and acceptable manner, within the available resources and in line with national guidelines. While providing the HTS, quality of the service should address the major parameters of quality: safety, efficiency, effective (correct result) and provided with reasonable time. It should be easily accessible services that meet the need of clients and providers, should be conducted in line with national guidelines. To ensure the quality of HTS, all service providing sites should:

- Provide standardized conventional HTS procedures at all time
- Ensure all service providers undergo on the standardized HTS training.
- Conduct RTKs quality control for all newly received kits.

- Conduct regular mentoring and periodic supervision sessions to provide technical assistance to improve quality of HTS.
- Introduce HTS quality assurance system to monitor performance of service providers on a regular basis.
- Conduct proficiency testing every four months.

Quality assurance

Quality assurance (QA) for HIV counseling and testing refers to periodic assessments of factors that affect the quality of HIV testing. Issues that need to be addressed while conducting QA:

- Have the counselors/service provider received basic training packages approved by MOH?
- Is there enough physical space to provide that ensures privacy of the clients and point of care testing?
- Are basic supplies and provider support tools available to provide HTS?
- Is the service accessible and affordable to the clients?
- Are clients satisfied with the services?
- Are counseling and testing sessions conducted according to nationally approved protocols?

- The QA Should be conducted by well-trained program supervisors on regular basis to ensure HIV testing service qualities. The roles and responsibilities of the supervisors are:
- To determine if counselors/service providers received standard trainings and refresher courses.
- To monitor how well counselors/service providers follow the counseling and testing protocol.
- To monitor whether clients feel that their confidentiality is protected and satisfied with the services they are provided.
- To make sure that HIV test results are given in person during the post-test counseling session.

Supervision for quality assurance of HIV testing:

The supervisory visits for HIV testing standards and biosafety should assure that:

- The national testing protocol is consistently being followed.
- The laboratory operating procedures are being observed.
- Infection prevention practice is in place.
- Proficiency testing is in place including quality control testing in central laboratory.
- A standardized laboratory logbook is available and being used.
- Technical support for QA is in place.

Quality control (QC) is a procedure or set of procedures intended to ensure that a performed service adheres to a defined set of quality criteria or meets the requirements of the client. It is an important means of verifying for the test kit and the procedures used are performing according to the manufacturer's intended specifications.

Quality control of HIV testing

Only test kits validated by the Ethiopian Public Health Institute should be used by counseling and testing sites. Training and supervision of laboratory staff, accurate testing materials that are well stored and have not expired, and good maintenance of laboratory records are essential to ensure the quality HIV testing.

External quality assessment (EQA)

For EQA, proficiency testing samples (dried tube specimen), which was prepared by the higher laboratory tier, i.e., national reference laboratory (NRL), regional reference laboratory (RRL) or EQA centers must be done. This method evaluates individual's performance in testing procedure, results reporting, capabilities of laboratories and testing points on performing HIV rapid testing. Sites failing the proficiency tests need to receive additional technical supervision and support. In addition, onsite evaluation must be done at least twice a year.

Internal quality control (IQC)

For IQC, trained laboratory technicians in the facility should regularly perform the IQC to verify the quality before the test kits are being used by the testing units.

3.9. Linking People Diagnosed with HIV Infection

Linkage is defined as a process of actions and activities that support people testing for HIV and people diagnosed with HIV to engage with prevention, treatment, and care services as appropriate for their HIV status. For people with HIV, it refers to the period beginning with HIV diagnosis and ending with enrolment in care or treatment. It is critical for people living with HIV to enroll in care as early as possible. This enables timely initiation of ART as well as access to interventions to prevent the further transmission of HIV, prevent other infections and co-morbidities and thereby to minimize loss to follow-up.

Linkage to HIV treatment, prevention, care, support, and other relevant services is the primary responsibility of HIV testing services and the testers and providers delivering HIV testing services. Multiple factors may hinder successful linkage to care, including distance from services, transport costs, long waiting times at the facility and, for those testing positive, stigma and disclosure-related concerns.

Linkage to HIV care should be improved through interventions that support people in the initial steps in the continuum of care. Such interventions may vary based on the local context, including: the health-care delivery systems, geography, and target population. A combination of interventions is needed to improve linkage to prevention, care, and treatment for specific groups at risk.

Post-test counselling messages remain key. They should be concise, addressing the needs of the client and focusing on supporting linkage to care. Post-test counselling messages need to be tailored to specific populations and their situations and whether their test results are HIV-positive, negative, inconclusive or they already know their status and need to engage, re-engage in care. Messages need to provide clients with the latest information, including:

- The personal health benefits of early ART initiation
- People living with HIV receiving ART who achieve and maintain Undetectable Viral Load result has minimum risk of transmission to their partners
- The benefits of voluntary provider-assisted referral for people living with HIV.

All people with HIV-positive diagnoses should be offered a package of support interventions that ensure timely linkage to care including ART initiation with in/outside the health facility, friendly and flexible services designed to suit specific population groups and digital platform. Providers should note that people who are HIV-negative but at ongoing risk also need to be linked to effective prevention services.

The following interventions have demonstrated benefit in improving timely linkage of PLHIV after an HIV diagnosis:

1. Efficient interventions to reduce time between diagnosis and engagement in care, including support for HIV disclosure, tracing and training staff.

2. Peer support and navigation approaches for linkage

3. Quality improvement approaches using data to improve linkage.

Good practices of linkage to care at site level

The recommended good practices to improve linkage of HIV positive person to care and treatment services after the person is found positive should be implemented at all sites.

A. Implement standardized service delivery system that will improve referral and linkage between HIV chronic care through:

- Prepare standard operating procedure (SOP) for inter- and intra- facility service outlets referral linkage system
- Strengthen referral system between health facility and community by using SOP.
- Establish site level support groups to improve escorting/accompanied referral and feedback practices for intra-facility referral.
- Establish facility and catchments area level regular referral linkage auditing system to ensure that all new HIV infected clients are linked to ART.
- Map and establish network between available, chronic care, and other support services in the area (linkage service directory).
- Preparing, avail and utilize service directory which can be availed in soft copy or other forms.

- Ensure that a referring and accepting health care facilities or sites are accountable for assurance of the client's referral is successful.

- Strengthen Post-test counseling in such a way that the client understands the benefits of ART; develop trust and confidence on the provider and reaches to informed decision on linkage.

- Promote health seeking behavior for service utilization

- Educate clients on benefits of early ART initiation and related care.

- Call or text to the client and remind him/her for linkage service.

B. Standardized documentation, reporting system and feedback practice through:

- Ensure the availability and sustainability of recording and reporting formats.
- Ensure a referral and linkage feedback mechanism in health facility.
- Ensure standardization of guidelines and training materials on referral and linkages issues.

C. Improve the engagement of Health Extension Workers (HEW), PLHIV in awareness creation activities:

- Support HEWs in their day-to-day information education and communication (IEC) / behavioral change communication (BCC) activities in relation to HIV.

- Establish and strengthen PLHIV associations and support groups to be involved in the facilitation of referral and linkage through escorting and other mechanisms.

D. Reduce stigma and discrimination through:

- Community involvement
- Identification and analysis of the root cause of stigma and discrimination.
- Development of IEC/BCC material and utilize media focusing on stigma and discrimination.
- Advocacy of gender inequality that predisposes to stigma and discrimination.
- Leadership role in community activities to address stigma and discrimination through contextual available values and norms of the community.
- Involvement of PLHIV to reduce stigma and discrimination and to be part of prevention and care services.

3.10. Ethical and Legal Recommendations for HIV Testing Service

- Knowingly transmitting HIV shall impose punitive legal measures.
- HTS shall be integrated into existing health and social welfare services and be available in all settings for KPP: government, non-governmental, private sector, workplace, and faith-based organizations.

- HTS shall be strengthened through effective networking, consultation, and collaboration among stakeholders.

- HTS shall be standardized nationwide and shall be supervised, supported, and regulated by appropriate government health authorities.

- Informed consent for testing shall be obtained in all cases unless it is a mandatory testing.

- Mandatory HTS should be permissible for the following conditions:

1. Screening purposes of blood and blood components for transfusion,
2. In cases of organ or tissue transplantation, and
3. Order of court.

- Adequate pre-test information needs to be given for ANC clients, TB and STI patients.

- All clients who are eligible for HTS based on HIV risk screening tool shall be offered pre-test information and post-test counseling.

- Test results, positive or negative, shall be declared to clients in person and must be provided with post-test counseling.

- HIV test results will not be provided in certificate form; however, referral will be offered to access prevention, care, and treatment services.

- Clients' confidentiality will always be maintained, however:
 - Results can be shared with other health care providers with those involved in clinical management of clients as part of shared confidentiality.
 - Clients can be referred for HIV services or other services while stating his/her HIV status if required or upon request.
 - Cases of altered state of consciousness and his/her partner or care giver are at risk of infection, based on the circumstances the endangered partner or care giver shall have the right to access the information regarding the sero-status of the index case.
- Couples shall be encouraged to be counseled, tested, and receive results together.
- Partner notification shall be encouraged in cases where one partner receives the results alone.
- In some special cases, such as child adoption, a counselor may refuse a testing request when this is not in the best interest of the child.
- Children who have been sexually abused and put at risk of HIV infection shall receive counseling, be encouraged to test for HIV and helped to access appropriate services.
- The result of HIV testing is the property of the child tested and shall not be disclosed to third parties unless clearly in the best interest of the child
- Youth-friendly counseling and testing services shall be made widely available for youth population based on HIV risk screening tool.
- People with physical disabilities and mental impairment require special care when providing counselling and testing services, particularly regarding communication shall accommodate the special needs of people with visual and hearing impairments by adopting appropriate medium of communication.
- Individuals under the immediate influence of alcohol or addictive drugs (substance use) shall not be offered HIV testing due to a mental inability to provide informed consent.
- For a mentally impaired individual requires the knowledge and consent of his/her guardian and should be for the benefit of the mentally impaired individual or patient.

Ethics in counseling

- All service providers shall abide by the rules, regulations and protocols contained in this document and other related national guidelines
- All service providers shall observe the ethical requirements of confidentiality, informed consent, proper counseling, anonymity and privacy.
- Shared confidentiality shall be promoted.

CHAPTER FOUR:

CARE AND TREATMENT OF PEOPLE LIVING WITH HIV INFECTION

4.1. General Care Packages for PLHIV

It is critical for people living with HIV to initiate ART as early as possible. This enables to shorten the time between HIV diagnosis and ART initiation hence significantly reducing HIV related morbidities and mortality, and transmission of HIV.

As all PLHIV are eligible for ART, enrolment in care provides an opportunity for close clinical and laboratory monitoring, early assessment and timely prevention and management of opportunistic infections and other comorbidities. Many interventions are relevant across the full continuum of care, including care for HIV-exposed individuals and PLHIV before initiation of treatment. The following critical interventions need to be addressed at initial encounter with the client:

- Confirm HIV status by retesting and enrolling into HIV care (including recording into HIV positive tracking and pre-ART registers).
- Ensure any OI and other clinical problems that may delay ART initiation are ruled out or addressed.
- Ensure initiation of ART within two weeks after TB treatment is started except when signs and symptoms of meningitis (both TB and/or cryptococcal meningitis) are present.

- Ensure that barriers to adherence and treatment continuity are assessed and addressed accordingly.
- Ensure client is fully aware and makes informed decision for early initiation and continuation of treatment

Key elements of chronic HIV care include:

- Retesting for verification
- Complete clinical assessment (history taking, complete physical examination and relevant lab tests)
- WHO clinical staging
- Prevention, screening and management of opportunistic infections and co-morbidities (see chapter 5)
- Rapid ART initiation
- Patient monitoring and follow up
- Support for disclosure and assisted partner notification
- Risk reduction counseling and combination HIV prevention approaches

- Screening for and managing mental health problems and substance use
- Adherence and psychosocial counseling and support
- Nutritional assessment and counseling
- Screening for other STIs
- Prevention screening and treatment of cervical cancer.
- Management of pain and symptoms.
- Pregnancy status, family planning and contraception.
- Document all relevant client information

4.2. Advanced HIV Disease

Definition of advanced HIV disease

For adults and adolescents, and children older than five years, advanced HIV disease is defined as CD4 cell count <200 cells/mm³ or WHO stage 3 or 4 event.

All children younger than five years old with HIV are considered as having advanced HIV disease.

Diagnosis of Advanced HIV Disease:

Diagnosis of advanced HIV disease is done through CD4 testing of clients at base line for those initiating treatment, (re-engaging with care after a period of interruption for >28 days) and targeting those who have interrupted ART treatment and with persistently unsuppressed VL (>1000 copies per ml). In addition to CD4 testing and when CD4 testing is unavailable, a clinical diagnosis of WHO stage 3 or 4 can also be used to diagnose advanced HIV disease.

Package of care for PLHIV with advanced disease

In Ethiopia among adults (ages 15-64 years) living with HIV, 35.8% (45.6% of men and 31.1% of women) had immunosuppression with CD4 count less than 350 cells per microliter (μL). Whereas among adults who reported that they were unaware of their HIV-positive status, 22.0% had severe immunosuppression which is a CD4 count less than 200 cells/μL (16.9% of men and 26.5% of women), (EPHIA 2017/2018). People with advanced HIV disease are at high risk of death, even after starting ART, the risk will increase with declining CD4 cell count. The most common causes of death are TB, severe bacterial infections and cryptococcal meningitis.

A package of interventions including screening, treatment and/or prophylaxis for major opportunistic infections, rapid ART initiation and intensified adherence support interventions should be offered to everyone presenting with advanced HIV disease including those

who are re-engaging with care after a period of interruption for >28 days. Baseline CD4 cell count testing for all PLHIV remains clinically important in order to identify those who have

advanced HIV disease and who should be offered the package of care as presented in table 4.1.

Table 4.1: WHO recommendations for components of packages of care for people with advanced HIV disease.

Areas for the package	Intervention	CD4 cell count	Adults & Adolescents	Children
Diagnosis	Sputum Xpert ® MTB/ RIF as the first test for TB diagnosis among symptomatic PLHIV	Any	Yes	Yes
	LF-LAM for TB diagnosis among people with symptoms and signs of TB	≤200 cells/mm ³ (inpatient) ≤100cells/mm ³ (outpatient) Or any CD4 count with TB symptoms or if seriously ill.	Yes	Yes
	Cryptococcal antigen screening	<100 cells/mm ³	Yes	No
Prophylaxis & preventive treatment	Co-trimoxazole prophylaxis	≤350 cells/ mm ³ or clinical stage 3 or 4	Yes	Yes ^z
	TB preventive treatment ^a	Any	Yes	Yes ^a
	Fluconazole pre-emptive therapy for cryptococcal antigen– positive people without evidence of meningitis ^b	<100 cells/ mm ³	Yes	NA
ART initiation	Rapid ART initiation as mentioned above ^c	Any	Yes	Yes
	Defer initiation if clinical symptoms suggest TB or cryptococcal meningitis	Any	Yes	Yes
Adapted adherence support	Tailored counselling to ensure optimal adherence to the advanced disease package, including home visits if feasible	<200 cells/mm ³	Yes	Yes

a. TB preventive treatment should be provided in accordance with current guidance.

b. When cryptococcal antigen screening is not available, fluconazole primary prophylaxis should be given to adults and adolescents living with HIV who have a CD4 cell count of < 100 cells/mm³.

c. People receiving a positive WHO four-symptom screen should initiate ART while being evaluated for TB if clinical signs and symptoms of meningitis are absent.

4.3. Preparing People Living with HIV for ART

Before initiating people on ART, assess client's willingness and their readiness to initiate ART, discuss in detail about the ARV regimen, dosage, the likely benefits and possible adverse effects, and agree on the required follow-up and monitoring visits. For children with HIV, this conversation should directly involve the parent/legal guardian and include discussion about disclosing their HIV status. Initiation of ART in children should consider nutritional status, any co-morbidities and potentially interacting medications for possible contraindications or dose adjustment.

After going through detailed discussion, quickly assess the readiness and offer rapid ART initiation. Check the following issues to assess readiness of the clients:

- Patient understands the benefits of ART, adherence and need for lifelong commitment and appointment schedules.
- Patient understands possible side effects of ARVs.
- Patient understands the importance of disclosure and family support.

If clients or caregiver declines initiating ART as early as possible, address the following issues in the subsequent visits with the aim of initiating ART as soon as possible:

- Identify and address barriers for starting ART.

- Reiterate the benefits of ART for the client's own health and for reduced risk of transmission to partners
- Consequences of delaying initiation.
- Encourage disclosure to partners/family.
- Encourage engagement in peer and community support.

NB: Clients who declined initiation need to be seen repeatedly on a weekly bases for continued counseling and support to initiate ART.

If there is mental health problem, substance use or psychosocial problems that are major barriers to initiation, appropriate support should be provided, and readiness to initiate ART should be reassessed at regular intervals. Utilize a range of patient information materials as well as community and peer support to help the person's readiness and decision to start therapy.

People starting treatment and care givers should understand that the first ART regimen offers the best opportunity for effective virological suppression and immune recovery, and that successful ART requires them to take the medications exactly as prescribed. They should be advised that many adverse effects are temporary or may be treated, or that substitutions can often be made when necessary. People receiving ART and care givers should also be asked regularly about any other medications that are taken, including herbal remedies and nutritional supplements.

People receiving ART should understand that; they should continue practicing safer sex (including condom use) and avoidance of other high-risk activities, such as sharing of injecting equipment, to prevent transmitting HIV to other people.

In conclusion the following principles should be considered:

- ART should be started rapidly based on a client's informed decision.
- Interventions should be implemented to address barriers to ART.
- HIV programs should promote treatment literacy among all PLHIV, including providing information on the benefits of early treatment, the lifelong commitment required, the risks of delaying treatment and available adherence support.
- Care providers should support shared decision-making.

4.4. When to Start ART

All HIV positive individuals are eligible for ART. The ideal time for ART initiation is at time of HIV diagnosis. Understanding of clients about HIV and the importance of life long treatment adherence need to be emphasized. All adherence barriers should be exhaustively assessed and addressed while ART is initiated.

Rapid ART initiation is defined as initiation of ART within seven days of HIV diagnosis, if there are no contraindications. Rapid ART initiation should be offered to all PLHIV following a

confirmed HIV diagnosis, clinical assessment, and assessment of client readiness except in the case of TB meningitis and cryptococcal meningitis. ART initiation should be offered on the same day (initiating ART on the date of HIV diagnosis), for people who are ready to start. Rapid ART initiation, including same-day increases the number of people starting ART, reduces mortality, and may further reduce both mother-to-child transmission and transmission to HIV-negative partners. This recommendation applies to all PLHIV at all age groups and is particularly important in people with very low CD4 cell counts who have an increased risk of death.

4.4.1. When to Start ART in Adults and Adolescents

Rapid treatment initiation is associated with clinical and HIV prevention benefits, improving survival, and reducing the incidence of HIV infection at the family and community level. Start ART rapidly, preferably same day, to all adults and adolescents with a confirmed HIV diagnosis who are ready and willing regardless of their WHO clinical stages and CD4 counts. Prenatally infected adolescents need special consideration and support as they may have readiness issues to accept HIV status and decision to start ART.

4.4.2. When to Start ART in Pregnant and Breast-feeding Women

Start ART rapidly, preferably same day, to all pregnant and breastfeeding women living with HIV regardless of their WHO clinical stages and CD4 counts. For women identified at labor and delivery, provide ART within the same hour of

HIV diagnosis with brief counseling and provide detailed counseling on ARV and adherence after delivery. Make sure to provide enhanced ARV prophylaxis for the infant immediately after birth.

4.4.3. When to Start ART in Children

Start ART rapidly, preferably same day, to all children living with HIV regardless of their WHO clinical stages and CD4 counts/percentage. Infants and young children infected with HIV have exceptionally higher morbidity and mortality. Up to 52% and 75% of children die before the age of two and five years respectively in the absence of any intervention.

For HIV infected infants diagnosed with the first DNA PCR result, initiate ART and take DBS specimen for confirmatory DNA PCR. Continue ART if the second DNA PCR confirms positive results; whereas if the second DNA PCR turns negative, without holding the ART, make the 3rd DNA PCR test. (Refer the algorithm, Figure 3.3)

For HIV infected infants and younger children who need particular care and support, ensure the readiness and understanding of their parents and care givers. Counseling on dosage and administration should be provided for parents/ care givers in case of HIV infected infants.

4.4.4. When to start ART in adults, adolescents, and children with TB

Start ART in all TB patients living with HIV as soon as possible within 2 weeks following initiation of anti-TB treatment regardless of their CD4 count except when there is TB meningitis. If a patient has TB meningitis, delay ART for at

least 4 weeks and initiate within 8 weeks after treatment of TB meningitis is initiated.

4.4.5. When to start ART in adults, adolescents, and children with drug resistant TB

Antiretroviral therapy is recommended for all patients with HIV and drug- resistant TB (both MDR/XDR-TB) requiring second-line anti-tuberculosis treatment, irrespective of CD4 cell count, as early as possible (within the first two weeks) following initiation of anti-tuberculosis treatment.

4.4.6. When to start ART in HIV/HBV co-infected patients

Start ART rapidly for all HIV/HBV co-infected patients regardless of their CD4 count.

4.4.7. When to start ART in HIV/ Cryptococcal meningitis co-infected patients

ART should be delayed by 4-6 weeks following initiation of treatment for cryptococcal meningitis. Earlier ART is associated with more severe adverse event and increased mortality with cryptococcal meningitis.

ART should be initiated for all individuals (children, adolescents, and adults) living with HIV rapidly, preferably same day, (within an hour for laboring mother) after confirming HIV diagnosis, regardless of WHO clinical stage and CD4 cell count except for TB and cryptococcal meningitis.

4.5. What ART regimen to start with (first-line ART)

Using simplified, less toxic, more effective, and convenient regimens as fixed-dose combination is recommended for first-line ART.

Table 4.2. Summary table for what ART regimen to start with (first-line ART)

Population	Preferred first-line regimens	Alternative first-line regimens	Special circumstances ^a
Adolescents(10 to 19 years OR weight ≥30 kg), adults, pregnant, childbearing and breast feeding women including those with TB/ HIV- co infection	TDF+3TC+DTG ^b (FDC)	TDF + 3TC + EFV AZT + 3TC + DTG AZT + 3TC + EFV	AZT+ 3TC + ATV/r TDF+ 3TC+ ATV/r ABC ^c +3TC+DTG
Children > 4weeks and ≥3kg but less than 10 years	ABC + 3TC + DTG ^{b*}	ABC+ 3TC+LPV/r AZT+3TC+DTG	ABC+3TC+EFV ^d AZT+3TC+EFV AZT+3TC+LPV/r ^a

^a ABC or boosted PIs (ATV/r, LPV/r) can be used in special circumstances for those clients who could take neither DTG nor EFV due to contraindication and/or side effects.

^b In case of TB-HIV co-infection, the dose of DTG should be 50mg BID.

^{b*}In case of TB-HIV co-infection, the dose of DTG should be doubled depending on body weight of the child.

^c **For PLHIV with renal insufficiency and anemia**

^d **EFV is for children 3 years and older.**

4.5.1. First-line regimen for adults and adolescents

The preferred first-line regimen for adults and adolescents (>10 years of age or >30 kg body weight) including pregnant and breast feeding women is TDF+ 3TC+DTG as a once-daily dose.

4.5.2. First line ART for children

The preferred first-line regimen for children > 4weeks and ≥3kg but less than 10 years is ABC+3TC+DTG. Availability of a new generic formulation of 10mg DTG in 2021 has created the opportunity to use DTG based regimen for children living with HIV (CLHIV) who are at least 4weeks of age and weight 3kg or more.

4.5.3. Consideration for first line ART regimen for PLHIV on TB treatment

In case of TB-HIV coinfections in adults and adolescents, including pregnant/breast feeding women and children >20kg body weight, the dose of DTG should be 50mg BID. For children less than <20kg, the dose of DTG depends on the exact body weight of the child.
(Refer Annex 8)

4.5.4. Consideration for alternative and special circumstance first line ART regimens

If the preferred regimen is contraindicated or not available, alternative regimen is used. These regimens are effective and tolerable but have potential disadvantages when compared with the preferred regimens. An alternative regimen may be a preferred regimen for some clients.

Special circumstances may include situations where preferred or alternative regimens may not be available or suitable because of significant toxicities, anticipated drug-drug interactions.

4.6. Monitoring response to ART

Monitoring of patients on ART should start from the day of initiation. Although taking ART is a lifelong commitment, the first six months of therapy are especially important.

4.6.1. What to expect in the first months of ART and how to manage them

Clinical and immunological improvement and virological suppression are expected when individuals adhere to ART, but care providers need to be alert as opportunistic infections and/or immune reconstitution inflammatory syndrome (IRIS) may develop, as well as early adverse drug events, such as drug hypersensitivity, in the first three months of ART.

ART significantly decreases mortality and HIV related illnesses; however, mortality can be higher in the first three to six months of ART initiation among people who started ART with advanced HIV disease with existing co-infections and/or co-morbidities, severely low hemoglobin, low body mass index (severe malnutrition) and/or very low CD4 counts.

In most adults and children, when ART is initiated, immune recovery starts and CD4 cell counts rise. Generally, this increase occurs during the first year of treatment, and then continues to rise further during the second year. However, severe immune-suppression may persist in some individuals who do not

experience a significant rise in CD4 cell count with treatment, especially those with a very low CD4 cell count at the time of ART initiation. Consideration should be given to continue prophylaxis for OI such as CPT till patients recover immunologically.

Immune Reconstitution Inflammatory Syndrome (IRIS)

Immune Reconstitution Inflammatory Syndrome (IRIS) is a spectrum of clinical signs and symptoms thought to be associated with immune recovery brought about by a response to ART. In patients with advanced HIV disease the risk of IRIS is high, it even goes higher when CD4 count gets below 50 cells/mm³. IRIS may present in two different ways: paradoxical IRIS, when an opportunistic infection or tumor diagnosed before ART initially responds to treatment but then deteriorates after ART starts; or unmasking IRIS, in which initiating ART triggers disease that is not clinically apparent before ART. It should be considered only when the presentation cannot be explained by a new infection, expected course of a known infection or drug toxicity.

The clinical spectrum is diverse, and IRIS has been reported for many different infections, tumors and non-infectious conditions. The most serious and life-threatening forms of paradoxical IRIS are due to TB, Cryptococcus, Kaposi's sarcoma and hepatitis. BCG vaccine-associated IRIS (localized and systemic) may occur in some HIV infected infants. IRIS is generally self-limiting, and interruption of ART is rarely indicated, but people may need to be reassured in the face of protracted symptoms to prevent discontinuation of or poor adherence to ART.

Timing of ART in people with opportunistic infections requires balancing a greater risk of IRIS after early initiation against continuing high mortality if ART is delayed.

Health care providers need to give due consideration for patients with signs and symptoms of TB meningitis and cryptococcus infection before initiating ART. Providers should give close attention to patients with risk factors for IRIS, including those presented with CD4 count <50 cells/mm³, disseminated opportunistic infections or tumors.

The most important steps to reduce the development of IRIS include:

- Earlier HIV diagnosis and rapid initiation of ART
- Improved screening and management of TB and Cryptococcus before ART
- Optimal management of opportunistic infections

The diagnosis of IRIS can be challenging, and typically has the following criteria:

- A low pretreatment CD4 count (often less than 100 cells/mm³) except in tuberculosis. IRIS secondary to preexisting M. tuberculosis infection may occur in individuals with CD4 counts >200 cells/mm³

- The absence of evidence of alternate diagnoses, including drug-resistant infection, bacterial super infection, drug allergy or other adverse drug reactions, and reduced drug levels due to non-adherence drug-drug interactions or mal-absorption. ;
- The presence of clinical manifestations consistent with an inflammatory condition; and
- A temporal association between antiretroviral therapy (ART) initiation and the onset of clinical features of illness- usually within the first 3 months.

Management of IRIS

- Patients should generally be treated for the underlying OI as soon as possible.
- Continuation of ART when IRIS occurs.

Role of anti-inflammatory agents:

Anti-inflammatory agents may be particularly helpful in the setting of obstructive mass lesions (e.g. expanding cervical lymph node). Use of anti-inflammatory agents, particularly corticosteroids, must be weighed against potential risks and side effects. When a decision is reached to treat with corticosteroids, initiate therapy with prednisone at a dose of 1 mg/kg/day (maximal dose 60 to 80 mg) followed by a rapid taper over a 10 to 14-day period. IRIS in closed spaces (e.g. CNS OI) should be managed promptly or referred to appropriate center to avert significant morbidity and mortality.

IRIS is not indicative of treatment failure or drug side effect. It is a transient phenomenon and is not a reason to stop ART or change regimen. The OI should be treated using standard guidelines and in critically sick patients short course of corticosteroid might be indicated to control severe symptoms.

4.6.2. Clinical and laboratory monitoring

Standardized clinical assessment of patients and, when available baseline CD4 count, are important to determine the severity of immunosuppression and decide on initiation of prophylactic therapies. Patients shall be thoroughly evaluated at baseline and at periodic follow up visits to monitor for response to treatment, IRIS, adverse effects, and treatment intolerance or non-adherence. Patient readiness before ART initiation and treatment adherence at subsequent visits should always be assessed, and necessary support should be provided when barriers are identified. Opportunistic infections including TB, cryptococcal infection, and other co-morbidities including non-communicable diseases (NCDs) should be regularly assessed and managed.

Table 4.3. Baseline and follow-up assessment

Baseline assessment, week 0 (First encounter) Objective: to conduct initial assessment.	
Assess	Act
■ Check/Assess if retesting for verification is done and documented. ■ Assess: socio-economic status, any HIV related illnesses in the past, advanced HIV disease, symptom screen for TB, other OI, co-morbidities, pregnancy, past and current medication. ■ Determine WHO staging. ■ Clinical records: fill intake form, follow up form and registers (positive tracking, pre ART or ART registers). ■ Counselling and education: adherence, treatment readiness, disclosure, and address adherence barriers. ■ BP and BMI	■ Do retesting for verification if not done Determine treatment readiness. ■ Start CPT and TPT if clinically indicated Consider provision of prophylaxis for other OIs (e.g. Fluconazole) ■ Treat OIs. ■ Manage co-morbidities ■ Start ART for those who are ready and have no adherence barriers; and give appointment to return after two weeks. If not, give appointment to return within one week. For those who still defer ART continue to counsel on benefit of rapid ART initiation and start them on ART as early as possible. (N.B. Continue follow up counselling sessions until the patient is initiated on ART). ■ Patients who are not initiated on ART due to meningitis (TB/cryptococcus), (Refer section 4.4.7.) ■ Continue ART for transfer-ins. ■ Screen for STI ■ Screen for cervical cancer for women between 15-49 ■ Refer if necessary.
2nd visit: 2 weeks after initiation Objective: To determine toxicity/intolerance, adherence, and IRIS	
■ Clinical assessment for: IRIS, toxicity etc. ■ Assess and support adherence, addressing adherence barriers. ■ Provide counseling and education including prevention of HIV transmission. ■ Lab tests if necessary. ■ Support disclosure if not done. ■ BP and BMI	■ Manage toxicity as indicated. ■ Treat OI if diagnosed. ■ Give appointment to return in 2 weeks. ■ Screen for cervical cancer for women between 15-49 if not done ■ Screen for STI ■ Refer if necessary.

3rd visit: 4 weeks after initiation

Objective: Same as second visit

- | | |
|---|---|
| <ul style="list-style-type: none">■ Same as 3rd visit.■ Hgb if patient is on AZT.■ BP and BMI | <ul style="list-style-type: none">■ Refill ART and other medicines as necessary for one month.■ Treatment of OI and comorbidities if identified.■ Manage drug toxicity and intolerance.■ Assess and provide adherence support and patient education including HIV prevention.■ Do other lab tests as indicated■ Screen for STI■ Screen for cervical cancer for women between 15-49 if not done■ Refer if necessary.■ Appointment to return after 4 weeks. |
|---|---|

4th visit: 8 weeks after initiation

Objective: same as 3rd visit

- | | |
|---|--|
| <ul style="list-style-type: none">■ Same as 4th visit.■ BMI and BP | <ul style="list-style-type: none">■ Refill ART and other drugs as necessary for 1 month.■ Treatment of OI and co-morbidities.■ Manage toxicity and intolerance.■ Provide adherence support and patient education including HIV prevention.■ Screen for STI■ Screen for cervical cancer for women between 15-49 if not done■ Refer if necessary.■ Appointment to return after 4 weeks. |
|---|--|

5th visit: 12 weeks after initiation

Objective: Same as 4th visit.

- | | |
|---|--|
| <ul style="list-style-type: none">■ Same as 5th visit.■ BP and BMI | <ul style="list-style-type: none">■ Refill ART and other drugs as necessary for 1 month.■ Treatment of OI and other co-morbidities.■ Manage toxicity and intolerance.■ Assess and provide adherence support and patient education including HIV prevention.■ Screen for STI■ Screen for cervical cancer for women between 15-49 if not done■ Refer if necessary.■ Appointment to return after 4 weeks |
|---|--|

6th visit: 16 weeks after initiation

Objective: Same as 5th visit

- | | |
|---|---|
| <ul style="list-style-type: none">■ Same as 6th visit.■ BP and BMI | <ul style="list-style-type: none">■ Refill ART and other drugs as necessary for 1 month.■ Treatment of OI and other co-morbidities.■ Manage toxicity and intolerance.■ Assess and provide adherence support and patient education including HIV prevention.■ Screen for STI■ Screen for cervical cancer for women between 15-49 if not done■ Refer if necessary.■ Appointment to return after 4 weeks. |
|---|---|

7th visit: 20 weeks after initiation

Objective: Same as 6th visit

- | | |
|---|--|
| <ul style="list-style-type: none">■ Same as 6th visit.■ BP and BMI | <ul style="list-style-type: none">■ Refill ART and other drugs as necessary for 1 month.■ Assess and provide adherence support and patient education including HIV prevention.■ Treat OI and other co-morbidities.■ Manage toxicity and intolerance.■ Screen for STI■ Screen for cervical cancer for women between 15-49 if not done■ Refer if necessary.■ Appointment to return after 4 weeks. |
|---|--|

8th visit: 24 weeks after initiation

Objective: Same as 7th visit

- | | |
|---|--|
| <ul style="list-style-type: none">■ Same as 6th visit.■ BP and BMI | <ul style="list-style-type: none">■ Assess and provide adherence support and patient education including HIV prevention.■ Determine viral load.■ Refill ART and other drugs as necessary for 3 months.■ Treat OI and other co-morbidities.■ Manage toxicity and intolerance.■ Screen for STI■ Screen for cervical cancer for women between 15-49 if not done■ Refer if necessary.■ Appointment to return after 3 months. |
|---|--|

Note:

- Newly started patients will be appointed every two weeks during the first month of treatment and every 4 weeks (every month) then after until 24 weeks of treatment. After the 24th week of initiation of antiretroviral therapy patients will be scheduled to return every twelve weeks.
- OIs prophylaxis, treatment for co-morbidities and SRH services should be provided in the context of/integrated with ART multi month dispensing.
- Patients should be encouraged to come at any time if they have concerns and can be seen out of the above schedule whenever necessary.
- For follow up schedule during COVID19 pandemic, refer service delivery section (Chapter 6).

Table 4.4. Recommended tests for HIV treatment monitoring and approaches to screening for co-infections and non-communicable diseases.

Management	Recommended	Desirable (if feasible)
Baseline/ART initiation	■ CBC including hemoglobin^d ■ CD4 cell count. ■ Cryptococcus antigen if CD4 cell count <100 cells/mm³. ■ FBS ■ Gene Xpert, LF-LAM if eligible	■ HBV (HBsAg)^a serology. ■ HCV serology. ■ Pregnancy test. ■ HPV-DNA if available ■ Lab investigations for major non communicable chronic diseases and comorbidities^b ■ Pregnancy test ■ Serum creatinine and estimated glomerular filtration rate (eGFR) for starting TDF^e

Management	Recommended	Desirable (if feasible)
Receiving ART	■ HIV viral load at 6 and 12 months after initiating ART and every 12 months thereafter ■ Viral load testing for pregnant mothers: ■ Newly diagnosed mothers: Conduct a viral load by three months after ART initiation to ensure that there has been rapid viral suppression. If viral load testing is expected to be undertaken near the planned viral load at 34–36 weeks of gestation, the first viral load test can be delayed until weeks 34–36 of gestation. ■ For women already on ART, conduct VL testing at the first contact at ANC (VL result conducted in the last 3 months before the first contact can also be used), at 34–36 weeks of gestational age or delivery at the latest, followed by three months after delivery and then every 6 months. ■ For those who are already on ART with previous VL test conducted more than three months back repeat VL test at first ANC contact / PMTCT visit, at 34–36 weeks of gestational age (or at the latest at delivery) and 3 months after delivery and every six months thereafter until MTCT risk ends. ■ For all pregnant women, regardless of ART initiation timing: conduct viral load testing at 34–36 weeks of gestation (or at the latest at delivery) to identify women who may be at risk of treatment failure and/or may deliver infants at higher risk of perinatal transmission ■ For all breastfeeding women, regardless of when ART was initiated: conduct a viral load test three months after delivery and every six months thereafter to detect viremic episodes during the postnatal period ■ FBS at 6 months after ART and every 12 months annually.	■ Serum creatinine and eGFR for TDFc ■ Pregnancy test, especially for women of childbearing age not receiving family planning. ■ CD4 cell count if indicated ■ HPV-DNA if available

^a If feasible, HBsAg testing should be performed at baseline to identify people with HIV and HBV coinfection and who should therefore initiate TDF-containing ART.

^b Consider assessing for the presence of chronic conditions that can influence ART management, such as hypertension, other cardiovascular diseases, diabetes and TB according to the WHO Package of Essential NCD interventions (PEN), mental health Gap Action Program (mhGAP) or national standard protocols

^c Monitoring may include a range of tests, including serum creatinine and estimated glomerular filtration rate (eGFR), serum phosphate and urine dipsticks for proteinuria and glycosuria. $eGFR = 140 - \frac{\text{age (years)} \times \text{body weight (kg)}}{72 \times \text{serum Cr in mg/dL}}$ for male and $eGFR = 140 - \frac{\text{age (years)} \times \text{body weight (kg)} \times 0.85}{72 \times \text{serum Cr in mg/dL}}$ for female.

^d Among children and adults with a high risk of adverse events associated with AZT (low CD4 or low BMI).

^e Among people with a high risk of adverse events associated with TDF: underlying renal disease, older age group, low body mass index (BMI), diabetes, hypertension and concomitant use of a boosted PI or potential nephrotoxic drugs.

N.B. If the recommended baseline tests are not available, it should not delay ART initiation

4.6.3. Monitoring drug toxicities and substitution of ARV

Guiding principles

- Establish whether the clinical condition is due to ARV toxicities, other drugs, or other illness including new OIs.
- Try to identify the responsible ARV drug.
- Assess the severity using toxicity grading matrix (Annex 10 and 11)

Major types of ARV toxicities

The major causes of drug discontinuation in the first 3-6 months after initiating ART are due to drug toxicities; and hence, they must be closely monitored.

They typically occur from few weeks to months after ART initiation or change. The most common side effects related to the ARV drugs recommended in these national guidelines are provided in the table below.

Table 4.5. Types of toxicities associated with first, second and third-line ARV drugs

ARV Drug	Major types of toxicity	Risk factors	Suggested management	Remark
ABC	Hypersensitivity reaction		Substitute with TDF or AZT	
ATV/r	Electrocardiographic Abnormalities (PR and QRS interval prolongation)	Pre-existing conduction system disease. concomitant use of other drugs that may prolong the PR interval	Use with caution in people with preexisting conduction disease or who are on concomitant drugs they may prolong the PR or QRS interval	
	Indirect hyperbilirubinemia (clinical jaundice)	Underlying hepatic disease, HBV and HCV co-infection Concomitant use of hepatotoxic drugs	This phenomenon is clinically benign; however, can be associated with social and personal discomfort and in some circumstances with stigma. Substitute with LPV/r only if adherence compromised	
	Renal stone	History of renal stone.	Substitute with LPV/r	
AZT	Anemia, neutropenia	Baseline anemia or neutropenia CD4 count ≤ 200 cells/mm ³ .	Avoid use of AZT for people with HIV and severe anemia at baseline (hemoglobin <7.0 g/dl) as first-line therapy. Substitute with TDF or ABC	
DRV/r	Hepatotoxicity	Underlying hepatic disease; Coinfection with HBV or HCV; Concomitant use of hepatotoxic drugs.	Substitute with ATV/r or LPV/r. When it is used in third-line ART, limited options are available.	
	Severe skin and hypersensitivity reactions.	Sulfonamide allergy.	For hypersensitivity reactions, substitute with another therapeutic class	
DTG*	Hepatotoxicity Hypersensitivity reactions	Coinfection with Hepatitis B or C, Liver disease.	Substitute another therapeutic class: EFV or boosted PIs	

ARV Drug	Major types of toxicity	Risk factors	Suggested management	Remark
	Insomnia Body weight gain or obesity	Older than 60 years Low CD4 or high viral load Female African ethnicity	Consider morning dose or substitute EFV, boosted PI Monitor body weight and promote anti-obesity measures (Such as diet and physical exercise). If significant increase despite measures, consider substituting EFV or boosted PI	
LPV/r	Electrocardiographic abnormalities (PR and QT interval prolongation, torsade's de pointes).	People with pre-existing conduction system disease; concomitant use of other drugs that may prolong the PR interval, Congenital long QT syndrome, Hypokalemia Concomitant use of drugs that may prolong the QT interval	Use with caution in people with pre-existing conduction disease or those on concomitant drugs that may prolong the PR or QRS intervals	
	Hepatotoxicity	Underlying hepatic disease HBV and HCV co-infection Concomitant use of hepatotoxic drug	If LPV/r is used in children, substitute with DTG or EFV according to age. ATV can be used for children older than 6 years. If LPV/r is used in second-line ART for adults, use ATV/r. If boosted PIs are contraindicated and the person has failed an NNRTI based first line ART use DTG or consult specialist	
	Pancreatitis	Advanced HIV disease, Alcohol class (INSTIs)	Substitute another therapeutic class (INSTIs)	
	Dyslipidemia	Cardiovascular risk factors such as obesity and diabetes	Substitute another therapeutic class (INSTIs)	
	Diarrhea	Risk factor unknown	Substitute with ATV/r or DTG	

ARV Drug	Major types of toxicity	Risk factors	Suggested management	Remark
NVP (For HEIs prophylaxis)	There are rare reports of Hypersensitivity and hepatotoxicity		Use AZT only	
EFV	Persistent central nervous system toxicity (such as abnormal dreams, depression or mental confusion)**	Depression or other mental disorder (previous or at baseline) Daytime dosing	DTG or boosted PI	
	Hepatotoxicity	Underlying hepatic disease – HBV and HCV co-infection Concomitant use of hepatotoxic drug		
	Convulsions	History of seizure		
	Severe skin Hypersensitivity reaction.	Risk factors unknown.	Use DTG or boosted PI	
	Gynecomastia	Risk factor unknown.	Use DTG or boosted PI	
TDF	Chronic kidney disease, acute renal injury, Fanconi syndrome.	Underlying renal disease; older age; BMI <18.5 (or body weight <50 kg); untreated diabetes mellitus; untreated hypertension; Concomitant use of nephrotoxic drugs or boosted PI.	Substitute with AZT or ABC. Do not initiate TDF at eGFR <50 mL/min, uncontrolled hypertension, untreated diabetes, or presence of renal failure. It is recommended to monitor growth in children taking TDF containing regimen.	
	Decreases in bone mineral Density.	History of osteomalacia and pathological fracture; risk factors for osteoporosis or bone loss.		
	Lactic acidosis or severe hepatomegaly with steatosis.	Prolonged exposure to nucleoside analogues; Obesity.		

*Hyperglycemia has been reported from different sites in Ethiopia and small-scale studies from other countries.

**Most CNS side effects will improve within 2-4 weeks after initiation

NB:

- For those patients with HBV and HIV co-infection suffering from TDF toxicity, consult or refer national viral hepatitis guideline.
- The clinical manifestations due to hypersensitivity reactions for some ARVs may be confused with IRIS

Drug substitutions for ARV drug toxicity

Drug regimen or single agent substitutions may be required for drug toxicity and to avoid drug interactions.

Clinical considerations

- Delaying substitutions or switches when there are severe adverse drug effects may cause harm and may affect adherence, leading to drug resistance and treatment failure.
- When drug interruptions are required, it is important to consider the various half-lives of ARV drugs. For example, when a NNRTI needs to be discontinued, a staggered approach should be used by prolonging the use of the NRTI backbone for two weeks except life threatening conditions (grade 4 conditions) where you have to discontinue all ARV drugs (see annexes 10 and 11 for details).

Strategies for managing adverse drug reactions:

Step 1: Establish whether the problem is due to antiretroviral drugs, other medications, OIs, non HIV related problems or clinical condition.

Step 2: Try to identify the responsible ARV drug.

Step 3: Assess the degree/severity of the Adverse Event using the ACTG/PACTG adverse events grading system.

Step 4: Manage the adverse event according to severity and decide whether to substitute or discontinue ARV drug based on common adverse events clinical grading system. Annexes 10 and 11.

Step 5: Report Adverse event through appropriate tools and channels (Refer chapter 6)

4.6.4. Drug interactions

Providers should be aware of all drugs that people with HIV are taking when ART is initiated and new drugs that are added during treatment maintenance.

Table 4.6. Key ARV drug interactions and suggested management

ARV drugs	Key interactions	Effect	Suggested management
AZT	Ribavirin and pegylated interferonalpha-2a		Substitute with TDF
EFV	Bedaquiline		Avoid the combination
	Amodiaquine, DHA/ piperazine		Use an alternative antimalarial agent or substitute EFV for DTG
	Artemisinin or lumefantrine		Use an alternative antimalarial agent or substitute EFV for DTG
			Risk of QT prolongation with ATV/r and LPV/r
	Methadone		Adjust the methadone dose as appropriate
	Hormonal contraceptives		Use alternative or additional contraceptive methods
	Amlodipine		Adjust the amlodipine dose as appropriate
	Simvastatin and atorvastatin		Adjust the statin dose as appropriate
Boosted PI (ATV/r, DRV/r, LPV/r)	Low-dose dexamethasone (COVID-19)		Double dose of dexamethasone
	Rifampicin		Adjust the PI dose or substitute with DTG
	Lovastatin and simvastatin	Increase concentration	Use an alternative dyslipidemia agent (for example pravastatin)
	Halofantrine and lumefantrine		Use an alternative antimalarial agent
	Estrogen-based hormonal contraception		Use alternative or additional contraceptive methods
	Methadone and Buprenorphine		Adjust methadone and buprenorphine doses as appropriate
	Astemizole and terfenadine		Use alternative antihistamine agent
TDF	Rifapentine	Reduce serum level	Do not provide 3HP with protease inhibitors and consider 6H in this case
	nephrotoxic drugs [e.g. aminoglycosides, amphotericin B, ganciclovir, pentamidine, vancomycin or interleukin-2]	Exacerbate nephrotoxicity	Avoid concurrent use

ARV drugs	Key interactions	Effect	Suggested management
	ritonavir boosted PIs		Closely monitor renal function
	Ledipasvir- or velpatasvir-containing Regimens		Monitor for TDF-associated adverse effects, including renal dysfunction, particularly when TDF is coprescribed with boosted HIV PIs
	Lithium		TDF: monitor renal function closely
DTG	Carbamazepine, Phenobarbital, and phenytoin		Use alternative anticonvulsant agent, (such as valproic acid or gabapentin) or if not possible substitute DTG with EFV and for children below 3 years substitute with boosted PIs
	Polyvalent Cation products containing Mg, Al, Fe, Ca, and Zn	Absorption of DTG is affected/ reduced	Use DTG at least 2 hours before or at least 6 hours after supplements containing polyvalent cations, including but not limited to – Fe-, Ca-, Mg-, or Zn-multivitamin supplements; mineral supplements, cation containing laxatives and Al-, Ca- or Mg- containing antacids. Monitor for virological efficacy.
	<i>Rifampicin,</i>	Increases metabolism of DTG, and hence reduces concentration of DTG in the blood.	DTG 50mg BID. For pediatrics DTG BID by weight band. Continue with twice daily dosing of DTG in children for 2 weeks after use of rifampicin has ended based on their weight band
	<i>Rifapentine</i>		No evidence that changes of dose of rifapentine or DTG is needed to achieve adequate exposures of DTG
	<i>Metformin</i>	DTG increases metformin levels, which may lead to hypoglycemia.	Start metformin at lowest dose and titrate based on glycemic control. Monitor for adverse events of metformin Or adjust metformin dosing if already on metformin When starting or stopping DTG in patients on metformin dose adjustment of metformin may be necessary to maintain optimal glycemic control and/or minimize adverse events of metformin.

Table 4.7: Drug-drug interactions (ARVs and anti-TB drugs for treatment of DR-TB).

Clinical Condition	Responsible ARV drug/s	Responsible anti TB drug/s	Recommendations
Bone marrow suppression	AZT	Lzd, R, H	Monitor blood counts regularly. Replace AZT if bone marrow suppression occurs. Consider suspension of Lzd. Also consider cotrimoxazole, if patient is taking.
Hepatotoxicity	EFV protease inhibitors, NRTIs	H, R, E, Z, Bdq, PAS, Eto/ Pto, FQ	When severe, stop both the ART and TB medications, and restart the TB medications first. Consider cotrimoxazole, if patient is taking. Also rule out viral hepatitis (Hepatitis A, B, C & Cytomegalovirus(CMV)).
Renal toxicity	TDF	Amino-glycosides, amikacin	TDF may cause renal injury. If possible, avoid TDF in patients receiving aminoglycosides or amikacin. If TDF is absolutely indicated, serum creatinine and electrolytes should be monitored (at least every two weeks). Even without the concurrent use of TDF, PLHIV have increased risk of renal toxicity secondary to aminoglycosides and amikacin. In the presence of renal insufficiency, ARV and anti-TB medications need to have their doses adjusted.
Central nervous system (CNS) toxicity	EFV	Cs, H, Eto/Pto, FQ	EFV has a high rate of CNS side effects in the first 2–3 weeks of use, but typically self-limited and resolves. At present, there are limited data on the use of EFV with Cs; concurrent use is the accepted practice with frequent monitoring for central nervous system toxicity. Psychosis can occur with Cs, but is rare with EFV alone; other causes should always be ruled out.
QTc Prolongation	Protease inhibitors (PIs) LPV/r	Bdq, Dlm, Mfx, Gfx, Cfz, Lfx, Ofx	PIs may result QTc prolongation. The additive effects of combining ART with known second-line anti-TB drugs on QTc prolongation is not known. Close monitoring is indicated.

Dysglycaemia (disturbed blood sugar regulation)	Protease inhibitors (PIs)	Gfx, Eto/Pto	Protease inhibitors tend to cause insulin resistance and hyperglycemia. Eto/Pto tends to make insulin control in diabetics more difficult, and can result in hypoglycemia and poor glucose regulation.
--	---------------------------------	--------------	--

For patients who are co-infected with HIV and MDR/XDR-TB, there is limited information on interactions of ARV drugs with new drugs such as bedaquiline and delamanid. Concomitant use of EFV and PIs with bedaquiline may interfere with drug concentrations and require close clinical monitoring; alternative ARV options should be considered, if possible.

N.B. For details of ARVs and anti-TB drugs for treatment of DR-TB interactions, please refer the latest TB guideline

4.7. Re-engaging with care after ART interruption

Interruption in treatment is defined as client has missed appointment for greater than 28 days from the last appointment date. People who interrupted their treatment need to be re-evaluated for possible adherence barriers and advanced clinical conditions.

For reengaged clients after missing appointment for greater than 28 days (interrupted treatment), the following essential elements should be part of the care.

- Exhaust all possible adherence barriers and provide an ongoing **Enhanced Adherence Support (EAS)**.
- **Resume the same (previous) ART regimen they used before interruption.**
- People interrupting treatment on NNRTI-containing regimen are at higher risk of drug resistance and should restart ART using a DTG-containing regimen if not contraindicated to DTG.
- Maintain close or frequent follow-up schedule.

- Monitor the viral load closely and determine the VL at 3 and 6 months after resuming ART. If the viral load is >1000copies/ml in the two consecutive measurements, switch to second or third line regimen and continue adherence support
- Those with low level viraemia, (50 –1000 copies/ml) need to be provided with enhanced adherence counselling and viral load testing after 3 months to promote more viral suppression.(see figure 4.1)

*For clients who missed their appointments less than 28 days, assess for the cause and strengthen adherence support. In addition, ensure the clients are on optimized regimen (DTG based treatment for those who are on first line).

4.8. Diagnosis and management of antiretroviral treatment failure

Monitoring individuals receiving ART is important to ensure successful treatment, identify adherence problems and determine whether and which ART regimens should be switched in case of treatment failure. Routine viral load testing is a more sensitive and

earlier indicator of treatment failure. Routine viral load testing should be done at 6 and 12 months of initiating ART and then every 12 months thereafter to detect treatment failure

proactively. Aside from the routine viral load testing schedule, viral load testing should be used whenever there is clinical or immunologic suspicion of treatment failure.

Table 4.8: Definitions of clinical, immunological and virological failure for the decision to switch ART regimens

Failure	Definition	Remark
Clinical failure	Adults and adolescents New or recurrent clinical event indicating severe immunodeficiency (WHO clinical stage 4 condition and certain WHO clinical stage 3 conditions (pulmonary TB and severe bacterial infections) may also indicate treatment failure) after 6 months of optimal adherence to appropriate ART regimen. Children New or recurrent clinical event indicating advanced or severe immune deficiency (WHO clinical stage 3 and 4 clinical condition with exception of TB) after 6 months of effective treatment.	The condition must be differentiated from immune reconstitution inflammatory syndrome occurring after initiating ART.
Immunologic failure	Adults and adolescents ■ CD4 count at or below 250 cells/mm³ following clinical failure or ■ Persistent CD4 levels below 100 cells/mm³. Children Younger than 5 years Persistent CD4 level below 200 cells/mm³ or <10% Older than 5 years Persistent CD4 levels below 100 cells/mm³.	Without concomitant or recent infection to cause a transient decline in the CD4 cell count. Persistent is to mean at least 2 CD4 measurements below the threshold. Current WHO clinical and immunological criteria have low sensitivity and positive predictive value for identifying individuals with virological failure.
Virologic failure	Viral load above 1000 copies/ml based on two consecutive viral load measurements 3 months apart with enhanced adherence support following the first viral load test	An individual must be taking ART for at least 6 months before it can be determined that a regimen has failed. VL testing should not be done when there is an acute infection/fever. (Refer algorithm on Figure 4.1.)

NB:

1. Viral suppression: is a viral load that is undetectable, equal to or less than 50 copies/ml.

2. Low-level viraemia: is any viral load results that are detectable (more than 50 copies/ml) but equal to or less than 1000 copies/ml.

3. Virological failure Viral load above 1000 copies/ml based on two consecutive viral load measurements in 3 months, apart with enhanced adherence support following the first viral load test.

Treatment failure threshold should remain at 1000 copies/ml. Viral suppression or undetectability, however, are defined as viral load equal to or less than 50 copies/ml. PLHIVs should be supported with adherence counselling to achieve viral suppression. However, treatment failure should be considered for those with a

repeat viral load result >1000 copies/ml after three months of EAS following the first viral load. Those with low-level viraemia at the first viral load test (50–1000 copies/ml) need to be provided with enhanced adherence support (EAS) and repeat viral load test after 3 months to promote viral suppression. If viral load is still 50-1000 copies/ml, maintain ARV drug regimen and continue viral load test every six months. In addition, continue the routine follow up support and link to community-based adherence support services.

Enhanced Adherence Support (EAS)

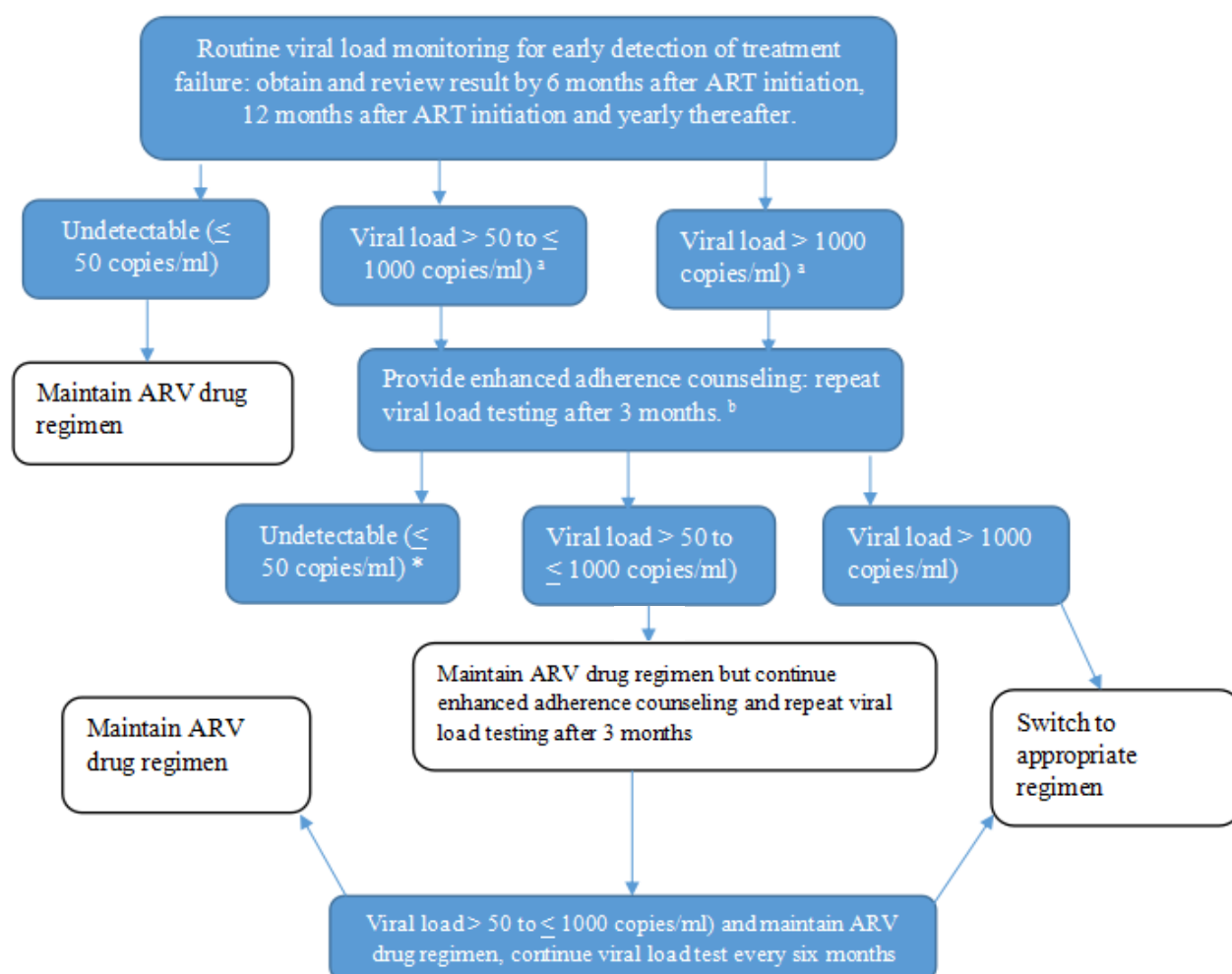
Enhanced adherence support is important for patients that have unsuppressed VL, persisted or new immunosuppression, developing new OI or have multiple adherence barriers. It shall be systematic and with documenting the interventions provided during the EAS period.

Table 4.9: Summary of components of EAS.

Enhanced adherence support sessions overview	
First Session (When the viral load result >50 is received; Day 0)	■ Review cognitive, behavioral, emotional, and socio-economic barriers to adherence: ■ Treatment literacy ■ Medications: dosage, timing, storage ■ Side effects ■ Motivation ■ Mental health screening (screen for depression and other common mental problems using national Mental Health Assessment tool (Annex 20)) Action: ■ Discuss risk reduction (e.g. for substance abuse) ■ Discuss patient's support systems ■ Assist patient to develop adherence plan to address the identified issues. ■ Referrals and networking when necessary

Second Session (30days after the first session)	■ Review adherence plan from the first session and discuss any challenge. ■ Identify other possible gaps and emerging issues. Action ■ Assist patient to modify the adherence plan to address the identified issues ■ Referrals and networking when necessary.
Third Session (60 days after the first session)	■ Review adherence plan from the first and second session and discuss any challenges. ■ Identify other possible gaps and emerging issues. Action ■ Assist patient to modify the adherence plan to address the identified issues. ■ Decision on repeat VL based on current adherence: ■ If the adherence is good, plan repeat VL testing after a month and explain possible ways forward, emphasizing the role of the patient and the health facility in terms of strengthening adherence. ■ If adherence challenges persist, link to appropriate client centered care.
Fourth Session (90 days from the first session)	■ Take the second VL sample. ■ All efforts should be employed to get the VL result as soon as possible. ■ As soon as the VL result is received, discuss the result with the client. Action 1.If VL is < 50 copies/ml, maintain the current regimen and encourage adherence 2.If VL is 50-1000 copies/ml, continue the current regimen and the monthly EAS for the next 3months then repeat the viral load testing. If the low-level viremia (50–1000 copies/ml) persists, maintain ARV drug regimen and continue viral load test every six months. In addition, continue the routine follow up support and link to community-based adherence support services. If VL is > 1000 copies/ml, switch to appropriate regiment (second or third line).

Figure 4.1. Treatment monitoring algorithm



a. Unsuppressed viral load results should be immediately communicated

b. Conduct same-day testing using point-of-care viral load testing for a repeat viral load test, where available, to expedite the return of results. If not available, viral load specimens and results for a repeat viral load should be given priority across the laboratory referral process (including specimen collection, testing and return of results).

Implementation considerations for treatment monitoring of pregnant and breastfeeding women

Whenever possible, use same-day point-of-care testing for viral load testing of pregnant and breastfeeding women to expedite the return of results and clinical decision-making. If this is not available, viral load specimens and results for pregnant and breastfeeding women should be given priority across the laboratory referral process (including specimen collection, testing and return of results).

For all pregnant women, regardless of ART initiation timing:

Conduct viral load testing at 34–36 weeks of gestation (or at the latest at delivery) to identify women who may be at risk of treatment failure and/or may deliver infants at higher risk of perinatal transmission.

In addition:

a) For pregnant women receiving ART before conception:

Conduct a viral load test at the first antenatal care visit (or when first presenting) to identify women at increased risk of in utero transmission.

- For women already on ART, conduct VL testing at the first contact at ANC (VL result conducted in the last 3 months before the first contact can also be used), at 34–36 weeks of gestational age or delivery at the latest, followed by three months after delivery and then every 6 months.
- For those who are already on ART with previous VL test conducted more than three months back repeat VL test at first ANC contact /PMTCT visit, at 34–36 weeks of gestational age (or at the latest at delivery) and 3 months after delivery and every six months thereafter until MTCT risk ends.

b) For pregnant women starting ART during pregnancy:

Conduct a viral load by three months after ART initiation to ensure that there has been rapid viral suppression. If viral load testing is expected to be undertaken near the planned viral load at 34–36 weeks of gestation, the first viral load test can be delayed until weeks 34–36 of gestation

For all breastfeeding women, regardless of when ART was initiated:

Conduct viral load test three months after delivery and every six months thereafter to detect viremic episodes during the postnatal period.

4.9. Management of treatment failure

4.9.1. Management of first-line treatment failure (switching to second-line ART)

Using a boosted PI + two NRTI combinations is recommended as the preferred strategy for second-line ART for adults, adolescents, and children. Two NRTI + DTG can be used as second line regimen if it is not used in the first line.

Guidance for changing ARV regimens for treatment failure

- Ensure diagnosis of treatment failure to avoid premature switching.
- Assess adherence and address barriers.
- Assess for and address drug interaction issues.
- Do not add one drug to a failing regimen.
- Consider resistance and cross resistance patterns.
- Get advice from experienced clinicians.
- At least two new drugs.
- Preferably one new drug class.

Once it is decided and patients are switched to second-line regimen, strong adherence support should be continued and viral load monitoring should be started after 6 months of second-line treatment and then every 12 month.

4.9.2. Management of second-line treatment failure (switching to third line ART)

Patients who are on second-line regimen and have high viral load level (>1000copies/ml) after 6 months of treatment need to go through the algorithm as described for first line treatment failure with enhanced adherence support and repeat test after three months to decide second line treatment failure. Once patients are confirmed to have second line failure, they should be referred to hospitals selected for third line ART service.

Treatment failure is indicated by virological non-suppression with or without immunologic and/or clinical deterioration. Before switching to third-line regimen, the issue should be discussed with experienced physician in HIV care and treatment. Treatment failure while on a second line PI regimen is often due to non-adherence.

Doctors and nurses should participate directly in the adherence assessments, and do not delegate the assessment to the adherence counselor alone (see enhanced adherence counseling section for detailed steps of adherence counseling for patients with adherence problems).

Before switching to third line regimen, health care providers should ensure the following.

- Two consecutive viral load measurements > 1000 copies/ml at least 3 months apart.
- First viral load measurement done at least 6 months after switching to second-line regimen.
- The repeat VL test should be done after 3 months of enhanced adherence support.

The approach in switching to third line should follow the guiding principles listed out for switching to second line drugs.

Table 4. 10: Summary of sequencing options for first line, second line and third line ART regimens for adults, adolescents, and children

Population	1 st line regimens	2 nd line regimens	3 rd line regimens a,b
Adults and adolescents 10 years & older	Preferred	TDF + 3TC + DTG	DRV/rC+TDF+3TC+EFV DRV/r+TDF+3TC+DTGe
	Alternative	TDF+3TC+EFV	DRV/r+TDF+3TC +DTGf
		AZT+3TC+ EFV	DRV/r+AZT+3TC+DTG
		AZT+3TC+DTG	DRV/r+AZT+3TC +EFV
	Special Circumstances	ABC+3TC+EFV	DRV/r+TDF+3TC +DTG
		ABC+3TC+DTG	DRV/r+TDF+3TC +EFV
		TDF+3TC+ATV/r or LPV/r	DRV/r+TDF+3TC +EFV
Children 4weeks to 10 years and >3kg	Preferred	ABC+3TC+DTG	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r+DTGg+ABC+3TC
	Alternative	AZT+3TC+ATV/r or LPV/r	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r+ABC+3TC+ EFV
		AZT+3TC+DTG	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r+DTG+AZT+3TC

	Special circumstances	ABC+3TC+EFV	AZT+3TC+DTG	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r +ABC+3TC+EFV
		AZT+3TC+EFV	ABC+3TC+DTG or LPV/r	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r +AZT+3TC+EFV or DRV/r+DTG+AZT+3TC
		AZT+3TC+LPV/r	ABC+3TC+DTG or EFV	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r+AZT+3TC+EFV or DRV/r+DTG+AZT+3TC

- Consider genotyping before constructing a third line regimen whenever accessible, however switching to third line should not be delayed while waiting for the result, if the TAT is prolonged.
- When constructing third line regimens for special circumstances in the absence of genotyping please consult senior experts on HIV
- In PI-experienced patients, the recommended DRV/r dose should be 600mg/100 mg twice daily; refer the weight band for pediatrics
- DRV/r should not be used in children younger than three years of age.
- DTG based 3rd line following use of INSTI must be administered with DTG BD.

4.9.3. Second line ART service in health centers

Patients who have treatment failure on first line ARV regimens in health centers can be switched to second line ART in the same facility based on the below mentioned criteria; also patients who were switched to second line ART in hospitals can be transferred to selected health centers for follow-up.

Eligible health centers to provide second line ART service

Second line ART service is available at health centers that have a patient load of 200 or more. That is a reasonably adequate patient load which could give rise to viable number of prospective clients who might need to be switched to a second line ART.

Eligible patients for second line ART service at health centers

The following criteria should be used to sort out treatment failure patients who will be switched to second line ART or will have follow up at the health centers:

- a. Patients who fulfill the criteria of treatment failure and with no signs of clinical failure
- b. Patients who are initiated on second line ART at hospital and became stable with no signs of OI or drug toxicity and transferred out to health centers having second line ART.

CHAPTER FIVE:

PREVENTION, SCREENING, AND MANAGEMENT OF COMMON CO-INFECTIONS AND COMORBIDITIES

ART has reduced mortality and morbidity associated with HIV and transformed HIV into a chronic disease requiring lifetime care. Coinfections and comorbidities, including physical and mental health conditions are common among people living with HIV. Comprehensive HIV care includes combination HIV prevention, the promotion of general health and well-being, maintaining quality of life, ART, the prevention and management of coinfections and comorbidities.

Opportunistic infections are the predominant causes of morbidity and mortality among HIV-infected patients. The level of immunity determines the occurrence and type of OIs. In general, milder infections such as herpes zoster and other skin infections occur early whereas serious life-threatening infections such as CNS toxoplasmosis and Cryptococcal meningitis occur later with severe immune-suppression. Some life-threatening infections, such as pneumonia and TB, may occur early as well as later. When TB occurs later it is atypical, more disseminated and more extra pulmonary. This chapter provides a brief overview of common and important concomitant conditions among people living with HIV. This includes information on co-trimoxazole prophylaxis, the diagnosis, prevention and treatment of TB, pneumonia, viral hepatitis, sexually transmitted infections, cervical cancer prevention and mental health.

Recommended general care for PLHIV

The following package of care and prevention intervention are nationally recommended for adults, adolescents and children living with HIV:

- Psychosocial counselling and support
- Disclosure and partner notification
- Co-trimoxazole prophylaxis
- TB counselling, screening and preventive therapy
- Preventing common fungal infections
- Preventing sexually transmitted infections and supporting reproductive health needs, including preventing and screening for cervical cancer
- Prevention of malaria
- Nutrition
- Family planning
- Prevention of mother-to-child HIV transmission;
- Water sanitation and hygiene
- Orphans and vulnerable children (OVC)

General strategies to prevent opportunistic infections are:

- Reduction of exposure
- Chemoprophylaxis (primary/secondary)
- Immunization and
- Rapid ART initiation

5.1. Co-trimoxazole preventive therapy (CPT)

Co-trimoxazole preventive therapy (CPT) should be implemented as an integral component of a package of HIV-related services. Existing recommendations cover initiation of CPT

among adults, adolescents, pregnant women, and children for prevention of pneumocystis pneumonia, toxoplasmosis, bacterial infections and diarrhea caused by *Isospora belli* or *Cyclospora* species, as well as benefits for malaria prophylaxis.

Table 5.1 CPT indication for primary prophylaxis

Age	Criteria for initiation	Criteria for discontinuation*	Monitoring approach
HIV exposed infants	In all, starting at 4–6 weeks after birth	Until the risk of HIV transmission ends or HIV infection is excluded	According to the national HEI follow up schedule
<5 years	In all	Discontinue when they become older than 5 years of age who are clinically stable, with evidence of immune recovery and/or viral suppression on ART**	Clinical at 3-month
Intervals ≥5 years, including adults	Any WHO stage and CD4 count ≤ 350 cells/mm ³ Or WHO stage 3 or 4, irrespective of CD4 level	Discontinue in those who are clinically stable (those individuals on ART for at least one year without any new WHO clinical stage 2, 3 or 4 events) with; ■ Evidence of immune recovery and/or viral suppression (CD4 count >350 cells/mm³, with viral load suppression) or ■ Two consecutive CD4 count > 350 cells/mm³ if no VL result	

Contraindications to co-trimoxazole preventive therapy:

- Severe allergy to sulfa drugs
- Severe liver disease

- Severe renal disease and
- Glucose-6-phosphate dehydrogenase (G6PD) deficiency

Table 5.2: Dosage of co-trimoxazole for adults, adolescents, children and infants.

Age(weight)	Suspension (240 mg/5ml co-trimoxazole)	Single strength tab (480 mg of Co-trimoxazole)	Double strength tab (960 mg of co-trimoxazole)
Up to 6month (5kg)	2.5ml/day	¼ tab/day	-
6 months to 5 yr (5-15 kg)	5 ml/day	1/2 tab/day	-
6-14 yr (15-30 kg)	10ml/day	1 tab/day	½ tab/day
>14 yrs (>30 kg)		2 tab/day	1 tab/day

Table 5.3: Adverse effects of CPT and management.

Toxicity	Clinical description	Recommendation
Grade 1	Erythema, pruritis	Prescribe antihistamine and continue CPT and close follow-up.
Grade 2	Diffuse maculopapular rash, dry desquamation	Prescribe antihistamine and continue CPT and close follow-up.
Grade 3	Vesiculation, minor mucosal ulceration	STOP CPT, manage and re-introduce after 2 weeks with observation (desensitize).
Grade 4	Exfoliative dermatitis Steven-Johnson syndrome or erythema multiforme, moist desquamation	STOP CPT NEVER RESTART CO-TRIMOXAZOLE

5.2. Tuberculosis

TB is the most frequent life-threatening OI and a leading cause of death among HIV infected people. TB remains the leading cause of mortality among people living with HIV, despite substantial scale-up of ART, accounting for 30% of the AIDS-related deaths reported in 2019. TB increases HIV replication through the process of immune activation leading to increased viral load and this results in a more rapid progression of HIV disease. On the other hand, HIV increases susceptibility to be infected with M. Tuberculosis, the risk of progression to TB disease and the incidence and prevalence of TB. The lifetime risk of HIV positive individuals to develop TB is 50%, the annual risk is 10%. The WHO recent estimation indicates the risk of developing active TB disease is 18 times higher in PLHIV. In Ethiopia, WHO estimation indicated that 5.3% of TB cases notified in PLHIV in 2019. Moreover, a national level data report indicated that the prevalence of TB in PLHIV is 7.3%. Thus, it is essential for both TB and HIV control programs to synergize their joint efforts and intensify the implementation of TB/HIV collaborative activities to mitigate the dual burden of TB/HIV in populations at risk or affected by both diseases. The rationale for the integration is that tuberculosis and HIV prevention and control programs share mutual challenges of high impact of TB on HIV and vice versa. Therefore, the two programs must collaborate to provide better service for the co-infected patients.

Nationally recommended TB/HIV collaborative activities

A. Strengthen the mechanisms for integrated TB and HIV services delivery

- Strengthen the coordination mechanism for integrated TB/HIV services at all levels;
- Conduct surveillance to determine HIV burden among TB patients and TB burden among HIV patients;
- Carry out joint TB/HIV planning for integrated TB and HIV services delivery; and
- Conduct monitoring and evaluation of collaborative TB/HIV activities.

B. Reduce the burden of TB in HIV infected people and initiate early antiretroviral therapy (the three I's i.e., Intensive case finding, TB preventive therapy and Infection control)

- Intensify TB case finding and ensure quality TB treatment;
- Initiate TB prevention with earlier initiation of ART and TB Preventive therapy (TPT); and
- Ensure tuberculosis infection control in healthcare and congregate settings.

C. Reduce the burden of HIV in patients with presumptive and diagnosed TB.

- Provide HIV testing and counseling to presumptive and confirmed TB patients.
- Provide HIV prevention services for presumptive and confirmed TB patients;
- Provide co-trimoxazole preventive therapy for HIV positive TB patients.
- Ensure HIV/AIDS prevention, treatment, and care for HIV positive TB patients; and
- Provide ART for HIV positive TB patients.

Table 5.4 Timing of ART for adults and children with TB

New HIV positive Patients with signs and symptoms suggesting (presumptive tuberculosis cases)	Patients with tuberculosis found to be HIV positive	HIV positive patients taking ART diagnosed with TB
■ Rapid ART initiation should be offered to all people living with HIV following a confirmed HIV diagnosis and clinical assessment and to people living with HIV with signs and symptoms suggesting TB. Except for central nervous system disease (meningitis), initiate ART while rapidly investigating for TB, with close follow-up within seven days to initiate TB treatment if TB is confirmed.	■ ART should be started in all TB patients, including those with drug-resistant TB, irrespective of the CD4 count. ■ ART should be started as soon as possible within two weeks of initiating TB treatment, regardless of CD4 count, among all people living with HIV. (Adult, adolescent and children), except when signs and symptoms of meningitis are present. ■ ART should be delayed at least four weeks (and initiated within eight weeks) after treatment for TB meningitis is initiated. Corticosteroids should be considered adjuvant treatment for TB meningitis. (Note. For people living with HIV with TB meningitis, immediate ART is associated with more severe adverse events compared with initiating ART two months after the start of TB treatment) ■ DTG should be used as the preferred drug in patients starting ART while on Anti-tuberculosis treatment but dose of DTG should be doubled (50 mg BID). ■ When second line is initiated LPV/r is preferable. Note that the dose of the ritonavir should be doubled to counter the effect of Rifampicin on the Protease inhibitors.	■ Start anti-TB ■ Modify ART regimen to avoid drug-drug interaction ■ Evaluate for treatment failure

The 3Is intervention

I. Intensify TB case finding and ensure quality TB treatment

Tuberculosis case finding should be intensified in all HIV testing and counseling services for HIV positive clients by using a set of simple questions for early identification of presumptive TB cases. HIV positive clients coming through HIV testing services should be informed about the advantages of being screened for TB. Once informed about the risk of developing active TB, they should undergo screening for it. Adults and adolescents living with HIV should undergo intensified tuberculosis case finding using symptom-based TB screening questions should be instituted at chronic HIV care and treatment clinics for all people living with HIV. Any HIV positive patient with TB screen positive result should undergo appropriate evaluation and investigation for TB. In addition to symptom based screening, to consider sensitive screening test like CXR for routine screening of TB in PLHIV with no sign and symptoms of TB.

Symptom based TB screening for all PLHIV

All clients living with HIV infection should be screened for active TB with any one of the following symptoms:

Adult and adolescents:

- Current cough
- Fever
- Weight loss or poor weight gain
- Night sweats

Children less than 10 years:

- Current cough,
- Fever,
- Poor weight gain
- Close contact with a person with TB disease

Clients with any one of the above symptoms should be investigated for active TB

Diagnosis of TB in HIV infected people

Although there have been significant improvements in the diagnosis and treatment of TB, there are still limitations in early detection and treatment of all forms of TB cases among PLHIV. A systematic review study that estimated the prevalence of TB in PLHIV studies reported on postmortem testing showed 46% of TB cases remain undiagnosed, which made TB the most prominent opportunistic infection among PLHIV. Moreover, of the total 815,000 estimated HIV-associated TB cases worldwide in 2019, only 56% were notified. This indicates that TB diagnosis is still a challenge in PLHIV, and it is rarely bacteriologically confirmed.

The conventional sputum microscopy is the cheapest and fastest method that is used to diagnose TB since 1882. However, the sensitivity of sputum smear microscopy for the diagnosis of TB is low in PLHIV due to poor quality sputum production and low bacillary concentration.

Particularly in severely immunocompromised patients, the sensitivity of sputum smear microscopy is significantly reduced. Other challenges that make TB diagnosis difficult

include non-specific clinical presentation of TB that is attributable to high prevalence of extra-pulmonary and disseminated forms of TB in individuals with advanced HIV disease.

Currently, rapid, and more accurate molecular technologies have been developed and available to diagnose TB in PLHIV. Polymerase chain reaction (PCR), real-time PCR, and loop-mediated isothermal amplification (LAMP) are the molecular techniques that are commercially available for the diagnosis of TB. However, the diagnostic performance of these technologies has not been fully evaluated for the diagnosis of TB in PLHIV.

Although rapid and accurate molecular assays have significantly reduced the gap in TB case detection among PLHIV, their accessibility continued to be a challenge, due to the infrastructure required and the cost of procurement and maintenance. Thus, access to rapid and accurate diagnostic tests at the point of care is significantly restricted in resource limited settings where the burden of TB/HIV co-infection is high. Moreover, interruption of electricity and inadequate laboratory infrastructure and inefficient sample referral mechanisms are the challenges that limit the accessibility of rapid and accurate molecular diagnostic tests.

To minimize the challenges that are associated to sputum based diagnostic tests, urine based rapid TB diagnostic tests are recommended to detect TB in PLHIV with Advanced HIV Disease. Alere lipoarabinomannan (LAM) assay is one of the urine based rapid diagnostic tests for TB detection in PLHIV. Evidence suggests the importance of LAM in the detection of TB at

high TB/HIV burden settings. It is recommended for use as a simple point-of-care test to assist TB diagnosis in HIV positive adult patients with signs and symptoms of TB and with low CD4 cell counts. Based on the generalization of the data from adults, this recommendation is also used in children living with HIV who have signs and symptoms of TB. However, there is limited evidence on the specificity of LAM test in children.

LAM test could decrease mortality through quicker diagnosis and early treatment commencement among PLHIV and severely sick.

Eligible PLHIV for LF-LAM Test

WHO 2019 policy update on LF-LAM recommends the currently available urinary LAM assays have suboptimal sensitivity and are therefore not suitable as general diagnostic tests for TB. However, unlike traditional diagnostic methods, they demonstrate improved sensitivity for the diagnosis of TB among individuals coinfecting with HIV. The estimated sensitivity is even greater in patients with low CD4 cell counts.

Therefore, HIV-positive adults, adolescents and children with the following criteria are eligible for Urine LF-LAM test to assist in the diagnosis of active TB:

For inpatient settings

- With signs and symptoms of TB (pulmonary and/or extra pulmonary)
- With advanced HIV disease or who are seriously ill

- Irrespective of signs and symptoms of TB, with a CD4 cell count of less than 200 cells/mm³

Seriously ill is defined based on four danger signs: respiratory rate of more than 30/minute, temperature of more than 39 °c heart rate of more than 120/minute and unable to walk unaided.

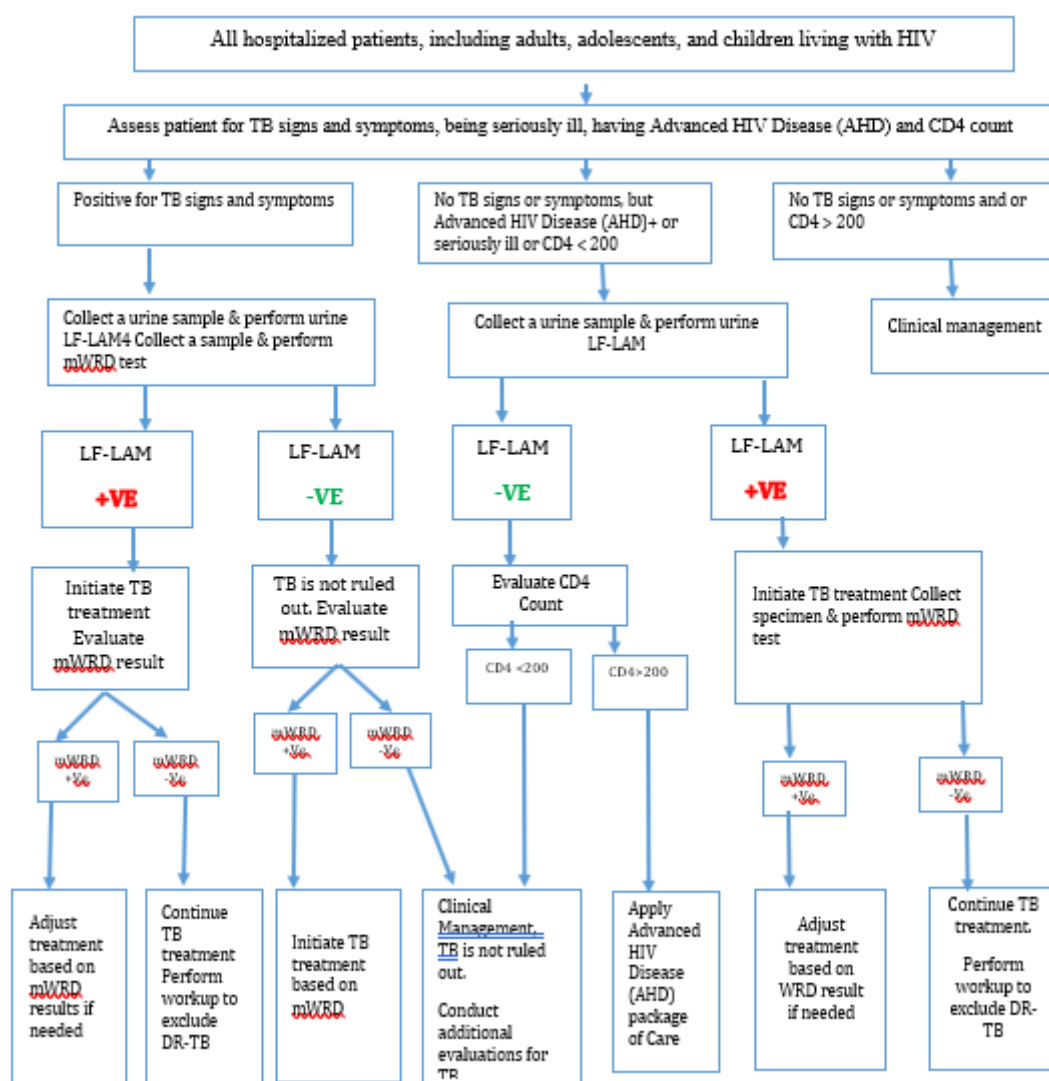
Advanced HIV disease: is defined as a CD4 cell count of less than 200 cells/mm³ or a WHO clinical stage 3 or 4 event at presentation for care for adults, adolescents and children greater than five years. All children with HIV who are

aged under 5 years should be considered as having advanced disease.

Figure 5.1. Algorithm for LF-LAM testing to aid in the diagnosis of TB among PLHIV in inpatient settings

For outpatient settings

- With signs and symptoms of TB (pulmonary and/or extra pulmonary) or seriously ill
- Irrespective of signs and symptoms of TB and with a CD4 cell count of less than 100 cells/mm³



In outpatient settings, WHO recommends against using LF-LAM to assist in the diagnosis of active TB in HIV-positive adults, adolescents, and children:

- Without assessing TB symptoms
- Without TB symptoms and unknown CD4 cell count or without TB symptoms and CD4 cell count greater than or equal to 200 cells/mm³
- Without TB symptoms and with a CD4 cell count of 100–200 cells/mm³

Remarks

a. All patients with signs and symptoms of pulmonary TB who can produce sputum should have as their initial diagnostic test at least one sputum specimen submitted for Xpert® MTB/RIF (Ultra) assay. This also includes children and adolescents living with HIV who can provide a sputum sample.

b. LF-LAM should be used as an add on to clinical judgement in combination with other tests. It should not be used as a replacement or triage test. More details are given in Algorithms for LF-LAM use.

Note for using LF-LAM test for children less than 5 years:

Latest WHO guideline consider all HIV infected children age <5 years old to follow as advanced HIV disease, besides nationally we have a limited number of under 5 children on

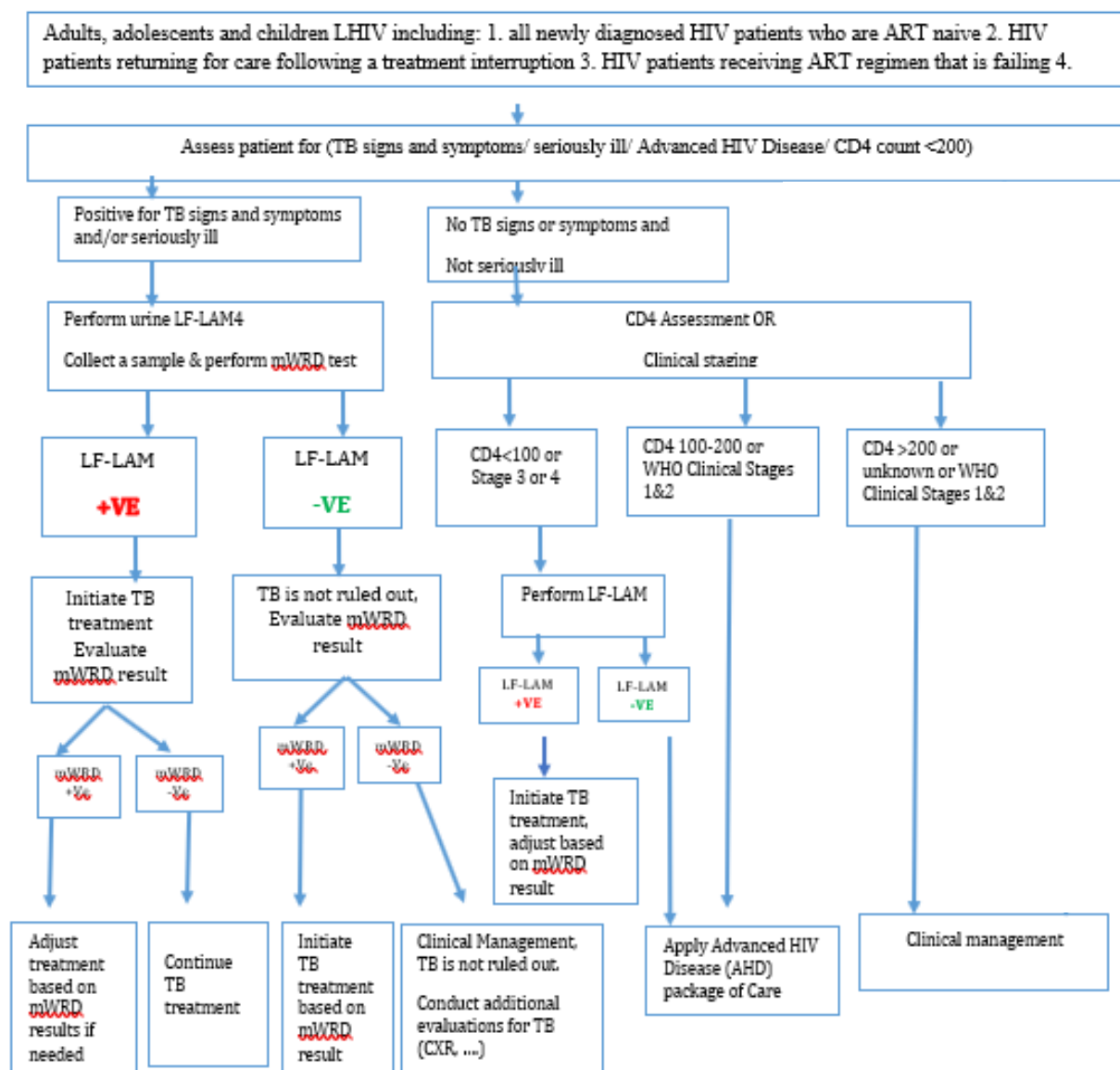
ART. Hence setting other criteria may limit the Urine LF-LAM utilization. Therefore LF-LAM test should be conducted for all HIV infected children under 5 years at least once as per below criteria:

To offer LF-LAM for all HIV infected <5 years children backlog/currently on treatment once irrespective of clinical stage or CD count/CD 4 percentage

All newly enrolled under 5 HIV infected children (at Enrollment) irrespective of clinical stage or CD4 count/CD4 percentage

Once cleared the backlog, offer LF-LAM at any time when children presented with TB symptoms, or serious illness or advanced HIV stage or CD4 percentage less than 15%.

Figure 5.2. National TB Diagnostic algorithm for PLHIV incorporating Urine LF-LAM testing in outpatient settings



Diagnosis of extra-pulmonary tuberculosis in PLHIV

Extra-pulmonary tuberculosis is more HIV-related than pulmonary tuberculosis. The accurate diagnosis of extra-pulmonary tuberculosis is complex and difficult, particularly in peripheral health facilities with limited

diagnostic capacity. Therefore, it is important for healthcare workers to have a high-index of suspicion and critically evaluate through clinical algorithms. It is also recommended to do organ specific investigations.

Note:

- Xpert MTB/RIF test is recommended as diagnostic test for CSF in patients presumed to have TB meningitis.
- One sputum sample for the facility which have Xpert test and two sample for sample referring facilities.

Antibiotic trial: Antibiotic trial has a role to treat concomitant bacterial infection for PLHIV with cough or serious illness. However, antibiotic trial is not helpful in the diagnosis of TB in PLHIV.

Table 5.5: Extra pulmonary TB diagnostic approaches in PLHIV.

Type of TB	Evidence Strongly Suggestive of EPTB	Investigations and recommendations
Lymph Node TB	2cm or more in size, Asymmetrical/localized; Painless swelling; Firm/fluctuated; Cervical location; patient with weight loss, night sweats, fever	LN Aspirate for AFB has 85% yield, if not possible to do FNAC of LN, start anti-TB.
Pleural effusion	Unilateral effusion: Aspirate of fluid is clear and straw colored and clots on standing in a tube without anti-coagulants or pleural fluid analysis shows protein >30g/L & >50% lymphocytes; Patients with weight loss, night sweats, fever, or evidence of TB elsewhere	Start anti-TB as soon as possible.
Tuberculosis Meningitis	Patients with weight loss, night sweats, fever; cerebrospinal fluid clear with high protein, low glucose, and lymphocytes; Cryptococcal antigen (or Indian Ink and fungal culture) negative in CSF, evidence of TB elsewhere	Admit patient, start anti-TB with steroids as soon as possible. Start treatment for cryptococcal meningitis based on clinical or lab evidence.
Pericardial Effusion	Hemodynamically significant pericardial effusion, often with pleural effusions, Lung fields clear and intra-thoracic lymphadenopathy. Usually patients with weight loss, night sweats, fever. N.B. 90% of pericardial effusions in HIV positive patients in high-TB burden areas is due to TB.	CXR, Echocardiograph or chest ultrasound; pericardiocentesis, and pericardial biopsy; routine TB Workup. Start anti-TB as soon as possible
Disseminated TB	Patients with weight loss, night sweats, fever, and cough; Abnormal CXR (which can include miliary pattern); Large spleen/liver, Anemia	Start anti-TB treatment (add antibiotics if critically ill)
TB of the Spine	Pain over localized area, children/adolescents –often thoracic vertebrae. Adults: frequently lumbar vertebrae.	Spinal imaging (e.g. X-Ray, MRI); FNA of vertebral lesions and /or Para spinous abscesses when feasible.

II. TB Prevention Therapy (TPT)

Latent tuberculosis infection (LTBI) is the state of persistent immune response to stimulation of mycobacterium tuberculosis antigens without evidence of clinically apparent active tuberculosis (TB). Estimates show a quarter of the global population is infected with tuberculosis, where most cases are asymptomatic and non-infectious. Studies show that, on average, 5-10% of these latently infected persons have lifetime risk of progressing to develop active TB, usually within the first five years of initial infection. However, risk of progression is dependent on immunological status.

Treatment of LTBI to prevent progression to active disease is one of the global key strategies to ending the TB epidemic. Increasingly, eligible targets and treatment options are expanding, with significant implications in the programmatic management of LTBI. TPT is the use of Isoniazid, rifapentine or other medications to sterilize latent TB infection. Screening for exclusion of active TB in HIV infected persons is the single most important step that should precede the decision to initiate TPT. Concerns regarding the development of INH resistance should not be a barrier to providing TPT.

The selection of treatment options for LTBI by programs and clinicians should consider the characteristics of the clients who are to receive treatment and acceptability of treatment for higher completion rate. The benefits of all the treatment options outweigh the potential harm. Shorter TPT regimens have been shown to be as effective and better tolerated than longer regimens, and thus may be preferable for individuals receiving treatment, clinicians providing treatment, and program managers.

For adults and adolescents living with HIV, TPT should be provided to those who are unlikely to have active TB based on appropriate clinical algorithm, irrespective of CD4 count, ART status, pregnancy status or history of treatment for prior episode of TB before THREE years. Children and infants less than 1 year of age should be provided preventive therapy only if they have a history of household contacts with pulmonary TB and active TB in the child has been excluded in investigation.

Clinicians should follow the algorithms for initiation and selection of regimen for the specific population groups eligible for TPT as indicated in section below.

Figure 5.3. Algorithm for initiating TPT in adults and adolescents ≥ 15 years living with HIV

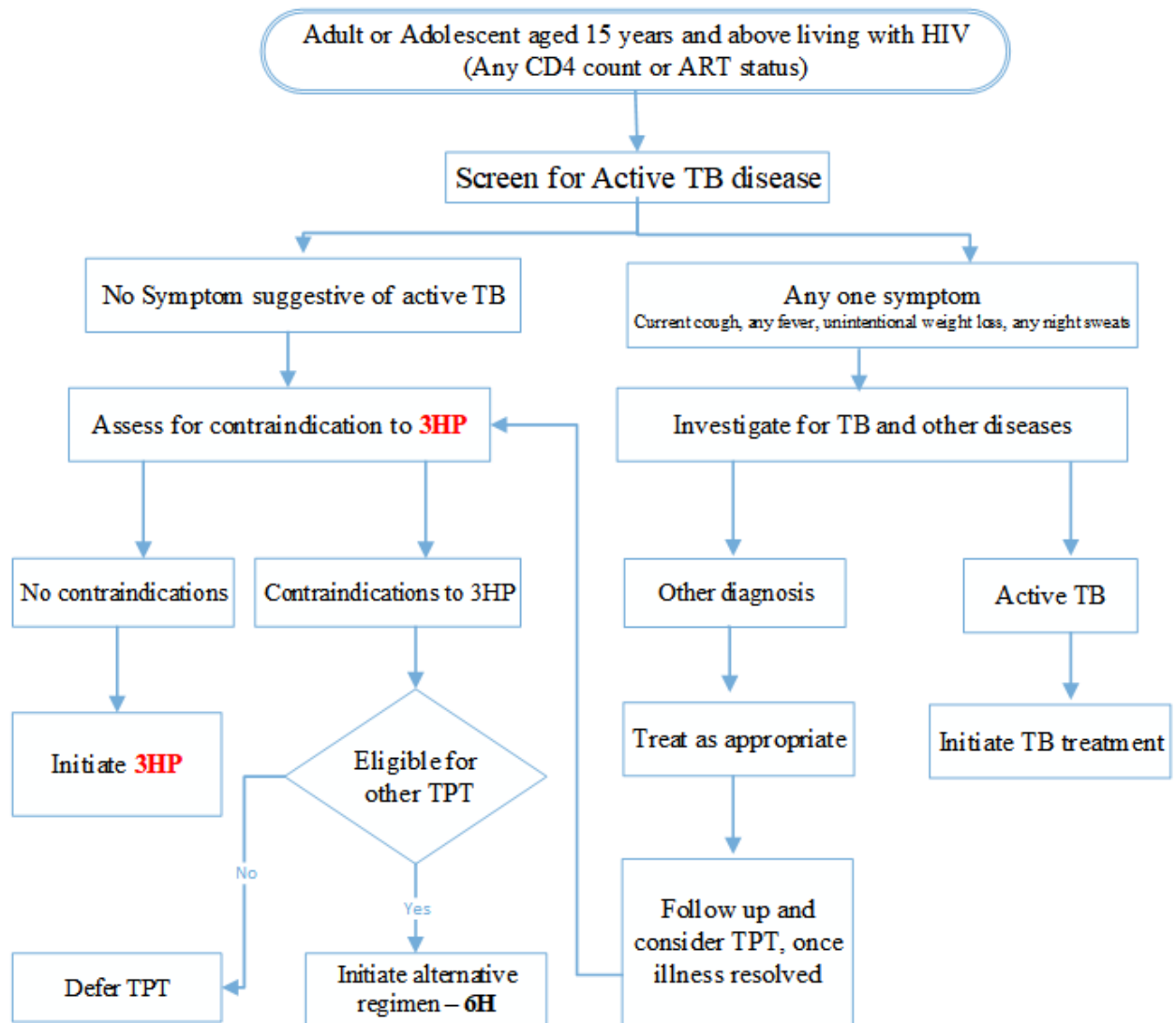


Figure 5.4. Children <15 years living with HIV and without household TB exposure

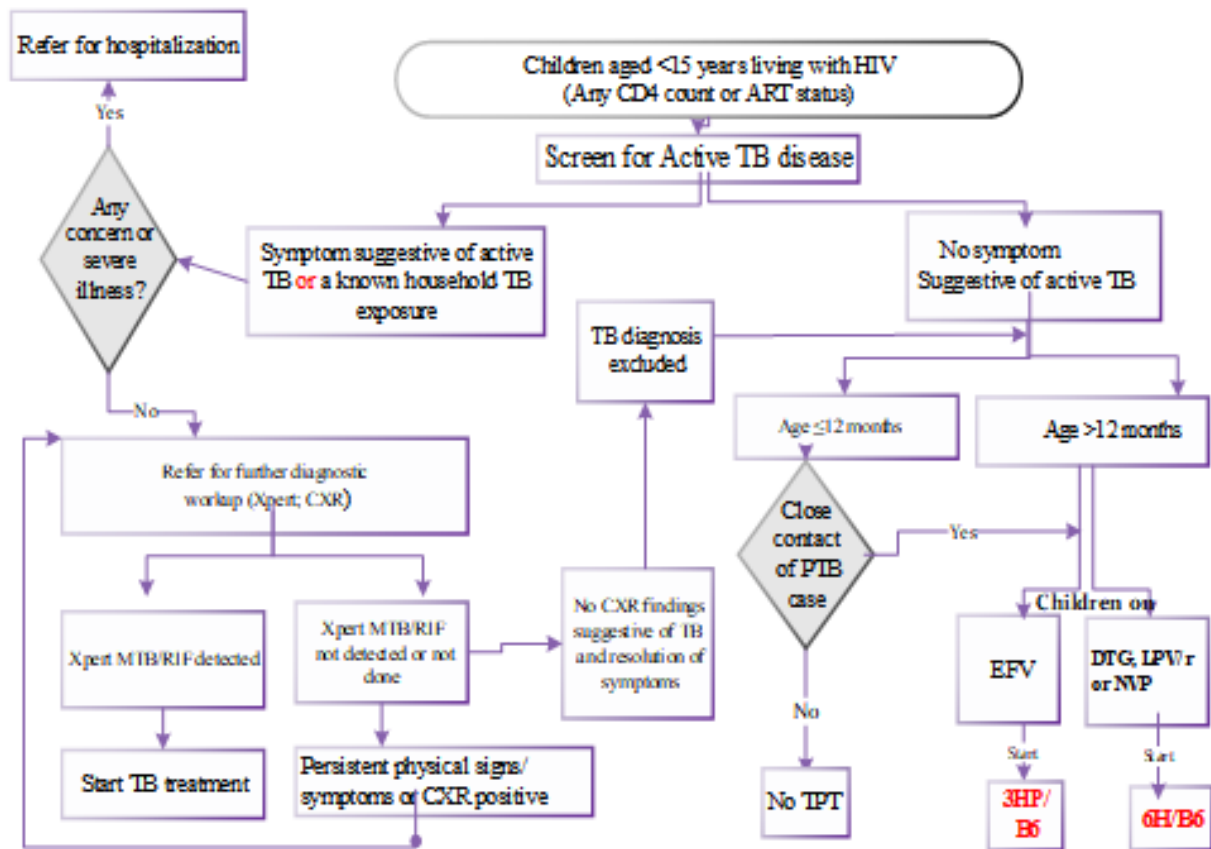
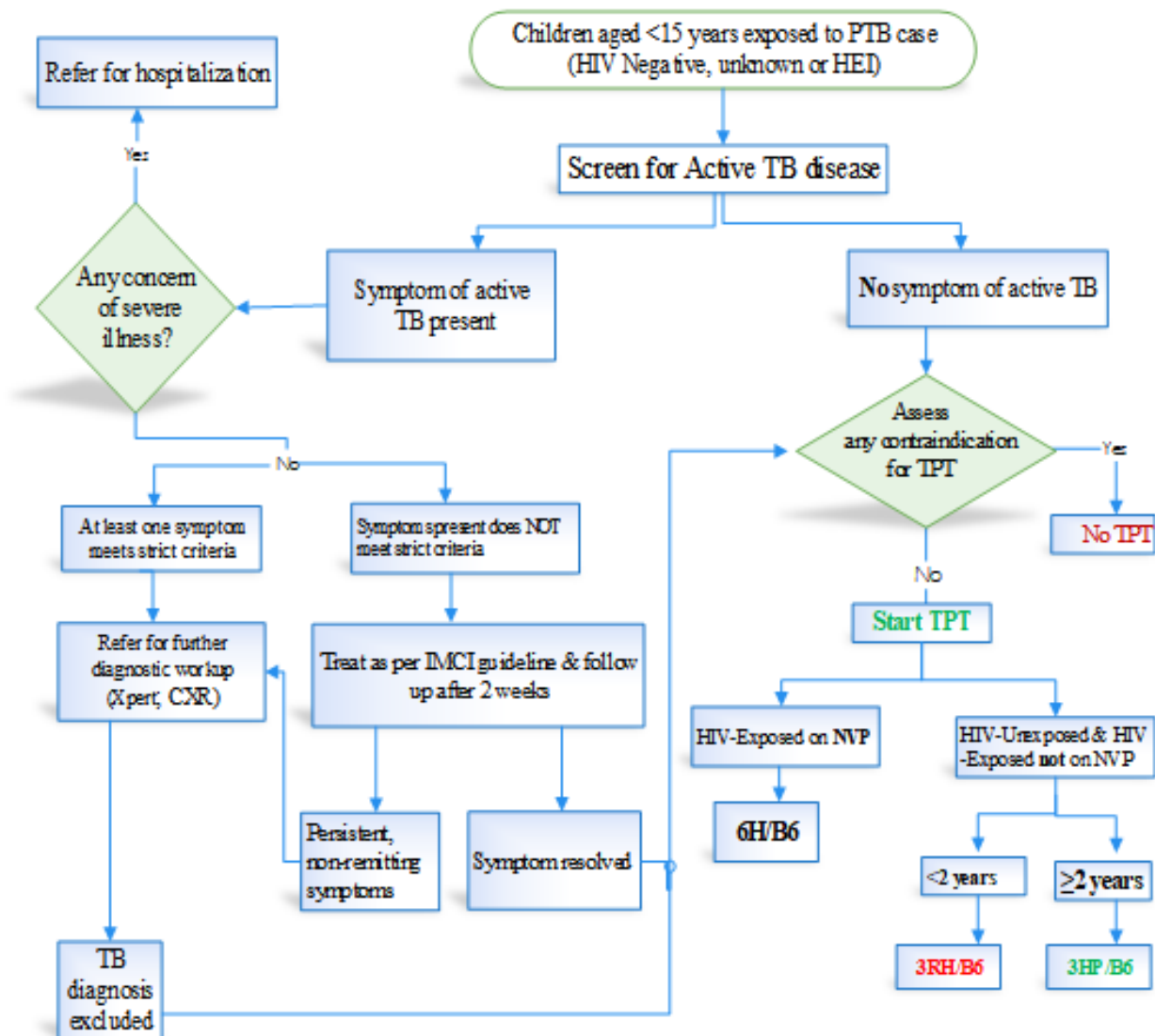


Figure 5.5. HIV-exposed Children and HIV negative children and adolescents <15 years of age with household exposure to PTB case



Regimen and dosage for treatment of LTBI

Table 5.6. Regimen for treatment of latent TB infection.

Population group	Age group and ART regimen	Selection of TPT regimen	
		Preferred regimen	Alternative regimen
Persons living with HIV		Preferred regimen	Alternative regimen
	Adults, adolescents, children, and infants of all ages taking a PI-based ART regimen	Daily isoniazid preventive treatment for 6 months (6H)	
	Children and adolescents aged <15 years taking a DTG-based ART regimen	Daily isoniazid preventive treatment for 6 months (6H)	
	Children and adolescents aged <15 years taking EFV-based ART regimen	Weekly isoniazid Plus rifapentine for 3 months (3HP*).	■ Daily isoniazid preventive treatment for 6 months (6H) ■ Daily rifampicin Plus isoniazid for 3 months (3RH).
	Adolescents and adults living with HIV (≥ 15 years of age) taking non- PI based ART regimen	Weekly isoniazid Plus rifapentine for 3 months (3HP).	■ Daily isoniazid preventive treatment for 6 months (6H)
HIV-negative persons	Infants and children <2 years of age)	Daily rifampicin Plus isoniazid for 3 months (3RH).	-Daily isoniazid preventive treatment for 6 months (6H)
	Eligible adolescents and children aged between ≥2 -14 years (refer to eligibility criteria specified above)	Weekly isoniazid Plus rifapentine for 3 months (3HP).	-Daily isoniazid preventive treatment for 6 months (6H) -3RH will be used as alternative to 3HP
*3HP: should be taken with food to prevent GI upset; if patients are unable to swallow tablets (due to age or illness), the tablets can be crushed and added to a small amount of semi-solid food			

Drug-drug interactions with antiretroviral medicines

Drug-drug interactions should be cautiously considered in using rifampicin and rifapentine in persons who are receiving antiretroviral treatment (ART). Both rifampicin and rifapentine should not be administered in persons who are receiving nevirapine or Protease Inhibitor (PI) based regimen. A three-month weekly rifapentine plus isoniazid can safely be used in patients receiving efavirenz-based antiretroviral regimen without the need for dose adjustment. Studies show that co-administration of rifapentine with raltegravir and dolutegravir based regimen is both safe and well tolerated in HIV infected adults and adolescents aged ≥ 15 and do not need any dose adjustment.

Monitoring adverse events

Routine clinical monitoring of persons receiving TPT is necessary to ensure adherence and continuity of care. Adverse effects, including those considered as medically “minor”, may be a barrier for adherence in a person who is otherwise well. TPT related adverse events (AEs), identified during treatment should be properly monitored, managed and reported per national recommendation, using health facilities reporting systems.

Management of adverse events (AEs)

Individuals receiving TPT do not have active disease and therefore their risk for adverse events during treatment must be minimized. Moreover, the regimen can be withheld while an AEs is assessed, and there is time for the regimen to be recommenced and completed

if safe to do so. Overall, 3HP is a safe and effective treatment for latent TB infection. Clinically significant drug reactions are rarely experienced by patients receiving 3HP, and even less commonly require discontinuation of treatment. Severe reactions are particularly rare. Nonetheless, healthcare workers should be familiar with the important drug reactions so that they can recognize rare occurrences and manage them appropriately.

Therefore, in general: If an AE occurs while a patient is receiving 3HP, they should be advised not to take any further doses and contact the ART providers and TB focal persons of the health facilities as appropriate.

Minor adverse events are likely to occur in a small proportion of individuals/patients. Rarely serious adverse events may occur, and hence both the health care provider and patient should be vigilant and manage such events rapidly. This can be achieved by careful assessment of the patient prior to commencing 3HP, and routine monitoring during treatment as indicated below:

Table 5.7. Baseline assessment and monitoring during TPT

Drug interactions	The most common drug reactions with 3HP are: ■ Liver toxicity (less common than for 6H) ■ Flu-like reactions (more common than for 6H) Drug reactions are usually mild and self-limiting, but occasionally they can be severe. Children usually tolerate 3HP very well and have much lower rates of drug reactions.
Baseline assessment	Active TB must be ruled out before commencing TPT. 3HP is currently not recommended in: ■ Pregnancy ■ Age <2years ■ Information on baseline liver function is important in the following: ■ HIV+ (done as part of ART assessment but not mandatory) ■ Daily alcohol consumption ■ Liver disorders including viral hepatitis ■ Postpartum period (≤3 months after delivery) ■ Concomitant use of other hepatotoxic substances Individuals at higher risk of peripheral neuropathy should be offered vitamin B6 (pyridoxine) supplementation with 3HP; if B6 is not available, this should not delay starting a course of 3HP
Counselling for AEs	Red/orange discoloration of urine and other body fluids while receiving 3HP is normal and completely harmless. If patients experience any symptoms concerning for an AE: ■ Do not take any further doses of 3HP ■ Contact a healthcare provider for advice ■ Only continue receiving 3HP if advised to do so by a healthcare provider Individuals should be alert to the following symptoms: ■ Weakness, fatigue, loss of appetite, persistent nausea (early symptoms of hepatotoxicity) ■ Flu-like, or other acute symptoms appearing shortly after receiving a dose of 3HP ■ Symptoms of active TB (appendix1)

Adherence support and monitoring

Clients receiving TPT should be supported at home level, either by adherence case managers, HEWs or family supporter. They should have monthly scheduled follow up that is coordinated with other services, such as HIV care, child and maternal health services, as necessary. At each follow-up visit, the healthcare provider should:

- Educate clients and their families about the benefit of TPT, potential side effects, and importance of returning back to a health facility for new symptom/sign or any concern.
- Evaluate and counsel clients on importance of treatment adherence and completion. Case managers, adherence supporters, and HIV care and child health service providers should support treatment adherence and monitoring, as part of comprehensive HIV care and child health services.
- Evaluate and routinely monitor drug side effects, including hepatitis, peripheral neuropathy or rash. Stop TPT if serious adverse effect is identified and manage the patient.
- Evaluate for signs and symptoms of active TB, other opportunistic infections (OIs) or diseases.
- Stop TPT, if active TB is diagnosed, which requires immediate start of anti-TB treatment

Contraindications for TPT

Individuals with any one or more of the following conditions should not receive TPT:

- Symptoms compatible with tuberculosis even if the diagnosis is not yet confirmed;
- Active hepatitis (chronic or acute);
- Regular and heavy alcohol consumption;
- Prior allergy or intolerance to medicine(s) in the regimen; and
- Symptoms of peripheral neuropathy

In addition, rifapentine is not currently indicated for children below 2 years, PLHIV receiving PI or NVP based ART regimens, and pregnant and breastfeeding women.

Patient management after treatment interruption:

TPT is said to be completed when a person completed a full course of treatment within the specified period. Completion of TPT is defined as “completed” if a patient completed the full course of therapy within nine months period (i.e. the six months of doses should be finished in nine months). If a patient has interrupted TPT without medical personnel advice, the client should be traced by health extension workers (HEW) or through the index person or other family member who is already enrolled in care, and treatment must be resumed. It is important to identify barriers and support treatment adherence. However, concerns about adherence should not be a barrier to the use of TPT. If the

client discontinues isoniazid-based TPT for a period of less than three months: Resume the same course by adding for the missed doses at the end. If the client discontinues treatment for a period of more than three months, then re-initiate new course of treatment.

Completion for 3HP is defined when the patient took at least 11 doses of treatment in 16 weeks.

TB preventive treatment in Newborns

Once a pregnant woman with TB has been on treatment for TB for several weeks before delivery, it is less likely that the baby will become infected. The risk is highest if a mother is diagnosed at the time of delivery or shortly thereafter. If a pregnant woman is found to have pulmonary TB shortly before delivery, then the baby, and if possible, the placenta, should be investigated for evidence of congenital TB infection. If the result is positive, the newborn should be treated accordingly. If asymptomatic, the newborn should receive TB preventive treatment followed by BCG immunization. Breastfeeding can be safely continued during this period.

TPT service provision amid COVID19:

At TPT initiation, all PLHIV will receive standardized comprehensive education and counseling on benefits of TPT, importance of adherence, TPT-related AEs, with clear instructions to present for evaluation at onset of any sign or symptoms indicative of TPT-related AEs. All patients will be placed on three months TPT MMD (both for clients on daily isoniazid for 6 months [6H] and rifapentine and isoniazid weekly for 3 months [3HP]). TPT

MMD will be aligned with ART MMD to reduce unnecessary facility visits. Facility level follow up visit & adherence monitoring will be aligned with TPT dispensation at initiation and month three. Documentation of TPT completion will be aligned to clinical visit following expected TPT completion at month six (for 6H) and at month three (for 3HP).

III. TB Infection, prevention, and control

PLHIV are at increased risk of acquiring TB in health care facilities and congregate settings. Each health care facility should have a TB infection control plan for the facility that includes administrative, environmental, and personal protection measures to reduce the transmission of TB in health care and congregate settings and to provide surveillance of TB disease among workers. Health care workers with HIV should be provided with ART and TPT if they are eligible. TB can affect everyone, but specific population groups, including PLHIV and healthcare workers, have a higher risk of acquiring TB infection and progressing to disease once infected.

Summary of recommendations for key actions for infection control

Administrative (facility-level infection control committee and protocols):

A set of administrative controls is the first and most important component of any IPC strategy. These key measures comprise specific interventions aimed at reducing exposure and therefore reducing transmission of *M. tuberculosis*. These include:

- Triage and patient separation systems (i.e. management of patient flows to promptly identify and separate presumptive TB cases);
- Prompt initiation of effective treatment; and
- Respiratory hygiene including cough etiquette.

Health workers and care providers

- Surveillance and information
- Package of care for HIV-positive workers (ART and TB preventive therapy)
- Protective equipment (particulate respirator masks that meet or exceed N95 standards)
- Relocation for health care workers living with HIV to a lower-risk area

Environmental

To reduce the risk of transmission of *M. tuberculosis*, air can be made less infectious through the use of three principles: dilution, filtration and disinfection. Environmental controls are aimed at reducing the concentration of infectious droplet nuclei in the air. This is achieved by using special ventilation systems to maximize airflow rates or filtration, or by using germicidal ultraviolet (GUV) systems to disinfect the air. Ventilation systems can also be used to control the direction of airflow to reduce the spread of infection; for example, through the use of exhaust fans to generate negative pressure gradients. Environmental controls are used in combination with other IPC measures to help prevent the spread of *M. tuberculosis*

Respiratory protection

Respiratory protection controls are designed to further reduce the risk of exposure to *M. tuberculosis* (and other airborne pathogens) for health workers in special areas and circumstances. The recommendations given here are aimed at strengthening these controls, and preventing the inadequate implementation of respiratory protection programs that may lead to a false sense of security and therefore increase the risk to health care staff.

- Ensure presumptive and TB patients do not walk along rooms in which susceptible individuals are evaluated (Eg. ART clinic)
- Enforcing cough etiquette; health education is given to cover mouth and nose with soft paper, cloth or using one's arm while coughing or sneezing.
- Properly implementing cough etiquette to contain the dispersal of respiratory secretions and practicing respiratory hygiene maneuvers to properly dispose used tissues and masks are important preventive measures hence;
- HCWs should give health education on the prevention and control of TB.
- Posting of pictures showing precautions at waiting area is essential.
- Issuing face masks for MDR-TB patients and providing health education to overcome MDR TB- associated stigma is also important.

To reduce exposure in households:

- Houses should be adequately ventilated, particularly rooms where people with infectious TB spend considerable time (natural ventilation may be sufficient to provide adequate ventilation);
- Anyone who coughs should be educated on cough etiquette and respiratory hygiene so as to behave accordingly at all times; while infectious TB patients should:
 - Spend as much time as possible outdoors;
 - Sleep in an adequately ventilated room;
 - Minimize contact with children(< 5yrs) and immune-suppressed individuals; and
 - Spend as little time as possible in congregate and public transport market.
- Reduce the burden of HIV in patients with presumptive and confirmed TB
- Provide HIV testing and counseling to presumptive and confirmed TB patients

HIV testing is an entry point for HIV care and treatment services including ART. This equally applies to TB patients. Significant proportions of TB patients are co-infected with HIV. Among TB patients who are also HIV-infected, other OIs are significant causes of morbidity and mortality even with a successful treatment of TB. Hence, HIV testing and counseling should be routinely offered to all TB patients.

Presumptive treatment of TB for people living with HIV

The rationale for presumptive TB treatment is to prevent the death of people with HIV in situations where expedited diagnosis of TB is not possible or feasible due to the clinical condition of the patient or limited access to TB diagnostic services. While there is no case definition of presumptive TB, WHO algorithms include initiation of TB treatment for people with HIV in peripheral facilities based exclusively on clinical suspicion (without TB investigations) for seriously sick patients (with respiratory distress) based on the judgement of the clinician.

This approach is based on expert opinion and emphasizes that every effort should be made to confirm the diagnosis of TB after initiation of presumptive treatment and that treatment should be stopped only if there is bacteriological, histological or strong clinical evidence of an alternative diagnosis.

Introduce HIV prevention interventions for presumptive and confirmed TB patients

All clients attending TB clinics should be screened for STI using a set of simple questions. Those with symptoms of STI should be treated or referred to the treatment providers. (Refer to the National Guideline for the Treatment of STI using the Syndromic Approach.) Condoms should be made available in TB clinics.

Provide co-trimoxazole preventive therapy for HIV positive TB patients

Co-trimoxazole is a safe and broad-spectrum antibiotic with activity against a broad range of bacterial, fungal and protozoal infections that are common cause for morbidity and mortality in HIV infected populations. Therefore, CPT is recommended to all TB/HIV co-infected patients regardless of their CD4 Count. All PLHIV with TB and receiving CPT should be registered on the unit TB register, as well as the Pre ART/ ART Register. They should also be monitored monthly.

Ensure HIV prevention, treatment, and care for HIV positive TB patients

Referral linkages between TB and HIV services must be strengthened to provide comprehensive HIV prevention services for TB patients and their families. Tuberculosis control program should implement procedures for prevention of occupational and nosocomial exposure to HIV infection in their services. Health units should be equipped with protective materials and routinely follow standard precautions to prevent HIV transmission in the healthcare settings. Linkage should be ensured for pregnant and non-pregnant HIV positive clients to access services for prevention of mother to child transmission. TB clinics should establish linkage with HIV services to provide the continuum of care and support for HIV positives during and after completion of anti-TB treatment.

Management consideration for TB/HIV Co-infections

In the management of TB/HIV co-infection, treatment of tuberculosis is always given priority over ART. As soon as HIV is identified in a TB patient, the patient should be enrolled to HIV chronic care. Ideally, the HIV care can be delivered at the TB clinic for the duration of TB treatment.

In settings where this is not possible, there should be a strong referral system in place to link the TB/HIV co-infected patient to the HIV clinic promptly. Appropriate clinical, psychosocial and laboratory evaluations (including complete blood count, chemistry tests and CD4 count tests) as per the national guidelines for ART should be performed as soon as possible. Patients should be initiated on CPT and ART regardless of CD4 count or WHO clinical stage and ART regardless of CD4 count or WHO clinical stage. ART is recommended for all HIV infected TB patients regardless of CD4 count or WHO clinical stage. CD4 cells count shall preferably be determined for all HIV infected TB patients. The preferred regimen for TB/HIV co-infected adult patients is TDF+3TC+DTG (adjust dose of DTG to 50mg BID, for children use appropriate weight band), regardless of pregnancy status.

Multidrug-resistant TB and HIV

Multidrug-resistant TB (MDR-TB) is defined as TB that is resistant to at least isoniazid and rifampicin. Patients with both HIV and MDR-TB face complicated clinical management, fewer treatment options and poor treatment outcomes.

Outbreaks of MDR-TB among people with HIV have been documented in hospital and other settings, especially in Eastern Europe and in Southern African countries with a high HIV prevalence. People with HIV with suspected drug-resistant TB should be tested using Xpert MTB/RIF where possible, since this test is more sensitive for detecting TB among people with HIV and rapidly detects rifampicin resistance, thus greatly shortening the time to diagnose and treat MDR-TB.

The burden of MDR-TB can be reduced by strengthening HIV prevention, improving infection control and improving collaboration between HIV and TB control activities, with special attention to the groups at the highest risk of MDR-TB and HIV infection, such as people who inject drugs and those exposed in congregate settings.

5.3. Pneumonia

5.3.1. Bacterial Pneumonia

This can occur in immune-competent individuals but in HIV-infected patients, particularly those infected with *S. pneumoniae*; incidence of bacteremia accompanying pneumonia is increased compared with individuals who are not HIV infected. Bacterial pneumonia occurs during the whole spectrum of HIV disease but tends to be more severe and recurrent as the CD4 counts drops significantly. In addition, pneumonia can concomitantly present with sinusitis and/or bacteremia. If not treated promptly, extra pulmonary complications like empyema, meningitis, pericarditis, hepatitis and arthritis can follow. *Streptococcus pneumoniae* and *Hemophilus influenzae* are the most common etiologies of community acquired pneumonia.

Clinical manifestation

Typically, the patient presents with sudden onset of cough, sputum production, chest pain, fever and/or shortness of breath.

Diagnosis

The clinical suspicion is based on a history of acute symptoms presented over days to a few weeks and/or abnormal physical signs of systemic infection and consolidation in the affected lung/s. Radiologic imaging can assist in confirming the diagnosis of pneumonia.

Treatment

For adults

For less severe pneumonia,

- Amoxicillin 500mg TID OR erythromycin 500 mg QID OR doxycycline 100 mg BID for seven days (Avoid doxycycline in pregnancy).
- Alternative, azithromycin 500 mg per day for three days OR clarithromycin 500 mg twice daily for seven days.
- If not improving after three days with good adherence to the antibiotics, review and consider switching to IV regimen:
 - Ceftriaxone 1-2gm IV once per day plus erythromycin 500 mg oral (if not taken earlier) or IV four times a day for 7 days.

For pediatric age groups:

For less severe pneumonia Amoxicillin 40mg/kg divided in to three doses per day for seven days. In children and infants allergic to penicillin, use erythromycin; for infants aged one month up to two years, erythromycin 125 mg four times daily for seven days; for age groups 2-8 years, erythromycin 250 mg four time per day for seven days; for age groups above 8 years

give erythromycin adult dose. If not improving after three days and if patient is adherent to the antibiotic, review and consider switching to IV ceftriaxone 75-100gm per kg IV/IM once a day or equally divided dosed twice a day for 7 to 10 days. Maximum dose 2-4gm per day for seven days.

Table 5. 8. Management of less severe pneumonia in PLHIV

Management of less severe pneumonia			
Age	First line treatment	If allergic to penicillin	If not improving after 3 days and good adherence
All ages	Amoxicillin 40mg/kg in three divided doses for 7 days	-	Ceftriaxone IV/IM: 75 mg/kg, once a day or in two divided doses for 7-10 days (Max dose: 2-4 gm/day)
1month-2 years	-	Erythromycin: 125 mg, QID, 7 days	
2-8 years	-	Erythromycin: 250 mg, QID, 7 days	
Above 8 years	-	Erythromycin: 500 mg, QID, 7 days	

When the patient has presented with clinical evidence of severe pneumonia, admit for parenteral antibiotic treatment and supportive therapy or refer the patient if admission is impossible. Signs of severe pneumonia include:

Children:

- Chest in-drawing, grunting and presence of danger signs in children

Adults:

- Tachypnea, old age (>65 years), cyanosis, hypotension, systolic blood pressure <80mm Hg, multi-lobar involvement and altered mental status,

The respiratory rate in tachypnea is summarized as below:

S. No.	Age-group	Respiratory Rate (RR)
1	Birth to 2 months	>60/minute
2	2 months to 1 year	>50/minute
3	1 year to 5 years	>40/ minute
4	5 years and above	>30/minute

5.3.2. Pneumocystis pneumonia

Pneumocystis pneumonia is caused by *Pneumocystis jiroveci*, formerly known as *pneumocystis carinii* pneumonia, a ubiquitous organism that is classified as a fungus but also shares biologic characteristics with protozoa. It commonly occurs when patients have significant immune suppression (adults: CD4<200cells/mm³ or children: CD4 percentage < 14%).

Clinical manifestation

Typical clinical presentations are characterized by insidious onset of low grade fever, dry cough, and dyspnea exacerbated by exertion. Physical examination reveals fever, tachypnea, tachycardia and scattered rales in the lungs, but examination of the lungs can appear normal in some patients. In children highest incidence is seen between 2-6 months of age and is characterized by abrupt onset of fever, tachypnea, dyspnea and cyanosis.

Diagnosis

Presumptive diagnosis of PCP is based on clinical judgement and typical chest X-ray findings revealing a perihilar interstitial infiltration with tendency to spread outwards. Note that the chest X-ray can be normal in 20% of patients. Definitive diagnosis of PCP is based on demonstration of the organism from

an induced sputum sample using special stains like Giemsa or methylene silver stains, but these tests are not routinely done in Ethiopia.

Treatment

- Use Trimethoprim 15-25 mg/kg, three or four times daily for 21 days.
- Close monitoring is necessary during the initial five days of treatment and if patient grows sicker, administration of oxygen is useful.
- In severely ill patients with marked respiratory distress and extensive chest X-ray findings:
 - Children: add prednisolone - 2mg/kg per day for the first 7 - 10 days followed by a tapering regimen for the next 10 - 14 days.
 - Adults: add prednisolone - 80mg for the first five days, 40 mg until 11 days and 20 mg until completion of intensive co-trimoxazole therapy.
- Toxicity of co-trimoxazole, like skin rash, bone marrow suppression, hepatitis and renal failure can be troublesome in some patients with advanced HIV disease and requires close monitoring.

Secondary prophylaxis after completion of the course of treatment with co-trimoxazole should be continued. (Refer Table 4.1).

The alternative regimens for mild to moderate cases of PCP include:

- Clindamycin 600 mg qid plus primaquine 15 mg bid; or
- Clindamycin 600 mg qid plus dapsone 100 mg daily.

Consider spontaneous pneumothorax in patients with sudden deterioration in clinical condition.

5.3.3. Lymphoid Interstitial Pneumonitis

Lymphoid interstitial pneumonitis (LIP) is one of the most common chronic lower respiratory conditions occurring in up to 25% of children with HIV/AIDS.

Clinical manifestations

It ranges from asymptomatic disease with isolated radiographic findings to bullous lung disease with pulmonary insufficiency. Symptomatic children present with insidious onset of tachypnea, cough, and mild to moderate hypoxemia with normal auscultatory findings or minimal rales or wheezing. Progressive disease is accompanied by digital clubbing and symptomatic hypoxemia. Associated physical findings include generalized lymphadenopathy, hepatosplenomegaly and parotid enlargement.

Diagnosis

It is usually based on findings of clinical examination. Diffuse bilateral reticulonodular infiltrate on X-ray with mediastinal lymphadenopathy. It is important to exclude tuberculosis and other infectious etiology.

Treatment

Provide symptomatic treatment (hydration, oxygen). Use antibiotics if there is a superimposed bacterial infection. Bronchodilators may be helpful in mild to moderate disease. Corticosteroids are usually reserved for children with significant hypoxemia and symptoms of pulmonary insufficiency. Give prednisolone 1 – 2 mg/kg/24 hrs for 6 – 8 weeks and then taper as tolerated.

5.4. Hepatitis B and C

Introduction

Chronic Hepatitis B and C Virus infection are major global public health problems. WHO estimates that, in 2019, 71 million people had chronic HCV infection and 257 million people chronic HBV worldwide, and 820 000 people died from HBV and 290 000 from HCV, mainly from cirrhosis or hepatocellular carcinoma. In 2019, there were 1.5 million new chronic HCV infections. In Ethiopia, over 30 studies have been conducted since 1980's, to determine a sero-prevalence of various hepatitis viruses in the country. It is reported that 12% of the hospital admissions and 31% of mortality in medical wards were due to all causes of chronic liver

disease. The National sero-survey conducted by MoH and EPHI in 2017 shows the prevalence of HBsAg was estimated to be 9.4 among the general population aged 15 years and above with regional variations; the highest being in the Afar region (28.8%). The pooled prevalence of anti-hepatitis C virus antibody (anti HCV) was 3.1%. Unlike HBV, anti HCV virus prevalence in HIV infected individuals were higher, 5.5%.

Early identification of people with chronic HBV or HCV infection enables them to receive the necessary care and treatment to prevent or delay the progression of liver disease. Testing also provides an opportunity to link people to interventions to reduce transmission, through counselling on risk behavior and provision of prevention commodities and HBV vaccination.

In our set up systematic screening for viral hepatitis B and C is prioritized for the following population groups:

- People who have received medical or dental interventions in settings where infection control practices are substandard
- Pregnant women
- Children born to HBV or HCV positive mothers
- People with injecting drugs
- People living with HIV
- Health care workers exposed to biological fluids
- People who ever received blood or blood products

- Inmates of correctional facilities
- Household and sexual contacts of HBsAg positive people
- Female Sex workers
- History of multiple sexual partners or STIs
- Patients undergoing renal dialysis
- People needing immunosuppressive therapy

Chronic HBV is defined as persistence of HBsAg for more than six months and chronic HCV infection is HCV antibody positive with viremic HCV infection.

5.4.1.HBV/HIV co-infection management

HIV co-infection has been shown to have a profound impact on almost every aspect of the natural course of HBV infection and includes more rapid progression to cirrhosis and hepatocellular carcinoma (HCC), higher liver-related mortality, and decreased treatment response compared with persons without HIV coinfection.

HIV/HBV co-infected persons also demonstrate more rapid HIV disease progression compared to those who are HIV-infected alone, and have an impaired recovery of CD4 cells. HIV clients are among the high risk groups for HBV and should be given priority for screening. i.e all HIV clients should get screened for HBV and evaluated for chronic infection as per the national viral hepatitis prevention and control guideline. After test result, vaccination or treatment and care for reactive patients are recommended when resources permit.

Treatment options for patients with HIV/ HBV co-infection:

- ART should be started rapidly.
- If treatment is indicated for HBV, TDF+3TC + DTG as a preferred regimen.
- The use of lamivudine as mono-therapy in any of these diseases is contraindicated due to high YMDD resistance.
- When switching treatment in patients with HIV on ART failure, the second line regimen should contain TDF + 3TC to continue the treatment against HBV.
- If tenofovir-associated renal toxicity occurs, the dose of tenofovir should be adjusted according to the renal clearance.
- Note: treatment of HIV without the use of TDF in the regimen may lead to flares of hepatitis B due to ART-associated immune reconstitution.
- Similarly, treatment discontinuation, especially of 3TC, has been associated with HBV reactivation, ALT flares and, in rare cases, hepatic decompensation.

NB: For further reference please consult the national Viral Hepatitis prevention and control guidelines.

5.4.2.HCV HIV co-infection management

HIV patients are among high risk groups for HCV and should be given priority for screening. Therefore, all HIV patients should be screened and confirmation VL test should be done for HCV screened positives.

Anti-HCV rapid diagnostic test (RDT) or immunoassay (IA) can be used for screening and HCV RNA viral load test using either quantitative or qualitative PCR should be used to confirm chronic HCV infection.

Persons with HIV/HCV coinfection generally have more rapid disease progression than mono-infected persons. Even among persons in whom ART leads to successful control of HIV infection (i.e. undetectable HIV viral load), the risk of hepatic decompensation among coinfecting persons is higher than among persons with HCV mono-infection.

All chronic HCV infected individuals should be treated to eradicate the virus and achieve cure so that complications can be avoided. Follow up quantitative or qualitative HCV RNA viral load is required to confirm if the patient has achieved Sustained Virologic Response (SVR).

HCV treatment outcomes with DAAs are comparable in persons with HIV/HCV coinfection to those with HCV mono-infection. Because DAAs are safe and effective for people with HIV/HCV, there is no longer any need to consider them as a special or difficult-to-treat population. However, there are important drug-drug interactions and shared side effects like headache, fatigue and anemia with pan-genotypic HCV regimens and ART. Therefore,

checking for drug-drug interactions and side effects between HIV and HCV medications needs to be given appropriate attention. The HCV treatment for children is differed until 12 year of age and treatment with interferon-based regimens should no longer be used.

For adolescents (12–17 years), the treatment at this time still requires genotyping to identify the appropriate regimen. This include:

- Sofosbuvir/ledipasvir 12 weeks in genotypes 1, 4, 5 and 6
- Sofosbuvir/ribavirin 12 weeks in genotype 2
- Sofosbuvir/ribavirin 24 weeks in genotype 3

For adults without cirrhosis, the following pan-genotypic regimens is recommended for use in adults 18 years of age or older can be used:

- Sofosbuvir/velpatasvir for 12 weeks
- Sofosbuvir/daclatasvir for 12 weeks
- Glecaprevir/pibrentasvir 8 weeks³

For adults with compensated cirrhosis, the following pan-genotypic regimens can be used:

- Sofosbuvir/velpatasvir for 12 weeks
- Glecaprevir/pibrentasvir for 12 weeks
- Sofosbuvir/daclatasvir for 24 weeks

Refer the national viral hepatitis prevention and control guideline for further information

5.5. Sexual Transmitted Infection and Cervical Cancer

The epidemiological synergy between HIV and sexually transmitted infections is well established, and they frequently coexist. Most of these infections are asymptomatic, especially among women. However, even asymptomatic STIs can cause complications, be transmitted to sexual partners and enhance HIV transmission. STI services should be an integral part of comprehensive HIV care among adults and adolescents. Refer the revised STI guideline for syndromic management of STIs.

Prevention of cervical cancer:

Cancer of the cervix is the second most common cancer, among women in Ethiopia with over 6,200 new cases every year and with a mortality close to 5,000 (Globocan 2018). According to Addis Ababa Population Based Cancer Registry (2012-2015), the leading cause cancer type among women is breast cancer (32.3%) followed by cervical cancer (14.5%).

Cervical cancer is a preventable disease and is curable if diagnosed and treated early. 99% of cervical cancer is caused by high risk human papilloma viruses (HPV) types. Types 16 and 18 cause 70 percent of cervical cancer. Women living with HIV (WLHIV) are about six times more at risk of developing pre-cancerous lesions and invasive cervical cancer. The risk and persistence of HPV infection increases with low CD4 count and high HIV viral load.

Target Population:

Women Living with HIV (WLHIV) are at a much greater risk for developing cervical cancer and require a more frequent screening. Start screening at the time of HIV diagnosis, regardless of age, once sexually exposed (15-49 is recommended).

Screening Frequency:

WLHIV should be screened every two years as part of routine care for HIV-positive women, regardless of whether they are on antiretroviral. Following abnormal results and/or treatment, repeat screening at 6months. If follow-up screening is normal, return to screening every two years.

Screening methods:

Depending on the availability of resources, this guideline recommends the following types of screening methods (refer annex 19 and 20 for the algorithm).

1. Visual inspection: with acetic acid (VIA) or Lugol's iodine (VILI)
2. HPV DNA test
3. Cytology:
 - conventional (Pap smear)
 - liquid-based cytology (LBC)

Treatment of Precancerous Lesions:

a. Cryotherapy

Cryotherapy is the treatment of choice for precancerous lesions that meet eligibility criteria. Treatment is to be offered without requiring biopsy diagnosis (screen and treat) in a single visit approach whenever feasible. Only providers who have demonstrated clinical competencies in cryotherapy are permitted to perform the procedure. Treat using a double-freeze (three minutes freeze, five minutes defrost, three minutes freeze) technique to achieve a 3-5 mm ice ball around the cryo-tip. Follow-up screening with VIA for those VIA positive is in one year.

NB: Do not treat with cryotherapy during pregnancy. Reschedule the woman when she is more than 12 weeks postpartum.

b. Thermal ablation

Thermal ablation is another novel ablative treatment of option for precancerous lesions, and is sometimes called "cold coagulation" or "thermocoagulation". The equipment is fairly simple, and treatment is based on a 20–30 second application of a reusable metallic probe that is electrically heated to approximately 100 °C, leading to epithelial and stromal destruction of the lesion. The treatment time is shorter with thermal ablation and also the use of thermal ablation overcomes one major disadvantage of cryotherapy which is the need for a refrigerant gas (N₂O or CO₂). The gas containers are bulky and heavy to transport and some areas of LMIC may have supply issues. In addition, frequent refilling of freezing gas can be costly. The use of thermocoagulation for the treatment

of CIN is as effective as other methods, such as cryotherapy and LEEP, with the advantage of being rapid and is also associated with low occurrence of side effects

c. Loop Electrosurgical Excision Procedure (LEEP)

LEEP is reserved for precancerous lesions that are not eligible for cryotherapy or thermal ablation. It should be done by providers who have demonstrated clinical competence in the procedure. LEEP requires anesthesia and is to be performed only in settings that can handle potential urgent complications related to the procedure (e.g., heavy bleeding). Follow-up screening with VIA in one year.

d. Conization (Cone Biopsy)

Conization is recommended for the treatment of lesions that cannot be treated with cryotherapy

and unclear type of cervical abnormality to rule out invasive cervical cancer. It involves the removal of a cone-shaped area from the cervix, including portions of the outer cervix (ecto-cervix) and inner cervix (endo-cervix) and is usually done in the operating room under general or regional (spinal or epidural) anesthesia.

Linkage and Integration of services:

ART clinics should be closely linked with VIA clinics or sites providing cervical cancer prevention services, or ideally, provide the services at the unit.

Table 5.9. Healthcare delivery level by type of service and type of health care workers

Healthcare delivery level	Service provided	Type of health care workers
Tertiary hospital	VIA/cryotherapy/thermal ablation, PAP smear, colposcopy, LEEP biopsy, histopathology radiotherapy, radical surgery chemotherapy and palliative care	Nurse, midwives, health officer, MD residents and specialists (gyn/obn, gyn-oncologists)
Regional/general hospitals	VIA/cryotherapy/thermal ablation, PAP smear, colposcopy LEEP biopsy histopathology, radical surgery and palliative care	Nurse, midwives, health officer, MD residents and specialists (gyn/obs, gyn-oncologists)
District hospitals / primary hospitals	VIA/cryotherapy/ thermal ablation, biopsy, palliative care	Nurse, midwives, health officer, MD
Health centers	VIA/cryotherapy/ thermal ablation, palliative care	Nurse, midwives, health officer

For details on prevention, screening and treatment, refer to guideline for cervical cancer prevention and control in Ethiopia, 2021.

5.6. Non-Communicable Disease (NCD)

According to WHO estimates, non-communicable diseases, including hypertension, cardiovascular disease, renal disease, cancer, chronic respiratory disease, diabetes and mental health disorders, account for 63% of global deaths. Low- and middle-income countries bear 86% of the burden of non-communicable diseases.

Compared with the general population, PLHIV have increased risk of developing a range of chronic non-communicable diseases, including cardiovascular disease, hypertension, diabetes, chronic obstructive pulmonary disease, kidney disease and cancer.

Cardiovascular disease is now one of the leading causes of non-AIDS-related morbidity and mortality among PLHIV. Both HIV and non-communicable diseases require health systems that can deliver effective acute and chronic care and support adherence to treatment. Chronic HIV care provides the opportunity for assessing, monitoring and managing non-communicable diseases, especially through primary care.

Integrating interventions such as nutrition assessment, dietary counselling and support, smoking cessation, exercise promotion, blood pressure monitoring and when available cholesterol management as part of HIV care can help to reduce the risks of non-communicable diseases among PLHIV and improve HIV treatment outcomes. Therefore, screening of PLHIV for these comorbidities during every visit is a critical component of care and treatment package.

5.7. HIV and Mental Health Disorders

Globally, Mental Health Disorders (MHD) are more common among PLHIV than among the general population. Depression, Anxiety, PTSD, and Cognitive impairment are common types of MHD in patients following HIV diagnosis or during its progression to AIDS. The magnitude of common MHDs among PLHIV in Ethiopia has been studied and the following results show that the prevalence is significantly high. A meta-analysis has shown that prevalence of depression among PLHIV in Ethiopia was 36.65%, and the prevalence for common mental disorders among PLHIV have been found to be 33% in Hawasa and 24% in Debre Markos.

One of the reasons why mental health problem are more common among PLHIV is that having MHD is one of the risk factor for becoming infected with HIV. People with MHDs are more likely to be exploited by others and less able to negotiate safe sexual relationships including use of condoms with partners. They may be less likely to stay in the kind of steady, long-term relationships in which partners can protect each other from getting HIV.

Furthermore, mental, neurological and substance use disorders (MNS) are important in HIV care because of issues related to adherence, risk of HIV transmission, and overall outcome of HIV care. Evidence also suggests that treatment of co-morbid psychiatric conditions may improve adherence to ART, underlining the importance of recognition and treatment of psychiatric conditions. Evidence further pointed out that there is continued under-recognition and under treatment of these conditions among PLHIV. Factors contributing to the under-recognition of

MNS symptoms related to:

- The overlap between the symptoms of depression and symptoms of HIV disease.
- Related to recognition of certain psychiatric disorders, specifically depression and anxiety, there is a mistaken belief that these symptoms are expected in those diagnosed with HIV disease. Depression or anxiety symptoms that are more severe or that persist beyond the period of initial discovery of HIV infection should prompt evaluation and treatment.
- The neuropsychiatric side effects associated with some ARVs like Zidovudine, abacavir, and efavirenz have been associated with neuropsychiatric symptoms.

The other issue worth mentioning with regard to comorbid HIV and MNS problems is drug-drug interactions (DDIs) between the ARV drugs and psychotropic drugs. DDIs are important to consider in the co-treatment of the two health conditions because they make either group of drugs less effective or toxic.

Therefore, integration of MNS symptom screening into HIV services is important to improve treatment outcome of PLHIV. This has been implemented in Ethiopia using a task-sharing approach among lay healthcare workers (case managers) and clinicians. In this model, trained lay case managers proactively screened patients using a mental health screening tool and subsequently linked potential clients with trained clinicians working at HIV clinics for further diagnosis and treatment. A pilot project study conducted in selected sites in Amhara

and Tigray regions showed that, of the total PLHIV screened by case managers, clinicians confirmed that about 8% of them had some form of MHD. In this study, the concordance between the case managers' screening results and the clinicians' diagnoses was about 68% over the 15-month pilot implementation period. Therefore, the proactive screening of PLHIV for MHD helps to early identification and management of PLHIV with co-morbidities. Ten percent or more of people will experience one of the common mental disorders in the course of their life. It includes problems with low mood, excessive worry, or difficult behaviors that do get in the way of daily function.

The rest are categorized as severe mental disorders which are much less common, but much more serious. They include problems with thinking that severely limit function and may make the person a danger to themselves or others.

Below are the common MNS problems integrated into HIV clinic. Please refer to annex 20 for the brief mental health disorders symptom screening and referral tool.

Priority mental health disorders (WHO)

1.Psychosis: this is the collective name for a group of serious disorders characterized by changes in behavior (for example poor self-care, restlessness), strange thoughts or beliefs (for example believing that others wish to do the individual harm) and related dispositions.

2.Mania: a form of severe mental illness in which a person is excessively happy or irritable (experiences extreme mood swings), appears

over-active and sleeps poorly. People with mania have poor reasoning skills (they have difficulty understanding what is good and what is bad), and display excessive self-confidence.

3.Depression: this is the most common priority disorder and is characterized by excessive sadness, loss of interest, lack of energy and related symptoms.

4.Suicide: refers to the intentional ending of one's own life.

5.Abuse of alcohol and other substances: this is the excessive use of these substances to the detriment of one's health.

6.Childhood mental disorders: these are different types of mental disorders related to childhood developments.

7.Dementia: is characterized by memory problems and broader problems with thinking and understanding.

8.Epilepsy: this is a chronic or longstanding condition caused by abnormal electrical conduction in the brain. In its most obvious form, it is characterized by episodic loss of consciousness and repetitive jerky movements of the body.

On the other hand, ACMs/HCWs should prioritize the below PLHIV groups for MHD screening at ART and PMTCT clinic.

- Newly initiated on ART
- Patients with high viral load
- Patients re-engaged into care after treatment interruption

- Patients with problem of ART adherence
- Patients who refused to start ART
- Patients who are on ART preparation for >2 weeks
- Patients with advanced HIV disease
- Patients with other health problems

Mental health aspects of living with HIV

Reaction to the diagnosis

People have many different reactions to learning that they have HIV infection. They may react initially with anger, shock, or denial and these reactions can change or recur over time.

If people **feel guilty** for something that they feel led to their acquisition of HIV:

- Reassure that no one can foresee all the consequences of what they do
- Reassure that life offers many opportunities to do good and they will have many chances in the future feel better about themselves.

For feelings of loneliness and isolation

- Note that there are millions of people living well with HIV, both in Ethiopia and other countries
- Note that they are right to be cautious about who they tell, but that over time they will find many accepting people and develop many close relationships
- Remind them that the ART site and its many services will always be a place that they can turn to for support and care

For feelings of grief and loss

- Acknowledge that this is a big change, that life looks very different
- Be optimistic that the loss will heal with time
- Express your willingness to be of support
- Suggest self-care
- Remind them that this is a treatable problem, and that you and the ART staff will be here to evaluate their concerns.

Mental health side effects of ARV drug

Many of the medications used to treat HIV and related infections have mental health-related side effects. For most medications, these are

not common or severe. Some medications to think about in particular are:

- Efavirenz can cause mental health-related side effects in as many as half of patients taking it. Symptoms can begin on the first day of treatment but are likely to resolve after 2-4 weeks. Some can be very troubling, and include confusion, trouble concentrating, and vivid, odd dreams. Hallucinations and mood changes have also been seen, and depression has been reported. There is concern that these side effects can be made worse by alcohol or other substances. INH (used in the treatment of TB) can cause hallucinations, paranoia, and low mood.

Table 5.10. ART-psychotropic/antiepileptic drug interactions

Psychotropic drugs	Interactions
Carbamazepine	Decrease PI and NNRTI levels; use other anticonvulsant. Carbamazepine toxicity possible if combined with RTV.
Diazepam, bromazepam	All PI's can increase levels of diazepam and bromazepam, as can Efavirenz and nevirapine can decrease diazepam levels. Lorazepam, oxazepam, and temazepam are unaffected by ARVs.
Fluoxetine	No specific problems for PIs or NNRTIs. Can be some variation in the level of fluoxetine or the ARVs. But does interact with many other medications.
Haloperidol	PIs can increase haloperidol level; NNRTIs can decrease haloperidol levels.
Phenobarbital	Can decrease PI and NNRTI levels but is ok to use with NRTIs. Not recommended with LPV/r.
Phenytoin	Can decrease PI and NNRTI levels but is ok to use with NRTIs. Not recommended with LPV/r. If must use with LPV, may need higher doses of both medications.
Risperidone	Some PIs may increase risperidone levels. Avoid use with RTV.
Tricyclic antidepressants	OK with NNRTIs and NRTIs. OK with PI's except LPV/r, RTV, – those medications can raise the TCA level to toxic levels (use fluoxetine instead).
Valproic acid	No significant interactions

5.8.HIV related skin and oral conditions

HIV infection increases the prevalence and severity of skin and oral diseases, especially when the person's CD4 cell count declines below 200 cells/mm³. As a result, skin and oral conditions affect up to 90% of adults and children with HIV in resource-limited settings. Adverse drug reactions of the skin are also 100 times more common among people living with HIV than among the general population, and their prevalence increases as immunodeficiency worsens. Immune reconstitution inflammatory syndrome occurs in 10–25% of people living with HIV starting ART.

Different kinds of OI, such as herpes zoster, and other viral, fungal and bacterial infections occur in the skin. Manifestations of adverse drug reactions and non-infectious conditions also occur in the skin. In most instances diagnoses of skin disorders with HIV disease are made on clinical grounds. Most skin disorders in HIV disease can be cured or ameliorated, but a few fail to improve even with good general clinical and immunological responses to ART.

Pruritus is the most common dermatologic symptom in HIV infected patients. It can be localized indicating primary skin lesion, or generalized that may or may not indicate primary skin lesions. In many patients, pruritus may be severe and may not be amenable to available therapy. The most common skin conditions associated with pruritus in patients with HIV include the following:

- 1.Excessive dryness of the skin (Xerosis cutis)
- 2.Eczemas like seborrheic dermatitis or contact dermatitis
- 3.Folliculitis that may include infections by *Staphylococcus aureus* or hypersensitivity to insects
- 4.Drug eruptions
- 5.Scabies
- 6.Intertrigo (*Candida*, tinea, herpes simplex)

In most patients, diagnosis can be established by examining the lesions. However, as immune deficiency advances it may be useful to use investigations such as biopsy to diagnose specific dermatosis or use staining and culture to diagnose specific infections.

5.8.1.Etiological classification of skin disorders in HIV disease

Skin disorders in HIV infected patients can occur due to infections, neoplasm, and hypersensitivity to foreign agents including drugs, or to unknown causes. Nevertheless, infections are commonly seen in clinical practice; refer to the following table:

Table 5.11: Common skin infections in HIV disease.

Infections	Disease	Clinical Presentations	Treatment	Remark
Bacterial	Cellulitis	Poorly defined erythema. Pus and crust at the site plus signs of inflammation.	Adult: Amoxicillin 500 mg TID for 5-14 days based on clinical response or erythromycin 500 mg QID if allergic to penicillin. Children: 20-40mg/kg per day on TID basis	Mostly encountered lower extremities and often unilateral.
	Impetigo	Erythematous small papules usually limited to few lesions coalescing in to crusted plaques.	Use topical antibiotics: use Amoxicillin for extensive disease for 5-14 days (see dose above for adults and children) for 7 days	Usually a superficial lesion.
	Carbuncle	Nodular lesion with extensions to the deeper structure. Signs of inflammation present.	Adults: Cloxacillin 500 mg QID for ten days. Children: 50-100 mg/kg/day on QID basis	Involves the trunk as well as extremities.
Viral	Herpes simplex	Painful vesicular lesion around mouth or genitalia. Recurrent and extensive, difficult to eradicate during advanced immune deficiency.	Adults: Acyclovir 400 mg TID for ten days. Children 20 mg/kg/dose 4X/d	
	Herpes zoster	Painful and vesicular eruptions with dermatomal distribution. When healed, scar will remain.	Adults: Acyclovir 800 mg 5X per day for seven days. Monitor renal function. Children: 20-40 mg/kg per dose four times a day.	When it involves the eyes, it is a medical emergency. Do not give Acyclovir* if duration is >72 hours.

Infections	Disease	Clinical Presentations	Treatment	Remark
Viral	Warts / verrucae	Painless flat to raised warts over fingers or genitalia in advanced immune deficiency, they tend to be multiple and exophytic.	Podophyllin, Imiquimod, Cryotherapy, Consult experts	Premalignant and risk for cervical cancer.
	Molluscum Contagiosum	Umbilicated and raised facial lesions that tend to be very big during immunodeficiency state.	May not require therapy;	Contagious
Parasitic infestation	Scabies	Pruritic lesions ranging from pinpointed erythematous papules involving inter digital, axillae and groin areas to varying degrees of hyperkeratotic plaques associated with significant skin thickening and crusting.	BBL, lindane or permethrin to be applied to whole body. Ivermectin 200 microgram/kg stat orally.	Burrows are visible in mild infestations but in crust scabies may not be evident leading to misdiagnosis.
Fungus	Dermatophytosis	Superficial causing ringworm or athlete's foot	Topical antifungal for limited skin affected. Fluconazole for extensive lesion 100mg daily for ten days.	
	Thrush	White plaques on the buccal mucosa including the tongue that can be scraped off leaving red base (bleeding). Can be associated with candida paronychia or intertrigo.	Miconazole gel 2% apply BID Fluconazole 100 mg daily for ten days for recurrent or oropharyngeal thrush.	
	Deep Fungal Infection	Presentation varies from fungating nodules and tumors to ulcers and diffuse papulo nodular disease	Disseminated Cryptococcus can be confused for Molluscum contagiosum. Treat with amphotericin induction and/ or fluconazole maintenance.	

*If patient has ophthalmic involvement refer to ophthalmic specialist.

5.8.2. Pruritic Papular Eruption (PPE)

Pruritic papular eruption is common among HIV infected patients causing substantial morbidity in sub-Saharan Africa. Its prevalence ranges from 12-46% and it is uncommon in HIV negative patients (PPV of 82-87% and may play role in diagnosing HIV). The pathogenesis is unknown but it may be related to hypersensitivity to arthropod bites. In extreme form, eosinophilia and eosinophilic infiltrates of the skin are present. Severity of rash often correlates with CD4 count. The clinical manifestation is intensely pruritic, discrete, firm papules with variable stages of development and predilection for extremities, though it can involve trunk and face with significant itching marks. Excoriation results in pigmentation, scarring and nodules. Treat with topical steroid and oral antihistamines. However, it is often refractory to treatment and hence short course prednisolone may be used. ART is often effective.

5.8.3. Dysphagia and odynophagia

Dysphagia (difficulty in swallowing) and odynophagia (painful swallowing) are symptoms of esophagitis occurring at advanced stages of AIDS. They are usually caused by candida, HSV, CMV, and aphthous ulcers. Esophagitis seriously impairs the patient's nutritional status. Therefore, prompt diagnosis and treatment are mandatory to avert nutritional complications and inability to swallow prescribed medications. Children will present with reluctance to eat, excessive salivation, or crying while feeding. If thrush is associated with dysphagia, odynophagia, and/or retrosternal pain, consider oesophageal candidiasis but this can also occur in the absence of oral thrush.

Oral candidiasis presents with oral sores, change in the sense of taste, and when it involves the throat and oesophagus, pain on swallowing; however it can be asymptomatic in some patients. Diagnosis is established clinically when a curd-like membrane is visible on the surface of the tongue and buccal mucosa. Typically the base of the membrane bleeds upon scraping it.

Diagnosis

Diagnosis of oral candidiasis is established clinically when a curd-like membrane is visible on the surface of the tongue and buccal mucosa. Typically, the base of the membrane bleeds upon scraping it.

Oesophagitis is frequently made on clinical grounds, but when facilities are available upper GI endoscopy with or without biopsy or contrast imaging may be done.

Treatment

Oral candidiasis: Oral and pharyngeal thrush are treated with oral miconazole gel 2% applied twice daily; patients should not eat/drink for two hours after applying the gel. If this does not work or is unavailable, use fluconazole 100mg daily for 14 days. Nystatin can also be used in place of miconazole as follows: apply suspension 5 times daily (after each meal and between meals) for 7 days (or until 48 hours after lesion resolves).

Dysphagia and/or odynophagia are treated as oesophageal candida on clinical grounds, in particular when oropharyngeal candida is present. Patients are empirically treated with Fluconazole in presumptive oesophageal

candida. If the response is unsatisfactory, they should be referred or investigated if facilities are available, to rule out other causes.

- Drug of choice: Fluconazole 200 mg (3mg/kg/day in children) PO daily for 14 days.
- Alternative: Ketoconazole 200 mg(3-6mg/kg/day daily in children) twice daily for 4 weeks.

Risk of recurrence after completing treatment may be high. If the patient is on ART, s/he should be investigated for treatment failure. Take necessary precautions regarding drug interactions especially with ketoconazole. Patients may need hospital admission for supportive care till the oesophageal symptoms improve and necessary long term treatments are started. If diagnosis suggests HSV oesophagitis use acyclovir 400mg po five times for 14 to 21 days.

5.9.HIV related Neurological conditions

Neurological manifestations of HIV can occur at any time from viral acquisition to the late stages of AIDS. They are varied and may affect any part of the nervous system including the brain, spinal cord, autonomous nervous system and the peripheral nerves. HIV affects the nervous system in 70-80% of infected patients. The effect may be due to direct effect of the virus, opportunistic infections and/or malignancies. For certain neurological manifestations, a single aetiology is responsible while in others it is due to multiple causes.

Most life-threatening neurological complications of HIV occur during the severe immunodeficiency state and specific aetiological diagnosis in the

Ethiopian setting is often a major challenge. Thus, this unit attempts to guide the management of common opportunistic infections and other treatable conditions in the nervous system.

Neurological complications in HIV patients may be due to:

- HIV (HIV encephalopathy)
- OIs (toxoplasmosis, cryptococcal meningitis)
- Neurosyphilis
- Malignancies (primary CNS lymphoma); and
- Drugs (e.g. EFV, etc.)

Diagnosis of neurological disorders in HIV in our setting depends on the history and standard neurological examinations. In view of this, health care providers must be able to perform a physical examination to detect neurological abnormalities.

There can be single or multiple abnormal neurological findings in the same patient necessitating holistic neurological evaluation. Thus, the examination should include assessment of:

- Mental status comprising cognitive function, orientation and memory.
- Cranial nerves.
- Motor function including deep tendon reflexes (DTR).
- Sensation.

5.9.1. *Toxoplasma gondii* encephalitis

Toxoplasmic encephalitis (TE) is caused by the protozoan *Toxoplasma gondii*. Disease appears to occur almost exclusively because of reactivation of latent tissue cysts. Primary infection occasionally is associated with acute cerebral or disseminated disease. Sero-prevalence varies substantially in different communities; in Ethiopia, general prevalence is about 80%.

Clinical Manifestations

Among patients with AIDS, the most common clinical presentation of *T. gondii* infection is focal encephalitis with headache, confusion, or motor weakness and fever. Patients may also present with non-focal manifestations, including only non-specific headache and psychiatric symptoms. Focal neurological abnormalities may be present on physical examination, and in the absence of treatment, disease progression results in seizures, stupor, and coma.

Diagnosis

HIV-infected patients with TE are almost uniformly seropositive for anti-toxoplasma immunoglobulin G (IgG) antibodies. The absence of IgG antibody makes a diagnosis of toxoplasmosis unlikely but not impossible. Anti-toxoplasma immunoglobulin M (IgM) antibodies usually are absent. Quantitative antibody titres are not useful for diagnosis. Definitive diagnosis of CNS toxoplasmosis requires a compatible clinical syndrome; identification of one or more mass lesions by CT, MRI, or other radiographic testing; and detection of the

organism in a clinical sample. In the absence of imaging support, empirical treatment is justified when patients present with focal neurological findings and the CD4 count is < 200 cells μl . Failure to respond to conventional therapy, based on presumptive clinical diagnosis within a week or two of initiation of therapy, suggests the diagnosis to be unlikely.

With empirical treatment for toxoplasmosis, nearly 90% of patients will demonstrate clinical improvement within days of starting therapy. Radiological evidence of improvement is usual after 14 days of treatment.

Treatment

First line regimen in the Ethiopia:

- Adult: Trimethoprim/sulfamethoxazole 80/400, oral, 4 tablets 12 hourly for 28 days, followed by 2 tablets 12 hourly for 3 months in adults.
- Children: 10mg of trimethoprim + 50mg of sulfamethoxazole/kg per dose every 12 hours for 28 days followed by maintenance therapy at 50% reduced dosage for three months.

Secondary prophylaxis: use co-trimoxazole 960mg daily for adults and in children. Refer to Table 4.1

Alternative regimen

I. Sulfadiazine, 1-2 gm PO QID for six weeks or 3 weeks after resolution of lesion
Side effects: crystal urea, rash

Contraindication: severe liver, renal and haematological disorders; known hypersensitivity to Sulphonamides.

Dosage/form: 500 mg tablets,

PLUS

Pyrimethamine: loading dose of 200 mg once, followed by 50-75 mg/day.

Side effects: rash, fever and bone marrow suppression (neutropenia and thrombocytopenia).

Contraindication: folate deficiency

Dosage/form: 25 mg tablets

PLUS

Folinic acid (Leucovorin): 10-20 mg/d

Side effects: allergy

Dosage/form: 5 and 10 mg tablets

OR

II. Pyrimethamine and Folinic Acid (Leucovorin): (standard dose)

PLUS

Clindamycin: 600 mg QID

Side effects: toxicities include fever, rash, nausea and diarrhoea (including pseudomembranous colitis or diarrhoea related to *Clostridium difficile* toxin).

Adjunctive corticosteroids should be used for patients with radiographic evidence of midline shift, signs of critically elevated intracranial pressure or clinical deterioration within the first 48 hours of therapy. Dexamethasone 4 mg every six hours (0.15mg/kg/dose every 6 hours for children) is usually chosen and is generally tapered over several days and discontinued as soon as possible.

Anticonvulsants should be administered to patients with a history of seizures but should not be given routinely for prophylaxis to all patients with the presumed diagnosis of TE. Careful attention needs to be paid to any potential drug interactions.

5.9.2. Cryptococcal infection

Cryptococcal meningitis is one of the most important opportunistic infections and a major contributor to high mortality before and after ART is initiated. The main reasons for this high death rate include delayed presentation, together with poor availability and high cost of treatment. Most HIV-associated cryptococcal infections are caused by *Cryptococcus neoformans*. In HIV-infected patients, cryptococcosis commonly presents as a subacute meningitis or meningoencephalitis with fever, malaise, and headache. Classic meningeal symptoms and signs, such as neck stiffness and photophobia, occur in only one-quarter to one-third of patients. Some patients experience encephalopathic symptoms, such as lethargy, altered mentation, personality changes, and memory loss that are usually a result of increased intracranial pressure, thought to result from impaired CSF absorption, or yeast infection of the brain.

Diagnosis

1. Lumbar puncture (LP) and CSF analysis:

- The opening pressure may be markedly elevated.
- CSF analysis
 - Protein: 30-150 mg/dl

- WBC: 0-100 /mm³ (monocyte)
- Culture: positive 95-100%
- Indian ink: positive 60-80%
- Cryptococcal Ag (CrAg) > 95 % sensitive and specific
- If it is not possible or contraindicated to do LP, serum cryptococcal antigen can be used for diagnosis.

Management

Requires hospitalization and evaluation by physician

Phases of management:

1. Induction phase

The following is recommended as the **preferred induction regimen**.

- For adults, adolescents and children, a short-course (one-week) induction regimen with Amphotericin B deoxycholate (1.0 mg/kg per day) and flucytosine (100 mg/kg per day, divided into four doses per day) is the preferred option for treating cryptococcal meningitis among people living with HIV.

The following induction regimens are recommended as **alternative options**.

- Two weeks of Fluconazole (1200 mg daily, 12 mg/kg per day for children and adolescents) + Flucytosine (100 mg/kg per day, divided into four doses per day).

- Two weeks of Amphotericin B deoxycholate (1.0 mg/kg per day) + Fluconazole (1200 mg daily, 12 mg/kg per day for children and adolescents up to a maximum of 800 mg daily).

2. Consolidation phase (8 weeks)

Option A: Fluconazole 800 mg/day (In children 12mg/kg/day)

Option B: Fluconazole 400-800 mg/day

Maintenance treatment (or secondary prophylaxis)- Fluconazole 200 mg daily (in children 6mg/kg/day).

Additional points about cryptococcal meningitis

1) Management of elevated intracranial pressure (ICP):

Management of increased ICP is critical as >90% of deaths in the first two weeks and 40% of deaths in weeks 3-10 are due to increased ICP. Failure to manage elevated ICP is the most common and most dangerous mistake in management (since the ICP is non-communicating hydrocephalus there is no risk of CSF tapping within the recommended volume).

- Daily serial LP should be done to control increased ICP by drawing 20-30 ml of CSF based on patient's clinical response. Signs of ICP include headache, altered mental status, meningismus and changing in hearing or vision should be closely monitored, if possible opening pressure should be measured.

- There is no role for acetazolamide, mannitol, or corticosteroids to reduce intracranial pressure.

2) Discontinuation of maintenance treatment (secondary prophylaxis)

When patients are stable and adherent to ART and anti-fungal maintenance treatment for at least one year and have a CD4 cell count of greater than or equal to 200 cells/mm³ (two measurements six months apart).

3) Timing of ART initiation

- Immediate ART initiation is not recommended in HIV-infected patients with cryptococcal meningitis due to the high risk of IRIS, which may be life-threatening.
- ART initiation should be deferred until there is evidence of a sustained clinical response to anti-fungal therapy usually 4-6 weeks from the initiation of antifungal treatment.
- Poor prognostic signs
- Extra CNS manifestation (especially pulmonary)
- Altered mental status
- Low CSF WBC count less than 20cells/μL
- High CSF cryptococcal antigen titer

Prevention of cryptococcal disease

According to a pilot study conducted in Ethiopia from June 2015 to July 2016, in 22 high case load facilities in all regions, the proportion of newly enrolled clients with CD4 count less than 100 cells/mm³ was 25.88%. In the same study, the prevalence of clients screened positive for cryptococcal antigenemia was high (9.9%).

The use of routine serum or plasma CrAg screening in ART-naive adults followed by pre-emptive antifungal therapy if CrAg screening is positive to reduce the development of cryptococcal disease, should be considered prior to ART initiation:

- Where patients with a CD4 count less than 100 cells/mm³; and
- Where this populational so has a high prevalence (>3%) of cryptococcal antigenemia.

The following algorithm or decision-making guide shows how to decide whether a patient needs prophylactic fluconazole treatment to CrAg screening positive and asymptomatic patients with CD4 count less than 100 cells/mm³.

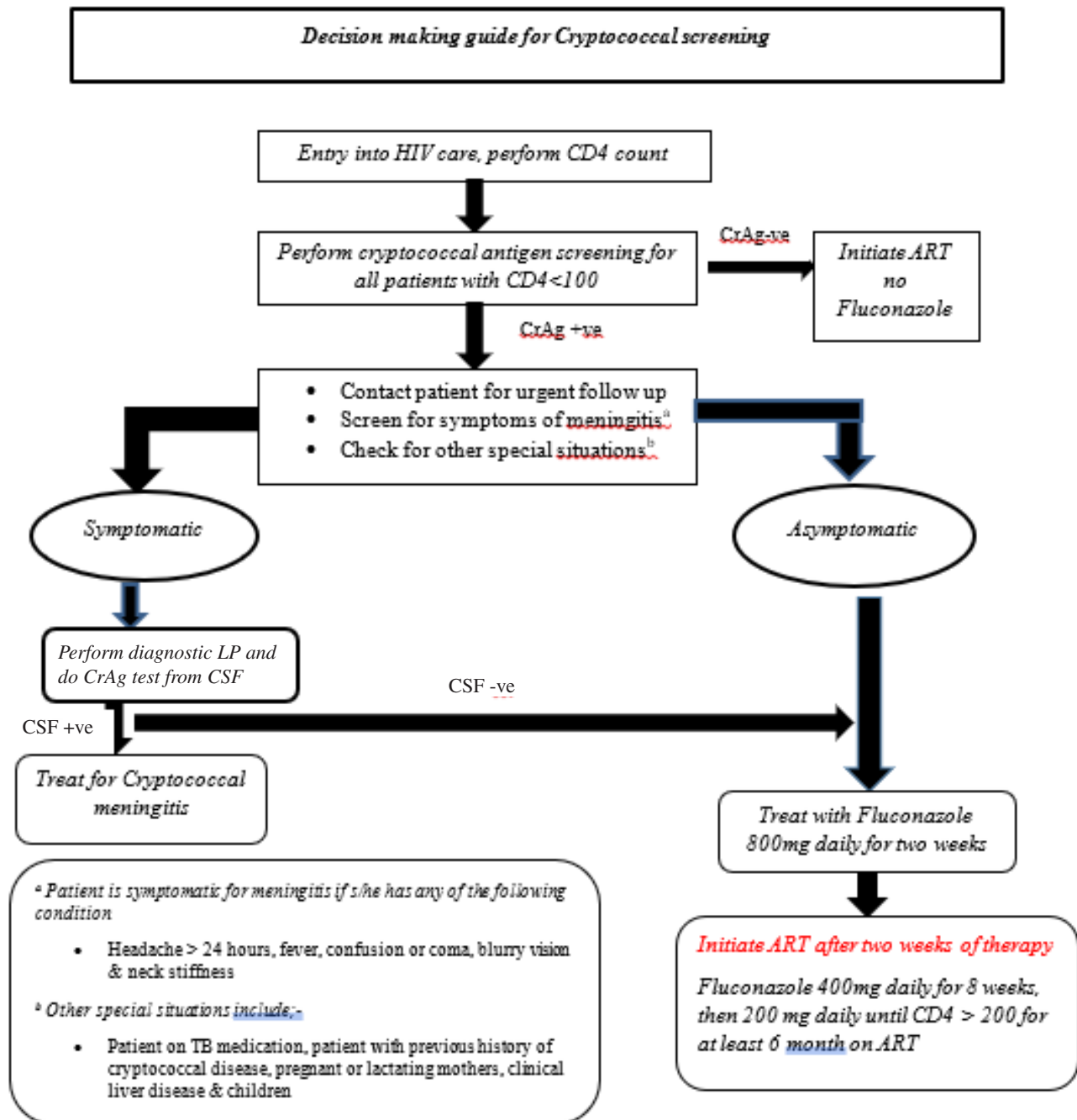


Figure 5.6. Decision-Making Guide for Cryptococcal Screening

5.9.3. Peripheral neuropathies

Peripheral neuropathies are among the most common causes of painful legs in HIV infection; they arise as a complication of HIV infection itself, of drug therapy, or of other metabolic or organ dysfunction or nutritional deficiencies.

Distal symmetrical sensory polyneuropathy is the most common presentation but mono-neuropathies can also occur. The neuropathies associated with HIV can be classified as:

- Primary, HIV-associated.
- Secondary causes related to medications (INH), OIs or organ dysfunctions.

Diagnosis

Peripheral neuropathy diagnosis in HIV-infected patients is based on the clinical picture presenting with pain, tingling sensations, paraesthesia or numbness. Physical examination can reveal depressed or absent ankle reflex, decreased sensitivity to different modalities of sensation and in severe cases, difficulty in walking. The feet and sometimes the hands are involved in symmetrical distribution. The diagnosis can be supported by electro diagnostic studies including electromyography (EMG) and nerve conduction studies (NCS) when available. Blood tests are frequently obtained to exclude other causes of neuropathy. In most instances, however, diagnosis is almost always clinical.

Treatment

- Avoid the offending agent if identified.
- Remove other drugs associated with peripheral neuropathy.
- Supplemental vitamin intake for all patients including concomitant administration of pyridoxine with INH.
- Adjuvants for pain management (such as Amitriptyline, carbamazepine) indicated for patients with pain and paraesthesia.

Monitoring of events

- Recognize presence of peripheral neuropathy.
- Assess severity at each clinical visit.
- Avoid drugs causing neuropathy.

5.10. Visceral leishmaniasis

Visceral Leishmaniasis (VL) is a systemic parasitic illness, transmitted primarily by the phlebotomine sand fly from animal or human reservoirs. Visceral Leishmaniasis is endemic in Ethiopia, with patchy distribution in the southern and north-western lowlands. The causative parasite is *L. Donovanii*. VL has emerged as a major OI associated with HIV. In HIV patients, VL represents reactivation of latent infection with *Leishmania* parasite.

Clinical features

The cardinal signs of VL in patients with HIV infection are unexplained fever, splenomegaly, and pancytopenia (anemia, leucopenia and thrombocytopenia). Presentation may not be typical. The bone marrow is packed with parasites but two-thirds of cases have no detectable anti Leishmanial antibodies. CD4+ cell count in co-infected patients is usually <300cells/ml.

Diagnosis

Parasite Detection-Visualization of the amastigote form of the parasite by microscopic examination of aspirates from lymph nodes, bone marrow or spleen aspiration. *Culture improves the detection of the parasites. However, this technique remains restricted to referral hospitals or research centers.

Antibody Detection-DAT and rK39 have been extensively evaluated and used for the diagnosis of VL in the field and in laboratory settings.

Antigen Detection Test-It is more specific than the antibody-based immunodiagnostic test. Evaluation of the performance of a urine latex agglutination (KATEX) at the Indian subcontinent and East Africa has shown that this test has a good specificity but only a low to moderate sensitivity.

Molecular Techniques-Compared to the other diagnostic techniques available, the molecular approaches remain expensive and technically highly demanding. Their applicability in the endemic areas is highly questionable.

Treatment:

Ambisome 40mg/kg, require longer treatment and more liable to relapse.

Treatment of relapsed patients: These are patients who are slower to respond and have a higher chance of further relapse and of becoming unresponsive to anti-monial drugs. Treatment is the same as above.

CHAPTER SIX:

SERVICE DELIVERY

Introduction

Most of the time health services are organized primarily to provide episodic acute care. As HIV begins to become a manageable chronic condition, program managers and care providers need to consider how current health delivery systems can be reorganized to provide chronic care.

Once people are diagnosed and enrolled in chronic care, follow-up visits should be scheduled and planned. Waiting until people present with symptoms or preventable complications is costly and inefficient. PLHIV require care that anticipates their needs at different stages of the care continuum. Compared with the acute care model, planned chronic care models provide opportunities for prevention, early identification of issues and timely intervention.

Chronic care requires broad support for PLHIV from their communities and health care teams to stay in care, adhere to treatment and cope with stigma. PLHIV and their families need to be informed about HIV infection and the anticipated side effects of medicines and supported to adhere to treatment. Health care teams play an important role in linking people living with HIV with community-level interventions, resources and support.

This chapter provides broad guidance on four operational and service delivery areas:

6.1. Differentiated care: addressing the diversity of needs of people in care.

6.2. Recommendations to strengthen the continuum of treatment and care:

- Disclosure
- Retention;
- Adherence;
- Task shifting;
- Decentralization;
- Integration;
- Adolescent-friendly health services.
- Improving the quality of HIV services

6.3. Pharmaceuticals supply and management system

6.4. Laboratory and diagnostic services.

6.1. Differentiated HIV service delivery

Differentiated care is a client-centered approach that simplifies and adapts HIV services across the cascade to reflect the preference and expectations of various groups of PLHIV while reducing unnecessary burdens on the health system. Continuing to provide ART to a large and growing number of individuals poses a significant challenge to health systems in resource limited setups where there is a shortage of clinical staff. The challenge is highlighted by substantial rates of attrition reported across ART programs. The pace of ART enrolment will likely further increase in the coming years with recommendations issued by WHO in 2016 to change the eligibility criteria for ART initiation as Ethiopia has also endorsed the recommendations.

The challenges of further scaling up ART to those in need and improving retention in care for those on ART require continued adaptations in the models of healthcare delivery to the reality of people's lives. As national, regional and district teams address the various challenges, lessons from innovative models of ART delivery can help shape the next stages of HIV care and treatment scale-up.

The previous guidance which was adopted from WHO was concerned that Differentiated service delivery for HIV treatment has focuses primarily on people who are clinically stable (established on ART). Subsequently, the need has been recognized to adapt services for those with advanced HIV disease, high viral load and comorbidities through simplified care packages

and differentiated models of service delivery; the principles of differentiating service delivery according to the needs of different groups has also been extended to improving the uptake of HIV testing and prevention.

The New WHO 2021 Guideline defines being established on ART should be applied to all populations, including those receiving second- and third-line regimens, those with controlled comorbidities, children, adolescents, pregnant and breastfeeding women and key populations. It also suggests, considering people's clinical needs, differentiated service delivery for HIV treatment should also consider the specific populations and contextual settings. There is also increasing experience of how such models have been adapted in settings with lower HIV prevalence, acute conflict or other emergency responses.

Differentiated service delivery for HIV treatment is based on four building blocks (Figure. 6.1.). In any given differentiated service delivery model for HIV treatment, the building blocks need to be defined separately for clinical consultations, ART refills and psychosocial support.

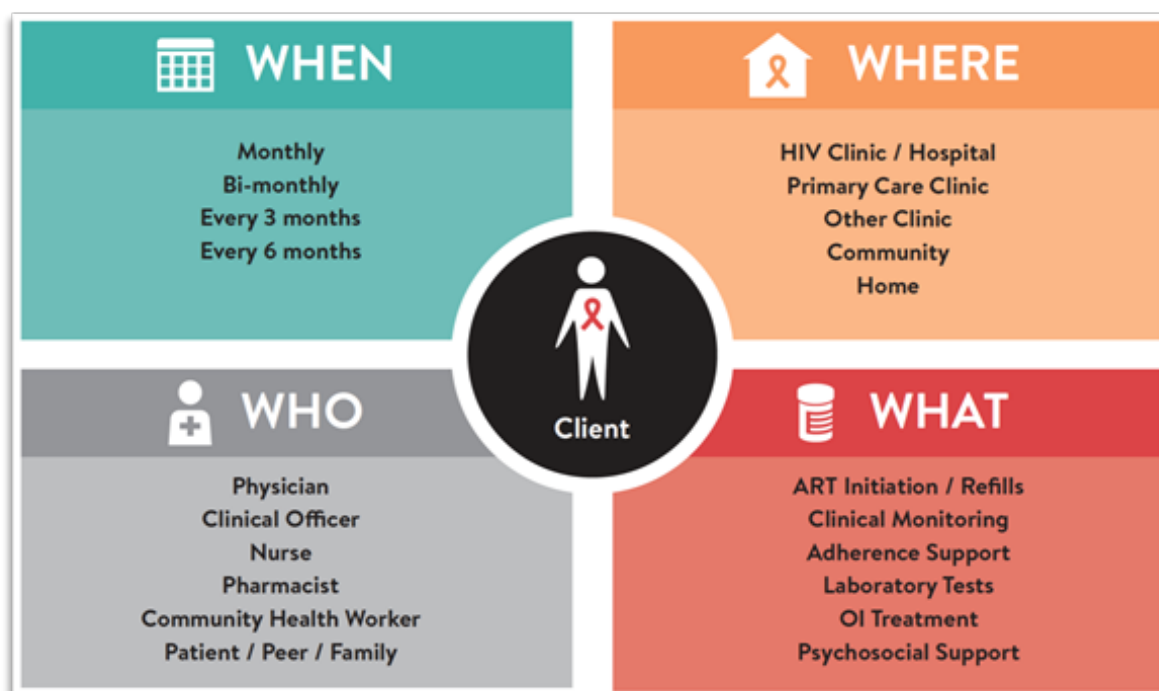


Fig. 6.1. The building blocks of differentiated service delivery for HIV treatment

To accommodate this, our country adopted Appointment Spacing Model/6MMD of HIV care as the first model of differentiated HIV service delivery in 2017. Ethiopia then scaled up other models such as Fast Track ART Refill, 3MMD, Health Extension Professional Managed Community ART groups (CAGs), Peer Led Community ART Group/ Distribution (PCAG/D), Adolescent ART group, Advanced HIV Disease (AHD), MCH and Key Population service delivery models.

All these approaches have reduced the burden for patients (reduced time and cost of travel to clinic and less income loss) and the health system (reduced clinic attendance), while maintaining high retention in care (more than 90% retained in care across multiple time points). Therefore, providers should ensure all required packages of care in the standard

are not missed during the clinical visits of the clients.

Differentiated care applies across the HIV continuum to all the 95-95-95 targets. Differentiated care includes models of testing people unaware of their HIV status to treatment and care to viral suppression of HIV clients enrolled in care. Based on the level of stability of the clients, DSD models can be based in the facility or in the community.

Eligibility criteria for less intensive DSD models

After appropriate classification based on the criteria below, clients should be informed and give verbal consent on the type of the service delivery model they prefer.

The following are the eligibility criteria for less intensive DSD models:

- Patients who are on ART at least six months
- No current illness, which does not include well-controlled chronic health conditions
- Good understanding of life long adherence; adequate adherence counseling provided (a patient with adherence of 95% for the last 6 months)
- Evidence of treatment success (at least one suppressed viral load result (i.e < 50 c/ml) and if no Viral Load result, a patient with rising CD4 cell count or CD4 cell count above 200 cells/millimeter cubes).

- Children with age greater than five years*
- A Patient who doesn't have current Opportunistic Infections
- A Patient with no adverse drug reactions and doesn't need careful clinical monitoring.
- A Patient who is willing or provide consent to get the ART service based on his/her preferred DSD models.

**WHO 2021 guidance did not put age limit for established DSD models but since all children < 5 years are considered to have AHD, age ≥ 5 years is considered for age limit.*

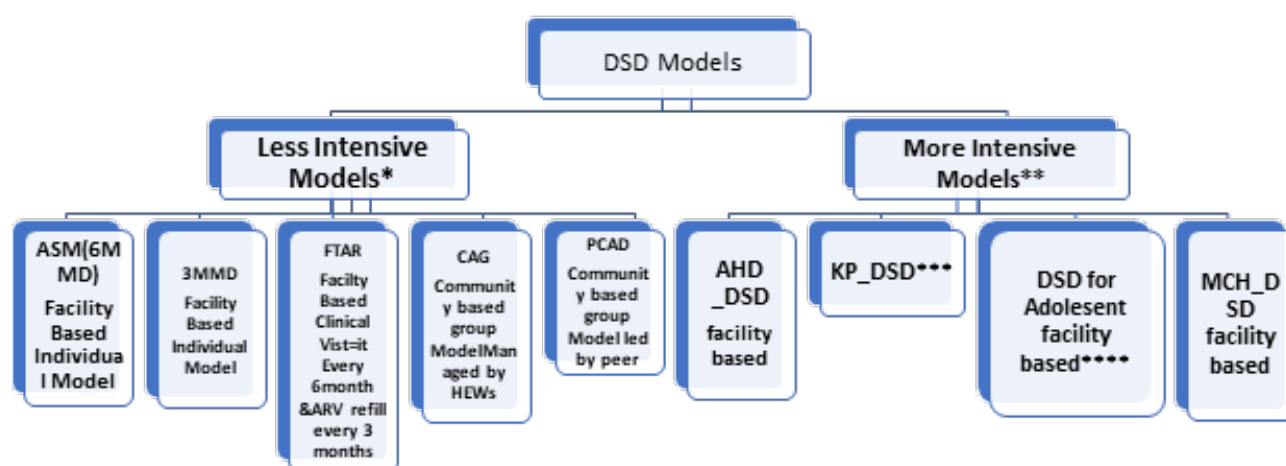


Figure 6.2. Differentiated service delivery model (DSD) framework in Ethiopia

*Less Intensive Model: This individual and group model is intended for clients who are established on ART. It includes both facility and community-based approaches. It requires less frequent clinic visits and focuses on education and empowerment of clients.

**More Intensive Model: This Model is intended for clients who need close follow up and frequent clinic visits. It includes clients with OI, unsuppressed viral load adolescents, pregnant women and those with psycho social barriers to adhere and retention.

*** KP _DSD: Can be implemented both at facility and community level

**** DSD for adolescent is categorized under more intensive models assuming that most of the adolescents require close follow up and adherence support. However, for stable adolescents providers may decide based on the above eligibility criteria and additional conditions.

These classification of DSD models are adapted from CQUIN documents

6.1.1 Less intensive DSD models

6.1.1.1 Appointment Spacing Model (ASM/6MMD)

In the appointment spacing care model, stable clients will be appointed every six months for clinical visit and medication refill. Clients in this model should get additional supports like arrangement of treatment supporter at home level among their family members and arrangement of adherence reminders like alarm and education on how to maintain the drug quality at home level. Clients should be counselled and encouraged to disclose their status and to participate in peer support groups or to be a member of PLHIV associations. Those clients who disclosed their status will get adherence support from PLHIV associations via home visits or telephone follow up.

6.1.1.2 Three Months ARV Dispensing (3MMD)

In the three months ARV dispensing model, clients who are eligible but not willing to be enrolled in 6 MMD (ASM) will be appointed every three months for both clinical visit and medication refill.

The MOH interim guidance for provision of HIV Services in the context of COVID-19 Pandemic clearly guides provision of 3 months ARV dose (3MMD) for PMTCT clients, children, newly identified clients including Key Populations (KPs), clients on second or third line ART and any other clients who do not need admission (Advanced HIV Disease, High Viral Load (HVL), those on EAC, etc).

For children taking 3MMD, the caregiver should be allowed to pick up the child's medication without bringing the child unless the child is due for a clinical visit. For clients requiring Co-trimoxazole, a 3-6-month supply should be provided at the same time as ART pickup. TPT should also be given in multi-month intervals at the same time as ART pickup. For children (especially those who are younger) starting a new medication, administration of the first dose should be done before the child and caregiver leave the ART site. **At least monthly virtual follow up shall be provided using phone calls and other communications mechanisms for children <30KG body weight for dose adjustment, for clients with co-morbidities, for clients with high viral load and other clients with concerns.** In addition, clinicians need to make every effort to align the visit dates and service delivery sites for children and their parents receiving ART.

6.1.1.3. Fast Track ARV Drugs Refill Model

Fast Track ARV drugs Refill (FTAR) is one of the facility based Differentiated Service Delivery Models of HIV care where patients categorized as stable make clinical visit once every six months but collect their medication every three months from pharmacy. The clinical visit at 6 months' interval is meant to ensure the standard package of care is delivered to the clients and to review if the patient still meets the stable criteria. The facility-based fast track system for ART refills can provide an opportunity for those ASM clients who, for various reasons, faced difficulty in taking all of the prescribed six-months of ARV drugs to their home at once from the ART pharmacy as in the ASM framework.

6.1.1.4. Health Extension Professional Managed Community ART refill group (HEP_CAG)

Community ART refill groups (CAGs) are groups comprising of stable clients on ART living in the same community/locality that have a shared understanding. This model is managed by health extension professionals (HEPs) who already have roles in HIV testing and other HIV service provision as one of their packages. The ART refill is for three months and each CAG will have one community refill in between the health facility visits that will happen every six months. Clients can return or referred to the facility at any point in the cycle for any issues that may arise between scheduled health facility visits.

6.1.1.5. Peer lead community based ART distribution/Group (PCAD/G)

The peer led Community based ART distribution (PCAD) groups are groups of PLHIV comprising of stable clients living in the same community/locality. In PCAD, group members will take turns to pick up ARVs at the health facility and distribute among the other group members in the community. Each client will get clinical evaluation and lab monitoring service as per the standard of care package. They will manage their own health and take action with the support of community and facility-based healthcare workers and this will help to align with the clinical consultation visits at health facilities as the recommended, in our set up is every six months.

6.1.2 More intensive DSD models

6.1.2.1. Health care worker managed DSD Model for adolescent living with HIV (DSD for ALHIV)

DSD for adolescents has three core elements which include ART refill, clinical consultation and psychosocial support. This model is coordinated by trained health care workers (HCWs), and regularly meet on weekends and share psychosocial supports. Young adults who already passed through psychosocial support program and transitioned out from Pediatric ART also facilitate the program voluntarily. During this peer session, adolescents interact, learn and share experience among their peers and from facilitators. Understanding the treatment, adherence to treatment and viral suppression are main parts of their discussion. Making the clinical, ARV refill and viral load monitoring service available over the weekends together with the psychosocial support program will make comprehensive HIV service a one stop shopping model for adolescents living with HIV.

Eligibility criteria for Adolescent clients on DSD model:

- Adolescent clients who disclosed their HIV status and willing or provide consent to get the ART service based on his or her preferred DSD models.
- Health care facility where there is functional pediatric psychosocial support program,
- Adolescents, 10- 19 years, fully disclosed, enrolled in the pediatric psychosocial support program,
- No restriction on stability- including both with suppressed and unsuppressed viral load test result

6.1.2.2 DSD for key population (for FSWs)

In Ethiopia there are efforts to make public and private facilities KP friendly by building the capacity of providers and arranging service delivery approach to match their needs. Confidentiality clinics and drop in centers (DICs) were established around hot spot areas in major towns in the country that provide comprehensive HIV services to female sex workers. The KP friendly services delivered at drop in centers have significantly improved HIV prevention, care and treatment service access to FSWs. Therefore, enhancing the coverage and quality of HIV prevention, care and treatment services for KP (FSWs) through initiating DSD for key population is deemed necessary.

6.1.2.3 DSD for Advanced HIV Disease and PLHIV at high risk Disease Progression

For adults and adolescents, and children older than five years, advanced HIV disease is defined as CD4 cell count <200cells/mm³ or 'WHO stage 3 or 4' event. All children under 5 who are not on effective ART are considered to have advanced disease because of high viremia and rapid disease progression with high mortality. For patients with advanced HIV disease, the frequency of clinical visit is recommended every month.

Frequency of visits and MMD for patients with advanced HIV disease in the era of COVID-19 pandemic

Careful consideration should be given to whether persons with AHD should be seen in

person; in some cases, the benefits of being seen in person may outweigh the risks of COVID-19 acquisition in health care facilities. As an alternative to in person visits, close virtual monitoring may also be considered.

Alarming symptoms that support in person clinical assessment include, but are not limited to, fever, cough, shortness of breath, weight loss, severe diarrhea intractable headache and inability to walk unaided. For children, alarming symptoms include fever, tachypnea, cough, lethargy, convulsions, poor oral intake, and persistent vomiting or diarrhea, and warrant in person health facility visit and evaluation by a clinician with pediatric training and experience. Further visits or decision to admit a client to health facility should be done by an experienced clinician and based on clinical and diagnostic evaluation findings.

In addition, extra effort should be taken to ensure that these patients with higher risk of severe outcomes from HIV infection have sufficient medications to avoid interruptions in treatment and unnecessary trips to a health facility. The national interim guidance for continuation of HIV services in the face of COVID 19 pandemic recommends 3 months dispensation of ART (3-MMD) for patients with advanced HIV disease. However, patients should be advised to look for concerning symptoms that warrant in person evaluation. In addition to ART, it is critical to ensure the availability and provision of other medications that have been shown to reduce mortality in Advanced HIV Disease, including cotrimoxazole and TB preventive therapy (TPT) once active TB has been ruled out. The general package of Care is outlined in Table 4.1.

6.1.2.4 MCH_DSD

Mothers living with HIV and their infants are important target population for differentiated service model (DSD). There are various models of care used to support MCH/HIV services. Some of the DSD models in Ethiopia includes family planning service integration to HIV care, point of care (POC) EID testing for HEIs and provision of 3 month ARV dispensing for HIV positive pregnant and breast feeding women during COVID 19 pandemic. For detail information refer to PMTCT guideline 2021.

The role of PLHIV and PLHIV associations in DSD

PLHIV and CSOs are meaningfully engaged in implementation, evaluation and oversight of DSD. Representatives from PLHIV are actively engaged as member of DSD technical working groups. Networks of PLHIV offer a key mechanism for enhancing support to those who are affected, and improving negative experiences of living with HIV. As programs encourage greater involvement of PLHIV, there is increasing engagement of affected communities in national responses to HIV, and PLHIV networks have the potential to maximize stakeholder contributions. In addition, approaches that increase engagement and empowerment can potentially shift the focus of PLHIV's roles from representation to the building of individual and community capacity to promote health.

6.2. Recommendations to strengthen the continuum of treatment and care

6.2.1 Disclosure

Disclosure is a process that involves ongoing discussions about the disease as the child matures cognitively, emotionally and sexually.

Commonly used Terminologies in Disclosure:

Full Disclosure to a person who is HIV positive means telling him/her that they have HIV. It can also include the reason for taking medicines. Example: You have HIV and we are giving you ARVs/medicines to help your body fight the virus.

Partial Disclosure to a person who is HIV positive means sharing information about his/ her illness but not using the word "HIV". Example: Your body is sick and we are giving you medicines to help keep your body healthy.

Complete Non-Disclosure to a person who is HIV positive means not telling a person anything about their condition. Example: A HCW may assess the person's health status and give him/ her medications without explaining anything about what the medicines are for or why the person must take them.

Deception is when a person who is HIV positive, is intentionally told something untrue. Example: You are taking medicines because you have TB.

6.2.1.1 Disclosure in Children

Pediatric disclosure is an ongoing process and even in the best of circumstances may be difficult. Adults struggle with the question of where, when or how to tell children that they have HIV, often agonizing over how to find the right words. All families are unique and there are no set rules regarding when and how to disclose to children.

Children react to HIV disclosure in different ways and it is not uncommon for relatives to disagree about disclosing HIV-related information to children. Even amongst the HIV/care team there may be disagreement on the best approach. Disclosure has to be individualized taking into consideration the particular child, parent/s, family, household and community.

HIV disclosure is not a topic that comes naturally for family discussion, especially when children are involved. The best way for a child to learn about his/her HIV status is through age-appropriate information shared by a loving and trusted caretaker. Disclosure to children should never happen casually, inadvertently or in the heat of anger or conflict. A child's maturity and cognitive capacity varies and is not only dependent on age. It is important to tailor the discussion to the child's cognitive level and to the child's personal and individual situation. It is important to assess readiness of the entire family for disclosure and address potential barriers to disclosure. It is also important to discuss benefits of disclosure which have both short and long term impact on the family.

Disclosure can:

- Help create a sense of closeness in the family.
- Help reduce feelings of anxiety and isolation on the part of the parents/ caregiver.
- Relieve the burden of living with the secret of being HIV-positive.
- Help build social support networks.
- Reduce the anxiety children experience when they suspect something is wrong; they will now have information to make better sense of the situation.
- May improve adherence in a non-adherent child.

Assessing readiness for disclosure

The child

- Is the child symptomatic? Taking medications?
- How old is the child?
- Is the child living with a sick parent or family member?
- Is the child asking questions about HIV?
- Does the child appear distressed, anxious or worried?
- Is the child sexually active and at risk of contracting or spreading HIV?

The parent or caregiver

- Has the parent or caregiver been tested for HIV?
- Is the parent or caregiver infected? Symptomatic? Taking medication?
- Is the adult ill? Is s/he in need of help from children in the household?
- Is the infected adult an important attachment figure for the child?

The family or household

- Are there any adults in the household with HIV infection? Who is aware?
- Are other children in the household HIV-infected? Who is aware?
- How many family members are taking HIV-related medications?
- Is the family unit cohesive, or characterized by separations and/or conflict?

The community

- Are testing and treatment generally available in the community?
- Are there people in the community who are open about their own HIV status? Does the child know anyone in the community who is open about his/her HIV status?
- How strong is the stigma surrounding HIV in the community?
- Are there risks to the family (isolation, discrimination) if unintentional disclosure occurs?
- Are there resources within the community for children – a youth group and/or trusted adults they can talk to?

6.2.1.1 Planning for disclosure

Disclosure is not an event or a one-off conversation. It is a **PROCESS** that takes time and constant communication in an age-appropriate manner. It is important to prepare adequately for disclosure. This involves preparation, education, planning and follow-up. Once the decision has been made to disclose to the child, it is important to understand that the topic will have to be visited over and over again. It is important to give a clear message and listen actively; take clues from the child and avoid lecturing; the emphasis should be on asking directly and indirectly and listening. The following examples can serve as a guide:

Age group of (2-6 years old):

These are too young to comprehend and may still be unable to explain the link between the cause and the illness. Younger children if symptomatic generally want to know what will happen to them. They do not need to know their diagnosis but the illness must be discussed with them. Young children may feel responsible for the parent's illness or just pretend nothing is happening. It is important to give reassurance and take cues from younger children.

- Use visual aids such as drawing, media or puppets in addition to stories to illustrate concepts
- Emphasize the concept of health, wellness and body concrete observation,

School aged children (7-10 years): These age group think systematically about multiple topics more easily. Child's illness descriptions progress from belief that disease is transferred through contact. Hence,

- Use visual aids such as drawings or media
- Emphasize the concept of health, wellness and body

Adolescents 11 years and older: These age groups view body as system, understands illness is within body and caused by external agent

- Visual aids like books
- Use slightly more complex language to convey interactions between health, illness and treatment ("HIV is a type of infection that destroys your immune system's CD4 cells")

Assuring children and adolescents that their HIV status and what they say is confidential, is also very important. Issues of disclosure to others should be discussed while allowing the adolescent to make his/her own decisions about.

6.2.1.2 Stages of disclosure

Stage 1 This is for children around the age of six. If they are symptomatic they want to know what will happen to them. They do not need to be informed of their diagnosis but the illness must be discussed with them.

Stage 2 This stage is for school age children (7-10 years). The illness can be discussed as a germ that can weaken their body soldiers without specifying the diagnosis.

Stage 3 Make sure the child and caregiver are ready for disclosure. Some may have difficulty coping with disclosure information leading to

changes in behavior (acting out in school, i.e. fights, low grades, truancy, anger, crying fits, and no expression of emotion). Others may have concerns that other children in the school or community will make fun of them. Encourage them to ask questions; do not be disappointed if they do not react in the manner you expected. At this stage words like HIV and CD4 should be introduced and check the understanding at each visit.

N.B: Full disclosure can be provided to most children over 10 years. Each steps of disclosure should be documented on the patient's chart.

Adolescents 11 years and older

Adolescents should know of their HIV status. They should be fully informed in order for them to appreciate consequences for many aspects of their health, including sexual behavior and treatment decisions. Be supportive and non-judgmental.

6.2.1.3 Post –disclosure assessments and follow-up

Disclosure is a process that does not end with telling an HIV-infected child the name of their illness or diagnosis. After the HIV diagnosis has been disclosed to the infected child, follow-up visits are needed to monitor the child and family's understanding of the illness and their emotional and psychological adjustment.

Once the diagnosis has been explained to a child, it needs to be reinforced or regularly discussed as the child develops because many children will not have understood the full implications of the disease or diagnosis at the time of disclosure. For example, pre-

adolescent children can cognitively understand the concepts about the virus but may be less likely to think of the future implications, such as transmission risks and safe sexual practices. As the child ages and matures, he/she will slowly understand and integrate the implications of the diagnosis into his/her life. Children's perception of self, health, illness, and death evolve as they mature through different developmental stages.

Some children who learn of their HIV status may experience guilt and shame and may isolate themselves as a result of the stigma and secrecy surrounding the disease. Changes in behavior and school functioning may occur in these children and may be symptoms of depression. Patients and families who have a difficult adjustment to HIV disclosure without progress over time should be referred for mental health services and additional support. In young adolescents it will be important to discuss about modes of HIV transmission, sexuality and reproductive health.

6.2.1.3 Disclosure in adults

Among other priorities, testing and counselling programs emphasize the importance of people with HIV disclosing their HIV status, particularly to sexual partners. Informing the sexual partners of an individual's HIV infection is not only an effective means of halting the transmission of HIV, but informing partners allows access to care and support as well as further prevention efforts among the client's partners and family.

Two main processes for informing partners of an individual's HIV infection are disclosure and partner notification. Disclosure refers to actions by individuals themselves to notify their partners of their HIV Sero-status. UNAIDS

and WHO strongly recommend disclosure, when appropriate, as this process is voluntary, respectful of the autonomy and dignity of the affected individuals and mindful of maintaining confidentiality. Providers of testing and counseling prefer that individuals use disclosure to inform those who need to know that they are infected. For the individual, his or her sexual partners, and family, disclosure allows for greater openness about HIV in communities and meets ethical imperatives. To encourage disclosure, it is needed to establish safe social and legal environments in which more people are willing and able to get tested for HIV and are empowered and encouraged to change their behavior according to the results. This can be done by expanding access to counseling, testing, care and treatment services; by protecting people from stigma and discrimination and removing legal barriers.

Disclosure can be difficult as people may be afraid of the consequences: for example, the threat of rejection and violence by partners and family or discrimination in the community and workplace. In some cases, people may have limited knowledge of their partners and how to locate them, or may not know the identity of their partners or where they can be located. Although evidence of effectiveness of partner notification is limited in resource-limited settings, it is advised that partner notification or ethical partner counseling be based on the informed consent of the source client, and maintain the confidentiality of the source client, and where possible, protect individuals from

physical abuse, discrimination and stigma that may result from partner notification. Ideally, partners of infected individuals should be encouraged to seek HIV testing and counseling, as this is a critical prevention and treatment tool in the control of HIV.

How to discuss disclosure in adults

- Ask the patient if he/she has disclosed his/her HIV test result or is willing to disclose the result to anyone.
- Discuss concerns about disclosure to partner, children, other family and friends.
- Assess readiness to disclose HIV status and to whom.
- Assess social support and needs (refer to support groups).
- Provide skills for disclosure (rehearsal can help).
- Help the patient make a plan for disclosure if now is not the time.
- Encourage attendance of the partner to consider testing and explore barriers. As couples may have different HIV status, partner testing is important.
- Reassure the patient that you will keep the result confidential and that disclosure is voluntary.

If the patient does not want to disclose the result:

- Reassure that the results will remain confidential.
- Explore the difficulties and barriers to disclosure. Address fears and lack of skills (help provide skills).
- Continue to motivate the client.
- Advise the client not to harm others.
- Offer to assist in disclosure (for example, talk with spouse).
- Offer another appointment and more help as needed (such as peer counselors or trained counselors).

For women, discuss benefits and possible disadvantages of disclosure of a positive result and involving and testing male partners. Men are generally the decision makers in most families and communities. Involving them will have greater impact on:

- Increasing acceptance of condom use, practicing safer sex and making appropriate reproductive choices.
- Helping to decrease the risk of suspicion and violence.
- Helping to increase support to their partners.
- Motivating to get tested.

- **Anticipated consequence of involving and testing the partner:** danger of blame, possible violence, abandonment. Health worker should assess the risk of violence or suicide and discuss possible steps to ensure the physical safety of patients. Health worker should try to counsel the couple together, when possible.

6.2.2. Retention

Ensuring retention for people living with HIV across the continuum of care is essential for optimal health outcomes. For people living with HIV who are receiving treatment, uninterrupted ART and continual monitoring are essential for sustained viral suppression and optimal treatment outcomes. Retention in ART program is more challenging in pediatric populations, postpartum women. Multiple factors may play a role in lost to follow-up, including distance to health facilities, lack of transport or inability to cover travel expenses, stigma and disclosure-related issues, being too sick and lack of understanding of the need for lifelong care.

6.2.2.1 Good practices to optimize retention

Optimizing retention in HIV care requires interventions at multiple levels of the health care system. Given the broad array of challenges and heterogeneity of barriers across settings, no single approach is likely to work for everyone in all settings. Improving the understanding of barriers and innovative strategies to address them are important.

Related transport costs and loss of income while seeking care serve as discouragements when health facilities are located far from the person's home. Bringing services closer to communities, where feasible, reduces the indirect costs of care for the PLHIV and their families and improves retention.

Waiting times at the facility during consultation are frequently high, especially in settings with a high burden of HIV infection. Reorganizing services, such as systems for appointment, triage, separating clinical consultation visits from visits to pick up medicine, integrating and linking services and family-focused care may reduce waiting times at the health facility.

Interventions harnessing social support have emerged as a promising approach to counteract the structural, economic, service delivery and psychosocial constraints that affect retention.

6.2.2.2 Specific population considerations

1) Pregnant and breastfeeding women

For pregnant women living with HIV, the transition between ART care and antenatal care/MNCH services is a potential point for lost to follow-up.

Interventions include:

- Monitoring approaches to ensure that registers are aligned and that women are tracked and followed across different ART service delivery points;
- Peer support, such as mothers-to-mothers program, may improve transition across service delivery settings, and

- Peer adolescent support groups for adolescent pregnant women living with HIV.
- They should have the choice to continue receiving their ART through the differentiated ART delivery model or to have their ART delivery integrated within their maternal, newborn and child health care.

2) Children

Caregivers are responsible for understanding the importance of ensuring retention for children in care, especially younger children. Disclosure to children typically occurs late, making it challenging to discuss the importance of follow-up. WHO recommends age appropriate disclosure to children. This is based on their judgement of the overwhelming benefits of the intervention in increasing retention in the testing to treatment cascade.

Interventions include:

- Supporting caregivers to attend for regular follow-up.
- Reinforcing to the caregivers the importance of the process of disclosing to the child; this can begin early with age-appropriate messaging and tools.

3) Adolescents

Frequent clinic visits, time spent waiting for services and having to miss school discourages adolescents' engagement in care. Negative health care provider attitudes, concerns regarding privacy and confidentiality and limited opportunity to discuss their concerns also act as barriers to retention for adolescents. Distance to facilities and out of pocket expenses may restrict their engagement.

Interventions include:

- Strengthen the adolescent DSD based at Health Facility level.
- Implementing adolescent friendly health service approaches to improve quality;
- Providing adolescent services at specific times or in separate areas with flexible appointment systems that accommodate school hours;
- Comprehensive services that address multiple needs, including psychosocial support and sexual and reproductive health, and

- Close monitoring of adolescents' engagement in care, rapid proactive follow-up and implementation of strategies for re-engagement.

4. Key populations

Specific consideration, especially in settings in which people are criminalized because of their behavior or identity, requires delivery community and peer-based options to support people to retain in care.

Table 6.1. Factors related to the health system and people receiving ART influencing retention and adherence with respective potential interventions.

Factors	Possible interventions
High direct and indirect costs of receiving care.	■ ART and related diagnostics and services free of charge at the point of care. ■ Decentralize ART where feasible. ■ Scheduled facility visits ■ Reduce waiting time at the facility level. ■ Client centered appointment system. ■ Separate clinical consultation visits from appointments for picking up medicines. ■ Link, integrate and coordinate care.
Stock-outs of ARV drugs.	■ Optimize pharmaceutical supply management systems to forecast, procure and deliver ARV drugs. ■ Ensure regular & accurate reporting and requisition system using RRF. ■ Use fixed-dose combinations to simplify forecasting and supply management systems.
Lack of a system for monitoring retention	■ Implement systems for patient monitoring across the continuum of care, including cohort analysis and patient tracking systems.

Factors	Possible interventions
Lack of a system for transferring people across different points of care.	■ Interlinked patient monitoring system across services for HIV, TB, maternal and child health and PMTCT; system for transitioning from pediatric to adolescent and adult services and from maternal and child health and TB services to chronic HIV care.
Pill burden and complex ARV drug regimens.	■ Use fixed-dose combinations to reduce the pill burden and simplify the regimens.
Lack of accurate information for patients and their families and peer support.	■ Engage and integrate community health workers, case managers, adherence supporters, volunteers, PLHIV in peer support, patient education and counseling, and community-level support.
Adherence support	■ Task shifting for involving case managers, adherence supporters and community health workers. ■ Linking with community-level interventions and resources such as peer adherence support (community adherence support groups). ■ Using known effect reminder methods ■ Peer support also provides opportunities for in-person reminders.
Gaps in relationship between patient and care provider.	■ Train health workers on how to reduce stigma; improve treatment preparedness, adherence and retention; provide adherence support and care for key populations; and provide simplified approaches for educating patients and their families.
Lack of time for educating people in HIV care.	■ Task shifting and sharing among clinic team members. ■ People living with HIV as expert patients and peer supporters. ■ A team approach to care.
Adverse drug effects	■ Preparedness and knowledge of how and when to self-manage adverse effects and when to return to the clinic. ■ Implement monitoring and report of adverse drug reactions through implementing pharmacovigilance system.

Factors	Possible interventions
Forgetfulness, life stress, stigma and discrimination.	■ Using reminder to keep patients engaged. ■ Peer and family support. ■ Link to community support group.
Comorbidity, substance and alcohol use disorders and mental health disorders.	■ Manage HIV with mental health disorders, alcohol and other substance use disorders and link with community and social support.
Patient knowledge and beliefs related to HIV infection, its course and treatment.	■ Integrate the education of patients and their families and counseling, broader community literacy and education and community engagement.
Trust in faith and other herbal remedies	■ Implement strong counseling system and engage faith based organizations in providing education in adherence and retention.

6.2.3. Adherence to ART

WHO defines treatment adherence as “the extent to which a person’s behavior – taking medications, following a diet and/or executing lifestyle changes – corresponds with agreed recommendations from a health care provider”. For ART, a high level of sustained adherence is necessary to suppress viral replication and improve immunological and clinical outcomes; decrease the risk of developing ARV drug resistance; and reduce the risk of transmitting HIV.

6.2.3.1. Barriers to adherence

Multiple factors related to the health care delivery systems, the medication and the individual taking ARV drugs may affect adherence to ART.

Individual factors: may include forgetting doses; being away from home; changes in daily routines; depression or other illness; a lack of interest or desire to take the medicines; and substance or alcohol use.

Medication-related factors: may include adverse events; the complexity of dosing regimens; the pill burden; and dietary restrictions.

Health system factors: may include requiring people with HIV to visit health services frequently to receive care and obtain refills; travelling long distances to reach health services; and bearing the direct and indirect costs of care. Lack of clear information or instruction on medication, limited knowledge on the course of HIV infection and treatment and adverse effects can all be barriers to adherence to ART.

Specific population groups face additional challenges to adherence, and these should be considered when implementing the recommended interventions.

Pregnant and postpartum women

The pregnancy and postpartum period presents significant biological, social and economic challenges that may affect treatment adherence. Pregnancy-related conditions such as nausea and vomiting may negatively affect treatment adherence. Other individual factors include suboptimal understanding of HIV, ART and PMTCT, lack of partner disclosure and support, and fear of stigma and discrimination. Service delivery barriers include poor-quality clinical practices, gaps in provider knowledge and training, poor access to services and health worker attitudes

Adolescents

Adolescents face specific challenges, including psychosocial issues such as peer pressure, the perceived need to conform and inconsistent daily routine. Adolescents are often left out of decisions and have limited opportunities to discuss their concerns, and there is limited availability of adolescent-specific treatment literacy and adherence counselling tools. For adolescents who are transitioning from pediatric to adolescent care, additional challenges may include assuming increased responsibility for their own care, issues relating to disclosure to peers or partners, difficulties in navigating the health-care system, lack of links between adult and pediatric services and inadequately skilled health workers.

Infants and young children

Successfully treating a child requires the commitment and involvement of a responsible caregiver. Parents and other family members of CLHIV may themselves be living with HIV, and suboptimal HIV care and treatment for family members could result in suboptimal care for the child. Other challenges include lack of nutrition support, limited choice of pediatric formulations, poor palatability of liquid formulations, high pill or liquid volume burden, large pill size, frequent dosing requirements and difficulties in swallowing tablets.

People with mental health conditions and substance use

People with HIV with uncontrolled depressive symptoms are more likely to have poor adherence to ART. Adherence is complicated by mental health comorbidity that results in forgetfulness, poor organization and poor comprehension of treatment plans. Similarly, use of alcohol and other substances may also contribute to poor adherence to ART. Alcohol and substance use can lead to forgetfulness, poor organization and diversion of monetary resources.

6.2.3.2. Supportive interventions

Several interventions may also be of value in addressing specific challenges that impact on adherence and/or viral suppression. Interventions to optimize adherence to ART includes using fixed-dose combination regimens for ART and strengthening drug supply management systems to reliably

forecast, procure, and deliver ARV drugs and prevent stock-outs. Efforts to support program-level interventions for improving adherence to ART include: avoiding imposing and maximize adherence should begin before ART is initiated. Developing an adherence plan and education are important first steps. Initial patient education should cover basic information about HIV, the ARV drugs themselves, expected adverse effects, preparing for treatment, and adherence to ART.

Patient education, counseling and peer support

Patient education and counseling are essential both when ART is initiated and throughout the course of treatment. Informing and encouraging people receiving ART and their families and peers are essential components of chronic HIV care.

Substance use and mental health interventions

Studies indicate that improving well-being by treating depression and managing substance use disorders improves HIV treatment outcomes.

Nutritional support

Nutrition assessment, counseling and support are essential components of HIV care. HIV programs should ensure that existing national policies on nutritional support are observed when it is necessary and feasible to maximize adherence to ART and achieve optimal health outcomes in food-insecure settings.

Reminder and engagement tools

Mobile phone calls can be considered as a reminder tool for promoting adherence to ART as part of a package of adherence interventions. Other patient reminder tools include alarms, phone calls, diaries and calendars can be used to as a reminders about the timing of ARV drugs, drug dosage and appointments.

6.2.3.3 Monitoring adherence to ART in routine program and care settings

Objective monitoring of adherence to ARV drugs is necessary for effective and efficient treatment planning and ongoing support. Each facility visit brings opportunity for assessing and supporting treatment adherence. Effective monitoring of adherence requires a combination of approaches based on human and financial resource capacity, acceptability to PLHIV and to health workers and the local context.

Viral load monitoring

Viral load monitoring is considered as a gold standard and recommended to diagnose and confirm treatment response and failure. Although treatment failure is often caused by lapses in adherence to ART, it may also result from other factors such as drug resistance, drug stock-outs, drug interactions or malabsorption. Viral load monitoring must therefore be combined with other approaches to monitoring adherence. These approaches should also be considered as a way to provide additional information about possible causes of virological failure or to support adherence monitoring in settings where viral load testing is not available.

Following an initial viral load result (>50-1000 & >1000 copies/ml), enhanced adherence intervention should be carried out prior to conducting a second viral load test. Viral load monitoring also has a high potential to motivate adherence.

Pharmacy refill records

Pharmacy refill records provide information on when PLHIV pick up their ARV drugs. When people obtain pharmacy refills at irregular intervals, this may indicate non-adherence to ART; however, in many routine care settings, people may pick up their medications when receiving care irrespective of their adherence level. This behavior could lead health care providers to overestimate adherence by solely using pharmacy refill records. In many settings, pharmacy refill records are already a part of national monitoring and evaluation frameworks and can also provide additional information on adherence to ART when used in combination with other tools.

Self-report

Asking people living with HIV or their caregivers how many doses of medication they have missed since the last visit (or within a specified number of days in the past) can help to estimate non-adherence.

However, although this method is commonly used, people may not remember missed doses accurately or may not report missed doses as they may want to be perceived as being adherent and to avoid criticism. Counselling on the importance of remembering and/or documenting ARV drug doses and an

environment that promotes and enables honest reporting of non-adherence are critical components of monitoring adherence to ART in routine care settings.

Pill counts

Counting the remaining pills in bottles may help to assess adherence. Pill counts usually take place at routine health care visits. However, some people may throw away tablets prior to health care visits, leading to overestimated adherence. Although unannounced visits at people's homes could lead to more accurate estimates, this approach poses financial, logistical and ethical challenges. Counting pills also requires health care personnel to invest significant time and may not be feasible in routine care settings. Pill count can perform better when combined with self-reported adherence.

6.2.4. Task shifting for HIV treatment and care

Reorganizing, integrating and decentralizing HIV treatment and care will require re-examining the roles and tasks of teams of health care providers involved in delivering chronic HIV care. Task shifting involves the rational redistribution of tasks among health workforce teams. With this approach, specific tasks are reassigned, where appropriate, from highly qualified health workers to health workers with shorter training and fewer complementary qualifications to efficiently and effectively use the available human resources. Task shifting should be implemented alongside other strategies designed to increase the total numbers and capacity of all types of health workers.

The quality of care in task shifting should be ensured by;

- a) Providing training, mentoring, supervision and support for nurses, non-physician clinicians and community health workers;
- b) Stating clear indications for patient referral;
- c) Implementing referral systems and
- d) Implementing monitoring and evaluation systems.

Both initial and ongoing training and mentoring, supportive supervision and administrative planning have been critical to the success of program that has implemented task shifting. Program need to train and establish a system for routine supportive supervision of health workers, including lay providers.

6.2.5 Decentralizing HIV treatment and care

Although rapidly scaling up HIV programs has significantly improved access to ART and increased the health and survival of PLHIV, it also poses significant challenges to health systems. Decentralizing ART to primary care settings will ease the burden of routine management on other parts of the health system and will improve equity by promoting access to ART in rural areas. Decentralizing HIV care and treatment could reduce the workload for health care personnel, thereby reducing waiting times for people with HIV and people receiving care at hospitals for other conditions and bring HIV services closer to people's homes. Decentralization and scale-up of HIV care and treatment services will continue based on need assessments to address community groups with high HIV prevalence.

6.2.6 Integrating and linking services

Chronic care requires integrating and linking related services to ensure comprehensive and consistent patient management over time including provision of related services in single settings, systems to share information and effective referrals across settings and providers.

Integrating and linking services are likely to reduce missed opportunities for initiating ART, enhance long-term adherence support and optimize patient retention. Programs for HIV, sexual and reproductive health, maternal and child health, mental health, non-communicable diseases, viral hepatitis and TB need to collaborate to successfully implement ART and related services at different levels of the health system.

Delivering HIV services in antenatal care, maternal and child health settings

HIV testing should be offered to all pregnant women through provider-initiated approaches as an essential component of MNCH services. It is also recommended to provide couple and risk screening based partner HIV testing for all pregnant women and their partners in maternal and child health care settings.

ART should be initiated and maintained in all pregnant and postpartum women and in infants at maternal and child health care settings, with linkage and referral to ongoing HIV care and ART, where appropriate.

Delivering HIV services in TB treatment settings

All confirmed and presumptive tuberculosis patients should be offered HIV testing services in TB clinics. TB patients co-infected with HIV should be initiated with ART through linkage to HIV care and ART services. All PLHIV enrolled to HIV care need to be screened for TB and those with no sign and symptoms of active TB should be provided with TPT. PLHIV who develop TB after the initiation of ART, should be linked to TB clinics for TB treatment.

Sexual Transmitted Infection, family planning and cervical cancer screening services in HIV care settings

Sexually transmitted infections, family planning and cervical cancer screening services should be integrated within HIV care settings. WHO recommends routine offer of HIV testing services for persons with an STI in all epidemic settings and for family planning clients in generalized epidemic settings. Likewise, all PLHIV should be screened for STIs and treated as per the National Guidelines for Syndromic Management of STI. Family planning services should be provided in HIV care/ART clinics to avoid unintended pregnancy among women living with HIV. Women living with HIV aged 15 years should be screened for cervical cancer as per the National Guidelines for Screening and Management of cervical cancer.

6.2.7. Adolescent-friendly health services

There is a growing cohort of adolescents living with HIV, which includes those infected from birth and those who have acquired HIV later in childhood and adolescence. While there is still a

lack of health outcome data for this age group, emerging evidence indicates that adolescents living with HIV are underserved by current HIV services including access to ART. Adolescents are at high risk of lost to follow-up both before and after ART initiation. Adolescents and youths aged 15–24 years and those attending services for the PMTCT of HIV are particularly at risk.

All adolescents, including those living with HIV, face significant barriers to accessing health services, due to inadequate health literacy, limited ability to navigate health services, legal requirements for parental or caregiver consent, and insufficient resources to pay direct and associated service costs. Adolescents face significant levels of stigma and discrimination.

Poor quality of services also limits adolescent engagement in health care. Adolescents often perceive health services as unacceptable due to concerns about confidentiality and negative health provider attitudes. Services are often not organized to accommodate adolescent needs, have inconvenient service schedules, inflexible appointments and unwelcoming environments. Without sufficient consideration and support, adolescents can be lost between paediatric and adult services. The rapid developmental and social changes that occur during adolescence exacerbate the impact of such barriers and can have a profound impact on the way adolescents engage with health services.

Due to their unique needs, adolescents living with HIV require quality and comprehensive services. These includes care & support access, continuity in treatment, adherence, psychosocial support, sexual & reproductive health and mental health care. According to the WHO quality of care framework, adolescent

friendly health service is defined as follows:

- Equitable: all adolescents, not just certain groups, are able to obtain the health services they need.
- Accessible: Adolescents are able to obtain the services that are provided.
- Acceptable: Health services are provided in ways that meet the expectations of adolescent clients.
- Appropriate: The right health services that adolescents need are provided.
- Effective: The right health services are provided in the right way and make a positive contribution to the health of adolescents.
- Adolescent-friendly HIV prevention, care and treatment service implementation considerations include:
 - Aligning approaches for HIV service delivery with national adolescent-friendly health service standards, protocols and activities;
 - Including implementation of adolescent-friendly approaches in HIV health service supervisory and monitoring systems;
 - Ensuring training, research and personal development opportunities for health service providers on adolescent HIV treatment and care;
 - Engaging service providers, adolescents and other key stakeholders to identify acceptable and feasible activities;
 - Implementing adolescent-friendly health service approaches in all HIV services used by adolescents, including antenatal care for pregnant adolescents living with HIV; and

Establishing linkages and referral pathways to ensure a comprehensive continuum of care, especially for the transition from paediatric to adult HIV services.

6.2.8. Improving the quality of HIV care services

HIV program should be innovative in addressing local challenges and aim to strengthen programme monitoring and the routine use of programme data to improve the quality of services. Quality of care emphasizes that services should be effective in achieving desired health outcomes and that health-care practices should be people-centred and safe. Strategies to improve the quality of HIV care services are needed both at the programme management level and at health facility and community levels where HIV care services are provided. If an intervention is to achieve the desired health outcomes, it needs to be evidence based, of high quality and achieve a level of coverage that brings desired outcomes at the population level.

Quality care means that people living with HIV receive the care they require to maintain their health and quality of life. For HIV program and health-care providers, quality HIV care optimizes programme effectiveness and efficiency. For policy-makers and funding agencies, quality care is an important requirement for maintaining health at the population level and ensuring the optimal use of available resources.

Quality of care should not be seen as an additional activity to routine HIV services or a short-term project to address implementation issues and gaps; it should be incorporated into daily activities at all levels, from service delivery to national programme management.

6.3. Pharmaceuticals Supply Management System

6.3.1. Supply Chain Management

The overarching objective of pharmaceuticals supply management system is to support national policy with the adequate and continuous availability of the most safe, effective, quality-assured ARVs and related pharmaceuticals at service delivery sites in the right quantity, at the lowest possible cost and in a timely manner.

The new recommendation that ART should be initiated in all PLHIV regardless of CD4 cell count will require an integrated national strategic response that considers the resources available and enables strong supply chain management systems at all levels of the health system. MOH, EPSA and other stakeholders need to work together to ensure that the national supply system is functioning well in accordance with the increasing national demand and achieve the 95-95-95 targets.

Ethiopian Pharmaceuticals Supply Agency (EPSA) is responsible for quantification, procurement, storage and distribution of pharmaceuticals to health facilities. Health facilities obtain HIV pharmaceuticals primarily through the Integrated Pharmaceutical Logistics System (IPLS), a single reporting and distribution system managed by EPSA. Pharmaceuticals used for HIV programs are also managed through IPLS.

Selection of ARV drugs and related supplies

Product selection is done based on existing national policies and ART guidelines. Products are selected from or become part of a National Essential Medicines List (EML) and are based on standard treatment guidelines. If a selected ARV drug is not on EML and/or not registered in the country, HIV program managers should coordinate with Ethiopian Food and Drug Administration (EFDA) and request that these drugs be put on the list and registered.

Quantification

National quantification of ARV and related pharmaceuticals is conducted by EPSA in collaboration with MOH, FHAPCO, other stakeholders and development partners. The national quantification is done every year covering a three-year forecast. Based on the forecast produced, supply plan for one year consumption will be prepared considering stock for procurement lead time and safety stock which can allow sufficient flexibility to accommodate minor consumption fluctuations due to unforeseen events.

Procurement

All ARV and related pharmaceuticals for use in the public and private sector should be procured at affordable prices, with assured quality and adequate shelf life, from a reliable supplier. ARVs and related commodities are procured by EPSA.

The procurement process follows the national and international procurement regulations, which is a competitive approach through international bidding.

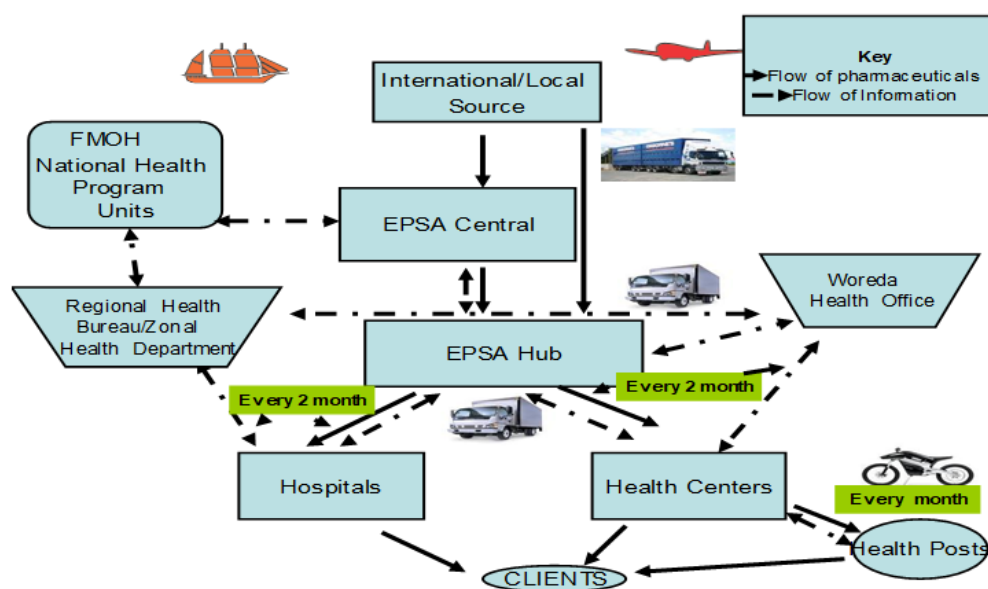
Storage and distribution

Proper storage facilities, including refrigeration, is critical to maintain the quality of the drugs and related supplies. Since the commodities required for HIV program are increasing, there should be adequate storage space, warehouse material handling equipment and distribution facilities need to be ensured at various levels.

Pharmaceuticals and health products distribution will follow the existing delivery

system and it extends from the central level to health facilities. Central EPSA will have a major role of delivering the products to its Hubs; subsequently the hubs distribute the commodities to health facilities based on orders placed by health facilities to EPSA hubs.

Figure 6.3 Flow of pharmaceuticals and information in the integrated pharmaceutical logistics system.



Inventory Control

Inventory control system is mandatory to maintain an appropriate stock level of products at all level of the supply chain to avoid shortage and wastage (due to overstock and/or expiry) of products. IPLS dictates the implementation of max-min inventory control system where there is a set of maximum and minimum months of stock to be kept at each level of the supply chain. Health Commodity Management

Information System (HCMIS) is an electronic inventory control system which is currently utilized at EPSA central, hubs and some health facilities whereas the remaining health facilities are utilizing a paper based system. HCMIS need to be strengthened to improve inventory control and data visibility at all levels of the supply chain. EPSA is responsible for collecting, validating, analyzing, and utilizing product information to ensure an uninterrupted supply of health products.

Quality Assurance

Mechanisms must be put in place to assure the quality of drugs entering the country through pre-procurement certification and post-marketing surveillance. Appropriate quality assurance mechanisms for pharmaceuticals is developed and implemented by EFDA. Quality standards should also define storage conditions at EPSA warehouses and health facility stores. The national laboratory must have the capacity to assure the quality of ARV pharmaceuticals. Quality assurance of drugs and supplies will be maintained using simple visual inspection methods and a First-Expiry-First-Out (FEFO) system to avoid expiration and ensure fresh supplies are available at all levels.

6.3.2. ART Pharmacy Services

The ART Pharmacy unit is determined to take responsibility and make all the necessary follow up to improve the ART pharmacy services. Pharmaceutical care provider who is working in ART Pharmacy is expected to play an important role in the follow-up of adherence, monitoring treatment outcomes, monitoring and preventing drug interactions, detecting and reporting ADEs, and other medication related issues. There is also a need for strict pharmacovigilance of ARVs since new ARVs are constantly entering the market.

Key activities and processes in ART pharmacy

The pharmacist assigned to provide ART pharmacy unit should minimally accomplish the following:

- Filling of patient information sheet for specific patient

- Confidential documentation system of patient information sheet
- Identifying ADR, drug interactions, adherence of ART clients using patient information sheet
- Record patient appointment date for easily tracking of patients for follow up
- Timely trace lost for follow up patient and make adherence support
- Register basic patient data on daily basis to summarize the data for informed decision making.
- Summarize data from registers to consumptions summary
- Generate monthly report and share to respective bodies.
- Analyze and interpret monthly report for decision making purpose
- Properly provide services for PEP and Emergency clients

Pharmaceuticals Management Information Systems

The management of patient related data serves two basic purposes among others: it enables the Pharmaceutical care provider to follow-up adherence, treatment outcomes, prevent adverse drug reactions, drug interactions and other medication related issues; and the information obtained is crucial for decision-making on the selection and quantification of ARVs. Accordingly, the system will allow

the effective and efficient documentation and reporting that is important in the management and monitoring of patient uptake and regimen profile at national, regional and health facility levels. Moreover, the overall prescribing and dispensing practices at facility level will be improved. It will also contribute to the identification of discrepancies in treatment protocols in an effort to improve adherence to latest ART guidelines. Hence, the information generated at health facilities is being utilized for decision- making throughout the drug supply management cycle of the country.

Tools and systems in ART pharmacy Services:

- Patient information sheet (PIS)
- Patient tracking chart (PTC)
- ARV drugs dispensing registers (Adult, Pediatric, PEP, Emergency)
- Drug Therapeutic Problem (DTP),
- Monthly consumption summary and
- Monthly activity report.

Medicine Use Evaluation

In order to sustain the benefits of ART, continuous availability of ARVs should go hand in hand with rational medicine use practices of prescribing, dispensing, and patient use. However, there are reports of non-adherence to treatment protocols. Irrational use of medicines contributes to drug resistance, treatment failure, and resource wastage. In

recent years, to optimize ART and achieve national treatment targets, new regimens and service delivery models have been introduced. Hence, ensuring the proper use of new regimens using methodologies such as drug use evaluation (DUE) should be given a priority. DUE is an ongoing, systematic, criteria-based program of medicine evaluations that helps ensure appropriate medicine use is provided. If therapy is determined to be inappropriate, interventions with providers or patients will be necessary to optimize pharmaceutical therapy. The drug and therapeutics committee (DTC) is responsible for identifying medicine use problems and implementing strategies to alleviate these problems. All DTCs should ensure that medicines are being used correctly so that patients receive the maximum benefit from their pharmaceutical therapy.

ARV drugs safety monitoring and reporting (Pharmacovigilance)

Pharmacovigilance is the science and activities relating to detecting, assessing, understanding and preventing adverse effects or any other possible drug related problems. Pharmacovigilance is a key component of comprehensive patient care and the safe use of medicines. Failure to monitor, understand and manage these events can result in poor adherence and treatment failure and can reduce confidence in antiretroviral therapy among both PLHIV and the care providers. Pharmacovigilance is required not only for long term antiretroviral therapy but also for the antiretroviral drugs used for PMTCT, PrEP and PEP.

Even though there are advances in HIV treatment, adverse drug events (ADEs) associated with antiretroviral medicines (ARVs) still poses a challenge to patient safety. Usually, ADEs occur in the context of comorbidities and pill fatigue due to the lifelong treatment regimens. Advances in HIV treatment would also mean continuous introduction of newer regimens whose safety profiles have not been well studied and established in various settings and population groups such as children, elderly, pregnant women, people with co-morbid conditions (TB, hepatitis and others). These facts call for strong prevention, early identification, management, and reporting of adverse events. And WHO recommends enhanced monitoring and surveillance of toxicity on transition to new ARV drugs in HIV programs. EFDA has established and coordinates a national pharmacovigilance center to strengthen this effort. Professionals involved in prescribing and dispensing as well as patients are encouraged to monitor and report ADEs to the center. The information obtained through the national pharmacovigilance system enables the HIV program to make informed decision on the safety profile of ARVs. Therefore, EFDA will avail reporting tools and build the capacity of providers through trainings and face to face discussions as appropriate.

The main pharmacovigilance methods include spontaneous reporting, cohort event monitoring and targeted spontaneous reporting.

- Spontaneous reporting system is the most widely used system. Health professionals and pharmaceutical manufacturers voluntarily submit suspected adverse drug reactions to the national regulatory authority. A serious limitation here is underreporting.

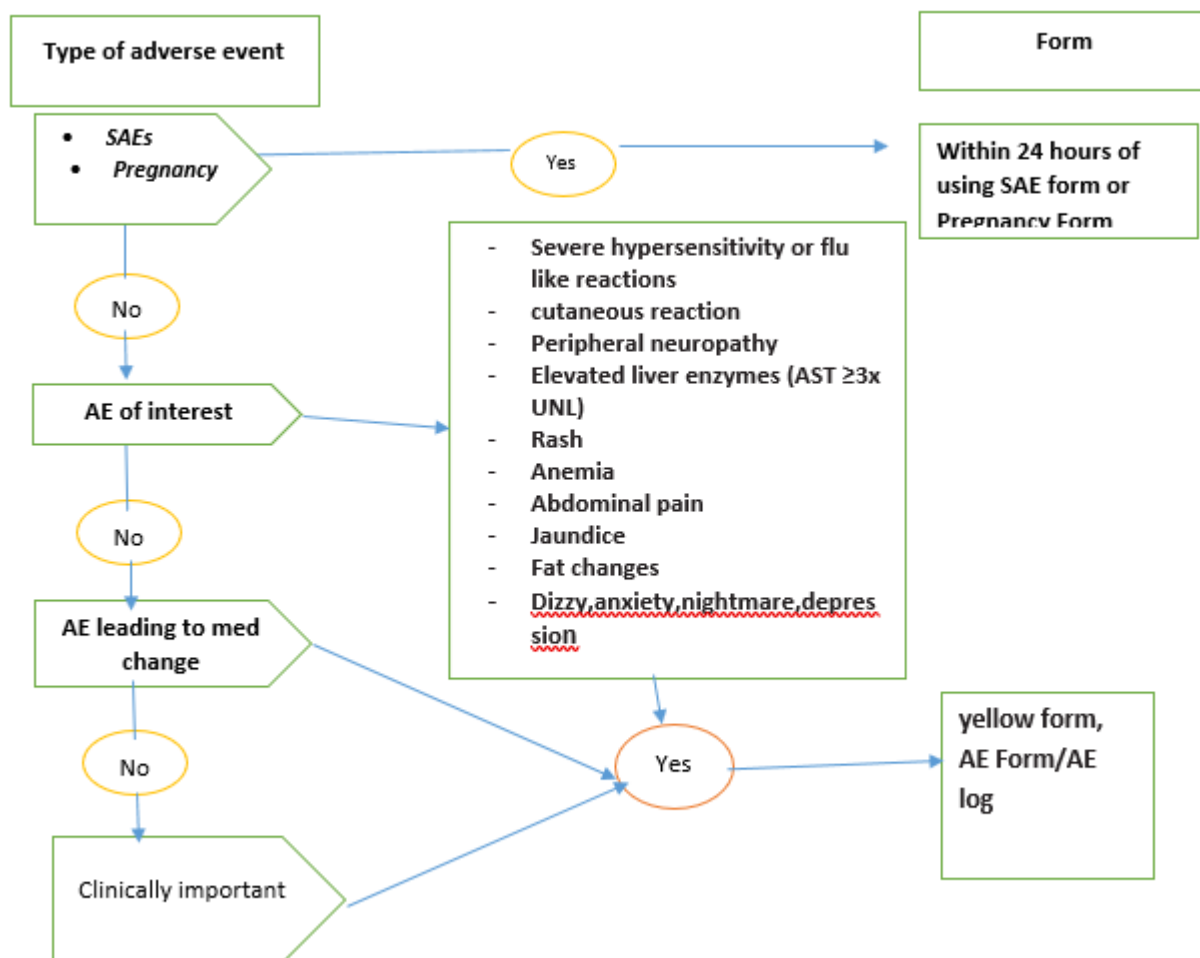
- Cohort event monitoring is a prospective observational cohort study of adverse events associated with one or more medicines. All adverse events occurring to a person taking antiretroviral drugs are collected regardless of the causality or relationship with the antiretroviral drugs.
- Targeted spontaneous reporting is a method that builds on the principles of both spontaneous reporting and cohort event monitoring. Targeted spontaneous reporting enables focus on a specific drug of interest (such as DGT, Tenofovir), a specific population of interest (such as women, children) or a specific adverse drug reaction (such as anemia).

Reporting of adverse events

Adverse drug events should be reported to the responsible authority body, Ethiopian Food and Drug Authority (EFDA). The adverse events that need to be reported includes medicine-related injuries, with at least a reasonable possibility to be caused by the direct pharmacological mechanism of a medicine, an individual's particular vulnerability, drug interactions, unexpected therapeutic ineffectiveness (e.g. resulting from drug interactions, product quality, problems or antimicrobial resistance), medication errors, and product quality defects.

In line with this, adverse events associated with the drugs should be reported using either the standardized adverse event reporting format (yellow form), using hot line 8482 (toll free line) or Telephone 01115523142(direct) or 0115524122(via operator), electronically on e-reporting of ADR available on the apps or website of EFDA (www.fmhaca.gov.et) and Using a mobile app Medsafety by health care providers and public health programs.

General Reporting approach and Timeline for active DSM



6.4. Laboratory and diagnostic services

The consolidated national HIV guidelines support an increased access to HIV care and treatment and laboratory diagnostic services. To ensure that laboratory services are accurate and reliable, relevant quality assurance systems need to be strengthened in multiple HIV diagnostic settings, as well as other laboratory testing services (eg. Viral Load, CD4 chemistry, CBC etc.). Therefore, strategic planning for proper placement and harmonization of HIV screening platforms and other laboratory

services should be carried out to ensure appropriate use and cost-effectiveness. To enhance implementations of essential and timely patient care, minimum packages of laboratory tests need to be provided as per the schedule or as needed. Minimum laboratory testing services packages for people with HIV are: treatment monitoring tests, HIV drug resistance tests, OI screening/co-infection assessment monitoring test and toxicity monitoring test.

Table 6.4. Summary Components of minimum Laboratory service packages for PLHIV

Laboratory tests	Period
Complete Blood Count	Baseline and every 6 months
CD4 cell count	Baseline and as needed
Kidney Function Tests: (Blood Urea Nitrogen and Creatinine)	Baseline and as needed
Liver Function Tests: ALT (alanine aminotransferase), AST (aspartate aminotransferase), Bilirubin, Alkaline Phosphatase, GGT (gamma glutamyl transpeptidase), LDH (lactic dehydrogenase)	
Blood Sugar: High blood sugar can be a side effect of the HIV protease inhibitors.	Baseline and as needed
HIV Viral Load tests	See ART monitoring section
HIV drug resistant testing	For second line treatment failure
AFB/MTB/RIF	As needed
LF-Urine LAM	For patients with ≤ 200 cells/mm ³ (inpatient) ≤ 100 cells/mm ³ (outpatient) Or any CD4 count with TB symptoms or if seriously ill At baseline and as required
Cryptococcal Antigen (CRAG) tests	for patients with CD4 cell count < 100 cells/mm ³
Others STIs (HBsAg, HCV, HPV)	As needed, for HPV DNA refer CxCa HPV DNA PCR testing

This guidance emphasizes the importance of leadership and governance to ensure implementations of minimum laboratory service packages, strengthen high-quality laboratory services, expanding testing services and developing the health workforce:

- To strengthen and expand laboratory testing services;
- To support a dedicated specimen referral system;
- To increase access to HIV viral load testing;
- To support the expansion of diagnostic services to include testing at the point of care;
- To train and certify health workers who perform the testing;
- To ensure high-quality laboratory services and plans for implementation, including quality assurance; and
- To ensure appropriate deployment of laboratory diagnostic technologies to increase their efficient and optimal use.

1) Strengthening and Expanding Laboratory Testing Services

The following areas are important to strengthen the network of laboratory services for implementing the national guidelines:

- Expanding and strengthening current laboratory networks with efficient specimen referral mechanisms to support and monitor the decentralization and integration of laboratory services or to provide access to laboratory tests which are available at limited number of sites (e.g. HIV viral load testing, DNA PCR, CD4 count etc.).
- Standardizing testing methods to streamline procurement, quality assurance and training.
- Incorporating new testing approaches and systems into national laboratory strategic plans and policies.
- Evaluating diagnostics for their performance and operational characteristics to validate testing algorithms (with back-up options) before introduction.
- Carrying out strategic planning for properly placing and harmonizing testing platforms to ensure appropriate use and cost-effectiveness.
- Allocating appropriate resources to ensure the availability of testing services, including human and financial resources.

2) Strengthen a dedicated Specimen Referral System

Laboratory referral systems and procedures for collecting and processing specimens need to be strengthened to increase access to viral load testing and other laboratory testing (for example, CD4 and early infant diagnosis). Providing for and strengthening a dedicated, efficient, safe and cost-effective specimen referral system requires reliable specimen transport with adequate conditions for whole blood, plasma and dried blood spot specimens (DBS) and rapidly and dependably reporting test results back to the referring sites to support for timely patient care.

3) Increasing Access to HIV Viral Load Testing

The guideline recommends the use of viral load testing to monitor treatment response and diagnose treatment failure and the use of dried blood spot samples for hard-to-reach sites for viral load testing. This will require strengthening the existing laboratory services and phased expansion of monitoring services into peripheral facilities and can include:

- Strengthening and leveraging existing specimen transport and result receiving networks;
- Ensuring that laboratories have adequate infrastructure, technical testing expertise and quality assurance and quality improvement programs;

- Ensuring an appropriate mix of high-volume centralized laboratory testing and point of care testing technologies. Point-of-care Viral load testing prioritized for the following populations groups:
- Pregnant and breastfeeding women
- Infants, children and adolescents
- People requiring a repeat viral load after a first elevated viral load
- People for whom treatment failure is suspected
- People presenting sick, living with advanced HIV disease or having a known opportunistic infection (TB, cryptococcal infection, etc.)
- First scheduled viral load test for people re-entering care
- The use of dried blood spots as a tool to increase access to viral load testing.

4) Planning for appropriate use of CD4 count testing as access to viral load testing increases

The country has formally quitted CD4 count-dependent treatment initiation thresholds and viral load monitoring replaces monitoring with CD4 cell count. However, CD4 cell count testing will continue as part of HIV programs for baseline assessment, monitoring OI prophylaxis and other clinical assessments, even in settings with full access to viral load testing.

5) Expanding point-of-care(POC) testing Technologies

Decentralizing laboratory services requires that all aspects of testing be in place before implementing services, including:

- Using only high-quality, evaluated and reliable diagnostic tests;
- Supervising and monitoring POC testing for quality and reliability;
- Implementing a strategy for managing supply chain and equipment service; and
- Establishing data management systems for timely identification of quality issues at regional and national data reporting.

6) Implementing Comprehensive Quality Management Systems

Implementing comprehensive quality management system including quality control, External Quality Assessment (EQA) and quality Improvement project is essential. The quality management system should:

- • Be implemented within the laboratory network and all remote testing sites
- Be incorporated into the routine testing procedures and periodically monitored
- Ensure that testing sites undertake quality control, as appropriate
- Ensure that testing sites are enrolled in

an external quality assessment scheme (proficiency testing program and on-site evaluation)

- Ensure the use of standard operating procedures for all processes, including specimen collection and processing, test methods, interpreting results and reporting
- Ensure the use of standardized logbooks or electronic data management and identifying errors and potential misclassification and
- Ensure the implementation of preventive and corrective measures for equipment

7) Providing guidance for developing health workers' capacity, including staff training and certification

Supplementing guideline for developing health care worker capacity building includes; qualification of personnel who will perform the laboratory tests, training requirements for specific tests and the process for certification and re-certification. The guideline also indicating that, all health workers assigned to perform POC and other laboratory testing services must be trained, proficient on the testing procedure, specimen collection and quality assurance before implementing these services.

CHAPTER SEVEN:

GUIDANCE FOR PROGRAM MANAGERS AND LEADERS

Policy development and review is a dynamic process. Policies change and/or revision like strategic shifts, evolvement of new initiatives as well as evidence and knowledge change over time at national, regional and global level are happening according to the lessons learnt during the implementation of the HIV program. Policies set early in the development of HIV program may negatively affect implementation and need to be revised. Policies, therefore, need to be able to respond to the current changes. Program managers and leaders should be cognizant of changes and challenges affecting the development and implementation of HIV/AIDS policies. These include political commitment, financial implications, global directions, administrative and structural reforms, community engagement, meaningful involvement of PLHIV and basic legislation.

The key benefits to the global and country HIV response have enabled governments, programmers, organizations like partners and CSOs, communities, and individuals to achieve epidemic control by addressing individual, community and societal level HIV fueling factors. These include matters related to sexual behavior, injecting drug use, socio-cultural practice, stigma associated with HIV/AIDS, gender inequality, poverty, political commitment, structural barriers, right to access HIV related services and human rights that need to be addressed.

7.1 Guiding principles

The following guiding principles are expected to lead the implementation of the national HIV program and services provision:

I. Ownership and commitment: An effective response to HIV/AIDS requires ownership and active involvement of the community and all other sectors. It also seeks strong leadership commitment at all levels.

II. Multi-sectoral: A multi-sectoral approach that includes partnership, consultations, and coordination with all stakeholders in the design, implementation, and service provision as well as monitoring and evaluation of the response are crucial to maximize the effectiveness of the response.

III. Gender Responsiveness: Gender sensitivity must be considered as a corner stone for the success of HIV/AIDS response with greater and sustained positive impact as the epidemic and its response is highly impacted by gender inequalities.

IV. Inclusiveness: An inclusive and people-centered approach that recognizes different prevention, care, and treatment options that individual may choose and benefited at different stages of their lives have to be designed and implemented.

V. Equity: Inequalities in health outcomes will be addressed through rights-based programming and through improving the understanding of, and response to, human rights and gender related barriers to accessing services.

VI. Resource mobilization and proper utilization: to sustain the gain in HIV response and attain epidemic control, mobilizing domestic and external resource, ensuring efficiency, transparency and accountability in use and proper allocation and effective utilization of resources are essential in the national response to HIV/AIDS epidemic at all levels.

VII. Service Integration and Linkage: HIV services must be gradually integrated with existing health services in the health facilities and effectively linked within the facility and to community based services.

VIII. Evidence Based Response: HIV program implementation will be based on coordinated joint planning, review, monitoring and evaluation and strategic information generation and use for decision with implementers and key stakeholders at all levels.

7.2. Overview of the HIV/AIDS National Strategic Plan 2021-2025

The 2021-2025 HIV/AIDS National Strategic plan was developed being informed by situation and response analysis and investment case analysis based on Goals Modeling using existing national data for geographic, population and intervention prioritization. The strategic plan aims to attain HIV epidemic control nationally by 2025, by reducing new HIV infections and AIDS mortality to less than 1 per 10,000 population.

The NSP has set the following impact targets to be achieved by the end of the 5-year period:

- Number of new HIV infections reduced to less than 1 per 10,000 population (Disaggregated by sex, age, region and population group)
- AIDS related deaths reduced to less than 1 per 10,000 populations.
- Incidence Mortality Ratio reduced to less than 1 (Target: From 1.08 to 0.9)
- Percentage of child HIV infections from HIV- positive women delivering in the past 12 months reduced from 16.9 % in 2019 to less than 5% by 2025; and less than 2% by 2030.

The Strategic Plan has Six Strategic Objectives to achieve the set goal.

1. Reach 90% of Key and Priority populations with targeted combination HIV prevention interventions by 2025.
2. Enhance HIV case finding to attain 95% of PLHIV knowing their HIV status and linked to care by 2025.
3. Attain virtual elimination of Mother to Child Transmission (MTCT) of HIV and Syphilis by 2025
4. Enroll 95% of PLHIV who know their status into HIV care and treatment and attain viral suppression to at least 95% for those on antiretroviral treatment.
5. Mobilize resources and maximize efficiencies in allocation and utilization.
6. Enhance generation and utilization of Strategic Information (SI) for an accelerated evidence-based response.

In addition, ten social and programmatic enablers are identified to maximize the reach and impact of Ethiopia's HIV/AIDS response.

- A. Gender and Gender based Violence.
- B. Stigma and discrimination
- C. The role of civil society, communities, PLHIVs and the private sector
- D. Embracing a Human rights approach to the HIV response
- E. Health Systems (supply and laboratory)
- F. Human resources for health/ HIV response
- G. Governance, leadership, coordination, and accountability
- H. Addressing policy and legal issues
- I. Partnership, Multi-sectoral Collaboration, Civil society, and the Private Sector
- J. Private for-Profit Sector Strategic Interventions

Program managers and leaders must understand the scope of NSP and its strategic approaches for the standardization of program approach across the country.

7.3. Program Performance and Response Analysis

All combination prevention programs require a strong community empowerment element and collective efforts to address legal and policy barriers, as well as the strengthening of health and social protection systems, plus actions to address gender inequality, stigma and discrimination.

Before deciding on a package of HIV prevention interventions for a specific HIV epidemic, a clear and evidence-informed picture of that epidemic

and its response is needed. This concept is known as the 'know your epidemic, know your response' approach, which is the starting point for combination prevention planning and programming, and is comprised of a series of exercises to help categorize an epidemic (such as whether it is 'generalized' – i.e., within the general population – or 'concentrated' within certain groups within the population, often referred to as 'key affected populations'). This involves looking at factors such as modes of HIV transmission, key affected populations, and key epidemiological trends as well as the response to the epidemic.

In the planning process, programmers, managers, leaders, and policy makers are recommended to follow is described below:

Inclusiveness of the planning process: Ensure the participation of all relevant stakeholders, including government officials, community leaders, civil society organizations, donors, and most importantly, people affected by HIV and AIDS.

Identify modes of transmission and the most affected populations: Identify factors that fuel HIV transmission and understand how HIV is spread, identify the most common modes of transmission, and the most affected populations, identify geographic variations in HIV prevalence, such as urban Vs. rural areas, know the size of key affected populations.

Ensure the availability and utilization of the appropriate tools to collect, review, analyze, monitor, evaluate and data use for program improvement.

Identify and understand structural barriers that might fuel HIV prevalence, analyze social, legal, economic, and cultural drivers of HIV epidemic. Upon completion of the 'know

your epidemic, know your response' planning process, a combined component of coordinated behavioral, biomedical, and structural HIV prevention interventions can be developed and implemented.

Determining whether current HIV prevention, care and treatment programs are adequate to address the needs that have been identified requires understanding of, who is currently accessing these services. Programmers should assess existing HIV prevention, case finding efforts, ART and viral load service coverage levels, linkage to care and adherence to treatment and viral suppression among the general population as well as key and priority populations. They also need to assess the overall outcome and impact of the response including decline in incidence and mortality. Disaggregated data for various groups enable assessment of HIV prevention, care and treatment needs and establishment of priorities for delivering services.

Data on case finding yield, linkage to care, adherence, retention, and viral load suppression are keys to assess the quality of the services provided. Surveillance of transmitted and acquired HIV drug resistance can also be instrumental in informing decisions on optimal regimen choices for ART program.

A review of epidemiological and programmatic data is incomplete without a deeper understanding of what drives HIV vulnerability and how various political, social, economic, and legal factors affect the ability and willingness of various groups to seek and access health services. Stigma, discrimination, poverty, gender inequality, education and migration status are key elements that should be considered to inform effective HIV programming.

7.4. HIV response in the context of COVID 19

Individuals living with HIV, especially those with co-morbid condition and/or advanced HIV disease may be at greater risk for COVID-19 complications and related impact. During the current pandemic, PLHIV that utilize health care institutions were often sicker than they would have been in the past, due to delays in seeking care. Thus, coordinated efforts from community stakeholders, civil society, public health agencies, healthcare providers and program managers is essential for addressing health inequities for PLHIV resulting from social stigma and poor access to Personal Protecting Equipment (PPE), preventative services and treatment. The present critical challenges for PLHIV that need to be addressed during the COVID-19 pandemic are access to their ART drug refills, maintaining of their psychosocial wellbeing, and sustaining their economic welfare.

Furthermore, access for HIV testing and prevention service for PLHIV partners and other KPPs still critical challenges amid COVID-19 pandemic

To ensure the health of PLHIV as well as maintaining the continuity of HIV services in the era of COVID-19 pandemic, program managers need to monitor the implementation of the national standard response guideline to be updated to prevent the HCPs and PLHIV from COVID-19 which includes COVID-19 testing, vaccine and treatment must be accessible for PLHIV.

7.5. Key parameters for decision-making

Ethics, equity, and human rights

Multiple legal, social, and normative obstacles have resulted in inequitable access to HIV prevention, testing, treatment, and care. Global and national commitments require providing HIV prevention, care, and treatment services to everyone in need, following the human rights principles of non-discrimination, accountability, and participation. National HIV strategies should be planned and implemented from the outset with the goal of delivering the full package of services and interventions recommended in these guidelines as soon as possible.

Key ethical principles of fairness, equity and urgency should also be observed in the process of reviewing and adapting guidelines. The design of effective and equitable policies implies that strategies should focus comprehensively on addressing barriers to access testing, prevention, and treatment services, particularly those faced by key and priority populations.

Impact and cost-effectiveness

Realizing positive impact for a population is an important goal of public health programs and policies. Examples of the impact of HIV programs include reduced HIV incidence, prevalence, morbidity and mortality and improved quality of life. Impact is often a result of a complex set of factors and a combination of diverse inputs and activities or processes, and it is often not attributable to a single intervention or program.

Cost-effectiveness analysis is one of the several economic evaluation tools used to measure the value of delivering services. Economic evaluation measures the costs and consequences of alternative programs, which are then compared to assess how the greatest health benefits can be generated. In cost-effectiveness analysis, impact is often measured using indicators related to a change in health status, such as disability-adjusted life-years gained, which includes the estimated number of deaths and infections averted.

As the experience of scaling up ART in low- and middle-income countries demonstrates, the cost-effectiveness of health interventions also changes over time, as costs fall because of gains in scale, improvements in technology or the design of more efficient delivery systems.

Although evaluating cost-effectiveness and health impact may be useful in systematically comparing various program interventions, they should be considered in the light of the ethical, equity and human rights implications associated with different courses of action, especially in settings in which not all eligible individuals currently have access to ART.

Investments in social and programmatic enabler programs (such as integrated treatment and rights literacy programs, legal services, stigma and discrimination reduction programs, training for health care workers and law enforcement, Gender, CSO, CBO and, PLHIVs involvement, Human rights, governance, leadership, coordination, and accountability, policy and legal issues, partnership, Multi-sectoral Collaboration, and the Private Sector

Interventions) can play a role in overcoming barriers to accessing treatment and other HIV-related services and keeping people connected to care. As such, these programs can contribute to overall cost-effectiveness, in addition to achieving other important objectives, such as reducing discrimination.

Opportunities and risks

The recommendations in these guidelines have the potential to further reduce HIV-related mortality, improve the quality of life, reduce the number of people acquiring HIV infection and enhance treatment effectiveness. The benefits accumulated from implementing them are likely to considerably outweigh the upfront investment needed and have the potential to fundamentally change the course of the epidemic. Nevertheless, domestic factors (such as budget cuts, shortage of ARV drugs, and attrition of trained health workers and emergence of drug resistance) and external contingencies (such as withdrawal of external financial support, political instability, and natural disasters) could negatively affect their implementation. It is essential to design strategies to mitigate such events so that continued service delivery can be assured, especially for those most in need.

Implementations

At the virtue of epidemic control with low prevalence among the general population, it is critical to identify opportunities to expand access to HIV prevention, testing, treatment, and care including case identification using high yield strategies focusing to key and priority populations.

Decentralization of services and task shifting in service provision is crucial to improve the access to HIV services in which both facility and community based interventions like targeted SBCC, and demand creation, HIV self-testing, and other differentiated service delivery models need to be strengthened. As coverage of ART increases and programs matures, expanding access to second line regimens and ARV drugs resistance monitoring increasingly becomes a programmatic priority. Access to third-line regimens for PLHIV who fail on second-line regimens should be considered too. Scaling-up viral load monitoring is vital to adequately identify treatment failure and to avoid switching unnecessarily to second-line or third line regimens. Viral load monitoring is also likely to play a central monitoring role in places in which ART is being broadly expanded to reduce HIV incidence.

7.6. Roles and responsibilities

As the response to HIV involves a wide diversity of actors, coordination at various levels of the system becomes important to ensure coherence and cohesion of efforts. HIV program managers should ensure effective coordination with other health programs, among the HIV activities in the health sector and those in other sectors, and between the different levels of the health system (National, regional and district).

Ministry of Health

The role of Ministry of Health is to lead and coordinate the health system which includes to provide policy guidance, regulation, ensuring accountability for health, capacity building,

resource mobilization (domestic and external) for HIV, health intelligence and building partnership across all health actors.

The Ministry of Health should also ensure that health services are quality and equitable; the HIV program managers should serve as a leader, manager, coordinator, facilitator, innovator, and liaise regularly with other health program and HIV actors.

The role of Ministry of health has been decentralized up to Woreda /district/ and community levels. For effective implementation and follow-up, national strategic plans must be linked and cascaded to all levels. RHB and Zonal/Woreda Health Offices are mandated to manage and coordinate the operation of primary health care services at the respective levels; they are responsible for planning, financing, monitoring, and evaluating of all HIV programs and service deliveries in the regions and zones/ Woredas and community levels.

Federal HIV/AIDS prevention and control office (FHAPCO)

FHAPCO has a role of leading the HIV primary prevention and overall coordination of the national HIV prevention and control program, domestic and external resource mobilization for HIV, ensure partnership with various government ministries, regional president's, non-governmental organizations, and people living with HIV, faith-based organizations, and private sector and development partners. Respective multi-sectoral response coordinating units at sub-national levels will play leading role in HIV primary prevention and over all coordination of the response.

Ethiopian Public Health Institute (EPHI)

EPHI will have a leading role in strategic information generation and ensure quality laboratory services for the national HIV/AIDS response. The institute has also greater role in monitoring and coordinating the third 95 target of the response through uninterrupted viral load monitoring services.

Ethiopian Pharmaceutical Supply Agency (EPSA)

EPSA has leading role in supply chain management of HIV related commodities and ensuring uninterrupted supply.

Ethiopian Food and Drug Authority (EFDA)

EFDA has the lead role in quality control of the HIV commodities and services.

Partners

The roles and responsibilities of local and international partners include:

- Technical and financial support for implementation of the newly adopted and existing innovative HIV interventions.
- Participate in the national and regional HIV program coordination mechanisms.
- Support the joint planning, monitoring and evaluation of the different program areas.

CSO/FBOs/CBOs

CSOs/FBOs/CBOs are responsible for implementation of HIV prevention, care and treatment interventions at grass-roots level. They are responsible for building community capacity and mobilizing communities to plan, implement, and monitor & evaluate the response. They also have key role in reaching KPPs with intervention articulated in this guideline and taking part in community led monitoring of the response.

7.7. Coordination mechanisms

Coordination in the health sector

High level coordination mechanisms such as the executive committee which is composed of the Minister, state ministers, and director generals of agencies, Joint Consultative Forum (JCF) meetings are held regularly between MOH and donors, Health Sector Joint Steering Committee, which involves the heads of the Regional Health Bureaus, oversee the performance of different programs in the health sector. The committee is chaired by senior officials in the Ministry of Health and provides general guidance to the health sector. Technical level coordination between health programs might occur through technical advisory/working groups.

Multi-sectoral Coordination

The HIV response needs multi-sectoral collaboration with different social and development sectors. The multi-sectoral

response against HIV/AIDS is coordinated by Federal HIV/AIDS Prevention and Control Office with leadership of MOH, the National AIDS council and Management board on top. The high level coordination mechanism includes National AIDS Council meeting, Joint Planning and Review Meeting with key sectors.

Donor coordination mechanisms

Some funding agencies, such as the Global Fund to fight AIDS, TB and Malaria, the President's Emergency Plan for AIDS Relief (PEPFAR) and other donors have own types of mechanisms for coordinating their in-country efforts. However, the health sector is involved and often a key member of these coordinating mechanisms and should always work to ensure consistency and harmonization.

Stewardship and advocacy for the HIV response in other sectors

The health sector can provide the evidence necessary to leverage action for HIV in other sectors. The Ministry of Health and Federal HHIV/AIDS prevention and control office have a crucial role in using its stewardship and advocacy power to ensure that HIV issues are addressed in all policies. This includes engaging ministries of education, social development, gender, transport, and other strategic sectors.

CHAPTER EIGHT:

MONITORING AND EVALUATION

8.1. Introduction to HIV/AIDS Program Monitoring & Evaluation (M&E)

Effective HIV prevention, care and treatment require standardized recording and reporting system. Recording and reporting is used to systematically monitor and evaluate progress of program performance.

The reporting of HIV prevention, care and treatment activities is integrated into the DHIS-2 and all forms and registers are standardized in line with it throughout the country. Health facilities are the primary sources of data. Any information concerning PLHIV should be recorded completely and correctly. Recording and reporting tools should be kept neat and maintained properly.

This guide on monitoring and evaluation of HIV in the health sector brings together the various elements of monitoring and evaluation systems for HIV program.

In order to establish a functional national HIV M&E system, the following key elements are important.

1. Presence of an M&E unit: Established M&E unit with qualified staff and enough budget. The unit also needs to build links with regions, sector ministries, research institutions, NGO, FBOs, civil associations and donors.

2. Clear goals and objectives of the program: It needs well-defined national program goals, objectives and targets where regular reviews/evaluations of the progress of the implementation of the National/Regional program is undertaken. Guidelines and guidance need to be put in place on the M&E to regions and sectors.

3. A core set of indicators and targets: It is important to identify priority/core indicators and additional indicators that cover program inputs,

activities/processes, outputs, outcomes and impact. In addition, selection of indicators needs to be through full participation of stakeholders and maintaining relevance and comparability.

The process needs also to utilize past and existing data collection efforts (e.g. DHS, BSS and Sentinel Surveillance) to assess national trends.

The main sources of data for HIV monitoring could be:

- Routine data sources such as DHIS2, CBS, EMR-ART
- Surveys like EDHS and EPHIA
- Estimations such as Spectrum, CSA estimates and research

4. Routine Monitoring: For HIV results at all levels to be measured, the entire spectrum of input, output, outcome and impact data are needed. Input and output monitoring data are important, as these answer questions about the resources and interventions needed and provided, and whether planned programs have been implemented. Input and output monitoring data are collected through routine monitoring systems. A plan for data collection and analysis: An overall national level data collection and analysis plan is important. The plan also has to address data collection and analysis systems at lower levels.

5. A clear plan for data dissemination and use: Establishment of an overall national level data dissemination plan is important.

8.2. Key Indicators

1. Number of individuals receiving Pre-Exposure Prophylaxis

Definition	Number of individuals, inclusive of those newly enrolled, that received oral antiretroviral pre-exposure prophylaxis (PrEP) to prevent HIV infection during the reporting period.
Formula	Number of clients that received Pre-exposure Prophylaxis during the reporting period
Interpretation	This indicator intends to measure number of clients who are taking PrEP within the reporting period. It counts the number of individuals that received PrEP at ANY point during the reporting period. It includes those who have been enrolled in the previous period and receiving PrEP and those who are newly enrolled in the reporting period. It excludes those who have been enrolled to PrEP but stopped taking it due to different reasons. Use of PrEP may cease once an individual is no longer at risk for HIV. Once they cease taking PrEP, they will not be counted.
Disaggregation	PrEP New and PrEP Current Age Sex Client Category: ■ Female sex workers (FSW) ■ Discordant Couples
Sources	PrEP Register and Tally Sheet
Reporting level	HC/ Clinic/ Hospital
Reporting frequency	Monthly

2. Number of persons provided with Post-Exposure prophylaxis

Definition	Number of persons provided with post-exposure prophylaxis (PEP) for risk of HIV infection through occupational and/or non-occupational exposure to HIV.
Formula	Number of persons provided with post-exposure prophylaxis (PEP) for risk of HIV infection as per the national guideline.
Interpretation	The indicator can be generated by counting the number of individuals receiving PEP for occupational and non-occupational purposes. And individuals should only be counted if they have received PEP drugs. PEP services for occupational exposure include a comprehensive package of services for occupationally exposed health care workers and patients. PEP services for non-occupational exposure include sexual violence. Individuals should be counted only if they have received PEP drugs (in accordance with national protocols). This indicator does not intend to capture the type and quality of PEP services provided. PEP services include first aid, counseling, testing, provision of ARVs, medical care, trauma counseling, linkages with police, and other follow-up and support. Simple monitoring of PEP availability through program records does not ensure that all PEP-related services are adequately provided to those who need them.
Disaggregation	Exposure type: ■ Occupational, ■ Sexual Violence ■ Other non-occupational
Sources	Post Exposure Prophylaxis Register
Reporting level	HC/ Clinic/ Hospital
Reporting frequency	Monthly

3. Percentage of people living with HIV who know their status.

Definition	Percentage of people living with HIV who know their status	
Formula	Number of people living with HIV who know their status	X 100
	Estimated Number of people living with HIV	
Interpretation	This indicator can be used as a proxy for the first 95 target of the 95-95-95 HIV targets. It is Critical to determine the proportion of people living with HIV who know their HIV status, as this knowledge is the entry point to the continuum of care for PLHIV. The three 95s are: ■ 1st 95 = 95% of all people living with HIV will know their HIV status. ■ 2nd 95 = 95% of all people with diagnosed HIV infection will receive ART ■ 3rd 95= 95% of all people receiving antiretroviral therapy will have viral suppression The numerator should be the sum of: ■ PLHIV who were reported as currently on ART in the previous reporting month ■ Total new HIV positives identified through HTS in the reporting period ■ Total number of lost in the reporting period. Limitation of this indicator: This indicator may miss those previously identified positives and who are alive and not started on ART. At onal, Woreda and facility levels, it is difficult to get estimates of PLHIV to compute the first 95. Therefore, these levels should monitor HCT uptake (Number of people tested for HIV) and its yield (Number of people tested positive for HIV).	
	Disaggregation	Age/Sex HIV test result: Positive, Negative Testing modality: VCT, PITC, ICT Population groups: FSW, IDUs, Long distance drivers, Mobile worker /Daily Laborers, Prisoners, OVC, Children of PLHIV, Partners of PLHIV, Other PPs (Widowed, Divorced, Separated, Re-Married), General population
Sources	PITC tally and VCT register, PMTCT Register, ART register	
Reporting level	HC/ Clinic/ Hospital	
Reporting frequency	Monthly	

4. Number of individual HIV self-test Kits Distributed

Definition	It is the number of HIV self-test kits distributed
Formula	Numerator: Number of individual HIV self-test kits distributed Denominator: N/A
Interpretation	This indicator aims to monitor trends in the distribution of HIV self-test kits at the lowest distribution point. HIV self-testing refers to a process in which a person collects his or her own specimen (oral fluid or blood), performs an HIV test, and then interprets the results. This is often done in a private setting, either alone or with a trusted person. HIV self-testing is a screening test and requires self-testers with a reactive (preliminary positive) result to receive further testing from a trained provider using a validated national testing algorithm. HIV self-testing approaches range from unassisted self-testing (with limited or no instruction provided) to directly assisted self-testing (where a testing provider demonstrates how to use the self-test kit). Self-test kits can be distributed in various ways (i.e., by providers or outreach workers, over the counter, etc.). Secondary distribution of HIV self-test kits may also occur (e.g., to partners of ANC attendees, or clients of FSWs). Data for the numerator should be generated by counting the number of individual HIV self-test kits distributed and NOT the number of individuals receiving an HIV self-test kit. Data is captured and reported at the lowest distribution point. This is to prevent double counting between the various higher supply chain levels.
Disaggregation	Type of self-testing: Directly assisted and Unassisted Number of Test kits distributed by Age/Sex ■ Total Number of clients reported having reactive self-test result by Age/Sex ■ Total Number of clients reported having reactive result and confirmed to conventional HIV testing service (HTS) using the national algorithm Age/Sex
Sources	Self-Test Register (KP Register)/HTS Logbook
Reporting level	HP/HC/Clinics/Hospital
Reporting frequency	Monthly

5. Proportion of STI cases tested for HIV

Definition	Proportion of STI cases tested for HIV in the reporting period.	
Formula	Number of STI cases tested for HIV in the reporting period	X 100
	Total number of STI cases in the reporting period	
Interpretation	This indicator is intended to provide information on the proportion of STI cases that are tested for HIV. It is helpful to measure the magnitude of the HIV and STI co-infection and to intensify the HIV prevention interventions. It also helps to track the number of STI cases. Additionally, the proportion of STI cases detected can be tracked by dividing the number of detected STI cases by the estimated number of STI cases in the catchment area.	
Disaggregation	HIV test result (Positive/Negative) by Age/Sex: <15M/F, >15 M/F By syndrome type/Age /Sex: <15M/F,>15 M/F	
Sources	PICT Tally, OPD and IPD registers	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

6. Number of newly identified HIV positive adults and children clients linked to treatment and care

Definition	Number of adults and children linked to care and treatment
Formula	Number of clients linked to care and treatment in the reporting period
Interpretation	Programmatically very important to monitor the status of newly identified HIV positive clients before initiate ART. In addition, helps to track variation between newly identified positive and those who started treatment in a specified period.
Disaggregation	By Age/Sex Pregnancy Status: pregnant, non-pregnant
Sources	Positive tracking Register
Reporting level	Health center /Clinic/ Hospital
Reporting frequency	Monthly

7. Percentage of people living with HIV receiving ART

Definition	Percentage of adults and children currently receiving antiretroviral therapy (ART)	
Formula	Number of adults & children receiving ART at the end of the reporting period	X 100
	Estimated number of people living with HIV	
Interpretation	This indicator measures the ongoing scale-up and uptake of ART and retention in ART programs as a critical step in HIV service provision and assesses progress towards coverage of ART. It also measures the progress towards providing antiretroviral therapy to all people living with HIV and the extent to which ART needs are met. Provision of Antiretroviral therapy has been shown to reduce HIV-related morbidity and mortality among those living with HIV, and onward HIV transmission. This indicator measures the 2nd 95 target.	
	Data for this indicator is generated by counting the number of adults and children who are currently receiving ART in accordance with the nationally approved treatment protocol at the end of the reporting period. Patients who have died, stopped treatment, transferred out, lost (patient not seen for 1 to 3 months from last visit) and dropped out (patient not seen for > 3 months from last visit) are NOT counted. Patients on ART who initiated or transferred in during the reporting period should be counted. Some people pick up several months of antiretroviral medicines (ARVs) at one visit, and efforts should be made to include these people in the numerator as receiving antiretroviral even if they do not attend the clinic in the last month of the reporting period.	
	As it will be difficult to get the PLHIV estimate or the expected number of individuals who know their status at the Zone/woreda and lower levels, this indicator will be calculated at these levels based on the target allocation during the planning phase. This indicator includes currently receiving ART clients at ART clinic and those currently receiving ART at PMTCT clinic. All PMTCT only sites are expected to report ART currently receiving clients on monthly basis.	
Disaggregation	Age/Sex by Regimen:	
	ARV Dispensing Quantity by Months of dispensing Age/Sex: ■ <3 months of ARVs (not MMD) dispensed to patient by: <15 F/M, 15+ F/M, ■ 3-5 months of ARVs dispensed to patient by: <15 F/M, 15+ F/M, ■ 6+ months ARVs dispensed to patient by: <15 F/M, 15+ F/M, Pregnancy Status: pregnant, non-pregnant	
Sources	ART Register, PMTCT register, ART regimen tally, EMR-ART Software	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

8. Number of adults and children with HIV infection newly started on ART

Definition	Number of adults and children newly enrolled on antiretroviral therapy (ART) in the reporting period
Formula	Number of adults and children newly enrolled on antiretroviral therapy (ART) in the reporting period
Interpretation	The indicator measures the ongoing scale-up and up-take of ART program. This measure is critical to monitor along with number of patients currently on ART in relation to the number of PLHIV that are estimated to be eligible for treatment to assess progress in the program's response to the epidemic in specific geographic areas and population as well as at the national level. This indicator includes newly initiated clients at ART clinic and those newly started ART at PMTCT clinic. All PMTCT only sites are expected to report ART new initiation on monthly basis.
Disaggregation	Age/Sex by Regimen Pregnancy Status: pregnant, non-pregnant
Sources	ART Register, PMTCT register, ART regimen tally, EMR-ART Software
Reporting level	Health center /Clinic/ Hospital
Reporting frequency	Monthly

9. ART retention rate

Definition	Percentage of adults and children known to be on treatment 12 months after initiation of antiretroviral therapy (net current cohort)	
Formula	Number of adults and children who are still on treatment at 12months after initiating ART	X 100
	Total number of adults and children who started ART in the 12 months prior to the beginning of the reporting period	
Interpretation	This indicator measures the proportion of adults and children with HIV known to be on treatment 12 months after initiation of antiretroviral therapy and it is one important measure of program success and is a proxy for overall quality of program.	
	The Numerator: Number of adults and children still alive and on ART at 12 months after initiating treatment. A 12-month outcome is defined as the outcome (i.e. whether the patient is still alive and on ART, dead or lost to follow-up) 12 months after starting. The numerator does not require patients to have been on ART continuously for the 12-month period. Patients may be included in the numerator (and denominator) if they have missed an appointment or drug pick-up or temporarily stopped treatment during the 12 months since initiating treatment, as long as they are recorded as still being on treatment at month 12. On the contrary, those patients who have died, stopped treatment, or been lost to follow-up as of 12 months since starting treatment are not included in the numerator. The number of adults and children on ART at 12 months includes patients who have transferred in (and their initiation date is known) at any point from initiation of treatment to the end of the 12-month period and excludes patients who have transferred out during this same period to reflect the net current cohort at each facility.	
	The Denominator: Number of adults and children in the ART start-up groups initiating ART at 12 months prior to the end of the reporting period (The denominator is the total number of adults and children in the (monthly) ART start-up groups who initiated ART at a point 12 months prior to the beginning of the reporting period, regardless of their 12-month outcome. This includes all patients, both those on ART as well as those who are dead, have stopped treatment or are lost to follow-up at month 12. Again, the denominator includes patients that have transferred in (and their initiation date is known) and excludes patients that transferred out during the time period.	
	The net current cohort is the number of patients in the start-up group plus any transfers in, minus any transfers out.	
Disaggregation	Age/Sex by Regimen Pregnancy Status: pregnant, non-pregnant	
Sources	ART Register, PMTCT register, ART regimen tally, EMR-ART Software	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

10. Number of ART Clients Interrupted Treatment

Definition	Number of ART patients (who were on ART in the previous reporting month) and then had no clinical contact for greater than 28 days since their last expected contact
Formula	Number of ART patients (currently on ART) with no clinical contact or ARV pick-up for greater than 28 days since their last expected clinical contact or ARV pick-up
Interpretation	This indicator is intended to: (1) help better understand fluctuations or steady growth in Currently on Treatment over time, (2) encourage tracing of patients when a patient has had no clinical contact for greater than 28 days since their last expected contact and (3) promote timely identification of patient outcomes among patients known to have missed clinical visits or drug pickups. Serious and repeated attempts should be made to reengage any such patients and return them to treatment. In case of death, mortality data should be analyzed and investigated to determine causes of death, where possible.
Disaggregation	Outcome by Age/Sex Died by Age/Sex Lost after on Treatment for <3 months by Age/Sex Lost after on Treatment for >3 months by Transferred Out by Age/Sex Refused (Stopped) Treatment by Age/Sex
Sources	ART Register, PMTCT register, ART regimen tally, EMR-ART Software
Reporting level	Health center /Clinic/ Hospital
Reporting frequency	Monthly

11. Proportion of PLHIV currently on Differentiated Service Delivery Model (DSD)

Definition	Total number of eligible/stable Adults on 1st line regimen currently on DSD in the reporting period	
Formula	Total Number of Clients on DSD	X 100
	Total number of eligible/stable Adults on 1st line regimen	
Interpretation	Important for service follow up and to show the performance of differentiated HIV Service Delivery	
Disaggregation	Age/Sex: <15M/F,>15 M/F	
	By different models: ASM, FTAR, CAG, PCAD	
Sources	DSD Register and Tally Sheets	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

12. Viral load suppression

Definition	Percentage of patients on ART with a suppressed viral load (<50 copies/ml) in the past 12 months	
Formula	Number of ART patients with suppressed VL results (<50 copies/ml) documented within the past 12 months	X 100
	Number of ART patients with a VL test result documented within the past 12 months	
Interpretation	This indicator could provide information that can contribute to quality improvement activities designed to maximize rates of viral suppression in patients on ART and prevent the acquisition of HIV drug resistance. The viral load of patients receiving ART provides an indication of adherence to treatment, patient compliance with disease monitoring and the quality of care delivered. Measuring viral suppression is a key programmatic indicator related to effective treatment. It helps as a proxy indicator to monitor the third 95 of UNAIDS’ 95-95-95 treatment target, that 95% of people receiving ART will have viral suppression by 2030. For the numerator: It is the actual number of PLHIV on ART that have a document suppressed VL result at the end of the reporting period. It is expected that viral load testing should be routine rather than episodic. If there is more than one VL result for a patient during the past 12 months, the most recent result should be reported. For the denominator: It is the actual number of PLHIV on ART that have documented VL test. Only patients who have been on ART for at least 3 months should be considered. Note: Viral load tests for PMTCT clients should also be included in this indicator.	
Disaggregation	By Age/Sex By Pregnancy status: non-pregnant and pregnant or breastfeeding	
Sources	ART and PMTCT registers	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

13. Proportion of clinically undernourished People Living with HIV (PLHIV) who received therapeutic or supplementary food

Definition	The proportion of individuals receiving therapeutic or supplementary food among those whose nutritional status was assessed and found to be undernourished	
Formula	No. of clinically undernourished PLHIV on ART who received therapeutic or supplementary food	X 100
	No. of PLHIV on ART who were nutritionally assessed & found to be clinically undernourished.	
Interpretation	Provision of nutritional treatment, care and support for those undernourished PLHIVs is important to prevent morbidity and mortality. Under nutrition significantly increases mortality risk for HIV-infected individuals regardless of treatment status. Among the clinically undernourished PLHIVs, those with severely undernourished (SAM) cases will receive the Ready-To-Use Therapeutic Food (RUTF) and those with moderately undernourished (MAM) cases receive Ready-To-Supplementary Food (RUSF) based on availability of supplies.	
	Severe Acute Malnutrition (SAM): Adult: -BMI less than 16 kg/m2. Pregnant and lactating: -MUAC less than 19 cm Children: -under 5: MUAC <11cm or WFH (weight for height) <70% or median <- 3 Z score, 5-18 years of age: BMI -for-Age <-3 z-score Moderate Acute Malnutrition (MAM): Adult: BMI 16-18.49 kg/m2 Pregnant and lactating: MUAC 19-23 cm Children: -under 5: MUAC 11cm to <12cm or WFH (weight for height/length) <-3 Z or ≥ 70% to < 80% median or ≥ -3 Z to < -2 Z score 5-18 years of age: BMI-for-Age between -2 and -3 z-score Normal/No Under nutrition: Adult: BMI >18.5, or MUAC >23cm, Children: WHZ >-2 or WHM>80%, MUAC ≥ 12 cm, BMI for Age: 5-18 years >-2 Z- score.	
Disaggregation	Nutritional Status (MAM and SAM) by Age / Sex Pregnancy status: Non-pregnant and Pregnant,	
Sources	ART Register, PMTCT register, ART regimen tally, EMR-ART Software	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

14. Percentage of non-pregnant reproductive age women living with HIV on ART using a modern family planning method

Definition	Percentage of non-pregnant women living with HIV on ART using a modern family planning method	
Formula	Number of non-pregnant women on ART aged 15-49 Years reporting the use of any method of modern family planning	X 100
	Total number of non-pregnant women on ART aged 15-49 Years	
Interpretation	This indicator will be used to monitor HIV/FP integration at ART sites. This indicator is a subset of contraceptive prevalence rate but focuses specifically on HIV-infected women to measure progress in prong 2 (“prevent unwanted pregnancies among women living with HIV”) of the four prongs of PMTCT. Preventing unintended pregnancies in women living with HIV is a critical step towards reducing mother-to-child transmission and is a core component of the international standards for a comprehensive approach to PMTCT. Inherent within this indicator is the principle that integrated HIV/FP program activities must respect a client’s right to make informed decisions about her reproductive life. This means that clients should have access to an appropriate and comprehensive range of contraceptive options; and/or to safer conception/pregnancy counseling depending upon their fertility desire and intentions. All non-pregnant PLHIV women on ART reporting the use of modern contraceptive irrespective of where the service provided will be reported as using modern family planning method. All non-pregnant PLHIV women on ART reporting the use of modern contraceptive irrespective of where the service provided will be reported as using modern family planning method.	
Disaggregation	Age Method: Injectable, Implant, IUD, Other methods	
Sources	ART Follow-up form, ART Registers and EMR-ART	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

15. Proportion of patients enrolled in HIV care who were screened for TB

Definition	The proportion of patients on ART who were screened for TB during the reporting period	
Formula	Number of patients on ART whose TB status was assessed during the reporting period	X 100
	Total number of eligible patients on ART who had visit during the reporting period	
Interpretation	PLHIV enrolled in HIV care includes all those continuing in care and those newly enrolled during the reporting period. The numerator is taken from ART registers by counting the number of patients whose TB status was assessed during the reporting period. Any patient who started on ART during the reporting period should be counted in the ART register. The denominator is those currently on ART during the reporting period. It is taken from ART registers by counting the number of patients with a visit during the reporting period. This indicator is intended to provide information on the proportion of HIV positive patients in HIV care and treatment who are screened for TB at the last visit. It measures the burden of known TB co-morbidity among people in HIV care. It might be used in drug supply planning for ART drug substitution in people treated for TB. An increase in this indicator suggests that a higher proportion of HIV patients are being screened for TB and other increased efforts such as: developing a standard screening algorithm, training HIV staff, revising cards/ registers, etc. A decrease in this indicator suggests that a lower proportion of PLHIV are being screened for TB and alert the program to make the appropriate decision to improve the service.	
Disaggregation	Age/Sex: <15M/F, 15+ M/F Start of ART by Screen Result: ■ New on ART/Screen Positive ■ Previously on ART/Screen Positive	
Sources	ART Follow-Up form, ART Registers and EMR-ART	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

16. Proportion of ART patients who started on a standard course of TB Preventive Treatment (TPT)

Definition	Proportion of ART patients who started on a standard course of TB Preventive Treatment (TPT) in the reporting period	
Formula	Total number of adults and children previously enrolled in HIV care who start (are given at least one dose of) TPT during the reporting period.	X 100
	Total number of eligible adults and children previously enrolled in HIV care during the reporting period	
Interpretation	All clients eligible for TPT in the previous reporting period should be initiated on treatment.. It helps to determine TPT consumption /uptake /Backlog clearance of TPT among patients previously enrolled on ART and in filling gaps in determining national TPT coverage.	
Disaggregation	TPT Status by Age, Sex and Type of Prophylaxis: ■ TPT Started in the reporting period - INH<15 F/M, 15+ F/M 3HP<15 F/M, 15+ F/M ■ Enrolment status ■ Among New on ART ■ Among Currently on ART	
Sources	ART Register, Follow-Up form, individual record form, EMR-ART	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

17. Proportion of ART patients who completed Standard Course of TB Preventive Treatment (TPT)

Definition	Proportion of ART patients who started on a standard course of TB Preventive Treatment (TPT) in the previous reporting period and completed therapy		
Formula	Number of patients completed TPT in the reporting period		X 100
	Number of ART patients initiated on any course of TPT during the previous reporting period		
Interpretation	This indicator measures the performance of HIV programs in scaling up TPT, with the goal of preventing progression to active TB disease among PLHIV and decreasing ongoing TB transmission in this population. ART patients should be monitored for proper completion and the effectiveness of the TPT initiative in the HIV/AIDS program. As part of a cascade from treatment current to TB screening, this indicator will inform programs on the pace of scale-up, and the proportion will allow for monitoring of cohorts through completion of therapy. The numerator is generated by counting the number of PLHIV on ART from the previous reporting period who were documented as having received at least six months of IPT or having completed any other standard course of TPT (such as 3-HP). The denominator is generated by counting the total number of patients who were started on ART who were started on any course of TPT during the reporting period prior to the one being reported.		
Disaggregation	TPT Status by Age, Sex and Type of Prophylaxis: ■ TPT Started in the reporting period ■ INH<15 F/M, 15+ F/M ■ 3HP<15 F/M, 15+ F/M a. TPT Completed in the current reporting period ■ INH<15 F/M, 15+ F/M ■ 3HP<15 F/M, 15+ F/M		
Sources	ART Register, Follow-Up form, EMR-ART		
Reporting level	Heath center /Clinic/ Hospital		
Reporting frequency	Monthly		

18. Proportion of HIV positive women (15+) on ART screened for Cervical Cancer

Definition	Proportion of HIV- positive women on ART screened for Cervical Cancer during the reporting period	
Formula	Number of HIV-positive women on ART screened for Cervical cancer during the reporting Period	X 100
	Total Number of women (15+) on ART Eligible for cervical cancer screening during the reporting Period	
Interpretation	This indicator is vital for understanding and estimating the demand for screening services and forecasting and planning for the resources required to meet that demand and the resulting treatment needs. Disaggregation enhances sensitivity of this indicator in order to help identify the need for further outreach, as well as trigger further situational investigation at lower levels of the health system. Cervical Cancer screening and treatment should be analyzed together at the woreda or sub-regional level that includes sites where both screening and treatment would occur, in order to monitor the percentage of positive women who receive treatment while accounting for patient referrals between facilities. The term “Eligible” in the denominator refers to those HIV-positive women – only women being screened for the first time in their lifetime or re-screened after 2 years negative cervical cancer screening test result or post-treatment follow-up screening. For Visual inspection of cervix with acetic acid (VIA) the benchmark of 5%-25% screen-positivity for women (aged 30-60) screened for the first time should be used when monitoring performance.	
Disaggregation	Screening Visit Type and Result by Age: 1st time screened (Negative, Positive, Suspected Cancer) by age 45-49, 50+, Unknown Age ■ Re-screened after previous negative (Negative, Positive, Suspected Cancer) by age ■ Post-treatment follow-up (Negative, Positive, Suspected Cancer) by age	
Sources	Follow-Up form, Cervical Cancer Screening and Treatment Register, EMR-ART	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

19. Proportion of women (15+) on ART Cervical Cancer screen – positive who received Treatment

Definition	Proportion of cervical cancer screen-positive women who are HIV-positive and on ART eligible and who received cryotherapy, thermocoagulation or LEEP during the reporting period	
Formula	Number of HIV-positive women on ART cervical cancer screen-positive who received treatment during the reporting period	X 100
	Total 15+ women on ART who are cervical cancer screen positive during the reporting period	
Interpretation	This indicator is vital to capture the number of individual HIV-positive women on ART who required and received treatment for precancerous cervical lesions. Cervical cancer screening and treatment should be analyzed together at the woreda or sub-regional level that includes sites where both screening and treatment would occur, in order to monitor the percentage of positive women who receive treatment while accounting for patient referrals between facilities.	
	The globally accepted benchmark of at least 90% eligible for treatment of precancerous lesions receiving treatment should be used when monitoring performance.	
Disaggregation	Screening Visit Type and Treatment Type by Age:	
Sources	Follow-Up form, Cervical Ca Screening individual record form, EMR-ART	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

20. Proportion of PLHIV on ART Screened for Viral Hepatitis

Definition	Proportion of PLHIV on ART screened for Viral Hepatitis during the reporting period	
Formula	Number of PLHIV on ART screened for Viral Hepatitis during the reporting period	X 100
	Total Number of PLHIV on ART eligible for Viral Hepatitis screening during the reporting period	
Interpretation	PLHIVs are among high-risk groups for hepatitis infection. Higher prevalence of hepatitis infection is reported among HIV co infected individuals. This indicator measures the burden of Hepatitis infection among PLHIV. It also helps to generate data to enhance the compelling efforts in awareness creation and promotion of safer sex as part of the overall national HIV prevention and control efforts. It also helps to understand and estimating demand for hepatitis screening services, forecasting, and planning for the resources required to meet that demand and the resulting treatment needs. Screening and treatment should be analyzed at woreda /zonal or regional levels which includes sites where both screening and Treatment would occur. It also helps to assess Progress in implementing the health sector response to viral hepatitis. Eligibility and frequency of screening could be determined based on the national strategy for prevention and control of Viral hepatitis.	
Disaggregation	Status of Test result by Age/Sex: Pos _HBV: <15 M/F >=15 M/F Pos_HCV, <15 M/F >=15 M/F Neg: <15 M/F,>=15 M/F	
Sources	Hepatitis Testing and Treatment Register	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

21. Proportion of PLHIVs diagnosed with Hepatitis infection who received treatment

Definition	Proportion of PLHIV screened and diagnosed with Hepatitis B and C infection who received treatment in the reporting period	
Formula	Number of hepatitis screen - positive PLHIV on ART who received Treatment in the reporting period	X 100
	Total Number of hepatitis screen –positive PLHIV on ART in the reporting period	
Interpretation	PLHIVs are among high risk groups for hepatitis infection. Higher prevalence of hepatitis infection is reported among HIV-coinfected individuals. This indicator helps to monitor whether PLHIV requiring (eligible for) treatment for hepatitis B and C infection received treatment. It helps to assess Progress in implementing the health sector response to viral hepatitis	
Disaggregation	Status of Test result by Age/Sex: Pos _HBV: <15 M/F >=15 M/F Pos_HCV , <15 M/F >=15 M/F Neg: <15 M/F ,>=15 M/F	
Sources	Comprehensive Lab Register (Serology) and Treatment Register	
Reporting level	Heath center /Clinic/ Hospital	
Reporting frequency	Monthly	

22. Percentage of Key and Priority Population (KPP) members reached with HIV defined packages of services who were linked to HTS providing facilities

Definition	Percentage KPP members reached with HIV defined packages of services who were linked to HTS providing facilities	
Formula	Number of KPP members who were linked to HTS providing facilities	X 100
	Number of KPP members reached with HIV prevention defined packages of services	
Interpretation	This indicator can be used as a proxy for the quality of community-based HIV prevention services. It is critical to determine the proportion of KPP members reached with defined HIV prevention packages who are linked for HTS service as these are hard to reach population who can mostly be accessed through their peer service providers. The denominator should be the sum of: 1 KPP members reached with defined HIV prevention packages through peer service providers 2) KPP members reached with defined HIV prevention packages through facility-based modalities (e.g. KPP friendly clinics) 3) KPP members reached with defined HIV prevention packages through other community-based service provision modalities including DICs, CSOs The numerator should be the number of KPP reached with the HIV prevention packages and linked to HTS providing facilities. Please refer to key and Priority Populations HIV prevention Minimum Service Package (FHAPCO, January 2019). Limitation of this indicator: data incompleteness might be a problem because it is collected from multiple sources.	
Disaggregation	By Population groups: FSWs, PWID, Prisoners, High risk AGYW, Long distance drivers, Workers in hot spot areas, Widowed and Divorced men and women and partners of PLHIV	
Sources	Community HIV prevention SBCC Register	
Reporting level	Community based services providers/WoHO	
Reporting frequency	Monthly	

23. Percentage of KPPs tested HIV positive at community programs who were linked to care and treatment

Definition	Percentage of KPPs tested HIV positive at community programs who were linked to care and treatment	
Formula	Number of identified HIV positive KPP members linked to care and treatment during the reporting period	X 100
	Number of KPP members tested HIV positive at community programs during the reporting period	
Interpretation	This indicator can be used as a proxy for the quality of community-based HIV prevention services. It is critical to determine the proportion HIV positive KPP members identified positives at community level who are linked to care and treatment to be enrolled in ART.	
	Limitation of this indicator: data incompleteness might be a problem because it is collected from multiple sources.	
Disaggregation	By Population groups: FSWs, PWID, Prisoners, High risk AGYW, Long distance drivers, Workers in hot spot areas, Widowed and Divorced men and women and partners of PLHIV	
Sources	Community HIV Testing Register	
Reporting level	Community based services providers/ WoHO	
Reporting frequency	Monthly	

8.3. Data reporting, data flow and quality assurance

Routine HIV HMIS data are assembled and reported on a monthly, quarterly and annual basis. Facilities aggregate and review their data monthly and report to their respective facility and administrative offices quarterly. The administrative office aggregates the data it receives, adds its own administrative figures, and monitors its own performance based on these reports and self-generated data. It then forwards the DHIS-2 report to the

next level. Annual reports include additional data that are not collected quarterly. These reports follow the same line and principles of disaggregation as the quarterly reports. Data aggregation methodology is maintained throughout the reporting chain so that it is possible to disaggregate data by facility type and ownership even at the federal level. DHIS-2 Data flow from the facilities to the federal level is depicted below.

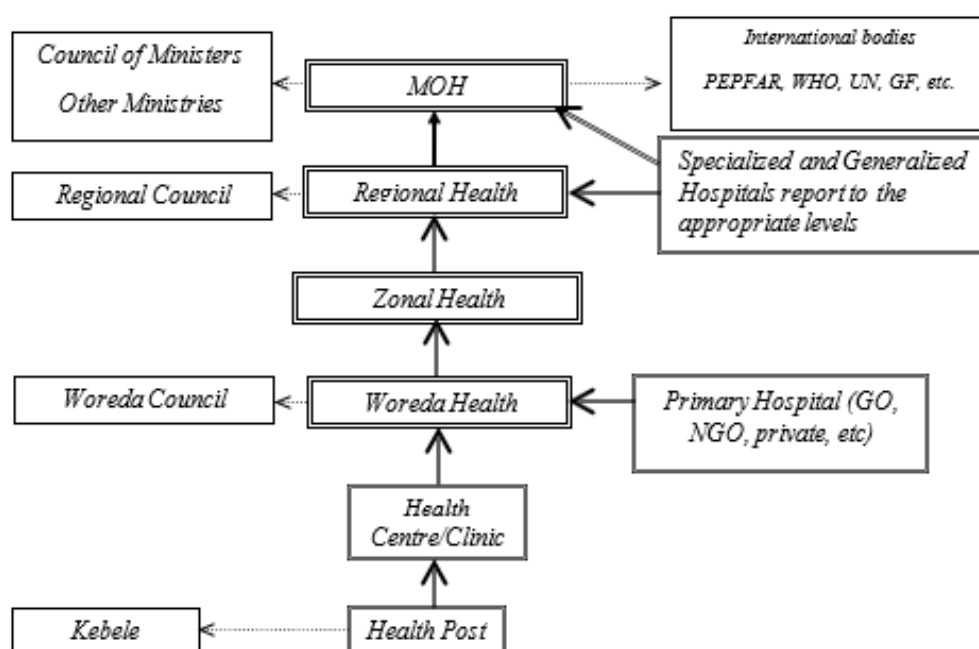


Figure 8.1 HMIS data flow and feedback loop

8.3.1. Data quality assurance

Data quality check is one of the components of the M&E system. Once data are collected, the data are checked for any inaccuracies and obvious errors at every level. The data quality assurance (DQA) is done at two levels: facility level and administrative level (district health offices). At facility level, such a mechanism is the Lot Quality Assurance Sampling (LQAS) methodology which is done on monthly basis. In this procedure randomly selected data

elements from the monthly reports are checked against the register or source of the report. The findings are then compared to a standard Data Accuracy Table. The same procedure is done at district health offices on quarterly basis before the data are sent to the next higher reporting unit. Hence, in HMIS all reports are quality checked at every level, from the healthcare institution to the federal level.

8.4. HIV/AIDS Program Monitoring

8.4.1. Supportive Supervision

Supervision aims at ensuring and improving quality, effectiveness and efficiency of services provided; it should also enhance competence and satisfaction of the staff at all levels. Supervision consists of observation, discussion, support and guidance. Since it is an essential tool in the management of staff and facilities, it should be done on a regular basis. The overall aim of supervision is the promotion of continuous improvement in the performance of the staff.

Supervisions at all levels are conducted in an integrated manner using standardized checklist clearly identified in the Integrated Supportive Supervision (ISS) guideline. It is done from every administrative level to the respective office and health facility. The ISS guideline shows the actual process of implementation, team composition and checklist.

Besides ISS, in-depth HIV program-specific supervisions using standardized HIV and TB/HIV supportive supervision tool can also be conducted whenever critical gaps that require intensive technical approach are identified during the ISS.

8.4.2. Program Performance Review

Review meetings organized at various levels create a very good opportunity to review the status of program implementation, achievements and challenges and come up

with workable solutions for the problems and challenges encountered. They are key elements for program management. Furthermore, review meetings are forums for exchange of ideas and experiences among the health professionals and program coordinators. In these meetings, program coordinators from the next lower levels will present activity reports of their respective area, including major achievements and challenges or constraints encountered during the period under review. Integrated review meeting is conducted on regular bases at every level. In this manner, activities taking place at all levels will then be brought forward to the respective review meeting sessions where HIV and TB/HIV program performance is reviewed as part of the overall review meeting.

8.5. Other Monitoring Considerations

Programs are increasingly moving beyond coverage indicators to focus on critical outcomes, such as viral load suppression and immune reconstitution, and on the broader impact of HIV treatment, including HIV-related mortality and HIV incidence. However, programs also need to measure potential unintended outcomes, such as HIV drug resistance and ARV-related toxicities. Periodic evaluations and implementation research are also central to reviewing programs.

Evaluation, including impact and program performance, and implementation research

Routine monitoring should be complemented by systematic evaluations and program reviews to assess the performance and effects of HIV programs, either comprehensively or with respect to specific priority areas. Social science and implementation research are important to assess perceptions and values of service recipients and communities along with barriers, facilitators and experiences in delivering and receiving services. Impact indicators, such as incidence, morbidity and mortality, are often difficult to measure.

Mathematical modeling is often undertaken to project various scenarios for program planning and evaluating impact. Ensuring the availability of robust data is especially important when estimating the prevention impact of ARV drugs at the population level, as multiple sources of information and uncertainty come into play. Specific data collection efforts and models for particular contexts may provide more accurate estimates.

8.6. Recording and reporting Tools in HIV/ AIDS Program

Different tools are used for routine recording and reporting of HIV /AIDS Program. List of recording reporting tools used includes : HTS Logbook ,PrEP register, PEP register , VCT register ,HIV Self-test Register , OPD /IPD registers , Positive Tracking Register , PMTCT register , ART register ,EMR-for ART , Follow-up forms , DSD Register, VL Register , Cervical Cancer Screening and Treatment Register, Hepatitis Screening and Treatment Register and Tally sheets(PITC/VCT/ICT/DSD/Clinical Care) .

CHAPTER NINE: ANNEXES

Annex 1: KP Minimum Service Package

KP Minimum Services Package		Frequency of Service Delivery	Resources (Equipment / Supplies)	Provider Support Tools	Formats and Registers	HIV Coordination	Remarks
I. Community based KP Outreach Services							
Providing peer education (home based) on HIV benefits of HIV Counseling and Testing including HIVST by Trained Peer Navigators.	Daily			HIVST Client information card	Peer Navigator Outreach activity monitoring Register (Amharic Version)		
Distributing of male condoms.							
Distributing IEC / BCC Materials							
Organizing and forming - up support groups (Positive and Negative FSWs)	Once		HIVST kits		Peer Navigator HIVST Register (Amharic Version)	Peer Navigators	
Conducting LTFU tracing (For HIV Positive FSWs who lost from treatment & declined to be enrolled)	Daily		Condoms	KP mapping tool			
Updating data on mapping of venues, FSWs hotspot sites	Quarterly		Penile model	Service directory	Peer Navigator one to one HE Template		
Conducting an update on FSWs distribution			Leaflet & brochures		Peer Navigator Monthly Reporting Template		
Following – up of the migration / movement of FSWs	Monthly						
Monthly outreach activity reporting to facility KP focal person							

KP Minimum Services Package	Frequency of Service Delivery	Resources (Equipment / Supplies)	Provider Support Tools	Formats and Registers	HIV Coordination	Remarks
II. Health Facility Based KP Friendly Clinical Services						
A. HIV Services Accessibility & Acceptability						
All FSWs have the right to choose receiving KP friendly service at public, private and NGO health facilities.					Clients	
All HIV prevention, testing, care & treatment services should be provided at KP-DSD friendly clinical service clinic.					HF CEO/ Medical Director	
KP-DSD services should provide by ART & KP trained nurse and/or health officer.						
All HIV prevention, testing, care & treatment services to be available during off working hours (Lunch break and Weekends).				Roles and Responsibilities of Health Care Providers	HF Medical director	
All FSWs & their sexual partners should be screened and Offered HIV testing & prevention services.	24 Hours /7 Days	Poster at the gate	KP Workflow Chart	Roles and Responsibilities of Peer Navigators	KP Service Providers & PNs	
All FSWs who doesn't wish to come HFs should offer directly assisted HIVST using PNs and Unassisted HIVST for those who doesn't want to be assisted.						
KP friendly community outreach service will be provided by trained Peer Navigators.						
KP friendly clinical service will be provided by trained health professional.					HF Medical director	

KP Minimum Services Package	Frequency of Service Delivery	Resources (Equipment / Supplies)	Provider Support Tools	Formats and Registers	HIV Coordination	Remarks	
B. HIV Testing & Prevention Services for HIV Negative FSWs							
Conduct Retesting of HIV Negative FSWs every six months	24 Hours / 7 Days	HIV rapid test & HIVST kits	Post Test Counseling Wall Chart	HIV Negative FSWs Follow Up Card	KP Service Providers		
Conduct Screening, diagnosing and managing STIs using the Syndromic Case Management approach		STI's Drugs	STI's Syndromic Management Wall Charts	PrEP screening tools for eligibility			
Conduct Screening, diagnosing and managing TB		Condom & FP methods		HIVST Register			
Promoting on and providing Family Planning Services including male condoms.		ARV drugs for PrEP & PEP	CEMs for PrEP	PrEP Register		KP Register for HIV Negative FSWs	
Promoting and providing Pre- Exposure Prophylaxis (PrEP) services							
Promoting and providing Post Exposure Prophylaxis (PEP) services							
Promotion and provision of Gender Based Violence (GBV) services							
C. HIV Prevention, Care & Treatment Service for HIV Positive FSWs							
Referring and linking HIV Positive FSWs for HIV Care and Treatment Services.	24 Hours / 7 Days	Chair	Post-Test Counseling Wall Chart	HIV Reporting Form			
Conducting initial and follow up adherence assessment		Guest chairs		IPV Risk Screening Tool			
Provide 3MMD ART for newly diagnosis FSWs and 6MMD ART for previously enrolled HIV Positive FSWs.		Tables	STIs Syndromic Management Wall Chart	Referral Forms			
Conduct Screening, diagnosing and managing STIs using the Syndromic Case Management approach.		Shelves Examination Coach		ART register	ART FU Card		
Conduct Screening, diagnosing and managing TB.		Speculum (medium & large size)	Linkage to Care Flow Chart	ASM Register			
Promoting and providing mental illness screening and management							

KP Minimum Services Package	Frequency of Service Delivery	Resources (Equipment / Supplies)	Provider Support Tools	Formats and Registers	HIV Coordination	Remarks
Promoting and providing cervical cancer screening and management	24 Hours / 7 Days	■ Examination gloves	■ LTFU Tracking SOP	■ VL Monitoring Register	ART & KP Service Providers	
Promoting and providing Family Planning Services including male condoms.		■ Penile model	■ ART Desktop Reference	■ HVL FU Register		
Promoting and referring HIV Positive pregnant FSWs to access PMTCT services		■ Condoms b(Males)	■ P & FB ICT Counseling Script	■ VL Request Form		
Promoting and providing Pre- Exposure Prophylaxis (PrEP) services for partners of HIV Sero-discordant couples.		■ HIV Testing kits including HIVST	■ P & FB ICT SOP	■ P & FB ICT Register		
Promoting and providing Post Exposure Prophylaxis (PEP) related services		■ Waste disposal items (for IP)		■ P & FB ICT Screening Tools		
Conducting intimate partner violence (IPV) risk screening and Post Violence care (PVC) services including referrals.				■ PrEP Register		
Complete HIV case based reporting form for assessing clients for eligibility for recency testing.				■ PrEP FU card		
Conducting HIV recency testing for all eligible FSWs.				■ HIVST Register		
Promoting and providing P & FB ICT services				■ Mental Health Illness Screening Tools		
Conducting HTS for all HIV Positive Index-FSWs elicited HIV unknown status contacts.				■ Referral Register		
Offering Unassisted HIVST for all HIV Positive Index-FSWs elicited HIV unknown status contacts who doesn't wish to come HF.				■ Retesting Register		
Conducting viral load test every 6, 12 and 12 months.						
Promoting and providing EAC sessions for HVL FSWs.						
Referral service for support groups for HIV Positive FSWs.						

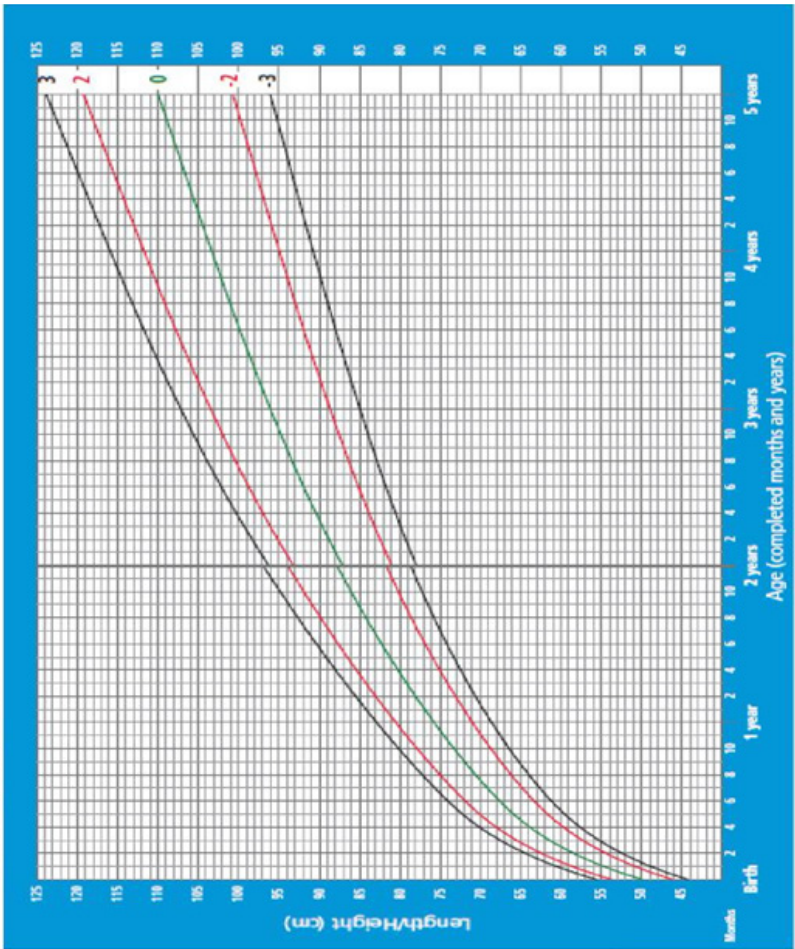
Annex 2: Immunization schedule for HIV exposed infants

S/NO	Vaccines	Target diseases to be prevented	Age	Route/Site of administration
1	Hep-B Vaccine Birth dose	Hep B Virus Infection	At birth or within 24 hours of birth. If child is born out of HFs or at home, track baby and vaccinate until the age of 14 days.	IM, left anterolateral Thigh
2	BCG	Severe forms of Tuberculosis	At Birth or as soon as possible after birth	
3	PCV	Meningitis and pneumonia associated with Streptococcus pneumonia bacteria	Weeks 6, 10 & 14	Intramuscular (IM), Right anterolateral thigh
4	OPV	Poliomyelitis	Birth (OPV0), weeks 6, 10 & 14	Oral drops
5	IPV	Poliomyelitis	Week 14	IM, right anterolateral thigh 2.5 cm apart from the injection site of PCV
6	DPT-Hib-Hep-B	Diphtheria, Pertussis, Meningitis, and pneumonia associated with Hemophilus influenzae bacteria and Liver disease due to Hepatitis B virus.	Weeks 6, 10 & 14	IM, Left anterolateral thigh
7	Measles containing vaccine	MCV	9 and 15 months	Subcutaneous (SC), Left deltoid
8	Rota virus vaccine	Rota virus associated gastro-enteritis.	Weeks 6 & 10	Oral only

Annex 3: Growth curves

Length/height-for-age BOYS

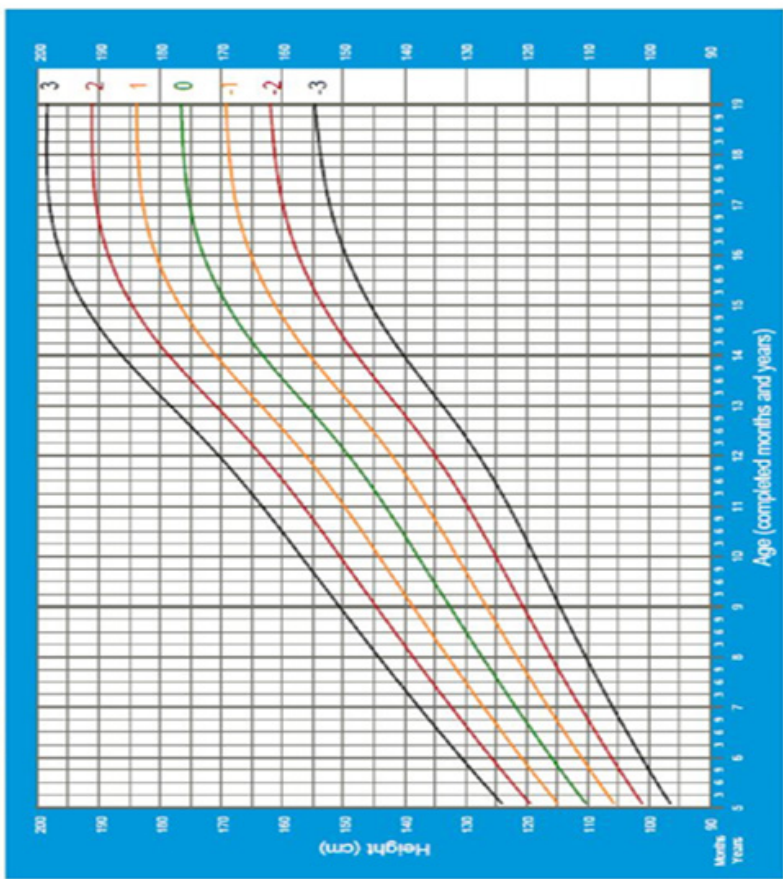
Birth to 5 years (z-scores)



WHO Child Growth Standards

Height-for-age BOYS

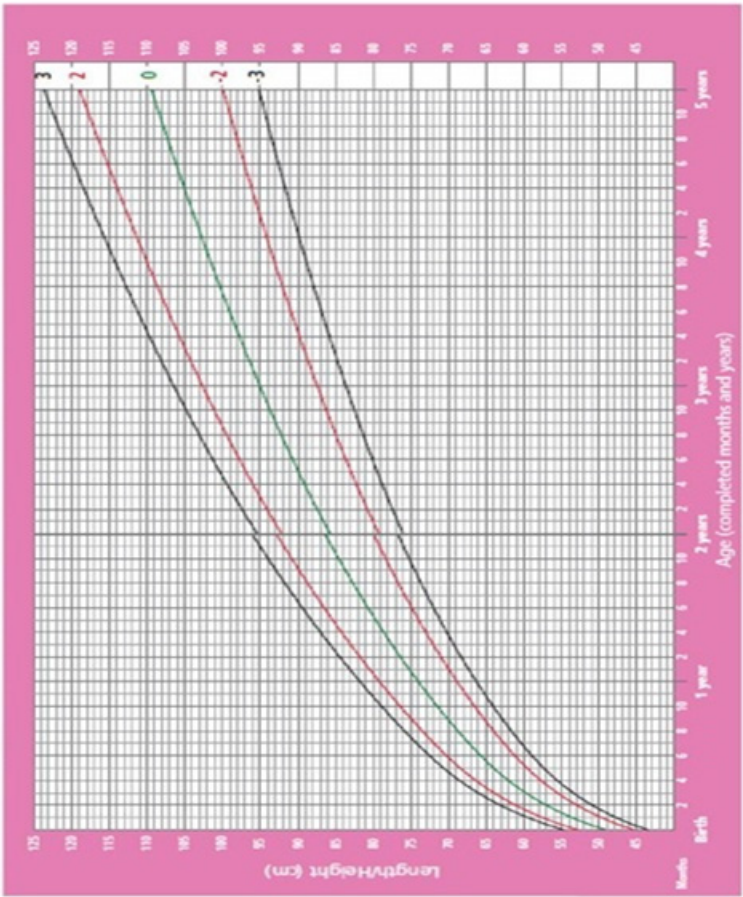
5 to 19 years (z-scores)



2007 WHO Reference

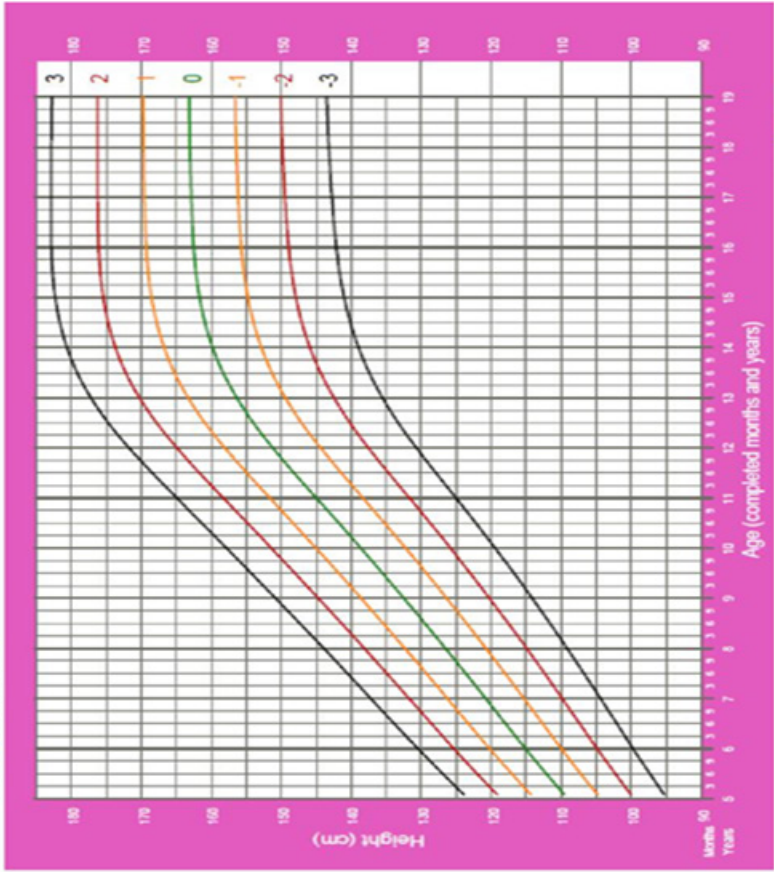
Length/height-for-age GIRLS

Birth to 5 years (z-scores)



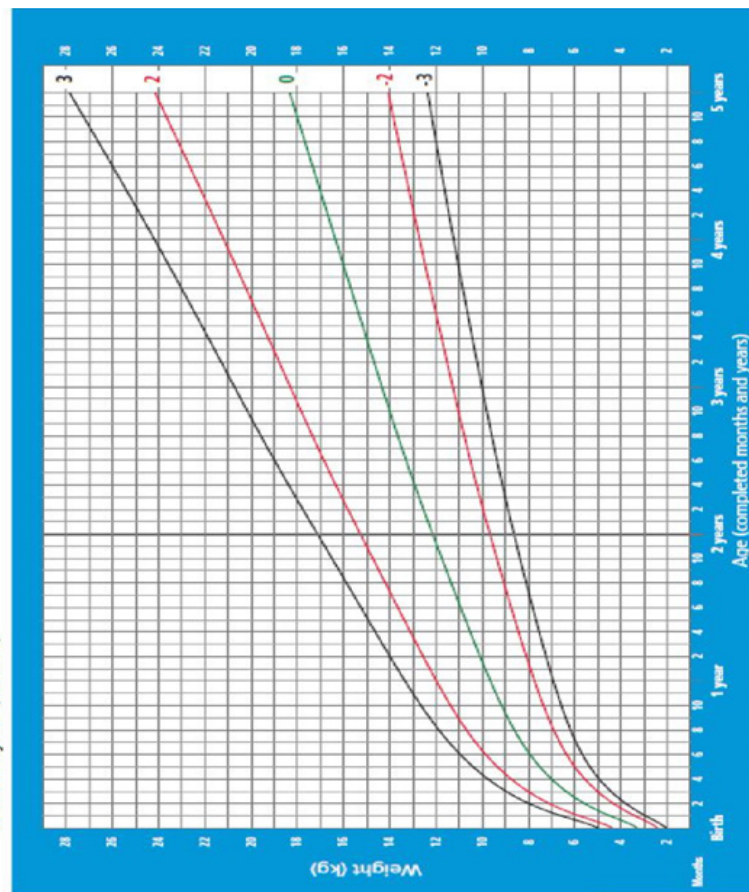
Height-for-age GIRLS

5 to 19 years (z-scores)



Weight-for-age BOYS

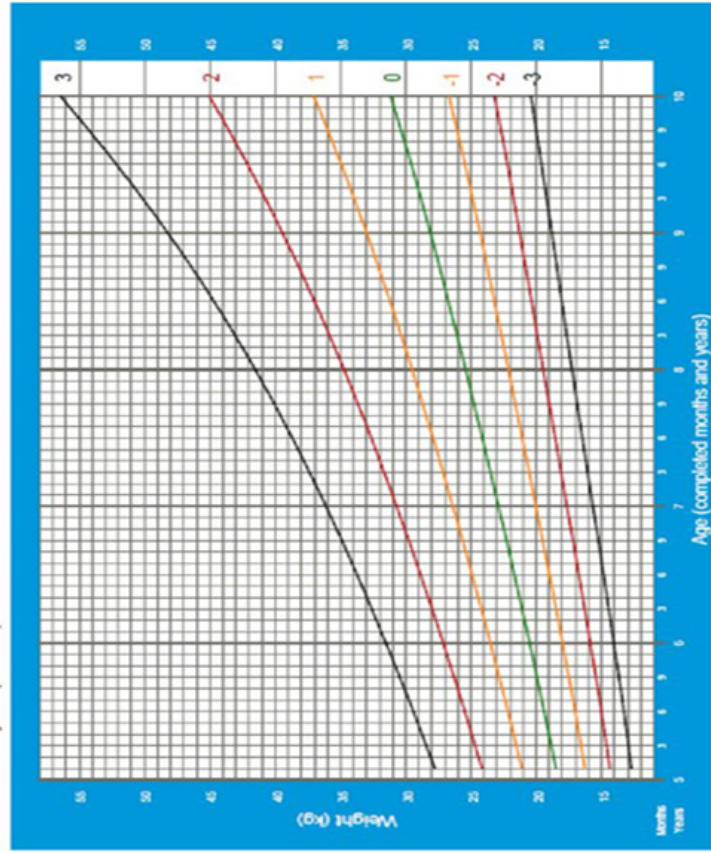
Birth to 5 years (z-scores)



WHO Child Growth Standards

Weight-for-age BOYS

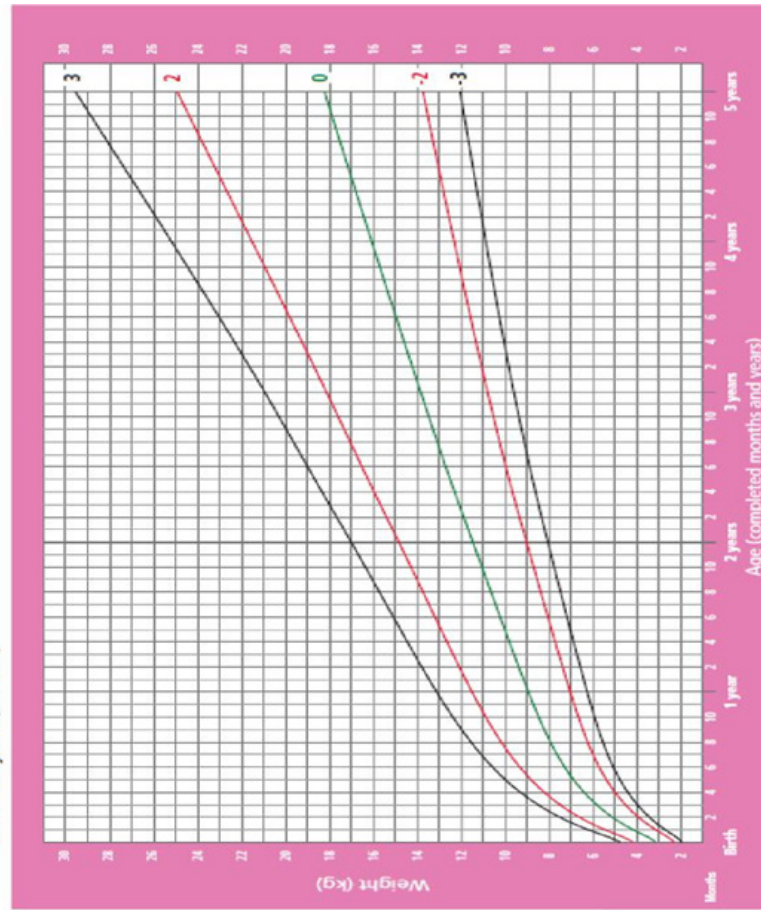
5 to 10 years (z-scores)



2007 WHO Reference

Weight-for-age GIRLS

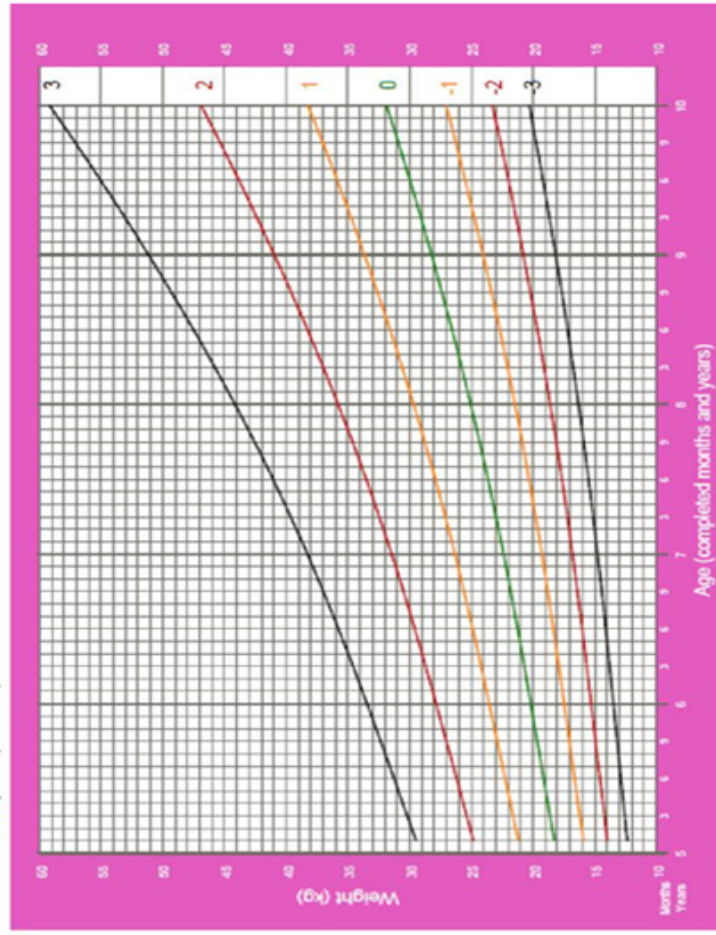
Birth to 5 years (z-scores)



WHO Child Growth Standards

Weight-for-age GIRLS

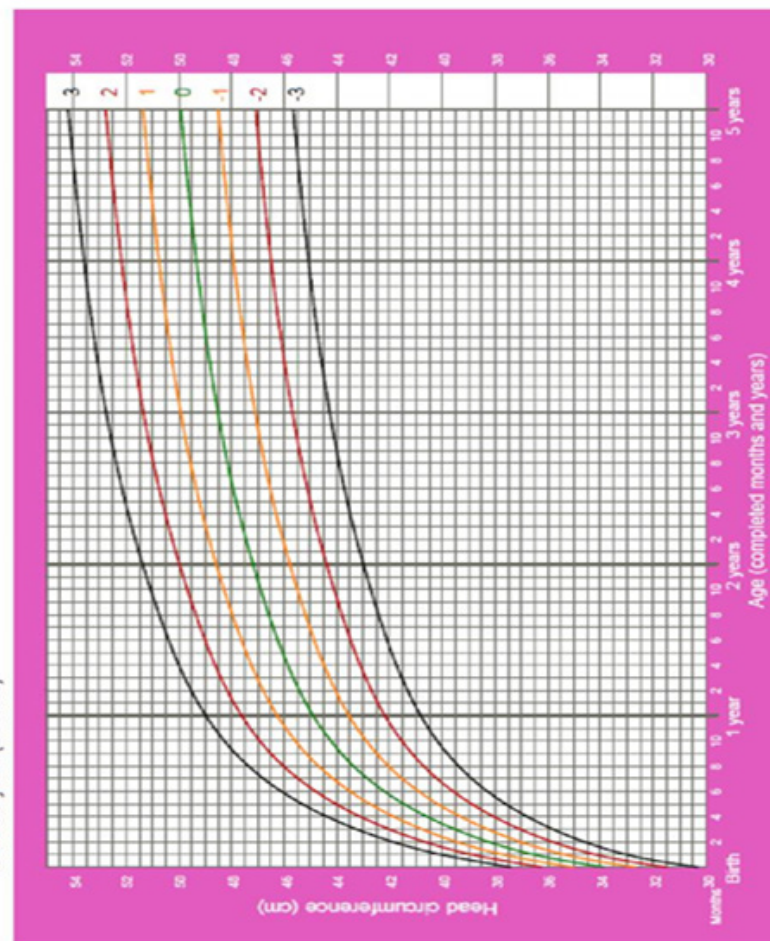
5 to 10 years (z-scores)



2007 WHO Reference

Head circumference-for-age GIRLS

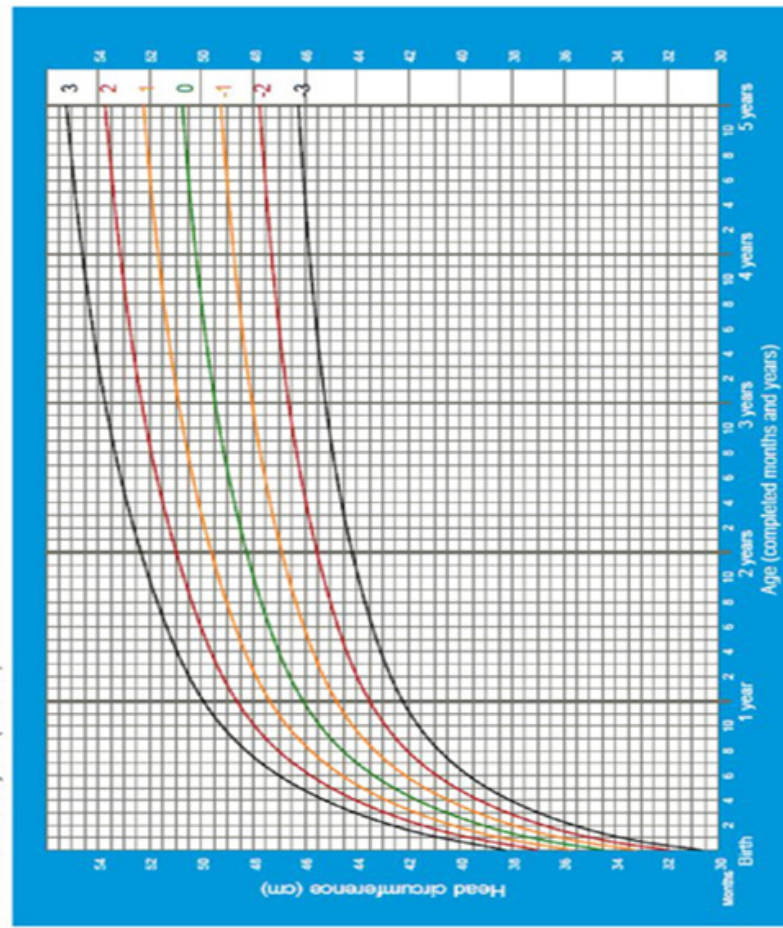
Birth to 5 years (z-scores)



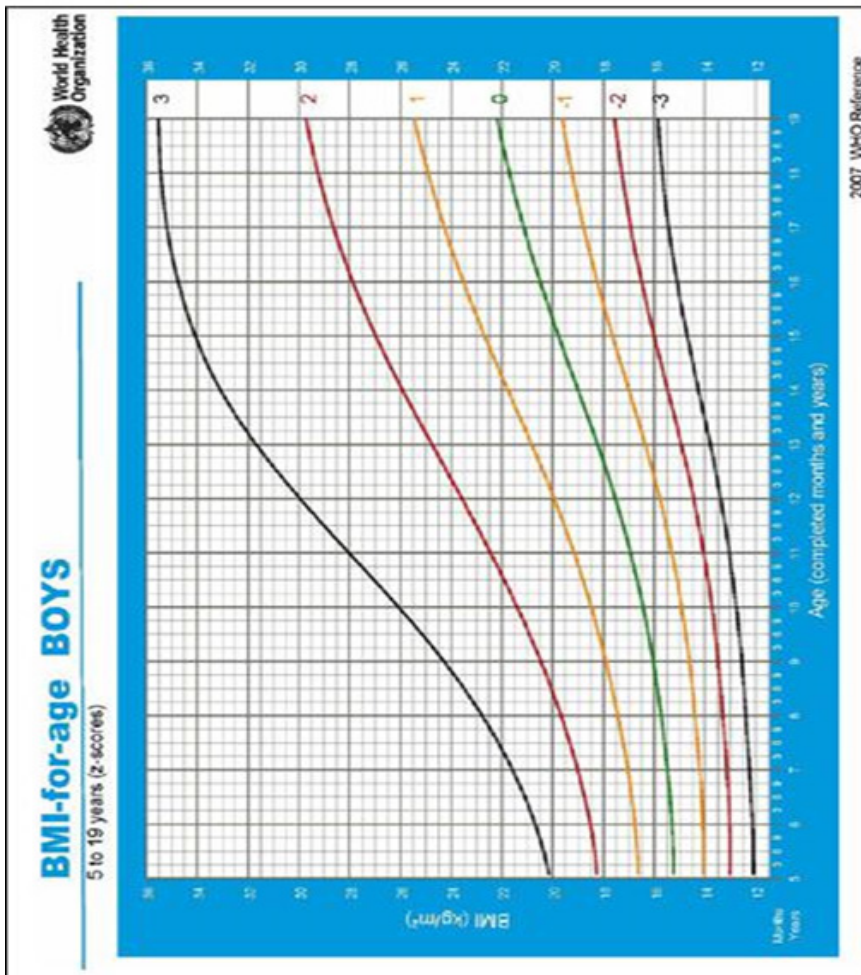
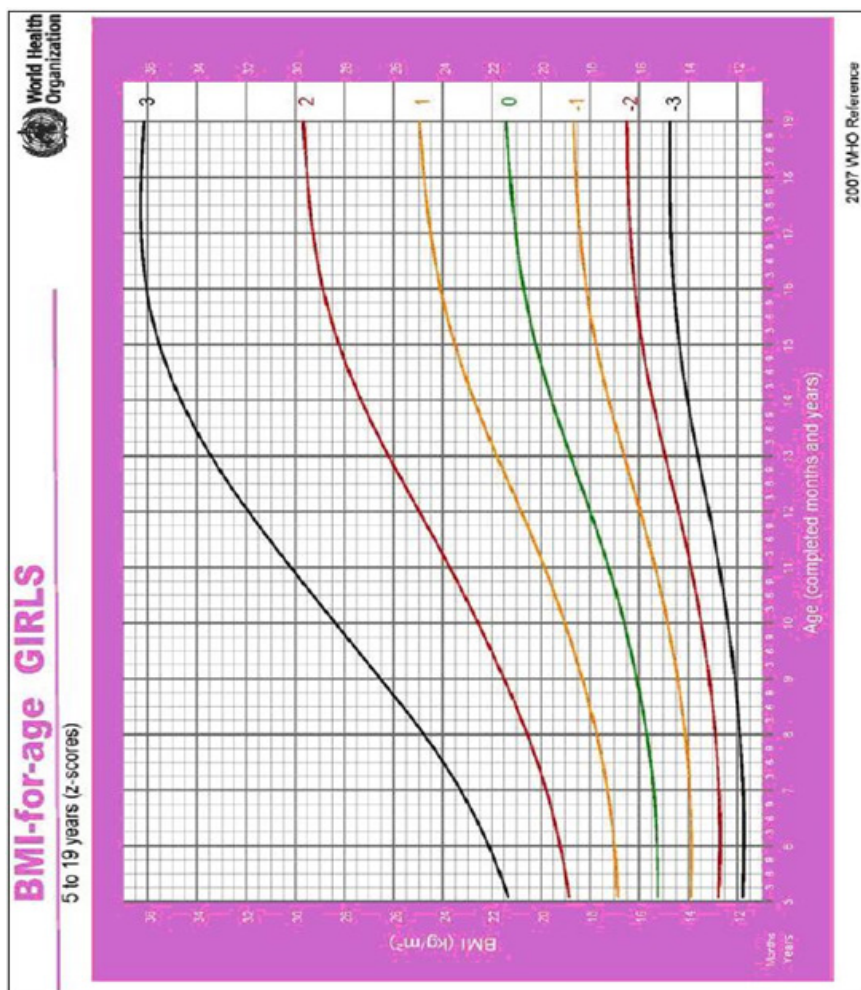
WHO Child Growth Standards

Head circumference-for-age BOYS

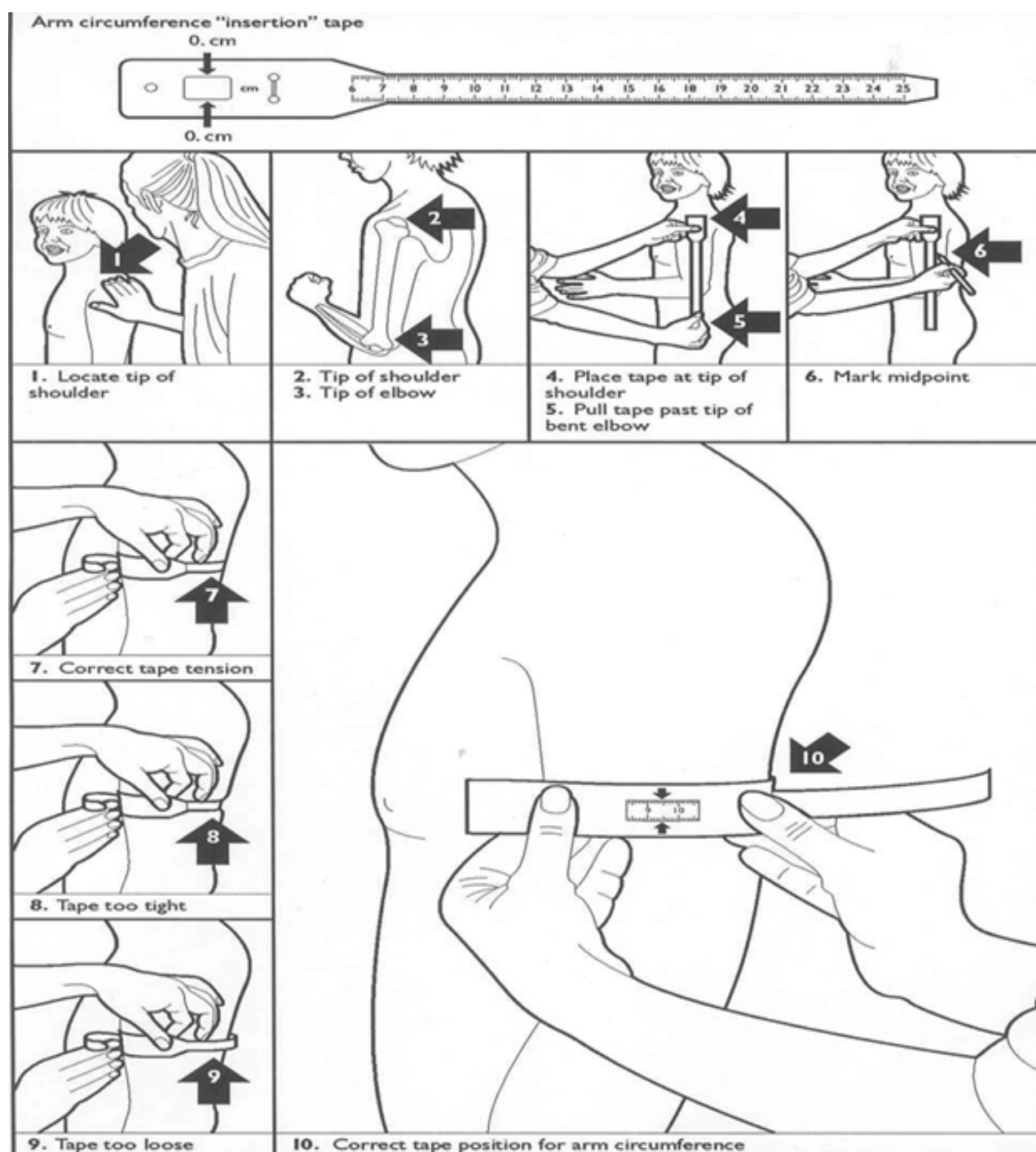
Birth to 5 years (z-scores)



WHO Child Growth Standards



Annex 4. Instruction for MUAC measurement



Source: How to Weigh and Measure Children: Assessing the Nutritional Status of Young Children, United Nations, 1986.

Annex 5a. HIV Risk assessment tool for children <15 years of age

Name _____ Age _____ Date: ____/____/____

For a child <18 months of age			
A. Mother's HIV status Known:	If Negative..... No action If Positive.....Test child unless mother is on PMTCT care and child/infant is on follow up.		
B. Mothers HIV status Unknown	Test mother and decide according to the above options ("A")		
C. If Child is orphan	Test the child		
For a Child > 18 months of age			
Step 1. Assess for Child's HIV Status	1. Unknown 2. Known Positive & on ART 3. Negative		
Step 2. If Child's HIV status is unknown, assess for the criteria below: If "Yes" to one of them, Child is eligible for HIV testing. Test the child.			
1. Are child's biological parents living with HIV?	Y ☐	N ☐	
2. Is the child Orphan or vulnerable with unknown parental HIV status?	Y ☐	N ☐	
3. Is the child diagnosed to have ANY ONE of the following? (If yes tick) (☐) ■ Confirmed or suspected TB (), ■ Recurrent lower respiratory infections (pneumonia) of > 2 in the past 6 months (), ■ Prolonged fever (> 2 weeks) (), ■ Chronic ear discharge (), ■ Chronic or recurrent diarrhea (), ■ Recurrent or extensive skin lesion (), ■ Severe malnutrition or failure to thrive (), ■ Developmental delay / regression (),			

4. Unexplained poor=health in the last three months.	Y ☐ N ☐
5. History of repeated admissions to hospital for medical illnesses.	Y ☐ N ☐
If the child is older than 10 years of age (10 to 14 years)	
6. Is the child having ANY ONE of the following? (If yes tick) (☐) ■ Is the child sexually active or sexually exploited? () ■ Is the child living on street? () ■ Is the child leading a family? () ■ Does the child use substance or alcohol? () ■ Is the child out of school or missed schools repeatedly? () ■ Does the child have history of STI? ()	
If Yes or (☐) to any one the above, child is Eligible for HIV testing. Test the child.	Eligible ☐ Non Eligible ☐
HIV test result	Positive ☐ Negative ☐

Annex 5b. HIV Risk assessment tool for adults and adolescents > 15 years of age

Direction: Evaluate clients using the tool and offer HTS services if a client is eligible. Note that a known HIV positive client should not be tested for HIV and a client who received HIV testing less than 3 months ago may not need HIV testing. Please refer to the current consolidated HIV Prevention, Care and Treatment guideline for details.

MRN: Age: Sex: Date:/...../.....
Exam room:

1. OI's & Vulnerability Status

- 1.1. Does the client have clinical signs/symptoms of HIV/AIDS? (Refer clinical S/S Job aid):
Yes ☐ No ☐ if yes offer HTS. If no, go to number 1.2.
- 1.2. Is the client a partner or biological child of an index case?
Yes ☐ No ☐ if yes offer HTS. If no, go to number two.

2. Marital Status

- 2.1. Is the client divorced or Widowed or widower?
Yes ☐ No ☐ if yes offer HTS.If no, go to number three.

3. KP/PP; Others

- 3.1. Does the client have Occupational HIV risk or other risks stated below?
Yes ☐ No ☐ if yes offer HTS.
{1. Female sex worker, 2. Prisoners, 3. PWID, 4. Long distance truck driver, 5. High risk workers in hotspot areas, 6. Daily laborer and mobile workers, 7. Others: GBV survivors, Homeless/street dweller; having multiple sexual partners; Substance and Alcohol abuse.}

KEY: if “yes” (✓) to one of the criteria, Patient/client will be eligible for HIV testing

Conclusion: **Eligible** **Non-Eligible**

Annex 6a. Clinical Conditions that make eligible for HIV Testing Service in adults and adolescents

■ Persistent Generalized Lymph adenopathy ■ Unexplained weight loss (> 10% of body weight) ■ Unexplained persistent hepatosplenomegaly ■ Unexplained Chronic diarrhea > 1 month ■ Recurrent Respiratory tract Infections (sinusitis, tonsillitis, otitis media , pharyngitis) ■ Herpes Zoster ■ Angular Cheilitis ■ Recurrent Oral ulcerations ■ Popular Pruritic Eruptions ■ Fungal nail Infections ■ Seborrheic Dermatitis ■ Unexplained Persistent Fever > 1 month ■ Persistent oral candidiasis ■ Oral Hairy Leukoplakia Pulmonary TB. ■ Recurrent /Severe Bacterial Infections (Pneumonia , Epyema, ■ Pyomyosities, Bone or joint Infections , meningitis , Bacteremia)	■ Unexplained pancytopenia (anemia < 8gm/dl, neutropenia< ■ 0.5*109 , or thrombocytopenia < 50*109 /L) ■ PCP (Pneumocystis (Jirowicini) Pneumonia) ■ Esophageal Candidiasis (candidiasis of Trachea , Bronchi or Lungs) ■ Extra Pulmonary Tuberculosis ■ Kaposi Sarcoma ■ CMV Infections (Retinitis or infection of Other Organs) ■ CNS Toxoplasmosis ■ Extra pulmonary Cryptococcosis (eg . Meningitis) ■ Disseminated Non tuberculosis mycobacterial Infections ■ PML (Progressive Multi focal Leukoencephalopathy) ■ Chronic Isosporiasis ■ Chronic Crypto Sporadiosis) ■ CNS Lymphoma (B_ Cell NHL) ■ Recurrent Septicemia (Including non-typhoid Salmonella) ■ Invasive Cervical Carcinoma ■ Visceral Leishmaniosis ■ Atypical Disseminated Leishmaniosis
---	--

Annex 6b. Clinical Conditions that make eligible for HIV Testing Service in children

■ Persistent Generalized Lymphadenopathy ■ Unexplained persistent hepatosplenomegaly ■ Recurrent or chronic upper respiratory tract infections (Otitis media, Otitis media, sinusitis, tonsillitis) ■ Herpes Zoster ■ Lineal gingival erythema ■ Recurrent oral ulceration ■ Popular Pruritic eruptions ■ Fungal nail Infections ■ Extensive wart virus infections ■ Extensive molluscum contagiosum ■ Unexplained malnutrition(moderate or severe) not responding to therapy ■ Unexplained persistent diarrhea (> 14 days) ■ Unexplained persistent fever (>1month) ■ Persistent oral candidiasis (after 6weeks of life) ■ Oral hairy leukoplakia ■ Lymph node TB Pulmonary TB	■ PCP (Pneumocystis jiroveci Pneumonia) ■ Recurrent severe bacterial infections (Empyema, Pyomyositis, Bone or joint Infections , meningitis , Bacteremia excluding pneumonia) ■ Esophageal Candidiasis (candidiasis of Trachea , Bronchi or Lungs) ■ Extra Pulmonary Tuberculosis ■ Kaposi Sarcoma ■ CMV Infections (Retinitis or infection of Other Organs with onset at age more than 1 month) ■ CNS Toxoplasmosis (after neonatal period) ■ Extra pulmonary Cryptococcosis (eg. Meningitis) ■ Disseminated Non tuberculosis mycobacterial Infections ■ PML (Progressive Multi focal Leukoencephalopathy) ■ Chronic Isosporiasis ■ Chronic Crypto Sporidiosis) ■ CNS Lymphoma (B_ Cell NHL) ■ Symptomatic lymphoid Interstitial pneumonitis ■ Acute necrotizing ulcerative gingivitis ■ Unexplained anemia , neutropenia or chronic thrombocytopenia ■ Severe recurrent pneumonia
---	---

Annex 6b. Clinical Conditions that make eligible for HIV Testing Service in children

- Excessive dryness of the skin (Xerosis cutis)
- Scabies
- STI (Sexually Transmitted Infections)
- Herpes Zoster
- Popular Pruritic Eruptions
- Fungal nail Infections (Dermatophytosis, thrush, deep fungal infections)
- Seborrheic Dermatitis
- Kaposi Sarcoma
- Bacterial Skin Infections (Cellulitis, Impetigo, Carbuncle)
- Viral Infections (Recurrent Herpes Simplex infections on the mouth, Genital Herpes Simplex Infection)
- Genital warts

Annex 7: Dosage of antiretroviral drugs for adults and adolescents

Nucleoside reverse-transcriptase inhibitors (NRTIs)	
Abacavir (ABC)	300 mg twice daily or 600 mg once daily
Emtricitabine (FTC)	200 mg once daily
Lamivudine (3TC)	150 mg twice daily or 300 mg once daily
Zidovudine (AZT)	250–300 mg twice daily
Nucleotide reverse-transcriptase inhibitors (NtRTIs)	
Tenofovir (TDF)	300 mg once daily
Non-nucleoside reverse-transcriptase inhibitors (NNRTIs)	
Efavirenz (EFV)	400–600 mg once daily
Etravirine (ETV)	200 mg twice daily
Proteases inhibitors (PIs)	
Atazanavir + ritonavir (ATV/r)	300 mg + 100 mg once daily
Darunavir + ritonavir (DRV/r)	800 mg + 100 mg once daily ^a or 600 mg + 100 mg twice daily ^b
Lopinavir/ritonavir (LPV/r)	400 mg/100 mg twice daily
	Considerations for individuals receiving TB therapy
	In the presence of rifabutin, no dose adjustment required. In the presence of rifampicin, adjusted dose of LPV/r: LPV 800 mg + RTV 200 mg twice daily or LPV 400 mg + RTV 400 mg twice daily).or, SQV/r (SQV 400 mg + RTV 400 mg twice daily), with close monitoring.
Integrase strand transfer inhibitors (INSTIs)	
Dolutegravir (DTG)	50 mg once daily
Raltegravir (RAL)	400 mg twice daily

a For individuals with no previous use of protease inhibitors.

b For individuals with previous use of protease inhibitors.

Annex 8: Dosage of antiretroviral drugs in children

Table 8a: Solid and oral liquid formulations for **once-daily dosing** for infants and children > 4 weeks of age

Drug	Formulations (tablet, capsule or oral liquid) and strength (mg/tab. or mg/ml for liquids)	Dose as number of tablets by weight band					Adult tablets & strength in mg	Dose as number of tablets or ml by weight band	
		3– 5.9Kg	6-9.9kg	10-13.9kg	14-19.9kg	20-24.9kg		25-29.9kg	30-34.9kg
DTG	50mg tablet	-	-	-	-	1	50mg	1	1 ^a
DTG**	10mg tablet (scored, dispersible)	0.5	1.5	2	2.5	Use the film coated DTG 50mg tablet, 1 tablet daily.			
DRV ^b	600mg tablet	-	-	-	1	1	600mg	1	1
	150mg tablet	-	-	-	4	4			
RTV ^c	25mg tablet				4	4	100mg	1	1
	50mg tablet				2	2			
ABC/3TC	120mg/60mg (scored, dispersible tablet)	1	1.5	2	2.5	3	600mg/300mg	1	1
ABC/3TC	60mg/30mg (scored, dispersible tablet)	2	3	4	5	6	600mg/300mg	1	1
EFV ^d	200mg tablet	-	-	1	1.5	1.5	200 tab	2	2
	50mg capsule	-	-	4	6	6	600mg		

Table 8b: Solid and oral liquid formulations **for twice-daily dosing** for infants and children > 4 weeks of age

Drug	Formulations (tablet, capsule or oral liquid) and strength (mg/tab. or mg/ml for liquids)	Dose as number of tablets/capsules or ml by weight band, morning (AM) and evening (PM)								Adult tablets and their strength in mg		Dose by weight band	
		3 – 5.9Kg		6-9.9 kg		10-13.9kg		14-19.9kg				25 – 34.9 KG	
		AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM
Solid formulations													
ABC/3TC	120mg/60mg (scored, dispersible tab)	0.5	0.5	1	1	1	1	1	1.5	1.5	600mg/300mg tab	0.5	0.5
AZT/3TC	60mg/30mg (scored, dispersible tab)	1	1	1.5	2	2	2	2.5	3	3	300mg/150mg tab	1	1
LPV/r ^{b,c}	100mg/25mg tablet	-	-	-	2	1	2	2	2	2	-	3	3
	40mg/10mg oral pellets per capsule	2	2	3	4	4	4	5	6	6	200mg/50mg	2	1
DRV	75mg tablet	-	-	-	-	-	-	5	5	5	-	-	-
RTV	25mg tablet	-	-	-	-	-	-	2	2	2	100mg tab	2	2
RAL	100 mg, chewable tablet	-	-	-	-	-	-	1	1	1.5	400mg tab	1	1
	25mg, chewable tablet	1	1	2	3	3	4	4	6	6		1	1
Liquid formulations													
AZT	10mg/ml	6ml	6ml	9ml	12ml	12ml	12ml	-	-	-		-	-
NVP ^d	10mg/ml	5ml	5ml	8ml	10ml	10ml	10ml	-	-	-		-	-
LPV/r ^e	80mg/20mg/ml	1ml	1ml	1.5ml	2ml	2ml	2ml	2.5ml	3ml	3ml		-	-
RTV	80mg/ml	0.5ml	0.5ml	0.5ml	0.5ml	0.5ml	0.5ml	-	-	-		-	-

^c RTV should only be used as a boosting agent in combination with DRV.^d Two EFV 50mg capsules is administered in combination with EFV 200mg tablet for children weighing 14-24.9KG.

Annex 9: Pediatric ARV drug formulations, side effects and special considerations in children

Drug / Formulation	Dosing recommendation and instructions	Side Effects
Nucleoside reverse transcriptase		
Lamivudine (3TC) : Formulations: Fixed-Dose Combination Tablets With Abacavir : 3TC60mg+120mgABC, scored and dispersible tablet With Zidovudine :	Instruction for dosing/administration: - Can be given with food. - For those who can't swallow whole tablet, dispersible tablets can be dissolved with 2 teaspoons (10ml) water in a small and clean container and taken immediately. - Scored tablets can be broken to administer lower doses for younger children as recommended in the national ART guidelines. Note: - Adult formulations are to be used for children older than 10 years with body weight > 35kg. - Please see dosing Chart for dose recommendation. - Storage: - Can be stored at room temperature	Side effects of 3TC: Common: head ache, nausea, abdominal pain. Less common: pancreatitis, neutropenia, increased LFTs. Hepatitis may exacerbate after discontinuation of lamivudine in the setting of chronic Hepatitis B virus infection. For those patients with HBV and HIV co-infection please consult "the national guideline on management of HBV infection-2021".
Abacavir (ABC) : Formulations: Fixed-Dose Combination Tablets With Lamivudine : ABC120mg+3TC60mg, scored and dispersible tablet	Instruction for dosing/administration: - Can be given with food. - For those who can't swallow whole tablet, dispersible tablets can be dissolved with 2 teaspoons (10ml) water in a small and clean container and taken immediately. - Scored tablets can be broken to administer lower doses for younger children as recommended in the national ART guidelines. Note: Adult formulations are to be used for children older than 10 years with body weight > 35kg. - Please see dosing Chart for dose recommendation. Storage: - Can be stored at room temperature	Side effects of ABC: Common: head ache, GI upset and rash. Less common: lactic acidosis, hepatomegaly with steatosis. Life threatening: potentially fatal hypersensitivity reaction (fatigue, rash, nausea and vomiting, sore throat, joint and muscle pain, cough, and dyspnea). Occurrence of hypersensitivity reactions requires immediate and permanent discontinuation of ABC. Advise patients/care givers what to do if hypersensitivity reactions occur at home. DO NOT re-challenge after hypersensitivity reaction.

Drug / Formulation	Dosing recommendation and instructions	Side Effects
Zidovudine (AZT or ZDV) Formulations: Oral solution, 50mg/5ml, 100 ml	Instruction for dosing/administration: ■ Can be given with food. ■ The syrup is for infant prophylaxis. ■ For those who can't swallow whole tablet, dispersible tablets can be dissolved with 2 teaspoons (10ml) water in a small and clean container and taken immediately. ■ Scored tablets can be broken to administer lower doses for younger children as recommended in the national ART guidelines. Note: ■ Adult formulations are to be used for children older than 10 years with body weight > 35kg. ■ Please see dosing Chart for dose recommendation. Storage: ■ Can be stored at room temperature ■ Syrup is light sensitive, store in a brown glass jar and protect from sun light.	Side effects of AZT: Common: neutropenia, anemia, granulocytopenia, macrocytosis, and headache. Less common: myositis, myopathy, mitochondrial disease.
Nucleotide reverse transcriptase inhibitor Tenofovir (TDF)	TDF is only approved for use for children 2 years and older. Target dose: 8 mg/kg or 200 mg/m² (maximum 300 mg). In case of TDF toxicity, substitute with AZT or ABC. Do not initiate TDF at eGFR <50 mL/min.	Chronic kidney disease Acute kidney injury and Fanconi syndrome Decreases in bone mineral density Lactic acidosis or severe hepatomegaly with steatosis

Drug / Formulation	Dosing recommendation and instructions	Side Effects
Non-nucleoside reverse transcriptase:		
Efavirenz (EFV) Formulations: Efavirenz 200 mg, scored tablet. Efavirenz 50 mg, tablet or capsule	Instructions for dosing/administration: ■ Used only for children > 3 yrs. ■ Can be given with food but avoid administration with a high-fat meal because of potential for increased absorption that leads to CNS toxicity. ■ Bedtime dosing is recommended, particularly during the first 2 to 4 weeks of therapy, to improve tolerability of central nervous system side effects. ■ Scored tablets can be broken to administer lower doses for younger children as recommended in the national ART guidelines. ■ Efavirenz can be swallowed as a whole capsule or tablet. For children who can't swallow the whole capsule/tablet, the capsule content can be administered by sprinkling on food to avoid/mask peppery taste of EFV. For sprinkling: ■ Hold capsule horizontally over a small container and carefully twist to open to avoid spillage. ■ Gently mix capsule contents with 1–2 teaspoons of an age-appropriate soft food (e.g. yogurt or banana or infant formula milk) at room temperature. ■ Administer the mixture that contains the sprinkle using a 10-mL syringe. ■ After administration, an additional 2 teaspoons of food or infant formula must be added to the container, stirred, and dispensed to the patient. Note: ■ Adult formulations are to be used for children older than 10 years with body weight > 35kg. ■ Please see dosing Chart for dose recommendation. Storage: ■ Can be stored at room temperature.	Side effects of EFV: Common: skin rash, dizziness, somnolence, insomnia, abnormal dreams, confusion, hallucinations, impaired concentration, psychosis, seizures, and suicidality. Less common: increased LFTs.

Drug / Formulation	Dosing recommendation and instructions	Side Effects
Nevirapine (NVP) Formulations: Nevirapine oral syrup, 50mg/5ml (10mg/ml)	Instructions for administration: Can be given with food. The syrup is for infant prophylaxis Storage: Store at room temperature.	Side effects of NVP: Common: skin rash, head ache, nausea, diarrhea. Patients should be warned of rash. Do not escalate dose if rash occurs. Less common: increased LFTs. Life threatening: Steven Johnsons syndrome, TEN, fatal hepatitis. For SJS and TEN discontinue drug and do not re-challenge.
Protease Inhibitors		
Lopinavir (LPV) Formulations: Combined with Ritonavir: Oral syrup, (LPV 80mg+ RTVLPV20mg)/ml Oral pellets, (LPV40mg+Ritonavir10mg) / capsule Tablet, (LPV100mg+Ritonavir25mg)	Instructions for administration: - Do not administer to neonates before a post-menstrual age of 42 weeks and a postnatal age of at least 14 days. - Should be taken with food. - For the capsules: open the capsules, add the pellets from the capsule content onto small soft or milk (not more than 2ml) and administer immediately. - Tablets should not be opened or crushed, swallow whole. Tablets are for older children. - Liquid has low volume but bitter taste. Caution: Multiple drug interactions. Note: Adult formulations are to be used for children older than 10 years with body weight > 35kg. Please see dosing Chart for dose recommendation. Storage: - For the syrup, store it in a refrigerator at least until dispensed. Can be stored at room temperature (250C) for 60 days. - For tablets and oral pellets, store at room temperature.	Side effects of LPV Common: diarrhea, head ache, nausea, vomiting,, increase in blood lipids Less common: pancreatitis, diabetes, hyperglycemia, hepatic toxicity, fat redistribution.

Drug / Formulation	Dosing recommendation and instructions	Side Effects
Ritonavir (RTV) Formulations: Combined with Lopinavir or Atazanavir:	Take with food to increase absorption and reduce GI side effects. Oral solution must be refrigerated. Can be kept at room temperature (250C) if used within 30 days. Bitter taste, coat mouth with peanut butter or chocolate milk.	Side effects of RTV: Common: N/V, diarrhea, headache, abdominal pain, anorexia Less Common: circumoral paraesthesia, increased LFTs, lipodystrophy, elevated cholesterol and triglycerides, hyperglycemia.
Danuravir (DRV) 75 mg tab; chewable tablets 25 mg; 100 mg/ml liquid	Substitute with LPV/r. When it is used in third-line ART, limited options are available. For hypersensitivity reactions, substitute with another therapeutic class.	Hepatotoxicity; severe skin and hypersensitivity reactions

Annex 10: Grading of toxicity in adults and adolescents

Item	Grade 1 (Mild toxicity)	Grade 2 (Moderate toxicity)	Grade 3 (Severe toxicity*)	Grade 4 (Severe life-threatening toxicity)
Peripheral neuropathy	■ Transient or mild discomfort, no limitation of activity. ■ No medical intervention/treatment required.	■ Moderate limitation of activity, some assistance might be needed. ■ Non-narcotic analgesia required.	■ Marked limitation in activity, some assistance usually required, medical intervention/therapy required, and hospitalization possible. ■ Severe discomfort and/or severe impairment (decrease or loss of sensation up to knees or wrists) narcotic analgesia required.	■ Life-threatening, extreme limitation in of activity, significant assistance required, significant medical intervention/therapy required, hospitalization/hospice care. ■ Incapacitating or not responsive to narcotic analgesia. ■ Sensory loss involves limbs and trunk.
Cutaneous/Rash/Dermatitis**	Erythema, pruritus	Diffuse, maculopapular rash or dry desquamation	Vesiculation or moist desquamation or ulceration*	Erythema multiforme or suspected Stevens-Johnson syndrome or Toxic Epidermal Necrolysis (TEN).
Management	Continue ARV; provide careful clinical monitoring; and consider change of a single drug if condition worsens.			Stop ARV and consult experienced physician.

Annex 11- Grading of adverse events in children

Parameter	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 Severe Life-threatening
Diarrhoea ≥1 year of age <1 year of age	Transient or intermittent episodes of unformed stools OR increase of ≤3 stools over baseline per day. Liquid stools (more unformed than usual) but usual in number.	Persistent episodes of unformed to watery stools or increase of 4-6 stools over baseline per day. Liquid stools with increased number of stools or mild dehydration.	Grossly bloody diarrhea or increase of ≥7 stools per day or IV fluid replacement indicated. Liquid stools with moderate dehydration.	Life-threatening consequences (e.g. Hypotensive shock). Liquid stools resulting in severe dehydration with aggressive rehydration indicated or hypotensive shock.
Nausea	Transient (<24 hours) or intermittent nausea with no or minimal interference with oral intake.	Persistent nausea resulting in decreased oral intake for 24-48 hours.	Persistent nausea resulting in minimal oral intake for >48 hours or aggressive rehydration indicated (e.g. IV fluids).	Persistent nausea with no or minimal oral intake resulting in dehydration with aggressive rehydration indicated.
Vomiting	Transient or intermittent vomiting with no or minimal interference with oral intake.	Frequent episodes of vomiting with no or mild dehydration.	Persistent vomiting resulting in orthostatic hypotension or aggressive rehydration indicated (e.g. IV fluids).	Life threatening consequences (e.g. hypotensive shock).
Acute systemic allergic reaction	Localized urticaria (wheals) lasting a few hours.	Localized urticaria with medical intervention indicated or mild angio oedema.	Generalized urticaria or angio oedema with medical intervention indicated or symptomatic mild bronchospasm.	Acute anaphylaxis or life-threatening bronchospasm or laryngeal oedema.

Pancreatitis	NA	Symptomatic and hospitalization not indicated (other than emergency treatment).	Symptomatic and hospitalization not indicated (other than emergency treatment).	Life-threatening consequences (e.g. Circulatory failure, haemorrhage, sepsis).
Rash	Localized macular rash	Diffuse macular, maculopapular, or morbilliform rash or target lesions.	Diffuse macular, maculopapular, or morbilliform rash with vesicles or limited number of bullae or superficial ulcerations of mucous membrane limited to one site.	Extensive or generalized bullous lesions or Stevens-Johnson syndrome or ulceration of mucous membrane involving two or more distinct mucosal sites or Toxic Epidermal Necrolysis.
Alteration in personality-behavior or mood	Alteration causing no or minimal interference with usual social and functional activities	Alteration causing greater than minimal interference with usual social and functional activities.	Alteration causing inability to perform usual social and functional activities and intervention indicated.	Behavior potentially harmful to self or others or with life-threatening consequences.
Altered Mental Status	Changes causing no or minimal interference with usual social and functional activities	Mild lethargy or somnolence causing greater than minimal interference with usual social and functional activities.	Onset of confusion, memory impairment, lethargy, or somnolence causing inability to perform usual social and functional activities.	Onset of delirium, obtundation, or coma.
Source: Antiretroviral therapy of HIV infection in infants and children in resource-limited settings. WHO 2006				

Annex 12. Laboratory grading of adverse events in adults and adolescents (ACTG)

Laboratory Test Abnormalities				
Item	Grade 1 toxicity	Grade 2 toxicity	Grade 3 toxicity	Grade 4 toxicity
Haemoglobin	8.0-9.4 g/dL	7.0-7.9 g/dL	6.5-6.9 g/dL	<6.5 g/dL
Absolute Neutrophil Count	1,000-1,500 mm3	750-990 mm3	500-749 mm3	<500 mm3
Platelets	-75,0000- 99,000	50,000-74,999	20,0000-49,999 mm3	<20,000
ALT	1.25-2.5 X upper normal limit	2.5-5 X upper normal limit	5.0-10 X upper normal limit	10 X upper normal limit
Bilirubin	1-1.5XULN	1.5-2.5 X ULN	2.5-5 x upper limits of normal	>5 x upper limits of normal
Amylase/lipase	1-1.5XULN	1.5-2 X ULN	2-5 x upper limits of normal	>5x upper limits of normal
Triglycerides *	200-399mg/dL	400-750 mg/dL	751-1200mg/dL	>1200mg/dL
Cholesterol *	1.0 –1.3 X Upper normal limit	1.3-1.6 X Upper normal limit	1.6-2.0 X Upper normal limit	2.0 X Upper normal limit
Management	Continue ARV Repeat test 2 weeks after initial test and reassess		Substitute responsible drug	Stop ARV and consult experience physician
	Lipid imbalances could be managed with diet, exercise and pharmacologically with the use of fibrates. ALWAYS SEEK EXPERT ADVICE IN CASE OF DOUBT			

Grade 1 (Mild reaction): are bothersome but do not require changes in therapy

Grade 2 (Moderate reaction): consider continuation of ART as long as feasible. If the patient does not improve in symptomatic therapy, consider single-drug substitution.

Grade 3 (Severe reaction): Substitute offending drug without stopping ART. Closely monitor using laboratory and clinical parameters.

Grade 4 (Severe life-threatening reaction): Immediately discontinue all ARV drugs, manage the medical event with symptomatic and supportive therapy, and reintroduce ARV drugs using a modified regimen (i.e. with an ARV substitution for the offending drug) when the patient is stabilized. Life-threatening toxicity includes severe hepatitis, pancreatitis, lactic acidosis or Steven-Johnson syndrome.

Annex 13 -Grading toxicities in children by selected laboratory findings

Parameter	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 Severe Life threatening
Haemoglobin (g/dL)	8.5 – 10.0	7.5 – <8.5	6.5 – <7.5	<6.5
ANC (mm ³)	750 – <1,000	500 – 749	250 – 500	<250
Platelets (mm ³)	100,000 – <125,000	50,000 – <100,000	25,000 – <50,000	<25,000 or bleeding
ALT (SGPT)	1.25 – 2.5 x ULN	2.6 – 5.0 x ULN	5.1 – 10.0 x ULN	>10.0 x ULN
AST (SGOT)	1.25 – 2.5 x ULN	2.6 – 5.0 x ULN	5.1 – 10.0 x ULN	>10.0 x ULN
Bilirubin (>2 weeks of age)	1.1 – 1.5 x ULN	1.6 – 2.5 x ULN	2.6 – 5.0 x ULN	>5.0 x ULN
Lipase	1.1 – 1.5 x ULN	1.6 – 3.0 x ULN	3.1 – 5.0 x ULN	>5.0 x ULN
Pancreatic amylase	1.1 – 1.5 x ULN	1.6 – 2.0 x ULN	2.1 – 5.0 x ULN	>5.0 x ULN
Cholesterol (fasting, <18 years old) mg/dL	170 – <200	200 – 300	>300	NA
Glucose, serum, Nonfasting (mg/dL)	116 – <161	161 – <251	251 – 500	>500
Glucose, serum, high: Fasting (mg/dL)	110 – <126	126 – <251	251 – 500	>500
Lactate	<2.0 x ULN without acidosis	2.0 x ULN without acidosis	Increased lactate with pH <7.3 without life threatening consequences or related condition present	Increased lactate with pH <7.3 with life threatening consequences (e.g. neurological findings, coma) or related condition present
Triglycerides: Fasting (mg/dL)	NA	500 – <751	751 – 1,200	>1,200

Source: Antiretroviral therapy of HIV infection in infants and children in resource-limited settings. WHO 2006

Annex 14: Pediatric TB screening tool

Children TB screening questions		Follow Up Visit			
		Date:	Date:	Date:	Date:
1. Current cough					
2. Fever					
3. Poor weight gain*					
4. Close Contact history with TB pt.					
*poor wt gain = reported wt loss, very low wt (<-3 Z-score), or underwt (< -2 Z-score), or confirmed wt loss (> 5%) since the last visit, or growth curve flattening					
Evaluation for positive TB screening					
Bacteriology: Gastric Aspirate/ Induced sputum/ Sputum for AFB	date ordered				
	date result received				
	result (+, -ve, Not Done)				
	date ordered				
Radiology: CxR, etc.	date received				
	result (Suggestive, inconclusive, other dx, Not Done)				
	date ordered				
Other: FNA, etc	date received				
	result (+, -ve, Not Done)				
TB diagnosis date: / / Circle type of TB: PTB :- smear pos, smear neg, EPTB TB Rx start date / /					
Is the child eligible for IPT? Yes___ No___ If no, reason If yes, start IPT and use the chart below					
Contraindications for IPT: Active TB, active hepatitis, allergy to INH, peripheral neuropathy					

Date INH collected	TB Symptoms [cough, fever, failure to gain wt or wt loss] (yes, no)	Hepatitis Sx [abd pain, nausea, vomiting, abnormal LFT] (yes, no)	Neurologic Sx [numbness, tingling, paresthesia](yes, no)	Rash (yes, no)	Adherence (≥95% =good; 85-94% = Fair <85% =Poor)	Remark
___/___/___						
___/___/___						
___/___/___						
___/___/___						
___/___/___						
___/___/___						

Outcome of IPT (Write Date): Completed ___/___/___ Defaulted ___/___/___ Died ___/___/___

Interrupted for any reason ___/___/___

Note:

- If there are symptoms suggesting TB during follow up, stop INH and work up for TB.
- If there are symptoms suggesting hepatitis, hold INH. Can resume when liver function normalizes.
- If there are neurologic symptoms, continue INH and give pyridoxine 50mg daily. This side effect is rare if the child is already on pyridoxine skin rash is very rare, if occurs and is extensive, discontinue INH, and give anti histamine.

Annex 15: Adult TB screening tool

Children TB screening questions		Follow Up Visit			
		Date:	Date:	Date:	Date:
1. Current cough					
2. Fever					
3. Weight lose					
4. Night seats					
Evaluate for TB if "yes" to anyone of the above (positive TB screening)					
Bacteriology: Sputum for AFB (+/- induced)	Done = yes/no				
	result (+, -ve, unknown)				
Radiology: CxR, etc.	Done = yes/no				
	Result (Suggestive, inconclusive, other dx, Not Done)				
FNA, culture, ultrasound etc	Done = yes/no				
	If done result				
TB diagnosed	Yes (write type of TB)/No				
Is patient eligible for IPT	Yes/No				
Contraindications for IPT: Active TB, active hepatitis, allergy to INH, peripheral neuropathy					
IPT start date _____					

Date INH collected	TB Symptoms [cough, fever, failure to gain wt or wt loss] (yes, no)	Hepatitis Sx [abd pain, nausea, vomiting, abnormal LFT] (yes, no)	Neurologic Sx [numbness, tingling, paresthesia](yes, no)	Rash (yes, no)	Adherence (≥95% =good; 85-94% = Fair <85% =Poor)	Remark
___/___/___						
___/___/___						
___/___/___						
___/___/___						
___/___/___						
___/___/___						

Outcome of IPT (Write Date): Completed ___/___/___

Defaulted ___/___/___ Died ___/___/___ Patient stopped ___/___/___

Stopped ___/___/___ Transferred out ___/___/___

Annex 16: Follow-up form for Clients Whose Viral Load Result >50 copies/ml

1. Patient Information			
Name: _____ Age: _____ Sex: _____ UAN: _____ MRN: _____			
2. ARV Information		Viral Load Results	
ARV Regimen (circle the regimen type)		1st result >50 copies/ml: _____ Copies/ml Date: ____/____/____	
1. 1st line	____/____/____	Date of 1st EAC1 given: ____/____/____	
2. 2nd line	____/____/____	Cohort Month: _____ (Refer High VL Register)	
3. 3rd line	____/____/____		
3. Communicate the viral load test result and explain the following to the client			
■ Viral Load is the number of HIV copies in the blood ■ High VL result can be due to poor adherence to medication or can be due to primary resistance. ■ When VL result is high in the blood, the CD4 count decreases, OIs will flare up and disease progresses. ■ High VL can be reduced as a result of good adherence to medication within three months. ■ Note: Enhanced Adherence Counseling(EAC) will be provided for 1st results >50 copies/ml			
4. Assess current adherence to treatment and document			
4.1. How many ARV dose/s do you take/day? Once or Twice			
4.2. Did you miss ARV doses in the past one month? Yes/No			
Adherence Rate		>95%, Good	85-94%, Fair
4.3. If yes, how many dose/s did you miss? ____ (select one)	Once per day	≤ 2 doses	3-4 doses
	Twice per day	≤ 3 doses	4-9 doses
		<85%, Poor	≥ 5 doses
			≥ 9 doses

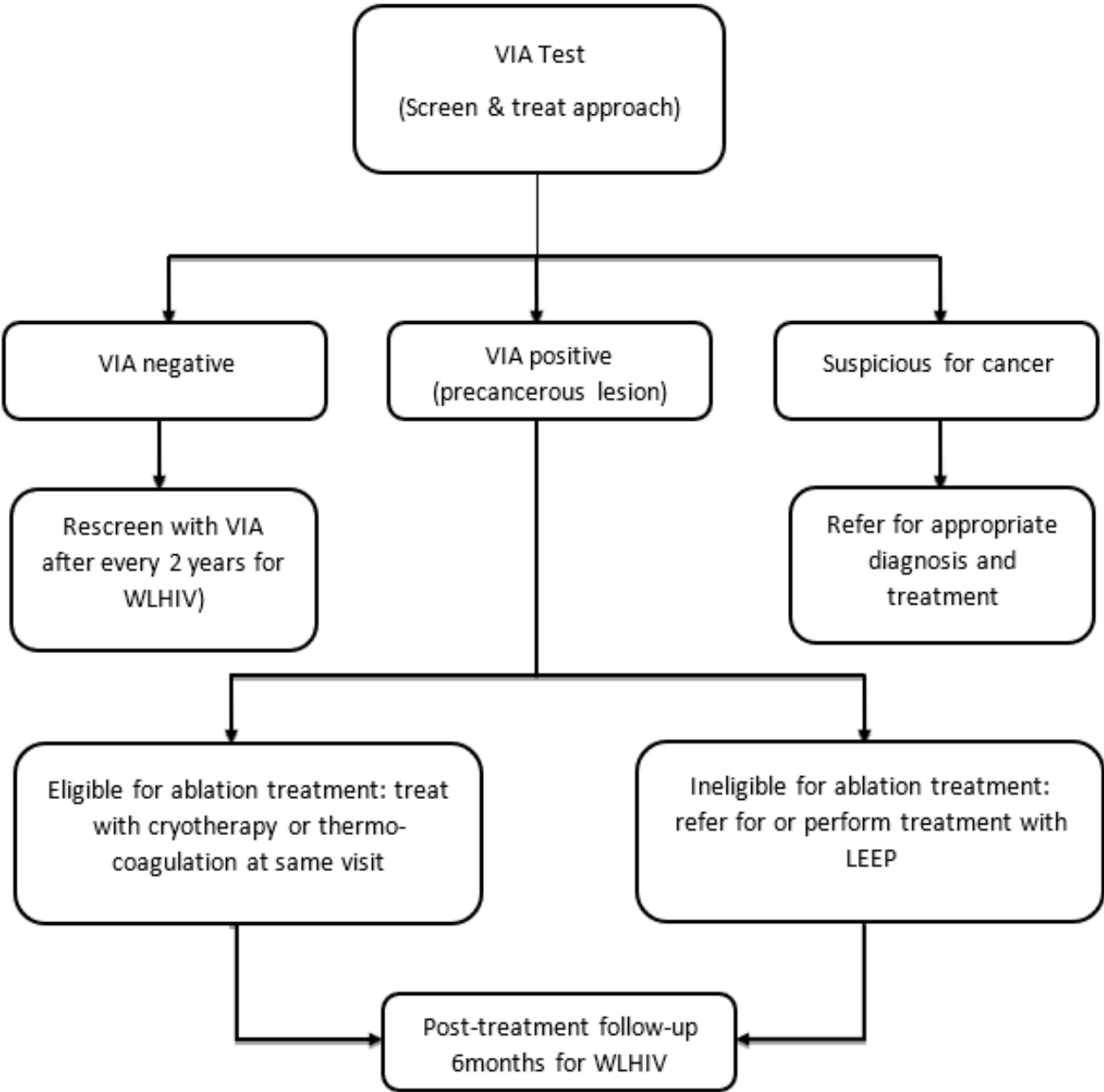
5. Explore Medical and Psychosocial Reasons for High VL		
5.1. Identify Medical Reasons for High Viral Load		EAC-1 Session
5.1.1. Did you take other drugs than ARVs without consulting your physician? (Yes, No), If Yes, identify the drug/s, review interaction with ARVs, counsel the client & take measure		
5.1.2 Have you ever developed recurrent OIs including cough, fever, weight loss, night sweat, diarrhea, and vomiting in the past? (Yes, No). If yes, investigate for TB and chronic diarrhea and manage		
5.1.3. Have you ever developed severe ARV drugs side effects in the past? (Yes, No), If Yes, investigate and manage for ARVs side effects.		
5.1.4 Have you ever taken ART/PMTCT prophylaxis in the past prior to ART initiation? (Yes, No), If Yes, suspect primary resistance and consult ART physician		
5.1.5. Did you discontinue your ARV in the past? (Yes, No), If Yes, identify reasons and develop treatment plan with the client.		
5.1.6 If the client is a child, check proper dosing for weight by reviewing chart and readjust ARV dosing for weight		
5.2 Identify Psychosocial Reasons for High Viral Load		
5.2.1 Cognitive barriers: understanding and expectation - counsel and explain expected outcomes What were the reason/s for missing your ARV dose/s in the past? Identify the reason/s, counsel and motivate the client to develop medication taking plan.		
5.2.2. Socio-economic barriers: lack of social support, disclosure, stigma, & poor living condition - Have you disclosed your HIV status to anyone? If No, provide disclosure counseling and encourage - Do you have treatment supporter? (Yes, No) If No, counsel to designate treatment supporter		
5.2.3 Behavioral barriers: attitude, motivation, confidence & skills educate, motivate, and empower client to manage medication taking and develop reminder - What do you do to remind yourself to take drugs on time? - Are you confident to take your ARV openly at home? If no, advise reminder mechanisms		
5.2.4 Psychological/Emotional barriers: common mental illness: depression, PTSD, substance abuse, and psychosis – link to psychiatric clinic for psychotherapy & treatment Have you ever felt sad for more than 2 weeks? Have you ever lost interest in activities that usually give you pleasure for more than 2 weeks? Have you regularly taken alcohol or chew Khat? If yes, counsel the client to avoid alcohol and/or Khat.		

6. Follow-up EAC Sessions																	
			Subsequent EAC Sessions						Repeat three EAC Sessions for clients whose confirmatory viral load > 50 and ≤ to 1000 copies/mL (Low Level Viremia)								
			Rate at EAC-2			Rate at EAC-3			Rate at EAC-1			Rate at EAC-2			Rate at EAC-3		
6.1. Assess current adherence to treatment and circle adherence rate			>95% Good	85 to 94% Fair	<85% Poor	>95% Good	85 to 94% Fair	<85% Poor	>95% Good	85 to 94% Fair	<85% Poor	>95% Good	85 to 94% Fair	<85% Poor			
6.2. If Yes, how many dose/s did you miss?			≤ 2	3-4	≤ 2	3-4	≥ 5	≤ 2	3-4	≥ 5	≤ 2	3-4	≤ 2				
			≤ 3	4-9	≤ 3	4-9	≥ 9	≤ 3	4-9	≥ 9	≤ 3	4-9	≤ 3				
7. Follow-up status of identified Reasons for High VL			EAC 2			EAC 3			EAC-1			EAC-2			EAC-3		
			Date: __/__/__			Date: __/__/__			Date: __/__/__			Date: __/__/__			Date: __/__/__		
7.1. Medical reason/s(y/N)																	
7.2 Psychosocial																	
7.2.1. Cognitive barriers(Y, N):																	
7.2.2. Socio-economic barriers(Y,N):																	
7.2.2. Behavioral Barriers(Y,N)																	
7.2.3. Psychological/emotional(Y,N):																	
8. Outcome of EAC Sessions																	
Did the patient attend all the appointments? (Yes No) If no, any reason? _____ Note: Client should have three consecutive good adherences before confirmatory(2nd) VL test 2nd Viral Load - Date ordered: __/__/__, 2nd Viral Load Result: _____ copies/ mL, Date Received __/__/__, Measure Taken 1. Continue 1st Line, 2. Continue 2nd line 3. Continue Third line 4. Switched to 2nd Line, 5. Switched to 3rdLine Comment: _____ ART Physician:/ART clinician _____ Date: __/__/__																	

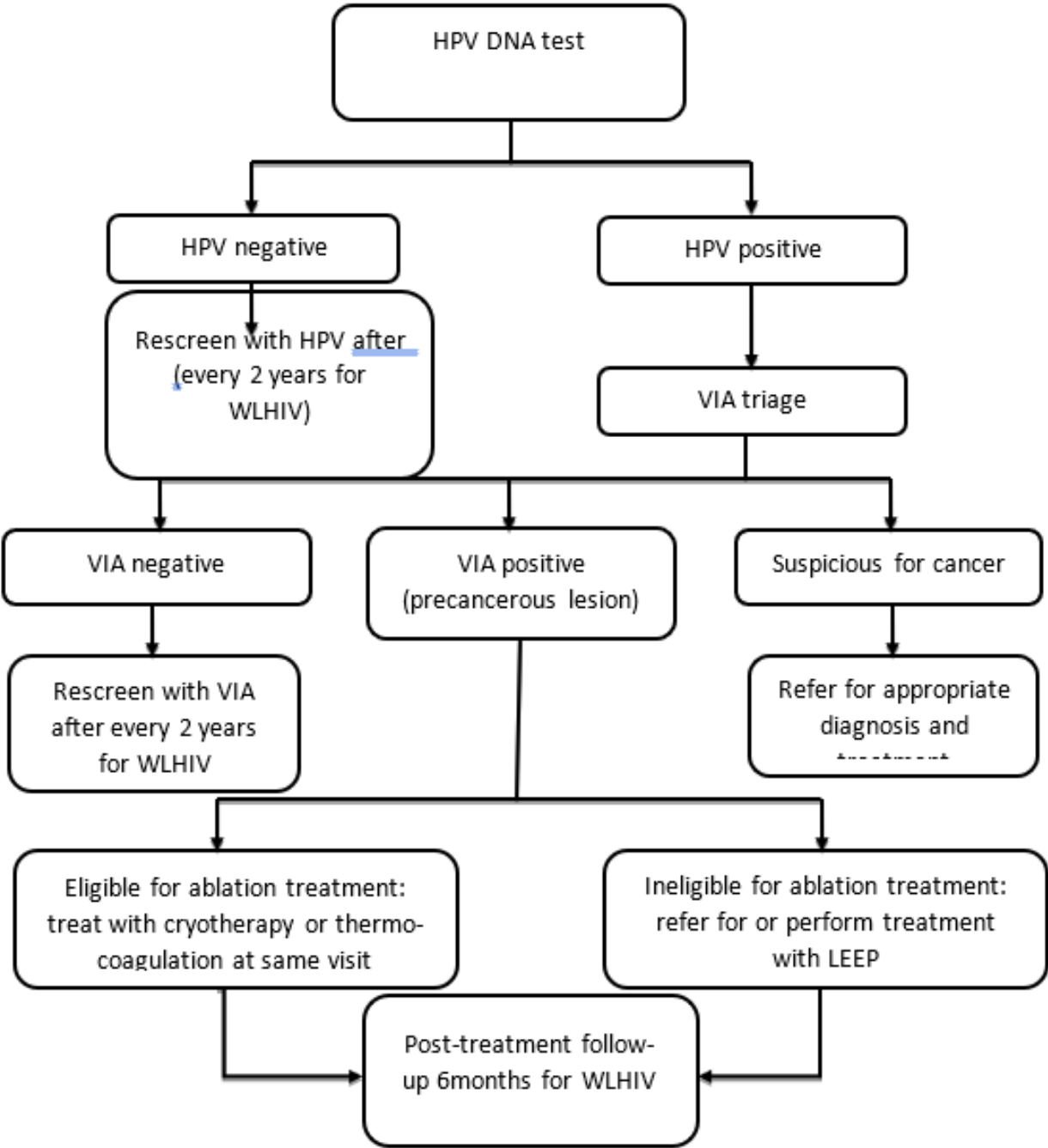
Annex 17 Cohort register for clients with unsuppressed viral load(>50copies/ml)

Cohort Register for client with Unsuppressed Viral Load(>50 copies/ml)																						
Facility Name: _____ Month: _____ Year: _____																						
SN	Name of Client	UAN	MRN	Age	Sex	Address (Kebele & Telephone)	PMTCT Clinic Only		Current Regimen (select one) 1 st Line 2 nd Line 3 rd Line	1 st VL Test		Code for 1 st VL result 1. >1000 c/ml 2. >50 to ≤1000 c/ml	EAC Session Date Three Consecutive EAC's	2 nd VL Test		Repeat Three Session For clients whose confirmatory viral Load >50 and ≤1000 copies /ml (Low Level Viremia)	Measure Taken (code) 1= Remain on 1 st line 2= Remain on 2 nd line 3= Remain on 3 rd line 4= Remain on 2 nd line 5= Remain on 3 rd line	Reason if measure not taken 1. TO 2. Lost 3. Died 4. Other, Specify	Measure Taken Date	Remark		
							Pregnant	Postnatal		Order Date	Recieved Date			Order Date	Recieved Date						Order Date	Recieved Date
							Gestational Age (Months)	Breast Feeding (Yes/No)		VL Result	VL Result			VL Result	VL Result						VL Result	VL Result
										___/___/___			EAC 1	___/___/___	EAC 1	___/___/___			___/___/___			
										___/___/___			EAC 2	___/___/___	EAC 2	___/___/___			___/___/___			
										VL=			EAC 3	___/___/___	VL=	___/___/___			___/___/___			
										___/___/___			EAC 1	___/___/___	___/___/___	___/___/___			___/___/___			
										___/___/___			EAC 2	___/___/___	___/___/___	___/___/___			___/___/___			
										VL=			EAC 3	___/___/___	VL=	___/___/___			___/___/___			

Annex 18: Screening with VIA testing using see and treat approach



Annex 19: Screening with HPV testing followed by VIA triage algorithm



Annex 20: Brief mental health disorders symptom screening tool and referral tool for PLHIV

Patient's Name:_____ Sex:_____ Age:_____ MRN:_____

This checklist is to assist you in assessing and making a timely referral of the client to the treatment team. All behaviors listed below are important and should be taken seriously; they are also designed to help you decide if you should refer the client to the treatment team for further assistance. An answer of "yes" to any one of the following questions should prompt further referral and evaluation by the treatment team or mental health professional. Please put a (√) to indicate a yes answer.

1. Questions to Identify Depression: In the past 3 months;

() Was there ever a time when you felt sad/hopelessness for more than 2 weeks in a row?

() Was there ever a time lasting more than 2 weeks when you lost interest in most things like hobbies, work, or activities that usually give you pleasure?

2. Questions to identify suicidal ideation: Since your last visit [or in the last 2 months];

() Have you wished you were dead, or wished you could go to sleep and not wake up?

() Have you had actual thoughts of killing yourself?

() Have you ever attempted to harm/kill yourself?

3. Questions to Identify

Anxiety: In the past 3 months;

() Did you ever have a period lasting more than 1 month when most of the time you felt worried and anxious?

() Did you have a spell or an attack when all of a sudden you felt frightened, anxious, or very uneasy when most people would not be afraid or anxious?

() Did you ever have a spell or an attack when for no reason your heart suddenly started to race, you felt faint, or you couldn't catch your breath?

4. Questions to Identify Mania: In the past 3 months;

() When not high or intoxicated, did you ever feel extremely energetic or elated or irritable and more talkative than usual?

5. Questions to Identify Substance Abuse.

() Have you ever felt the need to cut down on your use of alcohol or drugs?

() Has anyone annoyed you by criticizing your use of alcohol or drugs?

() Have you ever felt guilty because of something you've done while drinking or using drugs?

() Have you ever taken a drink or used drugs to steady your nerves or get over a hangover (eye-opener)? A total of ≥2 may be suggestive of a problem.

6. Questions to Identify Psychosis: Observe or ask families whether the patient (in the last 3 months);

() Talking & acting strangely or becoming very quiet and avoid talking.

() Claiming to hear voices or see things that other people don't.

() Being very suspicious, perhaps claiming that other people are trying to harm him/her.

7. Questions to Identify Dementia: Interview the patient or families whether the patient (in the last 3 months);

() Has trouble with memory.

() Has poor concentration.

() Has diminished executive function.

() Has diminished orientation to time, place & person.

8. Questions to Identify Epilepsy:

() Did you ever have partial or generalized fits [sharp, shaky movements] accompanied by frothing or loss of control of bowel or bladder function, sudden loss of consciousness, and stiff limbs?

Referred by: _____

Date: _____

9. () others

10. No identified mental health problem

Feedback(confirm the assessment)

The patient has:

() Mental Health Disorder (specify) _____

() Non Mental Health Disorder

Name of clinician: _____

Date: _____

The patient has:

() Mental Health Disorder (specify) _____

() Non Mental Health Disorder

Name of clinician: _____

Date: _____

Annex 21: DSD model of HIV care client classification tool/ Assessment form

A. Client's Current data(initial Assessment)

1. Client Name _____ Fathers' Name _____

Grand Fathers 'Name _____

For child only: - Mother's /Fathers Name: _____

2. Age: _____ years/months for < 1years Sex: ☐ M ☐ F

3. Address: Region _____ Zone/Sub city _____ Woreda _____

Kebele _____ House No. _____ Telephone _____

4. Marital Status: ☐ Never Married ☐ Married ☐ Divorced ☐ Widowed

5. Level of education:

☐ No education ☐ Primary ☐ Secondary ☐ Tertiary ☐ Other/specify _____

6. Occupation: _____

7. Client reside within the catchments area ☐ Yes ☐ No, If no; Counsel and encourage for referral to a nearby facility to his/her home

8. Date ART started(EC) _____ Month on ART _____

9. Unique ART ID: ____ / ____ / ____ / ____ MRN _____

10. Current ART regimen

☐ First Line Regimen: _____ ☐ Date (ET) Initiated ____/____/____ (dd/mm/yyyy)
☐ Second Line Regimen: _____ ☐ Date (ET) Initiated ____/____/____ (dd/mm/yyyy)
☐ Third Line Regimen _____: _____ ☐ Date (ET) Initiated ____/____/____ (dd/mm/yyyy)

11. Current ART Adherence ☐ Good >95% ☐ Fair (85 - 94%) ☐ Poor <85%

12. Is the client pregnant ☐ Yes ☐ No Breastfeeding ☐ Yes ☐ No

13. CD4 count (CD4% for <5 years): Most recent (<one Year) result ____cells/ul

Date (ET) ____/____/____

14. If VL test done: Most recent (<one year) result _____ copies/ml

15. Current Clinical observations/symptoms: WHO (Treatment) Staging:

☐ I ☐ II ☐ III ☐ IV

B. Inclusion criteria for less intensive model

- Patients who are on ART at least six months
- No current illness, which does not include well-controlled chronic health conditions
- Good understanding of life long adherence; adequate adherence counseling provided (a patient with adherence of 95% for the last 6 months)
- Evidence of treatment success (at least one suppressed viral load result (i.e < 50 c/ml) and if no Viral Load result, a patient with rising CD4 cell count or CD4 cell count above 200 cells/millimeter cubes).
- Children with age greater than five years*
- A Patient who doesn't have current Opportunistic Infections
- A Patient with no adverse drug reactions and doesn't need careful clinical monitoring.
- A Patient who is willing or provide consent to get the ART service based on his/her preferred DSD models.

C. Inclusion criteria for more intense

- Patients who are on ART for < 6 months
- A patient with Age <5 years*
- Evidence of treatment failure (at least one low level viral load result (i.e > 50 c/ml) and if no Viral Load result, a patient with falling CD4 cell count or CD4 cell count below 200 cells/millimeter cubes)
- Active current illness, which includes not well controlled chronic health conditions
- A Patient who does have current Opportunistic Infections
- A Patient with adverse Drug Reactions and does need careful clinical monitoring.
- A Patient with an adherence of less than 95% for the last six months

16. Assessed client:

1. Eligible for less intensive 2. Needs more intensive

17. Consented:

1. Yes 2. No if Consent is yes go to no. 18.

18. Enrollment TO DSD Models:

1. 3MMD 2. ASM/6MMD 3. FTAR 4. CAG 5. PCAD
6. DSD for Adolescent 7. DSD for KP DSD 8. DSD for MCH 9. DSD for AHD
10. Other Models Specify _____



ጤና ሚኒስቴር - ኢትዮጵያ
MINISTRY OF HEALTH-ETHIOPIA

የዜጎች ጤና ለሃገር ብልጽግና!
HEALTH FOR PEOPLE, FOR PROSPEROUS NATION